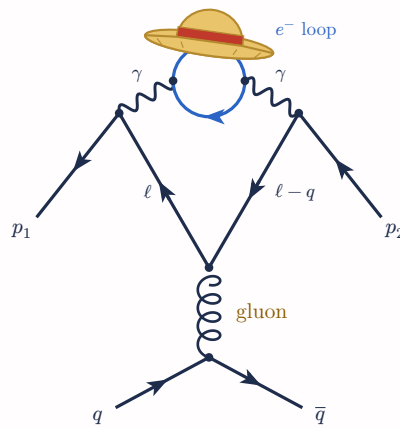


Feynman integrals at one loop

*Tadpole, Bubble, Integration by Parts, Differential Equations,
Feynman Parameters, Graph Polynomials, Mellin-Barnes,
Canonical Differential Equations, Sector Decomposition,
Expansion by Regions and Kira*



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THE TALK

An Introduction to Feynman Integrals. *An introduction to the evaluation of simple Feynman integrals at one-loop level, primarily, will be given with some explicit derivations in dimensional regularization. The use of some softwares will be mentioned which can be done during self-study based on Mathematica. Some explicit illustrations will be given to introduce the audience to integration by parts relations, differential equations and to graph polynomials and Mellin-Barnes techniques. A guide to the literature will be provided, which can be used by the ardent student for their edification. (Abstract of the talk.)*

These are my own notes of the two-lecture series (Part 1: the one-loop tadpole and bubble. Part 2: graph polynomials up to two loops). Everything starts from the basics, every step is written out and every formula was checked with an independent symbolic computation (Mathematica, FeynCalc 10.2, LiteRed from the FIRE distribution and Kira 3.1). Any mistake left in here is mine, not sir's.

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1 Notation used in these notes

The lecture used formulas from several books (Ramond, Peskin, Argeri-Mastrolia, Rauh) and each book has its own letters. Here one set is fixed and used everywhere, also inside the Mathematica code.

Symbol	Meaning
ℓ	loop momentum, the one we integrate over
p	external momentum that enters the diagram
$D = 2\omega = 4 - 2\varepsilon$	number of space-time dimensions, so $\varepsilon = 2 - \omega$
$\ell_E^2 = \ell_1^2 + \dots + \ell_D^2$	Euclidean square, always ≥ 0
$\ell^2 = \ell_0^2 - \ell^2$	Minkowski square, metric $(+, -, -, -)$
p_E^2 and s	Euclidean and Minkowski square of the external momentum, $s = -p_E^2$
$[d\ell]$	short for the Euclidean measure $d^D\ell/\pi^{D/2}$ (Rauh writes $[dl]$ for the Minkowski $d^d\ell/(i\pi^{d/2})$)
$x_{ij\dots k}$	short for $x_i + x_j + \dots + x_k$
I_n	$\int d^D\ell \frac{1}{(\ell_E^2 + m^2)^n}$, the tadpole with n propagator powers
$T(n)$	$\int [d\ell] \frac{1}{(\ell_E^2 + m^2)^n} = \pi^{-D/2} I_n$
$J(a_1, a_2)$	$\int [d\ell] \frac{1}{D_1^{a_1} D_2^{a_2}}$ with $D_1 = \ell_E^2 + m^2$, $D_2 = (\ell - p)_E^2 + m^2$
x	$p_E^2/(4m^2)$, the dimensionless variable of the bubble
β	$\sqrt{1 - 4m^2/s}$, speed of each particle in the centre of mass frame
μ	the arbitrary mass scale of dimensional regularisation
$+i\epsilon$	Feynman's prescription in the propagators, $\epsilon \rightarrow 0^+$ at the end (a different letter from the ε of $D = 4 - 2\varepsilon$; many books write $+i0$)
	a pink dot on a line: that propagator is raised to one more power

Table 1: Symbols used in these notes.

What $[d\ell]$ means. It is only a shorthand. $d^D\ell$ is the ordinary volume element of D -dimensional momentum space, $d\ell_1 d\ell_2 \dots d\ell_D$, just as $dx dy dz$ is the volume element of ordinary space. The square brackets tell us that we also divide by $\pi^{D/2}$:

$$[d\ell] = \frac{d^D\ell}{\pi^{D/2}} \quad (1)$$

Why this number? The simplest integral in D dimensions is the Gaussian and it splits into D copies of the one-dimensional one:

$$\int d^D\ell e^{-\ell^2} = \left(\int_{-\infty}^{\infty} d\ell_1 e^{-\ell_1^2} \right)^D = (\sqrt{\pi})^D = \pi^{D/2} \quad (2)$$

So with $[d\ell]$ the Gaussian gives exactly 1 and the $\pi^{D/2}$ that every loop integral produces is gone. For example the tadpole becomes a pure ratio of Gamma functions (Section 3),

$$\int [d\ell] \frac{1}{(\ell^2 + m^2)^n} = \frac{\Gamma(n - D/2)}{\Gamma(n)} (m^2)^{D/2 - n} \quad (3)$$

For pure mathematics this measure is the cleanest. For physics one needs $d^D\ell/(2\pi)^D$ and the two differ only by a constant:

$$\int \frac{d^D \ell}{(2\pi)^D} f = \frac{1}{(4\pi)^{D/2}} \int [d\ell] f \quad (4)$$

Constants like this never change an IBP relation or a differential equation, because every term in such a relation carries the same constant.

2 The toolkit, from zero

This chapter collects every piece of mathematics and physics that the lecture used without saying so. If you know a piece already, skip it. Nothing later assumes more than what is here.

2.1 What a Feynman diagram is

Take the simplest interacting quantum field theory, one real scalar field $\varphi(x)$ with Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)(\partial^\mu \varphi) - \frac{1}{2}m^2 \varphi^2 - \frac{\lambda}{4!} \varphi^4 \quad (5)$$

The first two terms describe a free particle of mass m . The last term lets four such particles meet at one point, with strength λ . A Feynman diagram is a picture of one way in which particles can travel and meet. Each piece of the picture stands for a factor in a formula:

line		propagator $\frac{i}{\ell^2 - m^2 + i\epsilon}$: amplitude for the particle to travel with four-momentum ℓ
vertex		vertex $-i\lambda$: four lines meet and momentum is conserved there
loop		a closed path: its momentum is not fixed, so we integrate $\int \frac{d^4 \ell}{(2\pi)^4}$

2.1.1 Momenta are four-vectors

Every line carries a four-momentum $\ell^\mu = (\ell^0, \ell^1, \ell^2, \ell^3) = (E, \ell)$. The index μ runs over time and the three space directions. Two four-vectors combine into a number with the Minkowski dot product,

$$\ell \cdot p = \ell^0 p^0 - \ell \cdot \mathbf{p}, \quad \ell^2 = \ell \cdot \ell = E^2 - \ell^2 \quad (6)$$

A real (on-shell) particle has $\ell^2 = m^2$. Inside a loop, ℓ runs over all values and ℓ^2 need not be m^2 (the particle is “virtual”).

Why do we carry the vector index μ around? Because the laws of physics do not depend on the direction in which we look or on how fast we move (Lorentz invariance). So the final answer for a diagram can only depend on the momenta through **scalars** such as p^2 or $p \cdot q$. When we need a derivative with respect to a momentum, we must differentiate with respect to each component p^μ and then contract with some vector to get a scalar back. For every independent momentum in the problem ($\ell, p, \dots$) there is one such vector available. This is what happens in integration by parts (Section 5.3) and in the differential equations (Section 10), where the board had $k^\mu \partial / \partial k^\mu$ and $\partial / \partial \ell^\mu \ell^\mu$.

2.1.2 Why a loop means an integral

At every vertex, momentum is conserved. What flows in flows out. In a tree diagram this fixes every internal momentum. In a diagram with a closed loop, one momentum is left over. We can send any ℓ around the loop and still conserve momentum at every vertex. Quantum mechanics says that we add the amplitudes of all possibilities. So we integrate over ℓ . This integral is the **Feynman integral** and these notes are about how to compute it.

Like the Survey Corps in Attack on Titan, the loop momentum goes beyond every wall: it runs over all values up to infinity. That is where the trouble (the divergence) comes from.

2.2 The Gaussian integral

DERIVATION: $\int e^{-x^2} dx = \sqrt{\pi}$

Call $G_1 = \int_{-\infty}^{\infty} e^{-x^2} dx$. Square it and write the second copy with the letter y :

$$G_1^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \iint e^{-(x^2+y^2)} dx dy \quad (7)$$

This is an integral over the whole plane. Go to polar coordinates, $x^2 + y^2 = r^2$, $dx dy = r dr d\theta$:

$$G_1^2 = \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2} dr = 2\pi \cdot \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi \quad (8)$$

So $G_1 = \sqrt{\pi}$. In D dimensions the integrand $e^{-(x_1^2+\dots+x_D^2)} = e^{-x_1^2} \dots e^{-x_D^2}$ is a product, so

$$\int d^D x e^{-x^2} = (\sqrt{\pi})^D = \pi^{D/2} \quad (9)$$

2.3 The Gamma function

For $\text{Re } z > 0$ define

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt \quad (10)$$

DERIVATION: THE THREE PROPERTIES WE USE

1. Recursion. Integrate by parts with $u = t^z$ and $dv = e^{-t} dt$:

$$\Gamma(z+1) = \int_0^{\infty} t^z e^{-t} dt = [-t^z e^{-t}]_0^{\infty} + z \int_0^{\infty} t^{z-1} e^{-t} dt = 0 + z\Gamma(z) \quad (11)$$

2. Factorials. $\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1$. Then $\Gamma(2) = 1 \cdot \Gamma(1) = 1$, $\Gamma(3) = 2 \cdot 1$ and in general $\Gamma(n) = (n-1)!$.

3. Half integers. Put $t = x^2$, $dt = 2x dx$: $\Gamma(1/2) = \int_0^{\infty} t^{-1/2} e^{-t} dt = 2 \int_0^{\infty} e^{-x^2} dx = \sqrt{\pi}$, using the Gaussian integral.

Why only $\text{Re } z > 0$? Near $t = 0$ we have $e^{-t} \approx 1$, so the integrand is about t^{z-1} . The piece $\int_0^1 t^{z-1} dt = 1/z$ is finite only if $\text{Re } z > 0$. For $z = -1/2$, say, the integrand is $t^{-3/2}$ near 0 and the area is infinite. So the integral simply does not give a value there.

How we still get a value. The recursion $\Gamma(z+1) = z\Gamma(z)$ was proved for $\text{Re } z > 0$. Read it backwards,

$$\Gamma(z) = \frac{\Gamma(z+1)}{z} \quad (12)$$

and use the right side as the **definition** of the left side wherever the integral fails. One example shows how it works. We want $\Gamma(-1/2)$. The integral cannot give it, but the right side needs only $\Gamma(1/2) = \sqrt{\pi}$, which the integral does give:

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma(1/2)}{-1/2} = -2\sqrt{\pi} \quad (13)$$

One more step gives $\Gamma(-3/2) = \Gamma(-1/2)/(-3/2) = 4\sqrt{\pi}/3$. Each use of the recursion moves us one unit to the left. Using it $k+1$ times in one go,

$$\Gamma(z) = \frac{\Gamma(z+k+1)}{z(z+1)(z+2)\dots(z+k)} \quad (14)$$

The numerator now needs $\text{Re}(z+k+1) > 0$, so this formula works for $\text{Re } z > -k-1$. Since k can be as large as we like, $\Gamma(z)$ is now defined for every complex z , except where a factor in the denominator vanishes: $z = 0, -1, -2, \dots$ These are the poles. This way of extending a function beyond the region where its first formula works is called **analytic continuation**. The extension is unique, because two analytic functions that agree on a region agree everywhere they are both defined (Figure 1).

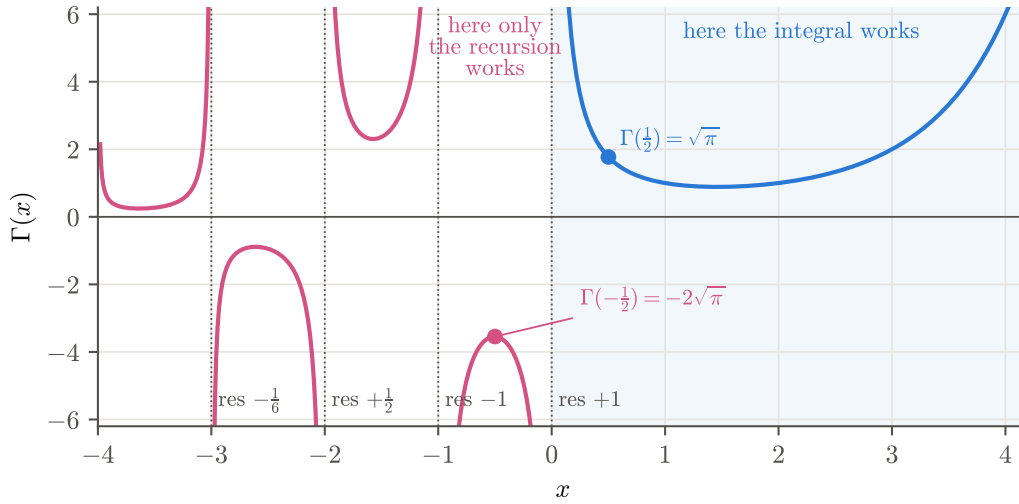


Figure 1: $\Gamma(x)$ on the real axis. In the shaded region ($x > 0$) the integral defines it. To the left only the recursion does, for example $\Gamma(-1/2) = \Gamma(1/2)/(-1/2) = -2\sqrt{\pi}$. At $x = 0, -1, -2, -3$ the function blows up, with residues $+1, -1, +1/2, -1/6$: the signs alternate, so the curve jumps from $-\infty$ to $+\infty$ (or back) at every pole.

How strong each pole is. Look at Equation 14 near $z = -k$ and write $z = -k + u$ with u small. Take the factors one by one.

- The numerator is $\Gamma(z + k + 1) = \Gamma(1 + u)$, which goes to $\Gamma(1) = 1$.
- The last factor of the denominator is $z + k = u$, which goes to zero. This is the pole.
- The other k factors, $z, z + 1, \dots, z + k - 1$, go to $-k, -k + 1, \dots, -1$. Their product is $(-1)^k k!$ (k minus signs times $1 \cdot 2 \cdots k$).

Put together,

$$\Gamma(z) \approx \frac{(-1)^k}{k!} \frac{1}{z + k} \quad \text{near } z = -k \quad (15)$$

The number in front of $1/(z + k)$ is the **residue**. For $k = 0$ it says $\Gamma(z) \approx 1/z$ (from $\Gamma(z) = \Gamma(z + 1)/z$ with $\Gamma(1) = 1$). For $k = 1$ the residue is -1 and for $k = 2$ it is $+1/2$, as in the figure.

This analytic continuation is the whole trick of dimensional regularisation. A Feynman integral will converge only for some range of the dimension D . There the answer is a Gamma function of D and the recursion then tells us what it is at $D = 4 - 2\varepsilon$, poles included.

The Gamma function near its poles is like Vegeta's scouter reading Goku: "It's over 9000!" Near $z = 0, -1, -2, \dots$ it simply blows up and the whole art is to find out by how much (the residue $(-1)^k/k!$).

The **digamma function** is the logarithmic derivative, $\psi(z) = \Gamma'(z)/\Gamma(z)$. Its value at 1 is minus the Euler constant, $\psi(1) = -\gamma_E$, $\gamma_E = 0.5772\dots$ Taking the log of the recursion and differentiating gives $\psi(z + 1) = \psi(z) + 1/z$, so $\psi(2) = 1 - \gamma_E$.

2.4 The Beta function

For $\text{Re } a, \text{Re } b > 0$,

$$B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (16)$$

DERIVATION: BETA IN TERMS OF GAMMA

Multiply two Gamma integrals:

$$\Gamma(a)\Gamma(b) = \int_0^\infty ds s^{a-1} e^{-s} \int_0^\infty dt t^{b-1} e^{-t} \quad (17)$$

Change variables to the total $w = s + t$ and the fraction $u = s/w$, so $s = uw$, $t = (1-u)w$. The Jacobian is

$$\left| \det \begin{pmatrix} \partial s / \partial u & \partial s / \partial w \\ \partial t / \partial u & \partial t / \partial w \end{pmatrix} \right| = \left| \det \begin{pmatrix} w & u \\ -w & 1 - u \end{pmatrix} \right| = w(1 - u) + uw = w \quad (18)$$

The ranges are $u \in [0, 1]$, $w \in [0, \infty)$. Then

$$\Gamma(a)\Gamma(b) = \int_0^1 du \int_0^\infty dw w(uw)^{a-1} ((1-u)w)^{b-1} e^{-w} = \int_0^1 u^{a-1} (1-u)^{b-1} du \int_0^\infty w^{a+b-1} e^{-w} dw \quad (19)$$

The last integral is $\Gamma(a+b)$, so $\Gamma(a)\Gamma(b) = B(a, b)\Gamma(a+b)$.

MATHEMATICA CHECK: GAUSSIAN, GAMMA AND BETA

Basic tools: Gaussian integral, Gamma and Beta functions, all symbolic. $\int_{-\infty}^\infty e^{-x^2} dx$

In [1]: `Integrate[Exp[-x^2], {x, -Infinity, Infinity}]`

Out[1]: $\sqrt{\pi}$

its square in polar coordinates: $\int_0^{2\pi} d\theta \int_0^\infty r e^{-r^2} dr$

In [2]: `Integrate[r Exp[-r^2], {r, 0, Infinity}, {th, 0, 2 Pi}]`

Out[2]: π

$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$

In [3]: `Integrate[t^(z - 1) Exp[-t], {t, 0, Infinity}, Assumptions -> z > 0]`

Out[3]: $\Gamma(z)$

$\Gamma(1), \Gamma(5), \Gamma(1/2)$

In [4]: `{Gamma[1], Gamma[5], Gamma[1/2]}`

Out[4]: $\{1, 24, \sqrt{\pi}\}$

$\Gamma(z+1)/(z\Gamma(z))$

In [5]: `FullSimplify[Gamma[z + 1]/(z Gamma[z])]`

Out[5]: 1

residues of $\Gamma(z)$ at $z = 0, -1, -2$

In [6]: `Table[Residue[Gamma[z], {z, -k}], {k, 0, 2}]`

Out[6]: $\left\{1, -1, \frac{1}{2}\right\}$

$B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du$

In [7]: `Integrate[u^(a - 1) (1 - u)^(b - 1), {u, 0, 1}, Assumptions -> a > 0 && b > 0]`

Out[7]: $\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$

$\Gamma'(1), \psi(1)$ and $\psi(2)$

In [8]: `{Gamma'[1], PolyGamma[1], PolyGamma[2]}`

Out[8]: $\{-\gamma, -\gamma, 1 - \gamma\}$

2.5 Complex numbers, the logarithm and $\pm i\epsilon$

Write a complex number as $z = |z| e^{i\theta}$. Then $\ln z = \ln|z| + i\theta$. But θ is only defined up to 2π : walking once around $z = 0$ adds 2π to θ and $2\pi i$ to $\ln z$. To make $\ln z$ a proper function we must forbid that walk. The standard choice is to cut the plane along the negative real axis and keep $-\pi < \theta < \pi$. Just above the cut $\theta \rightarrow \pi$, just below $\theta \rightarrow -\pi$:

$$\ln(-|z| + i\epsilon) = \ln|z| + i\pi, \quad \ln(-|z| - i\epsilon) = \ln|z| - i\pi \quad (20)$$

Here $\pm i\epsilon$ means “a tiny positive or negative imaginary part, sent to zero at the end” ($\epsilon \rightarrow 0^+$). This ϵ has nothing to do with the ε of $D = 4 - 2\varepsilon$. The two are written with different letters on purpose. In Feynman integrals the $+i\epsilon$ of the propagators decides which side of such a cut we are on. That is the origin of every imaginary part of a scattering amplitude (Section 15).

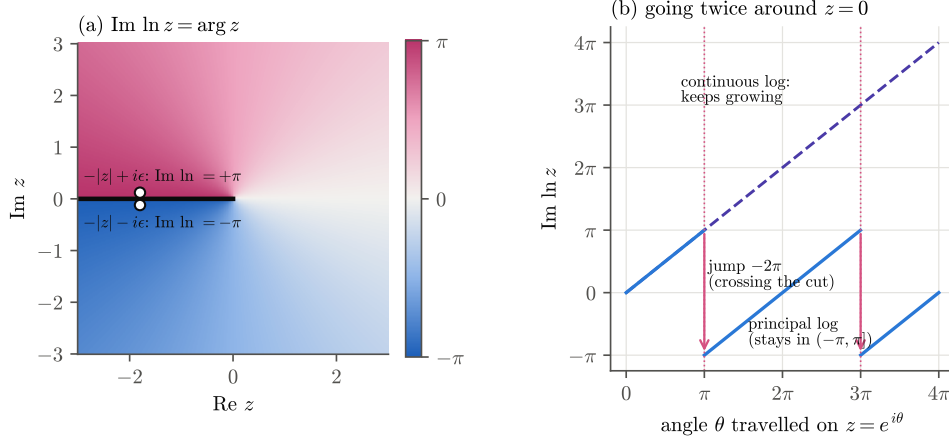


Figure 2: (a) The imaginary part of $\ln z$, which is the angle θ , over the complex plane. It jumps from $+\pi$ to $-\pi$ across the cut (thick line). The two white dots are $-|z| \pm i\epsilon$. (b) Walk twice around the unit circle, $z = e^{i\theta}$. If we follow the logarithm continuously (dashed) its imaginary part keeps growing, so after one full turn $\ln z$ does not come back to its old value. The principal logarithm (solid) stays between $-\pi$ and π by jumping down by 2π each time the path crosses the cut, at $\theta = \pi, 3\pi, \dots$. That jump is why a cut is needed.

Two more facts from complex analysis are used.

The residue theorem. If $f(z)$ has a simple pole at $z = a$, its residue is $\text{Res}_a f = \lim_{z \rightarrow a} (z - a)f(z)$. For a closed contour run anticlockwise, $\oint f dz = 2\pi i \sum \text{Res}$ of the poles inside. Run clockwise, there is an extra minus sign.

The Sokhotski-Plemelj formula. For a small $\eta > 0$,

$$\frac{1}{x \pm i\eta} = \frac{x}{x^2 + \eta^2} \mp i \frac{\eta}{x^2 + \eta^2} \quad (21)$$

Look at the two terms one by one, with Figure 3 in front of you.

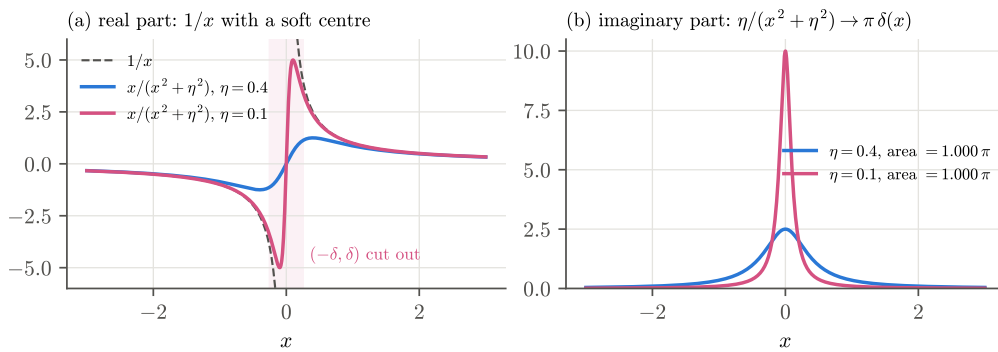


Figure 3: (a) $x/(x^2 + \eta^2)$ follows $1/x$ (dashed) except in a window of a few η around 0, where it is finite and odd. Removing a symmetric window $(-\delta, \delta)$ from $1/x$ has the same effect. (b) $\eta/(x^2 + \eta^2)$ for two values of η .

It gets taller and thinner, but its area (computed numerically in the legend) stays π .

The first term and the principal value. Far from the origin, $|x| \gg \eta$, the η^2 does nothing and $x/(x^2 + \eta^2) = 1/x$. Near the origin it is different. The function $1/x$ shoots to $\pm\infty$, but $x/(x^2 + \eta^2)$ stays finite. It is 0 at $x = 0$ and its largest value is $1/(2\eta)$ at $x = \eta$. So it is “ $1/x$ with a soft centre” (panel a).

Now take an interval $[a, b]$ that contains 0, so $a < 0 < b$ (for example $a = -1$ and $b = 2$) and ask what

$$\int_a^b \frac{f(x)}{x} dx \quad (22)$$

means. The area just to the right of 0 is $+\infty$ and the area just to the left is $-\infty$. Alone, each is meaningless. But $1/x$ is odd, so the two infinities are mirror images of each other and they can cancel if we treat both sides the same way.

That is the rule. Cut the interval $[a, b]$ into three pieces: $[a, -\delta]$, the small window $[-\delta, \delta]$ around 0 and $[\delta, b]$ (Figure 4). Keep the two outer pieces and throw the window away, because the window is where $1/x$ blows up. Only then send $\delta \rightarrow 0$, so the window shrinks to the single point 0:

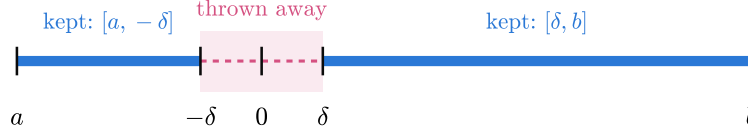


Figure 4: The principal value. The pieces $[a, -\delta]$ and $[\delta, b]$ are integrated, the window $[-\delta, \delta]$ is dropped and at the end $\delta \rightarrow 0$. The window must be symmetric about 0. The ends a and b can be anything.

In formulas,

$$\mathcal{P} \int_a^b \frac{f(x)}{x} dx = \lim_{\delta \rightarrow 0} \left(\int_a^{-\delta} + \int_{\delta}^b \right) \frac{f(x)}{x} dx \quad (23)$$

This is the principal value $\mathcal{P}$. There are only two integrals on the right because the middle piece is the one we threw away.

Why throwing the window away is allowed. Inside the window write $f(x) = f(0) + [f(x) - f(0)]$. The second part divided by x stays finite (it is close to $f'(0)$), so over a window of width 2δ it adds only about $2\delta f'(0)$, which vanishes as $\delta \rightarrow 0$. The first part is $f(0)/x$, which is odd. Its integral over the symmetric window is $+\infty$ from the right half and $-\infty$ from the left half, exactly the two infinities that were meant to cancel. The principal value says they cancel to zero. So dropping the window loses nothing finite. The only thing that could go wrong is an unequal window and we will see what happens then.

A small example makes it concrete:

$$\mathcal{P} \int_{-1}^2 \frac{dx}{x} = \lim_{\delta \rightarrow 0} [\ln \delta - \ln 1 + \ln 2 - \ln \delta] = \ln 2 \quad (24)$$

The $\ln \delta$ from the left side cancels the $\ln \delta$ from the right side. If we cut unequal windows, say $(-2\delta, \delta)$, the left side gives $\ln(2\delta)$ and the answer becomes $2 \ln 2$. So the word “symmetrically” matters.

The soft centre of $x/(x^2 + \eta^2)$ does exactly this job by itself. Inside a window of a few η the function is odd and finite, so that window adds $f(0)$ times the integral of an odd function, which is 0, plus a piece that vanishes with η . Outside the window it is $1/x$. So as $\eta \rightarrow 0$ it gives the principal value, with no extra rule needed.

The second term. $\eta/(x^2 + \eta^2)$ is a spike of height $1/\eta$ at $x = 0$ and width about η (panel b). Its area does not depend on η :

$$\int_{-\infty}^{\infty} \frac{\eta}{x^2 + \eta^2} dx = [\arctan(x/\eta)]_{-\infty}^{\infty} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi \quad (25)$$

A spike that gets thinner and taller but keeps area π is $\pi\delta(x)$. Hence

$$\frac{1}{x \pm i\epsilon} = \mathcal{P} \frac{1}{x} \mp i\pi\delta(x) \quad (26)$$

2.6 The divergence theorem, in four (and D) dimensions

This is the theorem behind integration by parts. It is the same statement in every dimension:

$$\int_M d^D x \frac{\partial}{\partial x^\mu} V^\mu(x) = \oint_{\partial M} dS n_\mu V^\mu \quad (27)$$

“the total source inside a region equals the total flux out through its boundary”. Here V^μ is a vector field with D components, $\partial_\mu V^\mu = \sum_{\mu=1}^D \partial V^\mu / \partial x^\mu$ is its divergence (a repeated index is summed), ∂M is the boundary

of M , n_μ the outward unit normal and dS the boundary “area” element. In Euclidean space upper and lower indices are the same thing.

- $D = 1$: M is an interval $[a, b]$, the boundary is the two end points and the theorem is $\int_a^b V'(x) dx = V(b) - V(a)$, the fundamental theorem of calculus.
- $D = 2$: a region in the plane, the boundary is a curve (Green’s theorem).
- $D = 3$: a region in space, the boundary is a surface (Gauss’s law of electrostatics: charge inside = flux of $\mathbf{E}$ out).
- $D = 4$: a region in four-dimensional Euclidean momentum space, the boundary is a three-dimensional “surface”.

WORKED EXAMPLE: THE THEOREM ON A 4-DIMENSIONAL BALL

Take M = the ball $|\ell| < R$ in four Euclidean dimensions and the vector field used in IBP,

$$V^\mu = \frac{\ell^\mu}{(\ell^2 + m^2)^n}, \quad \ell^2 = \ell_1^2 + \ell_2^2 + \ell_3^2 + \ell_4^2 \quad (28)$$

Left side, the divergence. By the product rule, for each μ ,

$$\frac{\partial}{\partial \ell^\mu} (\ell^\mu (\ell^2 + m^2)^{-n}) = (\ell^2 + m^2)^{-n} + \ell^\mu \cdot (-n)(\ell^2 + m^2)^{-n-1} \cdot 2\ell^\mu \quad (29)$$

(no sum yet. $\partial \ell^2 / \partial \ell^\mu = 2\ell^\mu$). Now sum over $\mu = 1, \dots, 4$. The first term appears four times and $\sum_\mu \ell^\mu \ell^\mu = \ell^2$:

$$\partial_\mu V^\mu = \frac{4}{(\ell^2 + m^2)^n} - \frac{2n\ell^2}{(\ell^2 + m^2)^{n+1}} \quad (30)$$

In D dimensions the 4 becomes D .

The boundary. The boundary is the 3-sphere $|\ell| = R$. The outward normal is $n^\mu = \ell^\mu / R$, so on the sphere $n_\mu V^\mu = \ell^2 / (R(R^2 + m^2)^n) = R / (R^2 + m^2)^n$, the same everywhere on the sphere. The area of the sphere of radius R in four dimensions is $\Omega_4 R^3 = 2\pi^2 R^3$ (derived in Section 4.1). So

$$\oint dS n_\mu V^\mu = 2\pi^2 R^3 \cdot \frac{R}{(R^2 + m^2)^n} = \frac{2\pi^2 R^4}{(R^2 + m^2)^n} \quad (31)$$

Check that the two sides agree. In spherical coordinates $d^4\ell = 2\pi^2 r^3 dr$ for a function of $r = |\ell|$ only. The volume side becomes

$$2\pi^2 \int_0^R r^3 \left[\frac{4}{(r^2 + m^2)^n} - \frac{2nr^2}{(r^2 + m^2)^{n+1}} \right] dr = 2\pi^2 \int_0^R \frac{d}{dr} \left[\frac{r^4}{(r^2 + m^2)^n} \right] dr = \frac{2\pi^2 R^4}{(R^2 + m^2)^n} \quad (32)$$

because $d/dr [r^4(r^2 + m^2)^{-n}] = 4r^3(r^2 + m^2)^{-n} - 2nr^5(r^2 + m^2)^{-n-1}$, which is the bracket times r^3 . Both sides agree.

What happens as $R \rightarrow \infty$. The flux behaves like R^{4-2n} . It goes to zero if $n > 2$. In D dimensions it is R^{D-2n} and goes to zero if $D < 2n$. Then the integral of the total derivative over all of momentum space is zero:

$$\int d^D\ell \frac{\partial}{\partial \ell^\mu} \frac{\ell^\mu}{(\ell^2 + m^2)^n} = 0 \quad (33)$$

In dimensional regularisation we first choose D small enough for this to be true and then continue in D . Both sides are analytic in D , so the zero stays zero. This is the starting point of IBP.

MATHEMATICA CHECK: THE DIVERGENCE THEOREM ON A D-DIMENSIONAL BALL

Divergence theorem on a D -dimensional ball of radius R , for $V^\mu = \ell^\mu / (\ell^2 + m^2)^n$. $\partial_\mu V^\mu$ at $|\ell| = r$. area times

In [1]: $n_\mu V^\mu$ on the sphere. $\int_{|\ell| < R} d^D\ell \partial_\mu V^\mu$

```
Omega[D_] := 2 Pi^(D/2)/Gamma[D/2];
divV = D/(r^2 + m2)^n - 2 n r^2/(r^2 + m2)^(n + 1);
```

```

volume = Omega[D] Integrate[r^(D - 1) divV,
  {r, 0, R}, Assumptions -> R > 0 && m2 > 0 && D > 0];
surface = Omega[D] R^(D - 1) (R/(R^2 + m2)^n);
FullSimplify[volume, R > 0 && m2 > 0 && D > 0]

```

Out[1]: $\frac{2 \pi^{D/2} R^D (m^2 + R^2)^{-n}}{\Gamma(\frac{D}{2})}$

volume integral minus the flux through the sphere

```
In [2]: FullSimplify[volume - surface, R > 0 && m2 > 0 && D > 0]
```

Out[2]: 0

the flux $\Omega_D R^{D-1} n_\mu V^\mu$

```
In [3]: surface
```

Out[3]: $\frac{2 \pi^{D/2} R^D (m^2 + R^2)^{-n}}{\Gamma(\frac{D}{2})}$

check in $D = 4$ with Cartesian components. $\partial_\mu (\ell^\mu / (\ell^2 + m^2)^n)$ in four dimensions

```
In [4]: V4 = {x1, x2, x3, x4}/(x1^2 + x2^2 + x3^2 + x4^2 + m2)^n;
div4 = Simplify[Sum[D[V4[[i]], {x1, x2, x3, x4}[[i]]], {i, 4}]]];
div4
```

Out[4]: $(m^2 + x_1^2 + x_2^2 + x_3^2 + x_4^2)^{-n-1} (4 m^2 - 2 (n - 2) (x_1^2 + x_2^2 + x_3^2 + x_4^2))$

What goes on inside the Walls is fixed by what crosses the Walls. Hange Zoë would approve of the divergence theorem. In dimensional regularisation nothing crosses the wall at infinity, so the inside must add up to zero.

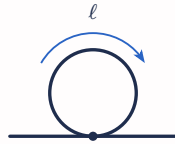
3 The simplest loop: the tadpole

3.1 Where the integral comes from

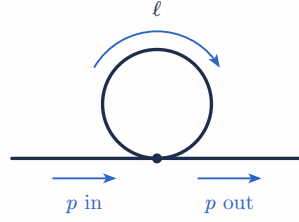
Take a real scalar field with a quartic self coupling,

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \varphi)^2 - \frac{1}{2} m^2 \varphi^2 - \frac{\lambda}{4!} \varphi^4 \quad (34)$$

At first order in λ the propagator gets a correction from the diagram where the φ^4 vertex joins the line to a closed loop:



Why no external momentum goes into the loop. The external line does carry a momentum. Call it p . Look at the single vertex, where four lines meet, the incoming line with p , the outgoing line with p and the two ends of the loop. Momentum conservation at the vertex says (total in) = (total out). If the loop leaves the vertex with momentum $\ell + k$ and comes back with ℓ , conservation reads $p + \ell = p + \ell + k$, so $k = 0$. Whatever flows into the loop at one end flows straight back in at the other end, at the same point. There is no second vertex where some of p could be handed over. So p passes straight along the external line and the loop only ever sees its own momentum ℓ :



That is why the tadpole integral depends on m but not on p . (In the bubble of Section 8 the loop touches the external line at **two** different vertices and then p is forced to flow through the loop.) The momentum ℓ running around the loop is not fixed by momentum conservation and quantum mechanics tells us to add all possibilities, so we integrate over it. With Peskin's Feynman rules (vertex $-i\lambda$, propagator $i/(\ell^2 - m^2 + i\epsilon)$, symmetry factor $1/2$) the diagram gives a correction M^2 to the mass,

$$-iM^2 = \frac{1}{2}(-i\lambda) \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\epsilon} \quad (35)$$

The constants in front multiply to $\frac{1}{2}(-i\lambda)(i) = \lambda/2$, because $(-i)(i) = 1$. So $-iM^2 = (\lambda/2) \int \dots$ and multiplying both sides by i (use $i \cdot (-i) = 1$),

$$M^2 = \frac{i\lambda}{2} \int \frac{d^4\ell}{(2\pi)^4} \frac{1}{\ell^2 - m^2 + i\epsilon} \quad (36)$$

Everything interesting sits in the integral. Strip off the constants and we get the integral the lecture started with,

$$I_1 = \int \frac{d^4\ell}{\ell_E^2 + m^2} \quad (37)$$

written in Euclidean space. The next section explains how the Minkowski integral in Equation 36 turns into this.

PHYSICAL PICTURE

What does the loop describe? Read the diagram from left to right. A φ particle comes in. At the vertex, the φ^4 interaction lets it create a second, **virtual** φ particle, which runs around the loop with some momentum ℓ and is absorbed again at the same point. Then the original particle goes on. The virtual particle is not a real, observable particle. It does not have to satisfy $\ell^2 = m^2$ and it exists only for the short time the uncertainty principle allows. Quantum mechanics says we must add up all such possibilities, every ℓ .

So a particle in an interacting theory is never alone. It keeps emitting and reabsorbing virtual quanta of its own field. This changes its energy and therefore its mass. Large ℓ means short-lived, short-distance fluctuations. All momenta up to infinity contribute, so the integral diverges and that is the first sign that the parameter m written in the Lagrangian is not the mass we measure.

3.2 From Minkowski to Euclidean: the Wick rotation

In words, the Wick rotation “changes the sign of m^2 , roughly”. Here it is done properly, because it is the source of every factor of i and $(-1)^n$ later.

DERIVATION, STEP BY STEP

Step 1, the energy integral. In Minkowski space $\ell^2 - m^2 = \ell_0^2 - E_\ell^2$ with $E_\ell = \sqrt{\ell^2 + m^2}$. The Feynman prescription $+i\epsilon$ moves the two poles of $1/(\ell_0^2 - E_\ell^2 + i\eta)$ off the real axis:

$$\ell_0^2 = E_\ell^2 - i\eta \Rightarrow \ell_0 = \pm \sqrt{E_\ell^2 - i\eta} \quad (38)$$

Since η is tiny, expand the square root, $\sqrt{1-u} \approx 1 - u/2$ for small u :

$$\sqrt{E_\ell^2 - i\eta} = E_\ell \sqrt{1 - \frac{i\eta}{E_\ell^2}} \approx E_\ell \left(1 - \frac{i\eta}{2E_\ell^2}\right) = E_\ell - i\eta', \quad \eta' = \frac{\eta}{2E_\ell} > 0 \quad (39)$$

So the two poles are at $\ell_0 = +(E_\ell - i\eta')$ and $\ell_0 = -(E_\ell - i\eta') = -E_\ell + i\eta'$. The positive energy pole sits just below the real axis and the negative energy pole just above it (see Figure 5).

Step 2, rotate the contour. The first and third quadrants of the complex ℓ_0 plane contain no poles. Take the closed contour made of the real axis from 0 to R , the quarter circle from R to iR and the imaginary axis back from iR to 0 (and the mirror image in the third quadrant). No pole is inside, so by Cauchy's theorem the integral around it is zero. On the quarter circle $|\ell_0| = R$ the integrand is of size $1/R^2$ and the arc length is $\pi R/2$, so that piece is at most of order $1/R \rightarrow 0$. Therefore “real axis” = “imaginary axis”, with both run in the same direction. In words. We may turn the contour anticlockwise by 90° without crossing any pole. Put $\ell_0 = i\ell_4$ with ℓ_4 running from $-\infty$ to ∞ :

$$d\ell_0 = i d\ell_4, \quad \ell^2 = \ell_0^2 - \ell^2 = -(\ell_4^2 + \ell^2) = -\ell_E^2 \quad (40)$$

Step 3, check it on one integral. Call $a = \sqrt{E_\ell^2 - i\eta} = E_\ell - i\eta' + O(\eta^2)$ (the root with positive real part), so $\ell_0^2 - E_\ell^2 + i\eta = \ell_0^2 - a^2 = (\ell_0 - a)(\ell_0 + a)$. Close the real axis with a big semicircle in the lower half plane. That semicircle again gives zero as $R \rightarrow \infty$ and the closed contour is run clockwise, so the residue theorem carries a factor $-2\pi i$. Inside it there is only the pole $\ell_0 = a$ (the other one, $-a$, is in the upper half plane). The residue there is

$$\text{Res}_{\ell_0=a} \frac{1}{(\ell_0 - a)(\ell_0 + a)} = \frac{1}{a + a} = \frac{1}{2a} \quad (41)$$

Hence

$$\int_{-\infty}^{\infty} \frac{d\ell_0}{\ell_0^2 - a^2} = -2\pi i \cdot \frac{1}{2a} = -\frac{i\pi}{a} \xrightarrow{\eta \rightarrow 0} -\frac{i\pi}{E_\ell} \quad (42)$$

Now do the same integral after the rotation. With $d\ell_0 = i d\ell_4$ and $\ell_0^2 = -\ell_4^2$,

$$\int_{-\infty}^{\infty} \frac{i d\ell_4}{-\ell_4^2 - E_\ell^2} = -i \int_{-\infty}^{\infty} \frac{d\ell_4}{\ell_4^2 + E_\ell^2} = -i \left[\frac{1}{E_\ell} \arctan \frac{\ell_4}{E_\ell} \right]_{-\infty}^{\infty} = -i \cdot \frac{\pi}{E_\ell} \quad (43)$$

Same answer, as it must be.

Step 4, the general rule. In D dimensions $d^D \ell = d\ell_0 d^{D-1} \ell$. Only ℓ_0 is rotated, so $d^D \ell = i d\ell_4 d^{D-1} \ell = i d^D \ell_E$, where $\ell_E = (\ell, \ell_4)$ is a Euclidean vector. Also $\ell^2 = -\ell_E^2$. So for any function of ℓ^2 that decays fast enough,

$$\int d^D \ell f(\ell^2) = i \int d^D \ell_E f(-\ell_E^2) \quad (44)$$

With $f = (\ell^2 - m^2)^{-n}$ we get $f(-\ell_E^2) = (-1)^n (\ell_E^2 + m^2)^{-n}$, which gives the rule in the box below.

RESULT: THE WICK ROTATION RULE

$$\int d^D \ell f(\ell^2) = i \int d^D \ell_E f(-\ell_E^2) \quad (45)$$

and for the propagator powers we meet in these notes

$$\int \frac{d^D \ell}{(\ell^2 - m^2 + i\epsilon)^n} = i(-1)^n \int \frac{d^D \ell_E}{(\ell_E^2 + m^2)^n} \quad (46)$$

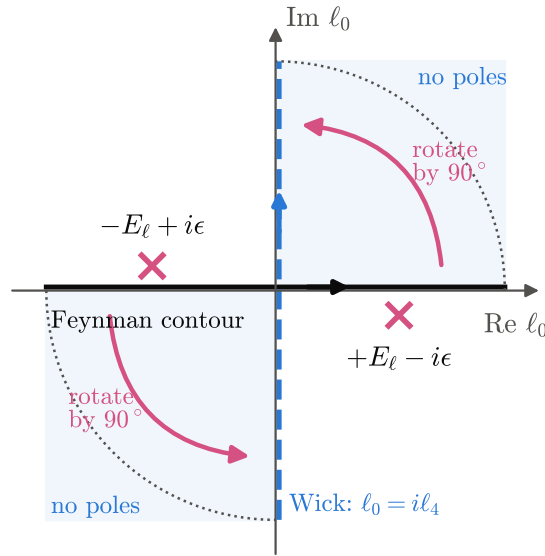


Figure 5: Complex ℓ_0 plane. The Feynman $+i\epsilon$ puts the poles (crosses) in the second and fourth quadrants. The shaded quadrants are empty, so the contour on the real axis can be swung to the dashed one on the imaginary axis. This is the Wick rotation $\ell_0 = i\ell_4$.

MATHEMATICA CHECK: THE ENERGY INTEGRAL AND THE WICK FACTOR

Minkowski energy integral with the Feynman $i\epsilon$ kept explicitly (called η in the code). $\int_{-\infty}^{\infty} d\ell_0 / (\ell_0^2 - E^2 + i\eta)$

```
In [1]: k0 = Integrate[1/(l0^2 - e^2 + I eta), {l0, -Infinity, Infinity},
Assumptions -> e > 0 && eta > 0];
k0
```

```
Out[1]: 
$$\frac{\pi}{\sqrt{-E^2 + i\eta}}$$

```

for $E > 0$ pull out iE : $\sqrt{-E^2 + i\eta} = iE\sqrt{1 - i\eta/E^2}$, then expand the square root. check: its square minus $(-E^2 + i\eta)$

```
In [2]: root = I e Sqrt[1 - I eta/e^2];
Simplify[root^2 - (-e^2 + I eta)]
```

```
Out[2]: 0
```

first two terms in η : $iE + \eta/(2E)$, so the pole moves just off the real axis

```
In [3]: Normal@Series[root, {eta, 0, 1}]
```

```
Out[3]: 
$$\frac{\eta}{2E} + iE$$

```

so the integral tends to $\pi/(iE)$

```
In [4]: Simplify[Pi/(I e)]
```

```
Out[4]: 
$$-\frac{i\pi}{E}$$

```

the same with exact numbers, pole at $a = 1 - i/10$ below the real axis. the integral and $-i\pi/a$

```
In [5]: {Integrate[1/((l0 - (1 - I/10)) (l0 + (1 - I/10))),
{l0, -Infinity, Infinity}], -I Pi/(1 - I/10)}
```

```
Out[5]: 
$$\left\{ \left( \frac{10}{101} - \frac{100i}{101} \right) \pi, \left( \frac{10}{101} - \frac{100i}{101} \right) \pi \right\}$$

```

after the Wick rotation $\ell_0 = i\ell_4$, $d\ell_0 = i d\ell_4$. $i \int d\ell_4 / (-(\ell_4^2 + E^2))$

```
In [6]: I Integrate[1/(-(l4^2 + e^2)), {l4, -Infinity, Infinity},
Assumptions -> e > 0]
```

Out[6]: $-\frac{i\pi}{E_t}$

full result: $\int d^D \ell / (\ell^2 - m^2 + i\epsilon)^n = i(-1)^n \pi^{D/2} \Gamma(n - D/2) / \Gamma(n) (m^2)^{D/2-n}$. Minkowski tadpole divided by the Euclidean one

In [7]:

```
IM[n_] := I (-1)^n Pi^(D/2) Gamma[n - D/2]/Gamma[n] m2^(D/2 - n);
IE[n_] := Pi^(D/2) Gamma[n - D/2]/Gamma[n] m2^(D/2 - n);
Simplify[IM[n]/IE[n]]
```

Out[7]: $i(-1)^n$

The first line keeps η finite. The answer is $\pi/\sqrt{-E^2 + i\eta}$. To see its small- η limit, pull out iE : $\sqrt{-E^2 + i\eta} = iE\sqrt{1 - i\eta/E^2}$ (the next two cells check that this is the right square root and expand it). With $\sqrt{1 - u} \approx 1 - u/2$ this is $iE + \eta/(2E)$, so the integral tends to $\pi/(iE) = -i\pi/E$, the result of Step 3. The line with exact numbers checks the same thing for $a = 1 - i/10$, a pole below the real axis. The rotated integral also gives $-i\pi/E$.

The last line says that the Minkowski tadpole with n propagators is $i(-1)^n$ times the Euclidean one. For $n = 1$ this is $-i$ and putting it in Equation 36,

$$M^2 = \frac{i\lambda}{2}(-i) \int \frac{d^4 \ell_E}{(2\pi)^4} \frac{1}{\ell_E^2 + m^2} = \frac{\lambda}{2} \int \frac{d^4 \ell_E}{(2\pi)^4} \frac{1}{\ell_E^2 + m^2} \quad (47)$$

The integrand on the right is positive, so for a cutoff regulator the mass shift is positive, as we expect for a repulsive φ^4 interaction ($\lambda > 0$). In pictures,

$$M^2 = \text{---}\bigcirc\text{---} = \frac{\lambda}{2} \int \frac{d^4 \ell_E}{(2\pi)^4} \frac{1}{\ell_E^2 + m^2} \quad (48)$$

If we start instead from the Euclidean path integral e^{-S_E} (as Ramond does), the rules are propagator $1/(\ell_E^2 + m^2)$, vertex $-\lambda$, same symmetry factors. The value of the diagram is then $\frac{1}{2}(-\lambda) \int d^D \ell_E / (2\pi)^D (\ell_E^2 + m^2)^{-1} = -M^2$ and the mass shift is minus the diagram value. Both routes give the same M^2 .

Naruto's Rasengan is a rotation that turns chakra into something you can hold. The Wick rotation is a 90° rotation that turns an oscillating Minkowski integral into a calm Euclidean one you can actually do.

3.3 Power counting: where does it diverge?

In D Euclidean dimensions go to spherical coordinates, $d^D \ell = d\Omega_D \ell^{D-1} d\ell$ (in $D = 4$ this is $\ell^3 d\ell$). For the tadpole with n propagators,

$$I_n = \int \frac{d^D \ell}{(\ell_E^2 + m^2)^n} = \Omega_D \int_0^\infty \frac{\ell^{D-1} d\ell}{(\ell^2 + m^2)^n} \quad (49)$$

where Ω_D is the total solid angle, worked out in Section 4.1. Only the radial integral can diverge. Look at its two ends:

- Large ℓ (ultraviolet, UV): for $\ell \gg m$ we have $\ell^2 + m^2 \approx \ell^2$, so the integrand behaves as $\ell^{D-1}/\ell^{2n} = \ell^{D-1-2n}$. To see whether the area is finite, stop the integral at a large momentum Λ (the **cutoff**) instead of at ∞ and watch what happens as Λ grows. The approximation $\ell^2 + m^2 \approx \ell^2$ is good only for large ℓ , so split the range at some fixed momentum L well above m (say $L = 10m$):

$$\int_0^\Lambda \frac{\ell^{D-1} d\ell}{(\ell^2 + m^2)^n} = \underbrace{\int_0^L \frac{\ell^{D-1} d\ell}{(\ell^2 + m^2)^n}}_{\text{finite, does not depend on } \Lambda} + \int_L^\Lambda \frac{\ell^{D-1} d\ell}{(\ell^2 + m^2)^n} \quad (50)$$

The first piece is a fixed number, the integral of a finite function over a finite range. Only the second piece knows about Λ and there the approximation holds:

$$\int_L^\Lambda \ell^{D-1-2n} d\ell = \frac{\Lambda^{D-2n} - L^{D-2n}}{D-2n} \quad (51)$$

The L term is again a fixed number. So all the Λ dependence sits in $\Lambda^{D-2n}/(D-2n)$, which stays finite as $\Lambda \rightarrow \infty$ only if $D-2n < 0$, that is $D < 2n$. For $n = 1$, $D = 4$ it grows like $\Lambda^2/2$ (quadratic divergence). For $D = 2n$ the power becomes ℓ^{-1} and the second piece is $\ln(\Lambda/L)$, which grows like $\ln \Lambda$ (logarithmic divergence). The choice of L changes only the fixed numbers, never the way the integral grows with Λ .

- Small ℓ (infrared, IR): with $m > 0$ nothing happens, the integrand behaves as ℓ^{D-1}/m^{2n} , which is harmless. If $m = 0$ the integrand behaves as ℓ^{D-1-2n} at small ℓ too and converges only if $D > 2n$.

So in four dimensions the massless tadpole with $n = 1$, $\int d^4\ell/\ell^2$, is perfectly fine at small ℓ (the integrand is $\ell^3/\ell^2 = \ell$ there). It has only the UV (quadratic) divergence. An IR divergence at the lower end first appears at $n = 2$, where $\int d^4\ell/\ell^4$ is logarithmically divergent at both ends.

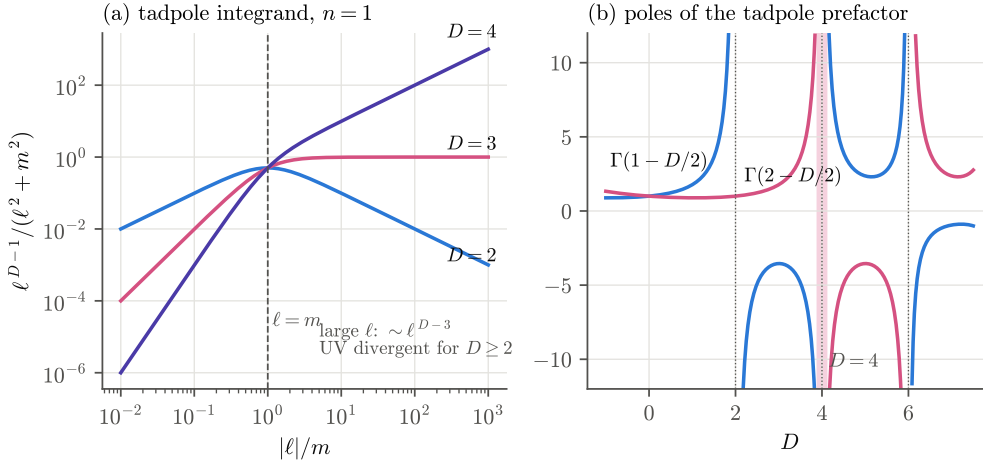


Figure 6: (a) The radial integrand of the tadpole with one propagator, $\ell^{D-1}/(\ell^2 + m^2)$, for $D = 2, 3, 4$. Beyond $\ell \sim m$ it goes like ℓ^{D-3} , so the area under it is infinite for $D \geq 2$. (b) After dimensional regularisation the answer is proportional to $\Gamma(n - D/2)$. The divergences have become poles at $D = 2n, 2n + 2, \dots$. The shaded strip is $D = 4$.

PHYSICAL PICTURE: SCALELESS INTEGRALS

For $m = 0$ the UV condition ($D < 2n$) and the IR condition ($D > 2n$) can never hold together. There is no value of D where the massless tadpole converges. This is because the integral has no scale at all: $\int d^D\ell/(\ell^2)^n$ has mass dimension $D - 2n$, but there is no mass around to carry that dimension. Dimensional regularisation sets every such scaleless integral to zero. One way to see it. Split the integral at some $\ell = \Lambda$ and do each piece where it converges.

$$\int_{\Lambda}^{\infty} \ell^{D-1-2n} d\ell = \left[\frac{\ell^{D-2n}}{D-2n} \right]_{\Lambda}^{\infty} = -\frac{\Lambda^{D-2n}}{D-2n} \quad (D < 2n) \quad (52)$$

$$\int_0^{\Lambda} \ell^{D-1-2n} d\ell = \left[\frac{\ell^{D-2n}}{D-2n} \right]_0^{\Lambda} = +\frac{\Lambda^{D-2n}}{D-2n} \quad (D > 2n) \quad (53)$$

Each answer is an analytic function of D , so we continue both to the same D and add. They cancel exactly, for every Λ . At $D = 2n$ it is “UV pole minus IR pole”, which is why one must be careful with massless integrals in practice.

4 Dimensional regularisation

Fullmetal Alchemist's law of equivalent exchange: “to obtain, something of equal value must be lost”. To obtain a finite answer, we give up $D = 4$ for a while.

The idea is simple to state. For $m > 0$ the integral Equation 49 converges for $0 < D < 2n$ (the upper limit from large ℓ , the lower one from the measure $\ell^{D-1} d\ell$ at small ℓ). In that region we can compute it exactly, as a function of D . The answer turns out to be a combination of Gamma functions, which make sense for any complex D . We then define the integral at $D = 4$ (or near it) by this analytic continuation. The divergence does not disappear, it shows up as a pole in $D - 4$. What we gain is that the pole is clean and shift invariance, Lorentz invariance and gauge invariance are all kept, which a momentum cutoff would spoil.

To use this we need two things, the area of a sphere in D dimensions and the radial integral.

4.1 Solid angle in D dimensions

DERIVATION: THE AREA OF THE UNIT SPHERE IN D DIMENSIONS

Step 1, spherical coordinates in D dimensions. Every point x is a radius $r = |x|$ times a direction (a point on the unit sphere). The volume of a ball of radius r must scale like r^D (stretch every axis by r), so write it as $V_D r^D$. A thin shell between r and $r + dr$ has volume

$$d(V_D r^D) = D V_D r^{D-1} dr \equiv \Omega_D r^{D-1} dr \quad (54)$$

where $\Omega_D = D V_D$ is the area of the unit sphere (the “total solid angle”). For a function that depends only on r we can therefore replace $d^D x \rightarrow \Omega_D r^{D-1} dr$. In three dimensions this is the familiar $4\pi r^2 dr$.

Step 2, compute the Gaussian in two ways. Call $G = \int d^D x e^{-x^2}$. In Cartesian coordinates it factorises and gives $G = \pi^{D/2}$ (Section 2.2). In spherical coordinates,

$$G = \Omega_D \int_0^\infty r^{D-1} e^{-r^2} dr \quad (55)$$

Step 3, turn the radial integral into a Gamma function. Put $u = r^2$, so $r = u^{1/2}$ and $dr = du/(2u^{1/2})$. Then $r^{D-1} dr = u^{(D-1)/2} du/(2u^{1/2}) = \frac{1}{2} u^{D/2-1} du$ and

$$\int_0^\infty r^{D-1} e^{-r^2} dr = \frac{1}{2} \int_0^\infty u^{D/2-1} e^{-u} du = \frac{1}{2} \Gamma\left(\frac{D}{2}\right) \quad (56)$$

Step 4, compare. $\pi^{D/2} = \Omega_D \Gamma(D/2)/2$, so

$$\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)}, \quad \text{or with } D = 2\omega: \quad \Omega_{2\omega} = \frac{2\pi^\omega}{\Gamma(\omega)} \quad (57)$$

Step 5, check. $D = 2$: $\Gamma(1) = 1$, so $\Omega_2 = 2\pi$, the length of the unit circle. $D = 3$: $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \sqrt{\pi}/2$, so $\Omega_3 = 2\pi^{3/2}/(\sqrt{\pi}/2) = 4\pi$, the area of the unit sphere. $D = 4$: $\Gamma(2) = 1$, so $\Omega_4 = 2\pi^2$, which we used for the 3-sphere in Section 2.6.

The formula never used that D is an integer. Ω_D is a perfectly good function of a real (even complex) D and that is the whole point of what follows.

MATHEMATICA CHECK

Solid angle Ω_D of the unit sphere in D dimensions, from the Gaussian trick. $\Omega_2, \Omega_3, \Omega_4$

```
In [1]: Omega[D_] := 2 Pi^(D/2)/Gamma[D/2];
        {Omega[2], Omega[3], Omega[4]}
```

```
Out[1]: {2 π, 4 π, 2 π^2}
```

radial Gaussian integral $\int_0^\infty r^{n-1} e^{-r^2} dr$

```
In [2]: radial = Integrate[Exp[-x^2] x^(n - 1),
        {x, 0, Infinity}, Assumptions -> n > 0];
        radial
```

```
Out[2]: Γ(n/2)
        2
```

$$\Omega_n \times (\text{radial}) - \pi^{n/2}$$

In [3]: `Simplify[Omega[n] radial - Pi^(n/2)]`

Out[3]: 0

4.2 The tadpole for any D

DERIVATION, STEP BY STEP

Start from Equation 49. Use the substitution $\ell = m \tan t$, so $d\ell = m \sec^2 t dt$ and $\ell^2 + m^2 = m^2 \sec^2 t$. The limits $\ell = 0, \infty$ become $t = 0, \pi/2$:

$$I_n = \Omega_D \int_0^{\pi/2} \frac{m^{D-1} \tan^{D-1} t \cdot m \sec^2 t}{m^{2n} \sec^{2n} t} dt = \Omega_D m^{D-2n} \int_0^{\pi/2} \sin^{D-1} t \cos^{2n-D-1} t dt \quad (58)$$

Two small steps were used here. The powers of m collect as $m^{D-1} \cdot m/m^{2n} = m^{D-2n}$. For the trigonometric part, one $\sec^2 t$ cancels against the denominator and then $\tan = \sin / \cos$, $\sec = 1 / \cos$:

$$\frac{\tan^{D-1} t}{\sec^{2n-2} t} = \frac{\sin^{D-1} t}{\cos^{D-1} t} \cos^{2n-2} t = \sin^{D-1} t \cos^{2n-1-D} t \quad (59)$$

The Beta function. Put $u = \sin^2 t$, so $du = 2 \sin t \cos t dt$, $\cos^2 t = 1 - u$ and $t : 0 \rightarrow \pi/2$ becomes $u : 0 \rightarrow 1$. For general exponents,

$$\sin^{2a-1} t \cos^{2b-1} t dt = (\sin^{2a-2} t \cos^{2b-2} t) (\sin t \cos t dt) = u^{a-1} (1-u)^{b-1} \frac{du}{2} \quad (60)$$

so

$$\int_0^{\pi/2} \sin^{2a-1} t \cos^{2b-1} t dt = \frac{1}{2} \int_0^1 u^{a-1} (1-u)^{b-1} du = \frac{1}{2} B(a, b) = \frac{\Gamma(a)\Gamma(b)}{2\Gamma(a+b)} \quad (61)$$

Here $2a - 1 = D - 1$ and $2b - 1 = 2n - 1 - D$, so $a = D/2$, $b = n - D/2$ and $a + b = n$. Put everything together:

$$I_n = \frac{2\pi^{D/2}}{\Gamma(D/2)} m^{D-2n} \frac{\Gamma(D/2)\Gamma(n-D/2)}{2\Gamma(n)} = \pi^{D/2} \frac{\Gamma(n-D/2)}{\Gamma(n)} (m^2)^{D/2-n} \quad (62)$$

The $\Gamma(D/2)$ of the solid angle cancels. The substitution $u = \ell^2/(\ell^2 + m^2)$ gives the same Beta function directly, if you prefer.

RESULT: THE EUCLIDEAN TADPOLE

$$I_n(m^2) = \int \frac{d^D \ell}{(\ell_E^2 + m^2)^n} = \pi^{D/2} \frac{\Gamma(n-D/2)}{\Gamma(n)} (m^2)^{D/2-n}, \quad (63)$$

$$T(n) = \pi^{-D/2} I_n = \frac{\Gamma(n-D/2)}{\Gamma(n)} (m^2)^{D/2-n}$$

In pictures, a loop with $n - 1$ dots:

$$T(n) = \text{diagram} (n=3), \quad T(1) = \text{diagram} = \Gamma\left(1 - \frac{D}{2}\right) (m^2)^{D/2-1} \quad (64)$$

With the usual physics measure, $\int d^D \ell / (2\pi)^D (\ell_E^2 + m^2)^{-n} = (4\pi)^{-D/2} T(n)$.

MATHEMATICA CHECK: RADIAL INTEGRAL AND THE TAN ROUTE

Euclidean tadpole $I_n = \int d^D \ell / (\ell^2 + m^2)^n$, done as a radial integral. radial integral $\int_0^\infty \ell^{D-1} d\ell / (\ell^2 + m^2)^n$

In [1]: `Omega[D_] := 2 Pi^(D/2)/Gamma[D/2];
radial = Integrate[l^(D-1)/(l^2+m^2)^n, {l, 0, Infinity},`

```

Assumptions -> {m > 0, 0 < D < 2 n}};
radial

Out[1]:  $\frac{\Gamma(\frac{D}{2}) m^{D-2n} \Gamma(n - \frac{D}{2})}{2 \Gamma(n)}$ 

 $I_n = \Omega_D \times (\text{radial})$ 

In [2]: Itad = FullSimplify[Omega[D] radial, m > 0];
        Itad

Out[2]:  $\frac{\pi^{D/2} m^{D-2n} \Gamma(n - \frac{D}{2})}{\Gamma(n)}$ 

the same with the substitution  $\ell = m \tan t$ .  $\int_0^{\pi/2} \sin^{D-1} t \cos^{2n-D-1} t dt$ 

In [3]: tanInt = Integrate[Sin[t]^(D - 1) Cos[t]^(2 n - D - 1),
        {t, 0, Pi/2}, Assumptions -> 0 < D < 2 n];
        tanInt

Out[3]:  $\frac{\Gamma(\frac{D}{2}) \Gamma(n - \frac{D}{2})}{2 \Gamma(n)}$ 

tan route minus radial route

In [4]: FullSimplify[Omega[D] m^(D - 2 n) tanInt - Itad, m > 0]

Out[4]: 0

```

Mathematica returns the radial integral as $m^{D-2n} \Gamma(D/2) \Gamma(n - D/2) / (2 \Gamma(n))$, the same Beta function we found. Multiplying by Ω_D cancels $\Gamma(D/2)$ and the tan route gives the same thing (difference 0).

PHYSICAL PICTURE: WHAT THE GAMMA FUNCTION IS TELLING US

Look at Figure 6 (b). For $n = 1$ the prefactor is $\Gamma(1 - D/2)$, with poles at $D = 2, 4, 6, \dots$. Each pole is one power of divergence. At $D = 2$ the integral is logarithmically divergent (the integrand goes like $1/\ell$) and at $D = 4$ it is quadratically divergent, but dimensional regularisation cannot tell a quadratic divergence from a logarithmic one. It only sees the logarithm hidden in it, as a simple pole. The power divergence Λ^2 is simply thrown away. We lose nothing by this, because power divergences depend on the regulator, while the logarithms carry the physics (they give the running of couplings). For $n = 2$ the prefactor $\Gamma(2 - D/2)$ is finite at $D = 2$ and has its first pole at $D = 4$, the logarithmic divergence of $\int d^4 \ell / \ell^4$.

4.3 Two ways to write D and the scale μ

Ramond writes $D = 2\omega$ and many other books write $D = 4 - 2\varepsilon$. They are related by $\varepsilon = 2 - \omega$. So “the pole at $\omega = 2$ ” and “the pole at $\varepsilon = 0$ ” are the same thing and $1/(2 - \omega) = 1/\varepsilon$. Some older books write $D = 4 - \varepsilon$, which changes every $1/\varepsilon$ by a factor 2. Always check the definition before comparing numbers.

In D dimensions the action $\int d^D x \mathcal{L}$ must be dimensionless. Then $[\varphi] = (D - 2)/2$ and $[\lambda] = 4 - D = 2\varepsilon$. To keep the coupling λ dimensionless we write it as $\lambda \mu^{2\varepsilon} = \lambda (\mu^2)^{2-\omega}$, where μ is an arbitrary mass. The physics cannot depend on μ and this requirement is what the renormalisation group equations express.

4.4 The expansion near $D = 4$

We need the Gamma function near its poles. The digamma function is $\psi(z) = \Gamma'(z)/\Gamma(z) = d \ln \Gamma(z) / dz$ and its value at 1 defines the Euler constant, $\psi(1) = \Gamma'(1) = -\gamma_E$ with $\gamma_E = 0.5772\dots$

DERIVATION: THE GAMMA FUNCTION NEAR ITS FIRST TWO POLES

Near $z = 1$. Taylor expand, using $\Gamma(1) = 1$ and $\Gamma'(1) = -\gamma_E$:

$$\Gamma(1 + \varepsilon) = 1 - \gamma_E \varepsilon + O(\varepsilon^2) \quad (65)$$

Near $z = 0$. From $\Gamma(z+1) = z\Gamma(z)$ with $z = \varepsilon$:

$$\Gamma(\varepsilon) = \frac{\Gamma(1+\varepsilon)}{\varepsilon} = \frac{1 - \gamma_E \varepsilon + \dots}{\varepsilon} = \frac{1}{\varepsilon} - \gamma_E + O(\varepsilon) = \frac{1}{\varepsilon} + \psi(1) + O(\varepsilon) \quad (66)$$

Near $z = -1$. Use the same rule once more, $\Gamma(\varepsilon) = (\varepsilon - 1)\Gamma(\varepsilon - 1)$, so

$$\Gamma(-1 + \varepsilon) = \frac{\Gamma(\varepsilon)}{\varepsilon - 1} = -\Gamma(\varepsilon) \cdot \frac{1}{1 - \varepsilon} = -\left(\frac{1}{\varepsilon} - \gamma_E + \dots\right)(1 + \varepsilon + \dots) \quad (67)$$

Multiply out and keep terms up to ε^0 : $(1/\varepsilon)(1) + (1/\varepsilon)(\varepsilon) - \gamma_E = 1/\varepsilon + 1 - \gamma_E$. So

$$\Gamma(-1 + \varepsilon) = -\left(\frac{1}{\varepsilon} + 1 - \gamma_E\right) + O(\varepsilon) \quad (68)$$

Why $1 - \gamma_E = \psi(2)$. Take the log of $\Gamma(z+1) = z\Gamma(z)$ and differentiate: $\psi(z+1) = 1/z + \psi(z)$. At $z = 1$: $\psi(2) = 1 + \psi(1) = 1 - \gamma_E$.

RESULT: EXPANSIONS WE USE AGAIN AND AGAIN

$$\Gamma(\varepsilon) = \frac{1}{\varepsilon} + \psi(1) + O(\varepsilon), \quad \Gamma(-1 + \varepsilon) = -\left(\frac{1}{\varepsilon} + \psi(2)\right) + O(\varepsilon), \quad a^\varepsilon = 1 + \varepsilon \ln a + O(\varepsilon^2) \quad (69)$$

The last one is just $a^\varepsilon = e^{\varepsilon \ln a}$ expanded. When a pole meets it,

$$\left(\frac{1}{\varepsilon} + c\right)(1 + \varepsilon \ln a) = \frac{1}{\varepsilon} + c + \ln a + O(\varepsilon) \quad (70)$$

so the pole leaves behind a finite $\ln a$. This is how every logarithm of a mass or momentum in a loop amplitude is born.

4.5 How bad can the poles get?

How strong can the poles be? The answer depends on the number of loops L . A one-loop integral in dimensional regularisation has at most $1/\varepsilon$ from the UV region. Each extra loop can add one more UV logarithm, so at L loops UV poles go up to $1/\varepsilon^L$. If there are massless particles, each loop can also give a soft and a collinear divergence at the same time, so with IR divergences the poles go up to $1/\varepsilon^{2L}$ (for example the one-loop massless box has a $1/\varepsilon^2$). In the massive examples of these notes only UV poles appear and we will always find at most one $1/\varepsilon$.

4.6 Soft and collinear poles

The UV pole comes from **large** loop momentum. Massless particles bring two more kinds of pole, both from **small** or special loop momenta. They are called infrared (IR) poles.

Soft pole. Let a massless particle run in the loop, a photon or a gluon with momentum k . Its propagator is $1/k^2$ and it blows up when the photon becomes very soft, $k \rightarrow 0$. In $D = 4$ the integral then diverges like a logarithm. In words, an electron can emit or absorb a photon of almost zero energy and this costs almost nothing. In dimensional regularisation this region gives a $1/\varepsilon$ pole. Smirnov's example (2006, Sect. 2.2, eq. (2.19)) is a massless line next to an on-shell massive line. At small k the integrand behaves like $1/[4k^2(q \cdot k)^2]$ and the integral diverges.

Collinear pole. Let an external line be massless and on shell, $p^2 = 0$, for example a massless quark. The propagator next to it is

$$\frac{1}{(p-k)^2} = \frac{1}{k^2 - 2p \cdot k} \quad (71)$$

and $(p-k)^2$ vanishes at $k \rightarrow 0$. It also vanishes whenever k is parallel to p , $k = xp$. In words, a massless particle can split into two massless particles moving in exactly the same direction and this also costs no energy. This region gives another $1/\varepsilon$. Rauh (Section 2.1) shows how it appears. After the energy integral the remaining angular integral diverges as $\cos \theta \rightarrow 1$, where θ is the angle between k and p . In D dimensions the Jacobian carries a factor $\sin^{D-4} \theta$ that regulates it. If the external particle had a mass m , the splitting would cost energy and the divergence would become a $\ln m$ instead of a pole.

Both at once. When the internal particle is soft **and** collinear the two regions overlap and one loop gives up to $1/\varepsilon^2$. This is the $1/\varepsilon^{2L}$ of the previous section. Two examples, computed with feynsage in the Package-X normalisation of Section 7:

- the massless triangle with two on-shell massless legs, $p_1^2 = p_2^2 = 0$ and $(p_1 - p_2)^2 = s$, has both:

$$C_0 = \frac{1}{s} \left[\frac{1}{\varepsilon^2} + \frac{1}{\varepsilon} \ln \left(\frac{\mu^2}{-s} \right) + \frac{1}{2} \ln^2 \left(\frac{\mu^2}{-s} \right) - \frac{\pi^2}{12} \right] \quad (72)$$

($-s$ means $-s - i0$, so above $s = 0$ the logarithms give an imaginary part). This is Smirnov's example of a collinear divergence (2006, Fig. 2.4 and eq. (2.21)) and Rauh's one-loop vertex (his eq. (2.10)). Rauh's eq. (2.13) gives the same result at $\mu = 1$. His measure leaves out the factor $e^{\varepsilon\gamma_E}$, so in his formula it appears in front as $e^{-\gamma_E\varepsilon}$

- the QED vertex triangle, two on-shell electrons of mass m with a massless photon between them (Smirnov's Fig. 4.5, masses $m, m, 0$ and $p_1^2 = p_2^2 = m^2$), has only the soft pole. The electron mass removes the collinear one. At $m = 1$ and $q^2 = -3$ feynsage gives 0.3419036 for the coefficient of $1/\varepsilon$ and no $1/\varepsilon^2$ (this number is our computation, not Smirnov's).

At two loops the worst case is $1/\varepsilon^4$. Smirnov (2006) gives the planar two-loop vertex with two massless legs, whose expansion starts with $1/(4\varepsilon^4)$ (his eq. (5.45)). This is the pole found by sector decomposition in Section 21.

PHYSICAL PICTURE: UV AND IR POLES ARE REMOVED IN DIFFERENT WAYS

- A UV pole comes from $k \rightarrow \infty$. It is removed by **renormalisation**, a counterterm in the Lagrangian.
- A soft or collinear pole comes from $k \rightarrow 0$ or $k \parallel p$. Renormalisation does **not** remove it. It cancels only when the loop is added to the process with one more real photon that is too soft or too collinear for the detector to see. This is the theorem of Bloch and Nordsieck (soft photons in QED) and of Kinoshita, Lee and Nauenberg (soft and collinear in general).

HOW TO TELL WHICH POLE IS WHICH

Give the photon a small mass λ . A soft pole turns into a logarithm of λ . For the QED triangle above C_0 falls by 1.574 each time λ shrinks by a factor of 10 and $0.3419 \times \ln(10^2) = 1.574$, so the soft $1/\varepsilon$ is exactly that coefficient times $\ln \lambda^2$. In the same way, giving the external fermion a small mass m turns a collinear pole into a logarithm of m .

5 Dots on lines and integration by parts

5.1 What a dot on a line means

In the lecture and in almost every paper on Feynman integrals, you will see diagrams with little dots drawn on the lines. The rule is:

RULE: THE DOT

A line with k dots stands for the propagator raised to the power $k + 1$:

$$\begin{aligned} \text{---} &= \frac{1}{\ell^2 + m^2} & \text{---} \bullet &= \frac{1}{(\ell^2 + m^2)^2} \\ \text{---} \bullet \bullet &= \frac{1}{(\ell^2 + m^2)^3} \end{aligned}$$

So the tadpole with n propagator powers, $T(n)$, is drawn as a loop with $n - 1$ dots. The dot does not change the momentum on the line, only the power.

A dot is Naruto's shadow clone jutsu for propagators: the same line, copied, with the same momentum. Three dots, three extra clones.

Why would anyone want higher powers? There are three reasons.

1. A dot is a mass derivative. Since

$$-\frac{\partial}{\partial m^2} \frac{1}{(\ell^2 + m^2)^a} = \frac{a}{(\ell^2 + m^2)^{a+1}} \quad (73)$$

differentiating a diagram with respect to the mass of one line puts a dot on that line (times the old power a):

$$-\frac{\partial}{\partial m^2} \text{---} = \text{---} \bullet \text{---}, \quad -\frac{\partial}{\partial m^2} \text{---} \bigcirc \text{---} = \text{---} \bigcirc \bullet \text{---} \quad (74)$$

2. A dot is a zero-momentum insertion. Write the mass term as a small change, $m^2 = m_0^2 + \delta$. A geometric series gives

$$\frac{1}{\ell^2 + m_0^2 + \delta} = \frac{1}{\ell^2 + m_0^2} - \frac{\delta}{(\ell^2 + m_0^2)^2} + \frac{\delta^2}{(\ell^2 + m_0^2)^3} - \dots \quad (75)$$

Each term is the same line with one more “mass insertion” (a two-point vertex that carries no momentum), drawn as a cross. Two propagators with the same momentum on either side of a cross is exactly one propagator squared:

$$\text{---} \times \text{---} = (-\delta) \times \text{---} \bullet \text{---}, \quad \text{---} \times \times \text{---} = (-\delta)^2 \times \text{---} \bullet \bullet \text{---} \quad (76)$$

So $n-1$ such zero-momentum insertions on the tadpole loop give n propagators with the same momentum, which is $T(n)$. In $\lambda\phi^4$ theory an insertion can also be a vertex where a **pair** of extra external legs with zero momentum attaches to the loop, since that vertex acts on the loop line like a two-point vertex. This is the meaning of “a tadpole with extra external legs” in the lecture: $n-1$ attachments, n propagator powers.

3. IBP produces dots. Differentiating $1/D_i$ with respect to the loop momentum gives $1/D_i^2$. So integration by parts (below) always mixes integrals with different numbers of dots.

5.2 Raising powers by differentiation

DERIVATION, STEP BY STEP

Differentiate $I_1(m^2) = \int d^D \ell / (\ell^2 + m^2)$ once with respect to m^2 . The derivative can go inside the integral:

$$\frac{dI_1}{dm^2} = \int d^D \ell \frac{\partial}{\partial m^2} (\ell^2 + m^2)^{-1} = - \int \frac{d^D \ell}{(\ell^2 + m^2)^2} = -I_2 \quad (77)$$

Each further derivative lowers the power by one more and brings down the old power as a factor:

$$\left(\frac{d}{dm^2} \right)^{n-1} (\ell^2 + m^2)^{-1} = (-1)(-2)\dots(-(n-1))(\ell^2 + m^2)^{-n} = (-1)^{n-1}(n-1)!(\ell^2 + m^2)^{-n} \quad (78)$$

Dividing by $(-1)^{n-1}(n-1)!$:

$$I_n(m^2) = \frac{(-1)^{n-1}}{(n-1)!} \left(\frac{d}{dm^2} \right)^{n-1} I_1(m^2) \quad (79)$$

Check with Equation 63. There $I_1 = \pi^{D/2} \Gamma(1 - D/2) (m^2)^{D/2-1}$. Each derivative of a power brings down the exponent and lowers it by one:

$$\left(\frac{d}{dm^2} \right)^{n-1} (m^2)^{D/2-1} = \left(\frac{D}{2} - 1 \right) \left(\frac{D}{2} - 2 \right) \dots \left(\frac{D}{2} - n + 1 \right) (m^2)^{D/2-n} \quad (80)$$

There are $n-1$ factors. Pull a minus sign out of each, $(D/2 - k) = -(k - D/2)$:

$$\left(\frac{D}{2} - 1 \right) \dots \left(\frac{D}{2} - n + 1 \right) = (-1)^{n-1} \left(1 - \frac{D}{2} \right) \left(2 - \frac{D}{2} \right) \dots \left(n - 1 - \frac{D}{2} \right) \quad (81)$$

Now use $\Gamma(z+1) = z\Gamma(z)$ repeatedly, starting from $z = 1 - D/2$:

$$\Gamma\left(1 - \frac{D}{2}\right) \left(1 - \frac{D}{2}\right) \left(2 - \frac{D}{2}\right) \dots \left(n - 1 - \frac{D}{2}\right) = \Gamma\left(n - \frac{D}{2}\right) \quad (82)$$

So $(-1)^{n-1}/(n-1)! (d/dm^2)^{n-1} I_1 = \pi^{D/2} \Gamma(n - D/2) (m^2)^{D/2-n} / (n-1)!$. The two signs $(-1)^{n-1}$ cancel and since $(n-1)! = \Gamma(n)$ this is I_n .

RESULT: DOTTED TADPOLES FROM THE PLAIN ONE

$$I_n(m^2) = \frac{(-1)^{n-1}}{(n-1)!} \left(\frac{d}{dm^2} \right)^{n-1} I_1(m^2), \quad \text{diagram} = \frac{1}{2} \left(\frac{d}{dm^2} \right)^2 \text{diagram} \quad (83)$$

MATHEMATICA CHECK: THE DERIVATIVE FORMULA

Raising the power of a propagator = differentiating in m^2 . a tempting but wrong variant. $(-1)^{n-1}/(n-1)!$

In [1]: $1!(d/dm^2)^{n-1} I_1$ divided by I_n , for $n = 2, \dots, 5$

```
I1 = Pi^(D/2) Gamma[1 - D/2] (m2)^(D/2 - 1);
Itad[n_] := Pi^(D/2) Gamma[n - D/2]/Gamma[n] (m2)^(D/2 - n);
correct[n_] := (-1)^(n - 1)/(n - 1)! D[I1, {m2, n - 1}];
variant[n_] := (-1)^n/n! D[I1, {m2, n - 1}];
Table[FullSimplify[correct[n]/Itad[n]], {n, 2, 5}]
```

Out[1]: {1, 1, 1, 1}

the variant $(-1)^n/n!(d/dm^2)^{n-1} I_1$ divided by I_n

In [2]: `Table[FullSimplify[variant[n]/Itad[n]], {n, 2, 5}]`

Out[2]: $\{-\frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}, -\frac{1}{5}\}$

The first line checks our formula for $n = 2, \dots, 5$ (ratio 1). The second line tries the variant $(-1)^n/n!$ and shows it is off by the factor $-1/n$, so the signs and factorials must be as above. The message. If we know one integral as a function of the mass, we know the whole tower of dotted integrals. This is the simplest example of the idea behind the method of differential equations.

5.3 Integration by parts (IBP)**5.3.1 Which vector do we put in?**

The divergence theorem (Section 2.6) says that the integral of a total derivative vanishes. In momentum space the derivative is $\partial/\partial\ell^\mu$. It carries an index μ , so on its own it does not give a number. We must hit it on something that also carries an index, a vector v^μ and sum over μ :

$$\int d^D\ell \frac{\partial}{\partial\ell^\mu} (v^\mu \text{integrand}) = 0 \quad (84)$$

Why is this zero? The integrand $v^\mu \times (\text{integrand})$ is a vector field in momentum space and we are integrating its divergence over all of momentum space. By the divergence theorem (Section 2.6) that equals the flux through a sphere of radius $R \rightarrow \infty$. The flux behaves like $R^{D-1} \times R^{1-2n} = R^{D-2n}$ (area of the sphere times the size of the field there), so for small enough D it goes to zero. This is the worked example of Section 2.6. Both sides are analytic in D , so the result stays zero after continuing to any D . So in dimensional regularisation there is effectively no boundary at infinity.

Which vectors are available? Only the momenta of the problem. For the tadpole there is no external momentum at all, so the only choice is $v^\mu = \ell^\mu$ and we get **one** identity. For the bubble there is also the external momentum p , so there are **two** choices, $v^\mu = \ell^\mu$ and $v^\mu = p^\mu$. In general, with L loop momenta and E independent external momenta, there are L derivatives and $L + E$ vectors, so $L(L + E)$ identities for each integral. This is why the board kept writing things like $\ell^\mu \partial/\partial\ell^\mu$ and $k^\mu \partial/\partial k^\mu$: every momentum in the problem gives one vector to use.

IBP is One For All from My Hero Academia: the power (information) of every integral is passed on to all the others, until only a few masters are left holding it.

5.3.2 The tadpole identity**DERIVATION: THE IBP IDENTITY FOR THE TADPOLE**

Take $v^\mu = \ell^\mu$ and the integrand $(\ell^2 + m^2)^{-n}$. By the product rule,

$$\frac{\partial}{\partial \ell^\mu} [\ell^\mu f(\ell)] = \frac{\partial \ell^\mu}{\partial \ell^\mu} f + \ell^\mu \frac{\partial f}{\partial \ell^\mu}, \quad f = (\ell^2 + m^2)^{-n} \quad (85)$$

The first factor is a trace: $\partial \ell^\mu / \partial \ell^\nu = \delta_\nu^\mu$ and setting $\nu = \mu$ and summing gives $\delta_\mu^\mu = D$, the number of dimensions. For the second, $\partial \ell^2 / \partial \ell^\mu = 2\ell_\mu$ and the chain rule gives

$$\frac{\partial f}{\partial \ell^\mu} = -n(\ell^2 + m^2)^{-n-1} \cdot 2\ell_\mu \Rightarrow \ell^\mu \frac{\partial f}{\partial \ell^\mu} = -2n\ell^2(\ell^2 + m^2)^{-n-1} \quad (86)$$

So

$$0 = \int d^D \ell \frac{\partial}{\partial \ell^\mu} \frac{\ell^\mu}{(\ell^2 + m^2)^n} = \int d^D \ell \left[\frac{D}{(\ell^2 + m^2)^n} - \frac{2n\ell^2}{(\ell^2 + m^2)^{n+1}} \right] \quad (87)$$

The second term has ℓ^2 upstairs, which is not a propagator. Add and subtract m^2 : $\ell^2 = (\ell^2 + m^2) - m^2$. Then

$$\frac{2n\ell^2}{(\ell^2 + m^2)^{n+1}} = \frac{2n(\ell^2 + m^2)}{(\ell^2 + m^2)^{n+1}} - \frac{2nm^2}{(\ell^2 + m^2)^{n+1}} = \frac{2n}{(\ell^2 + m^2)^n} - \frac{2nm^2}{(\ell^2 + m^2)^{n+1}} \quad (88)$$

Put this back and integrate term by term. The first term of the bracket gives DI_n . The second gives $-2nI_n + 2nm^2I_{n+1}$:

$$DI_n - 2nI_n + 2nm^2I_{n+1} = 0, \quad \text{that is} \quad (D - 2n)I_n + 2nm^2I_{n+1} = 0 \quad (89)$$

With $D = 2\omega$ this is $2(\omega - n)I_n + 2nm^2I_{n+1} = 0$, the form written in the lecture.

RESULT: TADPOLE IBP, IN FORMULAS AND IN DIAGRAMS

$$I_{n+1} = \frac{2n - D}{2nm^2} I_n, \quad I_2 = -\frac{D - 2}{2m^2} I_1 \quad (90)$$

$$\text{Diagram with a tadpole on a line} = -\frac{D-2}{2m^2} \text{Diagram with a tadpole on a line}, \quad \text{Diagram with two tadpoles on a line} = \frac{4-D}{4m^2} \text{Diagram with a tadpole on a line} = \frac{(D-2)(D-4)}{8m^4} \text{Diagram with a tadpole on a line} \quad (91)$$

Starting from the plain tadpole, the IBP relation gives every dotted one. We say the tadpole family has **one master integral**.

MATHEMATICA CHECK

IBP: $\int d^D \ell \partial / \partial \ell^\mu [\ell^\mu / (\ell^2 + m^2)^n] = 0$. In D dimensions $\partial_\mu \ell^\mu = D$ and $\ell \cdot \partial (\ell^2 + m^2)^{-n} = -2n\ell^2 / (\ell^2 + m^2)^{n+1}$; then $\ell^2 = (\ell^2 + m^2) - m^2$. write $\ell^2 = (\ell^2 + m^2) - m^2$. $(D - 2n)I_n + 2nm^2I_{n+1}$ with the closed form of I_n

In [1]:

```
Itad[n_] := Pi^(D/2) Gamma[n - D/2]/Gamma[n] (m2)^(D/2 - n);
ibp = D Itad[n] - 2 n (Itad[n] - m2 Itad[n + 1]);
FullSimplify[ibp]
```

Out[1]: 0

In [2]: $\frac{I_{n+1}}{I_n}$
FullSimplify[Itad[n + 1]/Itad[n]]

Out[2]: $2 - \frac{D}{2m^2}$

In [3]: $\frac{I_2}{I_1}$
FullSimplify[Itad[2]/Itad[1]]

Out[3]: $-\frac{D-2}{2m^2}$

PHYSICAL PICTURE: WHAT IBP BUYS US

IBP never computes an integral. It only says that many integrals are linear combinations of a few others (the master integrals), with coefficients that are rational functions of D , masses and momenta. For big multi-loop problems there are millions of integrals and only tens of masters. Programs like FIRE, LiteRed and Kira do nothing but solve these linear systems, with a lot of care for speed. The hard analytic work is then only needed for the masters.

5.4 The IBP relation as a difference equation

Argeri and Mastrolia (Section 5 of their review) take the tadpole IBP relation seriously as an equation in the integer n and solve it without knowing the answer. It is a good toy model for what happens in real calculations. Set $m = 1$ and define

$$U(n) = \pi^{-D/2} \int \frac{d^D \ell}{(\ell^2 + 1)^n} = T(n)|_{m=1} \quad (92)$$

The IBP identity Equation 90, divided by 2 and written for U , is

$$-\left(n - \frac{D}{2}\right)U(n) + nU(n+1) = 0 \quad (93)$$

In diagrams (with $m = 1$), for the first two values of n :

$$n = 1: -\left(1 - \frac{D}{2}\right) \text{ (tadpole)} + \text{ (tadpole with one dot)} = 0, \quad n = 2: -\left(2 - \frac{D}{2}\right) \text{ (tadpole with one dot)} + 2 \text{ (tadpole with two dots)} = 0 \quad (94)$$

Tokyo Ghoul: "What's 1000 minus 7?" Kaneki answers one step at a time, 993, 986, ... A difference equation works the same way: from $U(n)$ you get $U(n+1)$, then $U(n+2)$, forever. It just never tells you where to start.

This is a first order linear difference equation. It fixes $U(n+1)$ from $U(n)$ but leaves one constant free. Solving it by hand is easy. Write it as $U(n+1) = (n - D/2)/n U(n)$ and iterate from $n = 1$,

$$U(n) = \frac{1 - \frac{D}{2}}{1} \cdot \frac{2 - \frac{D}{2}}{2} \dots \frac{n - 1 - \frac{D}{2}}{n - 1} U(1) = \frac{(1 - D/2)_{n-1}}{(n-1)!} U(1) \quad (95)$$

where $(a)_k = a(a+1)\dots(a+k-1) = \Gamma(a+k)/\Gamma(a)$ is the Pochhammer symbol. So $U(n) = \Gamma(n - D/2)/(\Gamma(1 - D/2)\Gamma(n)) U(1)$. Mathematica's `RSolve` returns this, with C_1 in place of $U(1)$. The free constant is the one master integral and it must be fixed by something else.

Argeri and Mastrolia instead turn the difference equation into a differential equation and fix the constant from the behaviour at large n . The same trick works for harder families, so it is good to see it once in a simple case.

DERIVATION: FROM A DIFFERENCE EQUATION TO A DIFFERENTIAL EQUATION

Ansatz. Look for $U(n)$ in the form

$$U(n) = \int_0^1 dt t^{n-1} v(t) \quad (96)$$

for some unknown weight $v(t)$. This is a Mellin transform. (With $t = e^{-\tau}$ it becomes $\int_0^\infty d\tau e^{-n\tau} v(e^{-\tau})$, a Laplace transform.) Then $U(n+1) = \int_0^1 t^n v dt$ and

$$nU(n+1) - nU(n) = n \int_0^1 dt (t^n - t^{n-1})v(t) = n \int_0^1 dt t^{n-1} (t-1)v(t) \quad (97)$$

Integrate by parts in t . Use $nt^{n-1} = dt^n/dt$, with $u = (t-1)v(t)$:

$$n \int_0^1 t^{n-1} (t-1)v dt = [t^n (t-1)v(t)]_0^1 - \int_0^1 t^n \frac{d}{dt} [(t-1)v(t)] dt \quad (98)$$

The boundary term is zero at $t = 1$ as long as $(1 - t)v(t) \rightarrow 0$ there, so we do not need $v(1) = 0$. (The solution found below behaves like $(1 - t)^{D/2-1}$, so $(1 - t)v \rightarrow 0$ for $D > 0$.) At $t = 0$ it is zero when $t^n v(t) \rightarrow 0$, which we will check at the end. The derivative is $d/dt[(t - 1)v] = v + (t - 1)v'$, so

$$nU(n + 1) - nU(n) = - \int_0^1 t^n [v + (t - 1)v'] dt = \int_0^1 t^{n-1} [-tv - t(t - 1)v'] dt \quad (99)$$

Put it in the difference equation. $-(n - D/2)U(n) + nU(n + 1) = [nU(n + 1) - nU(n)] + (D/2)U(n)$ and $(D/2)U(n) = \int_0^1 t^{n-1} (D/2)v dt$. Adding,

$$0 = \int_0^1 dt t^{n-1} \left[-t(t - 1)v'(t) + \left(\frac{D}{2} - t \right) v(t) \right] \quad (100)$$

For all n . Call the bracket $g(t)$. We have $\int_0^1 t^{n-1} g(t) dt = 0$ for every $n = 1, 2, 3, \dots$, so also $\int_0^1 P(t)g(t) dt = 0$ for every polynomial P . Polynomials can approximate any continuous function on $[0, 1]$ as closely as we like (Weierstrass), so $\int_0^1 h(t)g(t) dt = 0$ for every continuous h . A function whose integral against every continuous h vanishes is zero (take h to be a narrow bump around any point). So $g = 0$:

$$-t(t - 1)v'(t) + \left(\frac{D}{2} - t \right) v(t) = 0 \Rightarrow \frac{v'}{v} = \frac{D/2 - t}{t(t - 1)} \quad (101)$$

Partial fractions. Write $(D/2 - t)/(t(t - 1)) = A/t + B/(t - 1)$. Multiply by $t(t - 1)$: $D/2 - t = A(t - 1) + Bt$. At $t = 0$: $D/2 = -A$, so $A = -D/2$. At $t = 1$: $D/2 - 1 = B$. Integrating $v'/v = (\ln v)'$,

$$\ln v = -\frac{D}{2} \ln t + \left(\frac{D}{2} - 1 \right) \ln(1 - t) + \text{const} \Rightarrow v(t) = v_0 t^{-D/2} (1 - t)^{D/2-1} \quad (102)$$

Back to $U(n)$. Multiply by t^{n-1} and integrate. The powers combine to $t^{n-1-D/2}$ and the integral is a Beta function with $a = n - D/2$, $b = D/2$:

$$U(n) = v_0 \int_0^1 t^{n-1-D/2} (1 - t)^{D/2-1} dt = v_0 B\left(n - \frac{D}{2}, \frac{D}{2}\right) = v_0 \frac{\Gamma(n - D/2)\Gamma(D/2)}{\Gamma(n)} \quad (103)$$

Now check the boundary term at $t = 0$: $t^n v \sim t^{n-D/2}$ goes to zero when $n > D/2$, the same condition under which the Beta integral converges.

Fix v_0 from large n . For large n the factor $(1 + \ell^2)^{-n} = e^{-n \ln(1 + \ell^2)}$ is tiny unless ℓ is small and for small ℓ , $\ln(1 + \ell^2) \approx \ell^2$. So

$$U(n) \approx \pi^{-D/2} \int d^D \ell e^{-n \ell^2} = \pi^{-D/2} \left(\frac{\pi}{n} \right)^{D/2} = n^{-D/2} \quad (104)$$

using the Gaussian integral with $x = \sqrt{n}\ell$. On the other side we need $\Gamma(n + a)/\Gamma(n) \approx n^a$ for large n and any fixed a . It follows from Stirling's formula $\ln \Gamma(z) \approx (z - 1/2) \ln z - z + \frac{1}{2} \ln(2\pi)$: the difference $\ln \Gamma(n + a) - \ln \Gamma(n)$ is $a \ln n + O(1/n)$. (For an integer $a = -k$ you can see it directly: $1/((n - 1)(n - 2) \dots (n - k)) \approx n^{-k}$.) With $a = -D/2$, $\Gamma(n - D/2)/\Gamma(n) \approx n^{-D/2}$. So the Beta answer behaves as $v_0 \Gamma(D/2) n^{-D/2}$. Matching, $v_0 = 1/\Gamma(D/2)$ and

$$U(n) = \frac{\Gamma(n - D/2)}{\Gamma(n)} \quad (105)$$

which is Equation 63 again, found without doing a single loop integral.

PHYSICAL PICTURE: THE TWO ENDS OF THE t INTEGRAL

The condition $n > D/2$ at $t = 0$ is the UV convergence condition $D < 2n$ from Section 3.3. The two conditions agree because large n probes small loop momentum and the region $t \rightarrow 0$ of the Mellin integral is where the large-momentum behaviour lives. Figure 7 shows the weight $t^{n-1}v(t)$ for $n = 3$. As D grows towards $2n = 6$ the weight piles up at $t = 0$ and at $D = 6$ its area becomes infinite. That is the UV divergence of $T(3)$ in six dimensions, seen from a completely different angle.

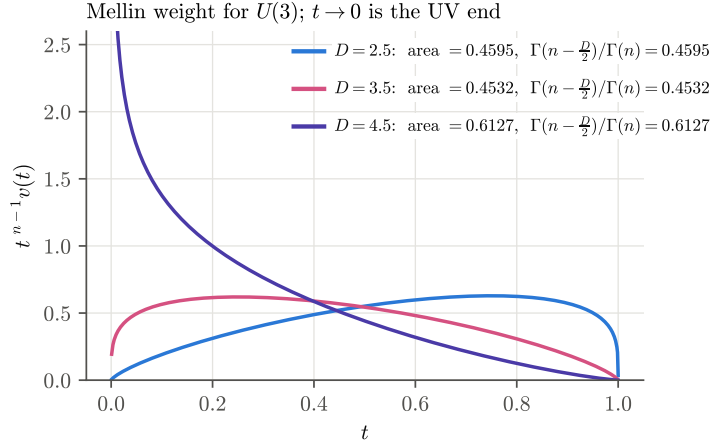


Figure 7: The weight $t^{n-1}v(t)$ with $v(t) = t^{-D/2}(1-t)^{D/2-1}/\Gamma(D/2)$ for $n = 3$. The area under each curve equals $\Gamma(n - D/2)/\Gamma(n)$, as printed in the legend.

MATHEMATICA CHECK: RSOLVE, DSOLVE AND THE BOUNDARY CONDITION

Argeri-Mastrolia: $U(n) = \pi^{-D/2} \int d^D \ell / (\ell^2 + 1)^n$. solve $-(n - D/2)U(n) + nU(n+1) = 0$

In [1]: `sol = RSolve[-(n - D/2) U[n] + n U[n + 1] == 0, U[n], n];`
`sol`

Out[1]: $\left\{ \left\{ U(n) \rightarrow \frac{c_1 \left(1 - \frac{D}{2}\right)_{n-1}}{(1)_{n-1}} \right\} \right\}$

Mellin ansatz $U(n) = \int_0^1 t^{n-1}v(t) dt$ gives an ODE for v . solve $-t(t-1)v'(t) + (D/2 - t)v(t) = 0$

In [2]: `vsol = DSolve[-t (t - 1) v'[t] + (D/2 - t) v[t] == 0, v[t], t];`
`vsol`

Out[2]: $\left\{ \left\{ v(t) \rightarrow c_1 e^{\frac{1}{2}((D-2)\log(1-t) - D\log(n))} \right\} \right\}$

$U(n)$ from $v(t) = v_0 t^{-D/2}(1-t)^{D/2-1}$

In [3]: `Un = v0 Integrate[t^(n - 1) t^(-D/2) (1 - t)^(D/2 - 1), {t, 0, 1},`
`Assumptions -> 0 < D < 2 n];`
`Un`

Out[3]: $\frac{v_0 \Gamma\left(\frac{D}{2}\right) \Gamma\left(n - \frac{D}{2}\right)}{\Gamma(n)}$

fix v_0 from large n : $U(n) \rightarrow n^{-D/2}$, $\lim_{n \rightarrow \infty} n^{D/2}U(n)$

In [4]: `Limit[n^(D/2) Un, n -> Infinity, Assumptions -> D > 0]`

Out[4]: $v_0 \Gamma\left(\frac{D}{2}\right)$

6 The mass correction in $\lambda\phi^4$, in both signatures

6.1 Two master formulas (Ramond's Appendix B)

Two formulas from Appendix B of Ramond's book ("Integration over arbitrary dimensions", his eqs. (B.12) and (B.13)) do most of the work. In our notation:

$$\int \frac{d^D \ell}{(\ell^2 + m^2)^A} = \pi^{D/2} \frac{\Gamma(A - D/2)}{\Gamma(A)} \frac{1}{(m^2)^{A-D/2}} \quad (106)$$

$$\int \frac{d^D \ell}{(\ell^2 + 2\ell \cdot p + b^2)^A} = \pi^{D/2} \frac{\Gamma(A - D/2)}{\Gamma(A)} \frac{1}{(b^2 - p^2)^{A-D/2}} \quad (107)$$

The first is Equation 63 with $n = A$. The second follows from the first in one line. Complete the square, $\ell^2 + 2\ell \cdot p + b^2 = (\ell + p)^2 + (b^2 - p^2)$ and shift $\ell \rightarrow \ell - p$. The shift is allowed because the integral is (dimensionally) regularised and $d^D \ell$ does not change under a shift. The same shift gives the tensor version,

$$\int d^D \ell \frac{\ell_\mu}{(\ell^2 + 2\ell \cdot p + b^2)^A} = -p_\mu \pi^{D/2} \frac{\Gamma(A - D/2)}{\Gamma(A)} \frac{1}{(b^2 - p^2)^{A-D/2}} \quad (108)$$

(Ramond's (B.14)) because after $\ell \rightarrow \ell - p$ the numerator ℓ_μ becomes $\ell_\mu - p_\mu$ and the odd part integrates to zero. (One can also get it by differentiating Equation 107 written for the power $A - 1$ with respect to p^μ .) All of these are Euclidean formulas. Here " $\ell \cdot p$ " is the Euclidean dot product.

6.2 Bare mass, physical mass and counterterms

The word “bare” appears again and again from here on, so let us be precise about it. Coleman (Lectures, Section 10.1) opens the subject with a story that makes it obvious.

Green's ping-pong ball. Hold a light ball under water and let go. Its mass is m_0 and it displaces water of mass $\rho V = 10m_0$. The forces are gravity $m_0 g$ down and buoyancy $10m_0 g$ up, so Newton's law with mass m_0 predicts an upward acceleration of $9g$. Nothing like that happens. George Green found the right answer in 1834 and Stokes explained it about ten years later: when the ball moves, the water around it has to move too, so pushing the ball means pushing some water along with it. The ball behaves as if its mass were

$$m = m_0 + \frac{1}{2}\rho V = m_0 + 5m_0 = 6m_0 \quad (109)$$

and its acceleration is $9m_0 g / (6m_0) = 3g/2$. If the whole universe were filled with water, nobody could ever take the ball out and weigh it dry. The only mass anyone could measure would be m . The number m_0 would be a parameter in the equations, never a measured quantity.

Lorentz's electron. The same thing happens to a charged sphere of radius r . Its Coulomb field carries energy ke^2/r and when you push the electron you push its field too. A scale weighs the electron together with its field: $m = m_0 + ke^2/(rc^2)$. For a point particle, $r \rightarrow 0$, the correction is infinite. The electron's measured mass is perfectly finite, so m_0 must be “minus infinity plus the measured mass” and that is no contradiction, because m_0 is never measured.

The Survey Corps never sees a Titan without its steam. In quantum field theory we never see a particle without its cloud of virtual quanta. The “bare” particle is a character who never appears on screen.

In our theory. The Lagrangian of Section 3 contains two numbers, which we now call m_0 and λ_0 :

$$\mathcal{L} = \frac{1}{2}(\partial\varphi)^2 - \frac{1}{2}m_0^2\varphi^2 - \frac{\lambda_0}{4!}\varphi^4 \quad (110)$$

- The **bare mass** m_0 is the coefficient in the Lagrangian. It would be the particle's mass if the interaction were switched off, like the dry mass of the ping-pong ball.
- The **physical mass** m_{phys} is what an experiment measures: the energy of the particle at rest. In the language of Feynman diagrams it is the position of the pole of the full propagator, $p^2 = m_{\text{phys}}^2$.
- They differ by the loops. At one loop the tadpole shifts the mass: $m_{\text{phys}}^2 = m_0^2 - \mathcal{T}$, where $\mathcal{T}$ is the value of the tadpole diagram, computed in the next section. The diagram has the pole $+\lambda m^2/(32\pi^2\epsilon)$, so the mass shift $-\mathcal{T}$ has the pole $-\lambda m^2/(32\pi^2\epsilon)$ and $m_0^2 = m_{\text{phys}}^2 + \lambda m^2/(32\pi^2\epsilon) + (\text{finite})$. Like the point electron, the bare mass blows up when the regulator is removed ($\epsilon \rightarrow 0$) and like the point electron, this does no harm, because m_0 is never measured.
- The same holds for the coupling. The bare coupling λ_0 in the Lagrangian differs from the coupling measured in a scattering experiment by the loop corrections of Section 13.

Counterterms. In practice we do not keep writing m_0 . We write $m_0^2 = m^2 + \delta m^2$, where m is a finite, measurable (renormalised) mass and δm^2 , the **counterterm**, is chosen to cancel the pole of the loop. The counterterm is part of the Lagrangian, so it is drawn as a vertex (a cross on the line) and it enters diagrams like any other term. The simplest choice, the $\overline{\text{MS}}$ scheme, puts into δm^2 only the pole together with the constants

$-\gamma_E + \ln 4\pi$ that always come with it. Then the renormalised mass m depends on the arbitrary scale μ . Point 2 of the box after the next worked example shows how. A slower first look at renormalisation, from the point of view of effective field theory, is in my blog post [Effective Field Theory, Starting From the Propagator You Just Learned](#).

6.3 Worked example A: Euclidean space

Now the mass correction of $\lambda\varphi^4$ theory, done slowly in Euclidean space.

DERIVATION, STEP BY STEP

The diagram. In pictures, $\mathcal{T} =$ .

Setting up. With the Euclidean rules (vertex $-\lambda$, propagator $1/(\ell^2 + m^2)$, symmetry factor $1/2$) and the coupling written as $\lambda(\mu^2)^{2-\omega}$, the value of the tadpole diagram in $D = 2\omega$ is

$$\mathcal{T} = \frac{1}{2}(-\lambda)(\mu^2)^{2-\omega} \int \frac{d^{2\omega}\ell}{(2\pi)^{2\omega}} \frac{1}{\ell^2 + m^2} \quad (111)$$

Do the integral with Equation 106, $A = 1$:

$$\int \frac{d^{2\omega}\ell}{(2\pi)^{2\omega}} \frac{1}{\ell^2 + m^2} = \frac{\pi^\omega}{(2\pi)^{2\omega}} \Gamma(1-\omega)(m^2)^{\omega-1} = \frac{\Gamma(1-\omega)}{(4\pi)^\omega} (m^2)^{\omega-1} \quad (112)$$

Pull out the four-dimensional factors. Write $(4\pi)^{-\omega} = (4\pi)^{-2}(4\pi)^{2-\omega}$ and $(m^2)^{\omega-1} = m^2(m^2)^{\omega-2}$:

$$\mathcal{T} = -\frac{\lambda m^2}{2(4\pi)^2} \left(\frac{4\pi\mu^2}{m^2} \right)^{2-\omega} \Gamma(1-\omega) \quad (113)$$

Expand near $\omega = 2$. Put $\omega = 2 - \varepsilon$, so $2 - \omega = \varepsilon$ and $1 - \omega = -1 + \varepsilon$. From Equation 69,

$$\Gamma(-1 + \varepsilon) = -\left(\frac{1}{\varepsilon} + \psi(2)\right) + O(\varepsilon), \quad \left(\frac{4\pi\mu^2}{m^2}\right)^\varepsilon = 1 + \varepsilon \ln \frac{4\pi\mu^2}{m^2} + O(\varepsilon^2) \quad (114)$$

Multiply the two, keeping everything up to ε^0 :

$$-\left(\frac{1}{\varepsilon} + \psi(2)\right) \left(1 + \varepsilon \ln \frac{4\pi\mu^2}{m^2}\right) = -\left(\frac{1}{\varepsilon} + \psi(2) + \ln \frac{4\pi\mu^2}{m^2}\right) + O(\varepsilon) \quad (115)$$

(The cross term $(1/\varepsilon) \cdot \varepsilon \ln(\dots)$ is the finite logarithm.) The prefactor is $-\lambda m^2/(2(4\pi)^2) = -\lambda m^2/(32\pi^2)$ and minus times minus is plus:

$$\mathcal{T} = \frac{\lambda m^2}{32\pi^2} \left[\frac{1}{2-\omega} + \psi(2) + \ln \frac{4\pi\mu^2}{m^2} + O(2-\omega) \right], \quad \psi(2) = 1 - \gamma_E \quad (116)$$

MATHEMATICA CHECK: THE ε EXPANSION

$\lambda\varphi^4$ tadpole, measure $\mu^{2(2-\omega)} d^{2\omega}\ell/(2\pi)^{2\omega}$, $D = 2\omega = 4 - 2\varepsilon$. expansion in ε

```
In [1]: tad = -lam/2 mu2^(2 - w) Gamma[1 - w]/(4 Pi)^w m2^(w - 1);
exp = Normal[Series[tad /. w -> 2 - eps, {eps, 0, 0}]];
Collect[Expand[PowerExpand[exp]], {eps, lam}, Simplify]
```

```
Out[1]: (lambda m^2)/(32 pi^2 epsilon) - (lambda m^2 (-log(4 pi mu^2) + log(m^2) + gamma - 1))/(32 pi^2)
```

minus Ramond's form $\lambda m^2/(32\pi^2)[1/\varepsilon + \psi(2) + \ln(4\pi\mu^2/m^2)]$

```
In [2]: FullSimplify[exp - lam m2/(32 Pi^2) (1/eps + PolyGamma[2] + Log[4 Pi mu2/m2]),
m2 > 0 && mu2 > 0]
```

```
Out[2]: 0
```

$\psi(2)$
In [3]: PolyGamma[2]
Out[3]: 1 - γ

The Mathematica output writes the finite part as $-(-1 + \gamma_E + \ln m^2 - \ln(4\pi\mu^2))$, which is $1 - \gamma_E + \ln(4\pi\mu^2/m^2) = \psi(2) + \ln(4\pi\mu^2/m^2)$ and the difference with Ramond's form is zero.

PHYSICAL PICTURE: WHAT THE ANSWER SAYS

1. The pole multiplies m^2 . A massless particle stays massless at this order. Dimensional regularisation has thrown away the Λ^2 piece that, with a cutoff, would push the mass up to the cutoff scale. The pole is removed by the mass counterterm $\delta m^2 \propto \lambda m^2/\epsilon$.
2. After removing the pole, the finite part depends on $\ln(\mu^2/m^2)$. Changing μ changes this log and to keep physics fixed the renormalised mass must change with μ . That is the running of the mass. To get the number, note that the mass shift $-\mathcal{T}$ has the pole $-\lambda m^2/(32\pi^2\epsilon)$, so the bare mass must be $m_0^2 = m^2(1 + \lambda/(32\pi^2\epsilon))$ to cancel it, while the bare coupling is $\lambda_0 = \mu^{2\epsilon}\lambda$. The bare quantities do not know about μ . From λ_0 fixed, $\mu d\lambda/d\mu = -2\epsilon\lambda + O(\lambda^2)$. From m_0^2 fixed, $0 = \mu \frac{dm^2}{d\mu} (1 + \frac{\lambda}{32\pi^2\epsilon}) + \frac{m^2}{32\pi^2\epsilon} \mu \frac{d\lambda}{d\mu}$ and inserting $-2\epsilon\lambda$ gives $\mu dm^2/d\mu = \lambda m^2/(16\pi^2)$ at this order.
3. The combination $1/\epsilon - \gamma_E + \ln 4\pi$ always appears together. The $\overline{\text{MS}}$ scheme removes this combination.

6.4 Worked example B: Minkowski space

Now the same integral in Minkowski space, as in Peskin and as in FeynCalc and LoopTools. The standard one-loop functions of FeynCalc and LoopTools are defined with a Minkowski measure and a particular normalisation (the Passarino-Veltman convention),

$$A_0(m^2) = \frac{(2\pi\mu)^{4-D}}{i\pi^2} \int d^D\ell \frac{1}{\ell^2 - m^2 + i\epsilon} \quad (117)$$

The factor $1/(i\pi^2)$ is there to cancel the i from the Wick rotation and the π^2 from the four-dimensional solid angle, so that A_0 is real and simple.

DERIVATION, STEP BY STEP

Wick rotate. By Equation 46 with $n = 1$,

$$\int d^D\ell \frac{1}{\ell^2 - m^2 + i\epsilon} = -i \int \frac{d^D\ell_E}{\ell_E^2 + m^2} = -i\pi^{D/2}\Gamma(1 - D/2)(m^2)^{D/2-1} \quad (118)$$

Put in the prefactor with $D = 4 - 2\epsilon$. Then $(2\pi)^{4-D} = (2\pi)^{2\epsilon}$ and $\pi^{D/2}/\pi^2 = \pi^{-\epsilon}$, so $(2\pi)^{2\epsilon}\pi^{-\epsilon} = (4\pi)^\epsilon$:

$$A_0(m^2) = \frac{-i}{i} (4\pi\mu^2)^\epsilon \Gamma(-1 + \epsilon)(m^2)^{1-\epsilon} = -m^2 \left(\frac{4\pi\mu^2}{m^2} \right)^\epsilon \Gamma(-1 + \epsilon) \quad (119)$$

Expand. $-\Gamma(-1 + \epsilon) = 1/\epsilon + 1 - \gamma_E + O(\epsilon)$ and $(4\pi\mu^2/m^2)^\epsilon = 1 + \epsilon \ln(4\pi) + \epsilon \ln(\mu^2/m^2) + \dots$ Multiply, keep terms to ϵ^0 and collect $1/\epsilon - \gamma_E + \ln 4\pi \equiv \Delta$:

$$A_0(m^2) = m^2 \left(\Delta + 1 - \ln \frac{m^2}{\mu^2} \right) + O(\epsilon), \quad \Delta = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi \quad (120)$$

RESULT: THE TWO EXAMPLES ARE THE SAME NUMBER

Since $\psi(2) = 1 - \gamma_E$, the Euclidean bracket above is $1/\epsilon + 1 - \gamma_E + \ln 4\pi - \ln(m^2/\mu^2) = A_0/m^2$. So

$$\mathcal{T}_{\text{Euclid}} = \frac{\lambda}{32\pi^2} A_0(m^2) \quad (121)$$

and the mass shift of Equation 36 is $M^2 = -\mathcal{T}_{\text{Euclid}} = -\lambda A_0(m^2)/(32\pi^2)$ (up to the overall $\mu^{2\epsilon}$ convention). The UV pole of A_0 is $m^2/\epsilon = -2m^2/(D-4)$, which FeynCalc confirms in Section 7.

A habit that saves time. When you meet a new Minkowski integral, do three things before anything else. Count the propagators n and write the factor $i(-1)^n$. Replace $\ell^2 - m^2$ by $-(\ell_E^2 + m^2)$ and every external p^2 by $-p_E^2$. At the very end, put back $p_E^2 = -s - i\epsilon$. The last step is where imaginary parts come from (Section 15).

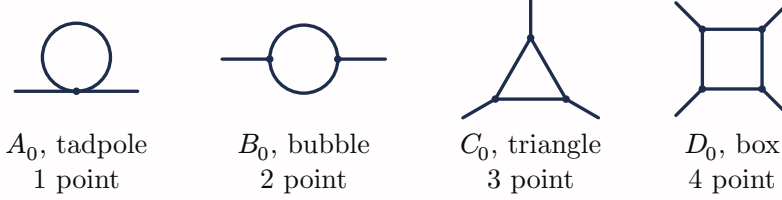
Two worked examples, one answer. Like Goku and Vegeta doing the fusion dance: two different routes, one Gogeta. The Euclidean and the Minkowski calculation must fuse into the same number and they do.

7 The basis of one-loop integrals

Any one-loop diagram, after the Dirac and Lorentz algebra is done, is a sum of integrals of the form

$$\int d^D \ell \frac{\ell^{\mu_1} \dots \ell^{\mu_r}}{D_1 D_2 \dots D_N}, \quad D_i = (\ell + q_i)^2 - m_i^2 + i\epsilon \quad (122)$$

with N propagators and r powers of loop momentum in the numerator (a rank r tensor integral). These tensor integrals reduce to scalar integrals with the same propagators. The method is called Passarino-Veltman reduction and below it is done by hand for the rank one bubble. Together with the reduction of five or more propagators to boxes, every one-loop integral comes down (up to terms of order ϵ) to a small set of scalar integrals. In the normalisation of Equation 117,



RESULT: THE ONE-LOOP BASIS IN FOUR DIMENSIONS

Every one-loop amplitude in $D = 4 - 2\epsilon$ can be written as

$$\mathcal{M} = \sum_i d_i D_0^{(i)} + \sum_j c_j C_0^{(j)} + \sum_k b_k B_0^{(k)} + \sum_l a_l A_0^{(l)} + R + O(\epsilon) \quad (123)$$

where the coefficients are rational functions of the external momenta and masses and R is a rational term (it comes from ϵ/ϵ , a D -dependent coefficient hitting a UV pole).

Why do we stop at the box? In four dimensions at most four vectors can be linearly independent. A pentagon has four independent external momenta and together with ℓ that makes five vectors in four dimensions, so there is a linear relation (a vanishing Gram determinant). Using it one can write $1/(D_1 \dots D_5)$ as a combination of boxes, up to terms of order ϵ . The same argument kills hexagons and everything beyond.

7.1 Tensor reduction on one example

Consider the rank one bubble in the Passarino-Veltman convention, $D_1 = \ell^2 - m_1^2$, $D_2 = (\ell + p)^2 - m_2^2$. By Lorentz invariance the only vector it can be proportional to is p^μ :

$$\frac{(2\pi\mu)^{4-D}}{i\pi^2} \int d^D \ell \frac{\ell^\mu}{D_1 D_2} = p^\mu B_1(p^2; m_1^2, m_2^2) \quad (124)$$

DERIVATION, STEP BY STEP

Contract both sides with p_μ . On the left we need $\ell \cdot p$ in terms of denominators. Since $D_2 - D_1 = 2\ell \cdot p + p^2 - m_2^2 + m_1^2$,

$$2\ell \cdot p = D_2 - D_1 - (p^2 + m_1^2 - m_2^2) \quad (125)$$

Therefore

$$p^2 B_1 = \frac{1}{2} \left[\int \frac{1}{D_1} - \int \frac{1}{D_2} - (p^2 + m_1^2 - m_2^2) \int \frac{1}{D_1 D_2} \right] \quad (126)$$

(with the prefactor understood). The first term is $A_0(m_1^2)$. In the second, shift $\ell \rightarrow \ell - p$ to get $A_0(m_2^2)$. The last is B_0 . So

$$B_1 = \frac{1}{2p^2} [A_0(m_1^2) - A_0(m_2^2) - (p^2 + m_1^2 - m_2^2)B_0(p^2; m_1^2, m_2^2)] \quad (127)$$

In pictures (a label on a line means that factor in the numerator):

$$\text{---}\bigcirc^{\ell^\mu}\text{---} = p^\mu B_1 = \frac{p^\mu}{2p^2} \left[\text{---}\bigcirc_{m_1}\text{---} - \text{---}\bigcirc_{m_2}\text{---} - (p^2 + m_1^2 - m_2^2) \text{---}\bigcirc\text{---} \right] \quad (128)$$

For equal masses A_0 terms cancel and $B_1 = -B_0/2$. This makes sense. For $m_1 = m_2$ the substitution $\ell \rightarrow -\ell - p$ swaps the two propagators and sends $\ell^\mu \rightarrow -\ell^\mu - p^\mu$, so $B_1 = -B_1 - B_0$.

The same game (contract with all external momenta and with $g_{\mu\nu}$, then solve the linear system) works for any rank and any number of legs. It fails only when the Gram determinant of the external momenta vanishes, which is a known source of numerical trouble in practice.

MATHEMATICA CHECK: FEYNCALC TENSOR REDUCTION

FeynCalc (Minkowski metric): tensor bubble $\rightarrow$ Passarino-Veltman functions. $\int d^D\ell \ell^\mu / ([\ell^2 - m_1^2][(\ell + p)^2 - m_2^2])$

```
In [1]: m2])
        tb = TID[FVD[l, mu] FAD[{l, m1}, {l + p, m2}], l, ToPaVe -> True];
        FCE[tb]
```

$$\text{Out}[1]: -\frac{i \pi^2 p^{\text{mu}} (-m^2)^2 + m 1^2 + p^2) \text{B}_0(p^2, m 1^2, (m^2)^2)}{2 p^2} - \frac{i \pi^2 p^{\text{mu}} \text{A}_0((m^2)^2)}{2 p^2} + \frac{i \pi^2 p^{\text{mu}} \text{A}_0(m 1^2)}{2 p^2}$$

 B_1 in terms of A_0 and B_0

```
In [2]: b1 = PaVeReduce[PaVe[1, {SPD[p]}, {m1^2, m2^2}]];
FCE[Simplify[b1]]
```

Out [2]:
$$-\frac{(-m^2)^2 + m^2 + p^2) B_0(p^2, m^2, (m^2)^2) + A_0((m^2)^2) - A_0(m^2)}{2 p^2}$$

B_1 for equal masses

```
In [3]: FCE[PaVeReduce[PaVe[1, {SPD[p]}, {m^2, m^2}]]]
```

Out[3]: $-\frac{1}{2} B_0(p^2, m^2, m^2)$

UV pole of A_0

```
In [4]: Simplify[PaVeUVPart[A0[m^2]]]
```

Out[4]: $-\frac{2m^2}{D-4}$

UV pole of B_0

```
In [5]: Simplify[PaVeUVPart[B0[SPD[p], m^2, m^2]]]
```

Out[5]: $-\frac{2}{D-4}$

Here **TID** does the tensor decomposition and **PaVeReduce** does the reduction. The first line is $i\pi^2 p^\mu$ times our formula for B_1 . **FeynCalc** prints the bare integral $\int d^D\ell$, while A_0 and B_0 carry the prefactor $(2\pi\mu)^{4-D}/(i\pi^2)$ of Equation 117. So strictly the bare integral is $i\pi^2(2\pi\mu)^{D-4}p^\mu B_1$ and **FeynCalc** leaves the scale factor $(2\pi\mu)^{D-4}$ implicit. The line **B1 in terms of A0 and B0** agrees with the formula derived by hand (**PaVe[0,{},{m^2}]** is A_0). The UV poles printed at the end are $-2/(D-4) = 1/\varepsilon$ for B_0 and m^2/ε for A_0 , the same as in Section 6.

7.2 Software and where each tool fits

The lecture mentioned several packages. I installed all of them for these notes (versions as of 29 September 2026) and here is what each one does.

Package	What it does	Used here for
FeynCalc 10.2 (Shtabovenko, Mertig, Orellana)	Mathematica package for Dirac algebra, tensor reduction (TID), Passarino-Veltman functions, Feynman parametrisation	Section 7, Section 11
FeynHelpers (Shtabovenko)	Bridge from FeynCalc to Package-X, FIRE, Kira, LoopTools, pySecDec	installed as FeynCalc add-on
Package-X 2.1.1 (H. Patel)	Analytic one-loop integrals up to boxes, with all ε poles and logs	Section 14, Section 21
FIRE 6.5 (A. V. Smirnov and collaborators)	IBP reduction with Laporta's algorithm; ships LiteRed 1.8 (R. Lee)	Section 8, LiteRed
Kira 3.1 (Maierhöfer, Usovitsch and collaborators)	Fast IBP reduction, standalone C++ program	Section 23

Table 2: Tools for one-loop (and multi-loop) integrals.

7.2.1 How I installed them

- FeynCalc was already in `~/Library/Wolfram/Applications/FeynCalc`. FeynHelpers (development branch, to match the FeynCalc development version) went into `FeynCalc/AddOns/FeynHelpers`. Load it with `$LoadAddOns = {"FeynHelpers"} before << FeynCalc``.
- FIRE was cloned from `gitlab.com/feynmanintegrals/fire` into `~/Library/Wolfram/Applications/FIRE`. Its Mathematica part runs without compiling anything. LiteRed was copied from `FIRE6/extra/LiteRed/Setup` to `~/Library/Wolfram/Applications/LiteRed`.
- Kira and its helper Fermat are Linux programs. The static binary runs on any x86-64 Linux machine. Details are in Section 23.
- Package-X 2.1.1 (patched version) and PVReduce come from the Package-X page on HepForge. Unzip both into `~/Library/Wolfram/Applications`. Then `<< X`` loads Package-X and FeynHelpers' `PaXEvaluate` can call it from FeynCalc.

Collecting all these packages felt like hunting for the Dragon Balls: seven of them, scattered over different websites and one (Package-X) hidden behind a gate that only opens for a real browser. With all seven, Shenron grants one wish: every result in these notes checked in more than one way.

8 The bubble: IBP reduction

8.1 The integral family

The next simplest diagram has two propagators. It appears in the vacuum polarisation of QED, in the self energy of a particle that decays into two and in the $\lambda\varphi^4$ four-point function (Section 13). We work in Euclidean space with equal masses, as Ramond does and use the family

$$J(a_1, a_2) = \int [d\ell] \frac{1}{D_1^{a_1} D_2^{a_2}}, \quad D_1 = \ell^2 + m^2, \quad D_2 = (\ell - p)^2 + m^2 \quad (129)$$

Here $[d\ell] = d^D \ell / \pi^{D/2}$ is the loop measure with the $\pi^{D/2}$ of the Gaussian taken out, so that a Gaussian integrates to 1 (Section 1).

the momentum p enters on the left,
 ℓ runs on the top line, $\ell - p$ on the bottom

Two common choices appear in the books. Ramond takes $m_1 = m_2 = m$ and Smirnov's book often takes $m_1 = m$, $m_2 = 0$ (in Minkowski space). The method below works for both. With equal masses the numbers are simpler, so we follow Ramond. Note two facts that we use again and again:

- Symmetry. The change $\ell \rightarrow p - \ell$ swaps $D_1 \leftrightarrow D_2$, so $J(a_1, a_2) = J(a_2, a_1)$.
- Subsectors. If one index is zero or negative, a propagator is missing and we get a tadpole: $J(a, 0) = J(0, a) = T(a)$ (for $J(0, a)$ shift $\ell \rightarrow \ell + p$).

8.2 Two IBP identities

Now there are two vectors we can put inside the derivative, ℓ^μ and p^μ . We need how the derivative acts on the denominators:

$$\ell \cdot \partial_\ell D_1 = 2\ell^2 = 2(D_1 - m^2), \quad \ell \cdot \partial_\ell D_2 = 2\ell \cdot (\ell - p) = D_1 + D_2 - 2m^2 - p^2 \quad (130)$$

The second one uses $\ell^2 + (\ell - p)^2 - p^2 = 2\ell \cdot (\ell - p)$.

DERIVATION: THE IBP WITH ℓ^μ

$$0 = \int d^D \ell \frac{\partial}{\partial \ell^\mu} \frac{\ell^\mu}{D_1^{a_1} D_2^{a_2}} = \int d^D \ell \left[\frac{D}{D_1^{a_1} D_2^{a_2}} - a_1 \frac{\ell \cdot \partial D_1}{D_1^{a_1+1} D_2^{a_2}} - a_2 \frac{\ell \cdot \partial D_2}{D_1^{a_1} D_2^{a_2+1}} \right] \quad (131)$$

Insert the two relations above. The a_1 term: $2(D_1 - m^2)/(D_1^{a_1+1} D_2^{a_2}) = 2J(a_1, a_2) - 2m^2 J(a_1 + 1, a_2)$ after integrating. The a_2 term: $(D_1 + D_2 - 2m^2 - p^2)/(D_1^{a_1} D_2^{a_2+1})$ splits into four pieces, D_1 cancels one power of D_1 (giving $J(a_1 - 1, a_2 + 1)$), D_2 cancels one power of D_2 (giving $J(a_1, a_2)$) and the constant $-(2m^2 + p^2)$ multiplies $J(a_1, a_2 + 1)$. Write each term as a member of the family:

$$0 = DJ(a_1, a_2) - 2a_1 [J(a_1, a_2) - m^2 J(a_1 + 1, a_2)] - a_2 [J(a_1 - 1, a_2 + 1) + J(a_1, a_2) - (2m^2 + p^2) J(a_1, a_2 + 1)] \quad (132)$$

At $a_1 = a_2 = 1$, with $J(0, 2) = T(2)$ and the symmetry $J(1, 2) = J(2, 1)$:

$$0 = (D - 3)J(1, 1) + (p^2 + 4m^2)J(2, 1) - T(2) \quad (133)$$

So the dotted bubble is not a new integral:

RESULT: THE DOTTED BUBBLE

$$J(2, 1) = -\frac{D-3}{p^2 + 4m^2} J(1, 1) + \frac{1}{p^2 + 4m^2} T(2), \quad T(2) = -\frac{D-2}{2m^2} T(1) \quad (134)$$

In pictures (a dot on the upper line):

$$\text{---} \bigcirc \text{---} = -\frac{D-3}{p^2 + 4m^2} \text{---} \bigcirc \text{---} + \frac{1}{p^2 + 4m^2} \text{---} \bigcirc \text{---} \quad (135)$$

The family has **two master integrals**, $J(1, 1)$ and $T(1)$. What this means in general is explained in Section 19.

DERIVATION: THE IBP WITH p^μ

Now use the other vector. We need $p \cdot \partial_\ell D_1 = 2\ell \cdot p$ and $p \cdot \partial_\ell D_2 = 2(\ell - p) \cdot p$. From $D_1 - D_2 = 2\ell \cdot p - p^2$,

$$2\ell \cdot p = D_1 - D_2 + p^2, \quad 2(\ell - p) \cdot p = D_1 - D_2 - p^2 \quad (136)$$

Since $\partial_\mu p^\mu = 0$ (p does not depend on ℓ), only the derivatives of the denominators appear:

$$\begin{aligned}
0 = & -a_1 [J(a_1, a_2) - J(a_1 + 1, a_2 - 1) + p^2 J(a_1 + 1, a_2)] \\
& -a_2 [J(a_1 - 1, a_2 + 1) - J(a_1, a_2) - p^2 J(a_1, a_2 + 1)]
\end{aligned} \tag{137}$$

At $a_1 = a_2 = 1$, with $J(2, 0) = J(0, 2) = T(2)$:

$$0 = -J(1, 1) + T(2) - p^2 J(2, 1) - T(2) + J(1, 1) + p^2 J(1, 2) = p^2 [J(1, 2) - J(2, 1)] \tag{138}$$

It only says $J(1, 2) = J(2, 1)$, the symmetry we already knew. At this level the p^μ identity gives nothing new, but for higher dots it is needed.

The master integrals are the One Piece. Every IBP identity is one more island of the Grand Line and at the end of the voyage only the treasure is left: here, two integrals.

A computer does all this bookkeeping for us. Here is LiteRed, which builds all IBP and Lorentz-invariance identities of the family, finds the masters and reduces anything we ask for.

MATHEMATICA CHECK: IBP REDUCTION WITH LITERED

Load LiteRed, the bubble basis $\{\ell^2 + m^2, (\ell - p)^2 + m^2\}$ and let it generate and solve all IBP relations (full

In [1]: code in the file). master integrals

```
MIIs[bub]
```

Out[1]: {j(bub, 0, 1), j(bub, 1, 1)}

$J(2, 1)$

In [2]: Collect[IBPReduce[j[bub, 2, 1]], _j, Factor]

Out[2]:
$$\frac{(d-2)j(\text{bub}, 0, 1)}{2m^2(4m^2 + p^2)} - \frac{(d-3)j(\text{bub}, 1, 1)}{4m^2 + p^2}$$

$T(2) = J(0, 2)$

In [3]: Collect[IBPReduce[j[bub, 0, 2]], _j, Factor]

Out[3]:
$$-\frac{(d-2)j(\text{bub}, 0, 1)}{2m^2}$$

$dJ(1, 1)/dp^2$

In [4]: Collect[IBPReduce[Dinv[j[bub, 1, 1], pp]], _j, Factor]

Out[4]:
$$\frac{j(\text{bub}, 1, 1)(dp^2 - 4m^2 - 4p^2)}{2p^2(4m^2 + p^2)} - \frac{(d-2)j(\text{bub}, 0, 1)}{p^2(4m^2 + p^2)}$$

LiteRed's $j[\text{bub}, 0, 1]$ is our $T(1)$. The line for $J(2, 1)$ is Equation 134 after putting $T(2) = -(D-2)T(1)/(2m^2)$. The last line is the derivative we need for the differential equation, already reduced to masters. We derive it by hand next.

9 The Gauss hypergeometric function

The bubble will turn out to be a **hypergeometric function**. Most students meet this function for the first time in a Feynman-integral course and it looks scary. It is not. It is a power series whose coefficients follow a simple pattern and a surprising number of familiar functions are special cases of it. This chapter builds it from scratch.

9.1 The Pochhammer symbol

For any number a and integer $k \geq 0$ define the rising factorial

$$(a)_k = a(a+1)(a+2)\dots(a+k-1), \quad (a)_0 = 1 \tag{139}$$

Some quick facts:

- $(1)_k = 1 \cdot 2 \dots k = k!$.
- Using $\Gamma(z+1) = z\Gamma(z)$ k times, $\Gamma(a+k) = (a+k-1)\dots(a+1)a\Gamma(a)$, so $(a)_k = \Gamma(a+k)/\Gamma(a)$.
- If $a = -N$ is a negative integer, $(a)_k = 0$ for $k > N$ (one factor is zero).
- For small ε , $(\varepsilon)_k = \varepsilon(1+\varepsilon)(2+\varepsilon)\dots(k-1+\varepsilon) = \varepsilon(k-1)! + O(\varepsilon^2)$ for $k \geq 1$.

9.2 The series

DEFINITION: THE GAUSS HYPERGEOMETRIC FUNCTION

$${}_2F_1(a, b; c; z) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{z^k}{k!} = 1 + \frac{ab}{c}z + \frac{a(a+1)b(b+1)}{c(c+1)} \frac{z^2}{2} + \dots \quad (140)$$

The “2” and “1” count the Pochhammer symbols upstairs (a, b) and downstairs (c) . The parameter c must not be 0, -1 , -2 , ...

Where the name comes from. In a geometric series each term is the previous one times z . Here each term is the previous one times

$$\frac{c_{k+1}}{c_k} = \frac{(a+k)(b+k)}{(c+k)(k+1)} z \quad (141)$$

a rational function of k . “Hypergeometric” means that the ratio of successive terms is a rational function of k .

Radius of convergence. If a or b is 0, -1 , -2 , ... the series stops after a few terms (a Pochhammer factor becomes zero) and ${}_2F_1$ is just a polynomial. Otherwise, for large k the ratio of successive terms tends to z and by the ratio test the series converges for $|z| < 1$ and diverges for $|z| > 1$. So the series alone does not define the function for $|z| > 1$. We will need a better definition (Euler’s integral, below).

9.3 Familiar functions in disguise

DERIVATION: THREE SPECIAL CASES

1. The binomial series. If $c = b$ the $(b)_k$ cancels against $(c)_k$:

$${}_2F_1(a, b; b; z) = \sum_k (a)_k \frac{z^k}{k!} = (1-z)^{-a} \quad (142)$$

(This is Newton’s binomial series, since $(1-z)^{-a} = \sum_k \binom{-a}{k} (-z)^k$ and $\binom{-a}{k} (-1)^k = (a)_k/k!$.)

2. The logarithm. With $a = b = 1$, $c = 2$: $(1)_k(1)_k/((2)_k k!) = k!k!/((k+1)!k!) = 1/(k+1)$, so

$${}_2F_1(1, 1; 2; z) = \sum_k \frac{z^k}{k+1} = \frac{1}{z} \sum_k \frac{z^{k+1}}{k+1} = -\frac{\ln(1-z)}{z} \quad (143)$$

3. The arcsine. With $a = b = 1/2$, $c = 3/2$ and argument z^2 one finds ${}_2F_1(1/2, 1/2; 3/2; z^2) = (\arcsin z)/z$ (compare the Taylor series $\arcsin z = z + z^3/6 + 3z^5/40 + \dots$).

So the logarithm, powers and inverse trigonometric functions all live inside ${}_2F_1$. This is why Feynman integrals, whose answers are made of logs, often come out as hypergeometric functions.

The hypergeometric function is a Death Note: write down four entries $(a, b; c; z)$ and a whole family of functions appears. Choose $(1, 1; 2; z)$ and you get a logarithm. Choose $(1/2, 1/2; 3/2; z^2)$ and you get an arcsine. Ryuk would be impressed.

9.4 The differential equation it solves

DERIVATION: THE HYPERGEOMETRIC EQUATION FROM THE SERIES

Use the operator $\theta = z d/dz$. It acts simply on powers: $\theta z^k = k z^k$. Write $F = \sum_k c_k z^k$ with $c_k = (a)_k(b)_k/((c)_k k!)$. The recursion $c_{k+1}(c+k)(k+1) = c_k(a+k)(b+k)$ reads, multiplied by z^{k+1} and summed,

$$\sum_k (k+1)(k+c)c_{k+1}z^{k+1} = z \sum_k (k+a)(k+b)c_k z^k \quad (144)$$

On the left, $(k+1)(k+c) = \theta(\theta+c-1)$ acting on z^{k+1} . On the right, $(k+a)(k+b) = (\theta+a)(\theta+b)$ acting on z^k . So

$$\theta(\theta+c-1)F = z(\theta+a)(\theta+b)F \quad (145)$$

Now expand using $\theta = z d/dz$ and $\theta^2 = z^2 d^2/dz^2 + z d/dz$. Dividing by z :

$$z(1-z)F'' + [c - (a+b+1)z]F' - abF = 0 \quad (146)$$

This second order ODE has three singular points, $z=0$, $z=1$ and $z=\infty$. Near each one, the solutions behave like simple powers (for example 1 and z^{1-c} near $z=0$). The function ${}_2F_1$ is the solution that is regular at $z=0$ with value 1 there. In Section 10 the bubble will obey a first order ODE with singular points at 0, -1 and ∞ in the variable x , which is the same picture after $z=-x$.

9.5 Euler's integral and why there is a cut

DERIVATION: EULER'S INTEGRAL REPRESENTATION

For $\text{Re } c > \text{Re } b > 0$,

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du \quad (147)$$

Proof. Expand $(1-zu)^{-a} = \sum_k \binom{a}{k} (zu)^k / k!$ (the binomial series, case 1 above). Each term gives a Beta function:

$$\int_0^1 u^{b+k-1}(1-u)^{c-b-1} du = B(b+k, c-b) = \frac{\Gamma(b+k)\Gamma(c-b)}{\Gamma(c+k)} \quad (148)$$

Multiply by the prefactor: $\Gamma(c)\Gamma(b+k)/(\Gamma(b)\Gamma(c+k)) = (b)_k/(c)_k$. Summing over k gives the series. $\square$

The integral makes sense for **every** z except where $1-zu=0$ for some u in $[0, 1]$, that is for real $z \geq 1$. So Euler's integral continues the function to the whole complex plane, cut along $[1, \infty)$. Across that cut the function jumps. In Section 15 this cut becomes the physical threshold of the bubble, where two real particles can be produced.

For the bubble we need the case $b=1$, $c=3/2$. Then $\Gamma(3/2)/(\Gamma(1)\Gamma(1/2)) = 1/2$ and, after putting $z=-x$,

$${}_2F_1\left(a, 1; \frac{3}{2}; -x\right) = \frac{1}{2} \int_0^1 (1-u)^{-1/2}(1+xu)^{-a} du = \frac{1}{2} \int_0^1 u^{-1/2}(1+x(1-u))^{-a} du \quad (149)$$

9.6 Transformations

The same function can be written in several ways. The one we need is **Euler's transformation**,

$${}_2F_1(a, b; c; z) = (1-z)^{c-a-b} {}_2F_1(c-a, c-b; c; z) \quad (150)$$

It follows from Euler's integral in two short steps.

DERIVATION: EULER'S TRANSFORMATION

Step 1: substitute $u = 1-v$ in Euler's integral. Then $u^{b-1}(1-u)^{c-b-1} \rightarrow (1-v)^{b-1}v^{c-b-1}$ and

$$1-zu = 1-z+zv = (1-z)\left(1-\frac{z}{z-1}v\right) \quad (151)$$

Pull out $(1-z)^{-a}$. What is left is again Euler's integral, now with b and $c-b$ exchanged and argument $w = z/(z-1)$:

$${}_2F_1(a, b; c; z) = (1-z)^{-a} {}_2F_1\left(a, c-b; c; \frac{z}{z-1}\right) \quad (152)$$

Step 2: use Step 1 once more, on the right-hand side. ${}_2F_1$ is symmetric in its two upper parameters, so write it as ${}_2F_1(c-b, a; c; w)$ and apply Step 1 with $a \rightarrow c-b$, $b \rightarrow a$:

$${}_2F_1(c-b, a; c; w) = (1-w)^{-(c-b)} {}_2F_1\left(c-b, c-a; c; \frac{w}{w-1}\right) \quad (153)$$

With $w = z/(z-1)$: $1-w = 1/(1-z)$ and $w/(w-1) = z$. So the argument comes back to z and the prefactors combine to $(1-z)^{-a}(1-z)^{c-b} = (1-z)^{c-a-b}$. This is Euler's transformation.

(The identity of Step 1 has its own name in books, the Pfaff transformation. It was not needed in the lecture and here it is only an intermediate step. The transform that was discussed in the lecture is the Laplace (or Mellin) transform of Section 5.4, which is a different thing. It turns a difference equation into a differential equation.) We use Euler's transformation with $a = (D-1)/2$, $b = 1/2$, $c = 3/2$: then $c-a = 2-D/2$, $c-b = 1$ and $c-a-b = (3-D)/2$.

9.7 Expanding in the parameters

For Feynman integrals the parameters depend on ε and we need to expand in ε . Take ${}_2F_1(\varepsilon, 1; 3/2; z)$. Since $(\varepsilon)_k = \varepsilon(k-1)! + O(\varepsilon^2)$ for $k \geq 1$ and $(1)_k = k!$,

$${}_2F_1\left(\varepsilon, 1; \frac{3}{2}; z\right) = 1 + \varepsilon \sum_{k=1}^{\infty} \frac{(k-1)!}{(3/2)_k} z^k + O(\varepsilon^2) \quad (154)$$

The sum at order ε is again a hypergeometric function, but with other parameters: shifting $k \rightarrow k+1$ and using $(3/2)_{k+1} = (3/2)(5/2)_k$ gives $(2z/3){}_2F_1(1, 1; 5/2; z)$. Mathematica sums it to $2(z - \sqrt{z(1-z)} \arcsin \sqrt{z})/z$, a combination of an arcsine and a square root, which for $z < 0$ becomes a logarithm. In general, expanding in the parameters produces logarithms and polylogarithms. This is how the logarithms of physics come out of the exact D -dimensional answer.

MATHEMATICA CHECK: SERIES, SPECIAL CASES, ODE, EULER'S INTEGRAL AND TRANSFORMATION

Gauss hypergeometric function: series, special cases, ODE, Euler integral, transformation. the series of

In [1]: ${}_2F_1(a, b; c; z)$ through z^2

```
Normal@Series[Hypergeometric2F1[a, b, c, z], {z, 0, 2}] // Simplify
```

Out[1]: $\frac{a(a+1)b(b+1)z^2}{2c(c+1)} + \frac{abz}{c} + 1$

${}_2F_1(1, 1; 2; z)$

In [2]: Hypergeometric2F1[1, 1, 2, z]

Out[2]: $-\frac{\log(1-z)}{z}$

${}_2F_1(1/2, 1/2; 3/2; z^2)$

In [3]: FullSimplify[Hypergeometric2F1[1/2, 1/2, 3/2, z^2], 0 < z < 1]

Out[3]: $\frac{\sin^{-1}(z)}{z}$

${}_2F_1(a, b; b; z)$

In [4]: Hypergeometric2F1[a, b, b, z]

Out[4]: $(1-z)^{-a}$

the ODE $z(1-z)F'' + (c - (a+b+1)z)F' - abF = 0$, checked as a series. residual through z^5

```
In [5]: F = Hypergeometric2F1[a, b, c, z];
ode = z (1 - z) D[F, {z, 2}] + (c - (a + b + 1) z) D[F, z] - a b F;
Simplify[Normal@Series[ode, {z, 0, 5}]]
```

Out[5]: 0

Euler integral $\frac{1}{2} \int_0^1 u^{-1/2} (1+xu)^{-a} du$, done term by term. $\frac{1}{2} \int_0^1 u^{-1/2} (1+xu)^{-a} du$

In [6]: `eul = Sum[Pochhammer[a, k] (-x)^k/k! Integrate[u^(k - 1/2),
{u, 0, 1}]/2, {k, 0, Infinity}];
Simplify[eul]`

Out[6]: ${}_2F_1\left(\frac{1}{2}, a; \frac{3}{2}; -x\right)$

Euler's transformation used for the bubble. $(1+x)^{(D-3)/2} {}_2F_1((D-1)/2, 1/2; 3/2; -x) - {}_2F_1(2-D/2, 1; 3/2; -x)$

In [7]: `Simplify[Normal@Series[Hypergeometric2F1[(dd - 1)/2,
1/2, 3/2, -x] (1 + x)^(dd - 3)/2 -
Hypergeometric2F1[2 - dd/2, 1, 3/2, -x], {x, 0, 5}]]`

Out[7]: 0

${}_2F_1(0, 1; 3/2; z)$ and the order- ε term of ${}_2F_1(\varepsilon, 1; 3/2; z)$

In [8]: `{Hypergeometric2F1[0, 1, 3/2, z], Sum[(k - 1)!/Pochhammer[3/2,
k] z^k, {k, 1, Infinity}]}`

Out[8]: $\left\{1, \frac{2(z - \sqrt{-(z-1)z}) \sin^{-1}(\sqrt{z})}{z}\right\}$

Every line is exact, the series coefficients, the three special cases, the ODE residual (zero order by order), Euler's integral summed term by term (Mathematica writes it with a and b swapped, which is the same since ${}_2F_1$ is symmetric in a, b) and Euler's transformation for the parameters of the bubble.

10 The bubble from a differential equation

10.1 Which variable?

After integration the bubble can depend on the vector p^μ only through the scalar p^2 . This is Lorentz invariance (in Euclidean space, rotation invariance). There is no other vector to form a scalar with. So $J = J(D, p^2)$ and the chain rule gives

$$\frac{\partial J}{\partial p_\mu} = \frac{\partial J}{\partial p^2} \frac{\partial p^2}{\partial p_\mu} = 2p^\mu \frac{\partial J}{\partial p^2} \Rightarrow p_\mu \frac{\partial J}{\partial p_\mu} = 2p^2 \frac{\partial J}{\partial p^2} \quad (155)$$

This is the key trick. It turns a derivative in the scalar p^2 (which we cannot take under the integral sign directly) into a derivative in the vector p^μ (which we can).

10.2 Deriving the equation

DERIVATION, STEP BY STEP

Differentiate under the integral. Only $D_2 = (\ell - p)^2 + m^2$ depends on p :

$$\frac{\partial}{\partial p_\mu} \frac{1}{D_2} = -\frac{1}{D_2^2} \frac{\partial D_2}{\partial p_\mu} = \frac{2(\ell - p)^\mu}{D_2^2}, \quad p_\mu \frac{\partial J}{\partial p_\mu} = \int [d\ell] \frac{2(\ell \cdot p - p^2)}{D_1 D_2^2} \quad (156)$$

Numerator in terms of denominators. $D_1 - D_2 = \ell^2 - (\ell - p)^2 = 2\ell \cdot p - p^2$, so $2\ell \cdot p - 2p^2 = D_1 - D_2 - p^2$. Then

$$p_\mu \frac{\partial J}{\partial p_\mu} = J(0, 2) - J(1, 1) - p^2 J(1, 2) = T(2) - J(1, 1) - p^2 J(2, 1) \quad (157)$$

Use Equation 155. Divide by $2p^2$:

$$\frac{dJ}{dp^2} = \frac{1}{2p^2} T(2) - \frac{1}{2p^2} J - \frac{1}{2} J(2, 1) \quad (158)$$

The dotted bubble is an “unwanted” integral and the IBP result Equation 134 removes it:

$$\frac{dJ}{dp^2} = -\frac{1}{2} \left[\frac{1}{p^2} - \frac{D-3}{p^2 + 4m^2} \right] J + \frac{T(2)}{2} \left[\frac{1}{p^2} - \frac{1}{p^2 + 4m^2} \right] \quad (159)$$

Write $T(2)$ through $T(1)$. $T(2)/2 = -(D-2)T(1)/(4m^2)$, so:

$$\frac{dJ}{dp^2} + \frac{1}{2} \left[\frac{1}{p^2} - \frac{D-3}{p^2 + 4m^2} \right] J = -\frac{D-2}{4m^2} \left[\frac{1}{p^2} - \frac{1}{p^2 + 4m^2} \right] T(1) \quad (160)$$

Dimensionless variable. Put $x = p^2/(4m^2)$, so $d/dp^2 = (4m^2)^{-1} d/dx$. Multiply by $4m^2$:

$$\frac{dJ}{dx} + \frac{1}{2} \left[\frac{1}{x} - \frac{D-3}{1+x} \right] J = \frac{T(2)}{2} \left[\frac{1}{x} - \frac{1}{1+x} \right] = \frac{T(2)}{2x(1+x)} \quad (161)$$

RESULT: THE BUBBLE DIFFERENTIAL EQUATION

$$\frac{dJ}{dx} + \frac{1}{2} \left[\frac{1}{x} - \frac{D-3}{1+x} \right] J = \frac{T(2)}{2x(1+x)}, \quad x = \frac{p_E^2}{4m^2}, \quad T(2) = \Gamma(2-D/2)(m^2)^{D/2-2} \quad (162)$$

In diagrams:

$$\frac{d}{dx} \text{---} \text{---} \text{---} \text{---} \text{---} + \frac{1}{2} \left[\frac{1}{x} - \frac{D-3}{1+x} \right] \text{---} \text{---} \text{---} \text{---} \text{---} = \frac{1}{2x(1+x)} \text{---} \text{---} \text{---} \text{---} \text{---} \quad (163)$$

A first order linear ODE, with the dotted tadpole as a source. The tadpole is the simpler integral of the subsector, so it is known already.

10.3 Solving it

DERIVATION: HOMOGENEOUS PART, VARIATION OF CONSTANTS, BOUNDARY

Homogeneous equation. Drop the right side: $J_0'/J_0 = -\frac{1}{2x} + \frac{D-3}{2(1+x)}$. The left side is $(\ln J_0)'$ and integrating each term gives $\ln J_0 = -\frac{1}{2} \ln x + \frac{D-3}{2} \ln(1+x) + \text{const}$. Exponentiate:

$$J_0(x) = A x^{-1/2} (1+x)^{(D-3)/2} \quad (164)$$

Variation of constants. Call $h(x) = x^{-1/2} (1+x)^{(D-3)/2}$ and try $J = h(x)\varphi(x)$. Then $J' = h'\varphi + h\varphi'$ and since $h' + \frac{1}{2}[1/x - (D-3)/(1+x)]h = 0$, the equation becomes simply $h\varphi' = T(2)/(2x(1+x))$. So $\varphi' = T(2)/(2x(1+x)h)$:

$$\varphi'(x) = \frac{T(2)}{2x(1+x)} \cdot x^{1/2} (1+x)^{-(D-3)/2} = \frac{T(2)}{2} x^{-1/2} (1+x)^{-(D-1)/2} \quad (165)$$

(Powers of x : $x^{-1}x^{1/2} = x^{-1/2}$. Powers of $1+x$: $(1+x)^{-1}(1+x)^{-(D-3)/2} = (1+x)^{-(D-1)/2}$.)

Integrate. Put $t = xu$, $dt = x du$, $u \in [0, 1]$:

$$\int_0^x t^{-1/2} (1+t)^{-a} dt = x^{1/2} \int_0^1 u^{-1/2} (1+xu)^{-a} du = 2x^{1/2} {}_2F_1\left(a, \frac{1}{2}; \frac{3}{2}; -x\right) \quad (166)$$

by Euler's integral of Section 9 with $b = 1/2$, $c = 3/2$ (prefactor $\Gamma(3/2)/(\Gamma(1/2)\Gamma(1)) = 1/2$). With $a = (D-1)/2$:

$$\varphi(x) = \varphi(0) + T(2) x^{1/2} {}_2F_1\left(\frac{D-1}{2}, \frac{1}{2}; \frac{3}{2}; -x\right) \quad (167)$$

Boundary condition from the equation itself. The homogeneous piece $x^{-1/2}$ blows up at $x = 0$. But the integral J is perfectly finite at $p = 0$: there both propagators are equal and $J(p=0) = \int 1/D_1^2 = T(2)$. So we must take $\varphi(0) = 0$ and

$$J = T(2)(1+x)^{(D-3)/2} {}_2F_1\left(\frac{D-1}{2}, \frac{1}{2}; \frac{3}{2}; -x\right) \quad (168)$$

Tidy up with Euler's transformation (Section 9). ${}_2F_1(a, b; c; z) = (1 - z)^{c-a-b} {}_2F_1(c - a, c - b; c; z)$ with $a = (D - 1)/2$, $b = 1/2$, $c = 3/2$, $z = -x$. Here $c - a = 2 - D/2$, $c - b = 1$ and $c - a - b = (3 - D)/2$, so the factor $(1 + x)^{(3-D)/2}$ cancels the prefactor $(1 + x)^{(D-3)/2}$ exactly:

$$J(D, x) = T(2) {}_2F_1\left(2 - \frac{D}{2}, 1; \frac{3}{2}; -x\right) \quad (169)$$

At $x = 0$ the hypergeometric function is 1, so $J(D, 0) = T(2)$, the boundary condition.

RESULT: THE EQUAL-MASS BUBBLE FOR ANY D

$$J(1, 1) = \int [d\ell] \frac{1}{(\ell^2 + m^2)((\ell - p)^2 + m^2)} = \Gamma\left(2 - \frac{D}{2}\right) (m^2)^{D/2-2} {}_2F_1\left(2 - \frac{D}{2}, 1; \frac{3}{2}; -\frac{p_E^2}{4m^2}\right) \quad (170)$$

MATHEMATICA CHECK: DSOLVE, REGULARITY AT $x = 0$ AND THE FEYNMAN-PARAMETER SUM

Solve the bubble differential equation in $x = p^2/(4m^2)$, $m^2 = 1$. $T(1)$ with the measure $[d\ell]$. $T(2) = -(D - 2)/2 T(1)$ from IBP. for $x > 0$ the exponential of logarithms is a product of powers. general solution

In [1]:

```
T1 = Gamma[1 - d/2];
T2 = -(d - 2)/2 T1;
ode = J'[x] + 1/2 (1/x - (d - 3)/(1 + x)) J[x] == T2/(2 x (1 + x));
gen = J[x] /. First@DSolve[ode, J[x], x];
gen = Collect[PowerExpand[gen], C[1], Simplify];
gen
```

Out[1]:

$$-\frac{1}{2} (d-2) (x+1)^{\frac{d-3}{2}} \Gamma\left(1 - \frac{d}{2}\right) {}_2F_1\left(\frac{1}{2}, \frac{d-1}{2}; \frac{3}{2}; -x\right) + \frac{c_1 (x+1)^{\frac{d-3}{2}}}{\sqrt{x}}$$

homogeneous piece

In [2]:

```
Coefficient[gen, C[1]]
```

Out[2]:

$$\frac{(x+1)^{\frac{d-3}{2}}}{\sqrt{x}}$$

residual of the ODE for $J = T(2) {}_2F_1(2 - D/2, 1; 3/2; -x)$, expanded around $x = 0$. residual, through x^6

In [3]:

```
Jreg = T2 Hypergeometric2F1[2 - d/2, 1, 3/2, -x];
res = D[Jreg, x] + 1/2 (1/x - (d - 3)/(1 + x)) Jreg - T2/(2 x (1 + x));
Simplify[Normal@Series[res, {x, 0, 6}]]
```

Out[3]: 0

$J(x=0)/\Gamma(2 - D/2)$

In [4]:

```
FullSimplify[(Jreg /. x -> 0)/Gamma[2 - d/2]]
```

Out[4]: 1

Feynman-parameter form: binomial expansion and Beta functions, summed to all orders.

In [5]:

```
fpSeries = Sum[(-1)^k Pochhammer[a, k]/k! (4 x)^k Beta[k + 1,
k + 1], {k, 0, Infinity}];
fpSeries
```

Out[5]:

$${}_2F_1\left(1, a; \frac{3}{2}; -x\right)$$

`DSolve` returns the general solution with a free constant `C[1]` multiplying the homogeneous piece $x^{-1/2}(1+x)^{(d-3)/2}$. Regularity at $x = 0$ sets `C[1] = 0` and what remains is our solution in its first form. The residual of the ODE, expanded around $x = 0$, vanishes order by order and $J(x = 0)$ divided by $\Gamma(2 - D/2)$

is exactly 1, the boundary value $T(2)$ at $m = 1$. The last line takes the Feynman-parameter form of the next chapter, expands $(1 + 4xy(1 - y))^{-a}$ binomially, integrates each term (a Beta function) and sums all orders. The result is ${}_2F_1(a, 1; 3/2; -x)$ (Mathematica writes a and 1 in the other order). Figure 8 shows the same agreement as curves over four decades of x , including $x > 1$ where the series itself no longer converges.

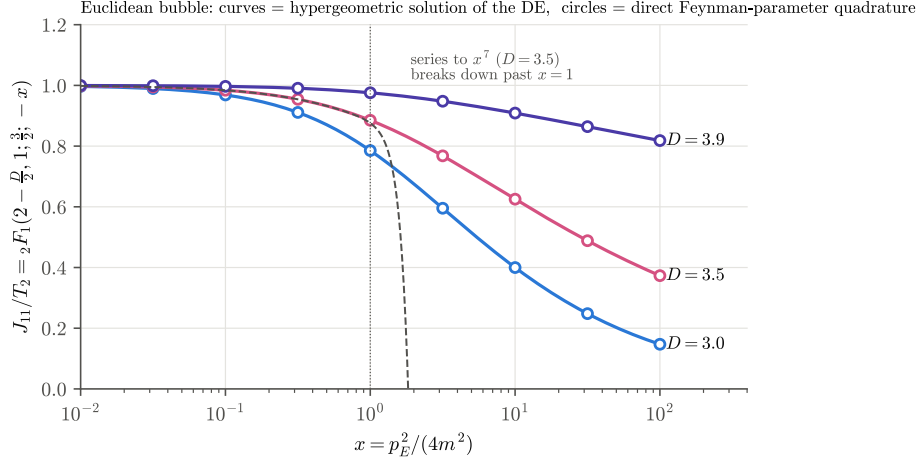


Figure 8: The Euclidean bubble divided by its value at $p = 0$. Solid lines: the hypergeometric solution Equation 170 of the differential equation. Circles: direct numerical integration of the Feynman-parameter form Equation 178. The dashed line is the power series of ${}_2F_1$ truncated at x^7 for $D = 3.5$. It follows the curve for $x < 1$ and then runs away, because the series converges only for $|x| < 1$.

PHYSICAL PICTURE: WHAT THE EQUATION KNOWS

- Kinematics in the argument, dimension in the parameters.** This was the highlighted remark of the lecture. The physical variable $p^2/4m^2$ sits in the argument of ${}_2F_1$, while D only enters the parameters $2 - D/2$. So expanding in $\varepsilon = 2 - D/2$ means expanding the hypergeometric function in its parameters, which produces logarithms and polylogarithms of x .
- The singular points of the ODE are the physical thresholds.** Equation 162 has singular points at $x = 0$, $x = -1$ and $x = \infty$. In Minkowski language $p_E^2 = -s$, so $x = -1$ is $s = 4m^2$: the energy at which two real particles of mass m can be produced. The point $x = 0$ is the “pseudo-threshold” $s = (m_1 - m_2)^2 = 0$. The ODE allows a solution singular there, $x^{-1/2}$, but the Feynman integral on the physical sheet does not have it and that is how we fixed the boundary condition.
- High energy.** At fixed $D < 4$ the bubble falls like a power, $x^{D/2-2}$, at large x . After expanding around $D = 4$ and removing the $1/\varepsilon$ pole, that power becomes a logarithm. The finite part decreases only like $-\ln x$. This slow logarithm is what makes the four-point coupling run with energy.

11 Feynman parameters and graph polynomials

The differential equation gave the bubble without ever doing the loop integral. The older and more direct way is to combine the denominators into one, so that the loop integral becomes a tadpole.

Feynman parameters are the Fusion Dance of Dragon Ball: two propagators, one careful ritual and out comes a single stronger denominator. Get the pose wrong (forget a factor $x_i^{a_i-1}$) and you get fat Gotenks instead.

11.1 The general formula

DERIVATION: FROM SCHWINGER PARAMETERS TO FEYNMAN PARAMETERS

Schwinger’s trick. For $A > 0$ and $a > 0$,

$$\frac{1}{A^a} = \frac{1}{\Gamma(a)} \int_0^\infty d\alpha \alpha^{a-1} e^{-\alpha A} \quad (171)$$

which is just the definition of $\Gamma(a)$ after $\alpha A = u$.

Product of k denominators. Apply it to each factor:

$$\prod_{i=1}^k \frac{1}{A_i^{a_i}} = \frac{1}{\prod_i \Gamma(a_i)} \int_0^\infty \prod_i d\alpha_i \alpha_i^{a_i-1} e^{-\sum_i \alpha_i A_i} \quad (172)$$

Separate the overall scale. Insert $1 = \int_0^\infty d\lambda \delta(\lambda - \sum_i \alpha_i)$ and write $\alpha_i = \lambda x_i$. Then $\prod d\alpha_i = \lambda^k \prod dx_i$, the delta function becomes $\delta(\lambda(1 - \sum x_i)) = \lambda^{-1} \delta(1 - \sum x_i)$ and with $a = \sum_i a_i$,

$$\prod_i \alpha_i^{a_i-1} = \lambda^{a-k} \prod_i x_i^{a_i-1} \quad (173)$$

Do the λ integral. $\int_0^\infty d\lambda \lambda^{a-1} e^{-\lambda \sum x_i A_i} = \Gamma(a) / (\sum x_i A_i)^a$. Collecting,

$$\prod_{i=1}^k \frac{1}{A_i^{a_i}} = \frac{\Gamma(a_1 + \dots + a_k)}{\Gamma(a_1) \dots \Gamma(a_k)} \int_0^1 dx_1 \dots dx_k \delta\left(1 - \sum_i x_i\right) \frac{\prod_i x_i^{a_i-1}}{(x_1 A_1 + \dots + x_k A_k)^{a_1 + \dots + a_k}} \quad (174)$$

MATHEMATICA CHECK: THE FORMULA AND THE SHIFT FOR THE BUBBLE

General Feynman-parameter formula, two denominators with powers a_1, a_2 . the formula for $a_1 = a_2 = 1$

```
In [1]: fp[a1_, a2_, A_, B_] := Gamma[a1 + a2]/(Gamma[a1] Gamma[a2]) *
        Integrate[x^(a1 - 1) (1 - x)^(a2 - 1)/(x A + (1 - x) B)^(a1 + a2),
        {x, 0, 1},
        Assumptions -> A > 0 && B > 0];
Simplify[fp[1, 1, A, B], A > 0 && B > 0]
```

Out[1]: $\frac{1}{AB}$

the formula for $a_1 = 2, a_2 = 3$, minus $1/(A^2 B^3)$

```
In [2]: Simplify[fp[2, 3, A, B] - 1/(A^2 B^3), A > 0 && B > 0 && A != B]
```

Out[2]: 0

the combined denominator of the bubble and the shift $\ell \rightarrow \ell + (1-x)p$. $x D_1 + (1-x) D_2$

```
In [3]: den = Expand[x (l^2 + m^2) + (1 - x) ((1 - p)^2 + m^2)];
den
```

Out[3]: $m^2 - p^2 x + p^2 + 2 p x \ell - 2 p \ell + \ell^2$

after the shift

```
In [4]: Expand[den /. l -> lp + (1 - x) p]
```

Out[4]: $m^2 - p^2 x^2 + p^2 x + (\ell')^2$

11.2 The bubble with one Feynman parameter

DERIVATION, STEP BY STEP

With $a_1 = a_2 = 1$ and $x_2 = 1 - x_1 \equiv 1 - x$:

$$\frac{1}{D_1 D_2} = \int_0^1 \frac{dx}{[x D_1 + (1-x) D_2]^2}, \quad x D_1 + (1-x) D_2 = \ell^2 - 2(1-x)\ell \cdot p + (1-x)p^2 + m^2 \quad (175)$$

Shift $\ell = \ell' + (1-x)p$ (allowed in dimensional regularisation). The linear term cancels:

$$\ell'^2 + m^2 + (1-x)p^2 - (1-x)^2 p^2 = \ell'^2 + m^2 + x(1-x)p^2 \quad (176)$$

Now the ℓ' integral is a tadpole with $n = 2$ and effective mass $M^2(x) = m^2 + x(1-x)p^2$. From Equation 63:

$$J(1, 1) = \int_0^1 dx \int [d\ell'] \frac{1}{(\ell'^2 + M^2)^2} = \Gamma\left(2 - \frac{D}{2}\right) \int_0^1 dx [m^2 + x(1-x)p_E^2]^{D/2-2} \quad (177)$$

RESULT: FEYNMAN-PARAMETER FORM OF THE BUBBLE

$$J(1, 1) = \Gamma\left(2 - \frac{D}{2}\right) \int_0^1 dx [m^2 + x(1-x)p_E^2]^{D/2-2} \quad (178)$$

In pictures, the two lines have fused into a tadpole with one dot and an x -dependent mass:

$$\text{---} \bigcirc \text{---} = \int_0^1 dx \text{---} \bigcirc^{\text{dot}} \text{---}^{M^2(x)}, \quad M^2(x) = m^2 + x(1-x)p_E^2 \quad (179)$$

DERIVATION: FROM THE FEYNMAN PARAMETER TO THE HYPERGEOMETRIC FUNCTION

Take $m = 1$ and put $a = 2 - D/2$. The integrand is $(1 + x(1-x)p^2)^{-a}$. Expand it with the binomial series $(1+w)^{-a} = \sum_k (a)_k (-w)^k / k!$ (valid for $|w| < 1$):

$$\int_0^1 (1 + x(1-x)p^2)^{-a} dx = \sum_{k=0}^{\infty} \frac{(a)_k (-p^2)^k}{k!} \int_0^1 x^k (1-x)^k dx \quad (180)$$

Each x integral is a Beta function, $B(k+1, k+1) = (k!)^2 / (2k+1)!$. Now use the identity

$$\left(\frac{3}{2}\right)_k = \frac{3}{2} \cdot \frac{5}{2} \cdots \frac{2k+1}{2} = \frac{3 \cdot 5 \cdots (2k+1)}{2^k} = \frac{(2k+1)!}{2^k \cdot 2^k k!} = \frac{(2k+1)!}{4^k k!} \quad (181)$$

(the odd numbers $3 \cdot 5 \cdots (2k+1)$ are $(2k+1)!$ divided by the even numbers $2 \cdot 4 \cdots 2k = 2^k k!$). So $(k!)^2 / (2k+1)! = k! / (4^k (3/2)_k)$ and the sum becomes

$$\sum_k \frac{(a)_k k!}{(3/2)_k k!} \left(-\frac{p^2}{4}\right)^k = \sum_k \frac{(a)_k (1)_k}{(3/2)_k} \frac{(-x)^k}{k!} = {}_2F_1\left(a, 1; \frac{3}{2}; -x\right), \quad x = \frac{p^2}{4m^2} \quad (182)$$

using $(1)_k = k!$. With the prefactor $\Gamma(2 - D/2)$ and restoring the mass (M^2 carries $(m^2)^{D/2-2}$ in front once m^2 is pulled out), this is Equation 170. The binomial series proves the equality for $|x| < 1$. Both sides are analytic in x for all Euclidean $x \geq 0$ (Euler's integral of Section 9 shows this for the right side), so the equality holds there too. Two completely different routes (differential equation and Feynman parameters) give the same function. The Mathematica box of Section 10 does this sum to all orders.

PHYSICAL PICTURE: WHAT THE FEYNMAN PARAMETER IS

The parameter x says how the external momentum is shared between the two lines. After the shift, line 1 effectively carries momentum $(1-x)p$ and line 2 carries $-xp$. The effective mass $M^2(x) = m^2 + x(1-x)p^2$ is smallest at the ends ($x = 0, 1$) and largest at $x = 1/2$, where the momentum is shared equally. In Minkowski space $p^2 \rightarrow -s$ and $M^2(x) = m^2 - x(1-x)s$ can become negative. That happens first at $x = 1/2$, when $s = 4m^2$: the threshold again.

11.3 Graph polynomials (Symanzik polynomials)

The same steps work for any one-loop integral and they produce two polynomials in the Feynman parameters that can be read off the picture of the graph.

DERIVATION: THE ONE-LOOP GRAPH POLYNOMIALS FROM COMPLETING THE SQUARE

Take N propagators $D_i = (\ell - q_i)^2 + m_i^2$ (Euclidean), with powers a_i and $a = \sum a_i$. Here q_i is the momentum that has flowed out of the loop before line i (fixed by momentum conservation). After the Feynman-parameter formula we need the combination

$$\sum_i x_i D_i = \left(\sum_i x_i \right) \ell^2 - 2\ell \cdot \left(\sum_i x_i q_i \right) + \sum_i x_i (q_i^2 + m_i^2) \equiv \mathcal{U} \ell^2 - 2\ell \cdot Q + J \quad (183)$$

Complete the square. $\mathcal{U} \ell^2 - 2\ell \cdot Q = \mathcal{U}(\ell - Q/\mathcal{U})^2 - Q^2/\mathcal{U}$. Shift $\ell \rightarrow \ell + Q/\mathcal{U}$:

$$\sum_i x_i D_i = \mathcal{U} \ell^2 + \frac{\mathcal{U}J - Q^2}{\mathcal{U}} \equiv \mathcal{U} \ell^2 + \frac{\mathcal{F}}{\mathcal{U}}, \quad \mathcal{F} = \mathcal{U}J - Q^2 \quad (184)$$

Do the loop integral. Rescale $\ell = \ell'/\sqrt{\mathcal{U}}$, so $d^D \ell = \mathcal{U}^{-D/2} d^D \ell'$ and use the tadpole formula with “mass” $\mathcal{F}/\mathcal{U}$:

$$\int [d\ell] \frac{1}{(\mathcal{U} \ell^2 + \mathcal{F}/\mathcal{U})^a} = \mathcal{U}^{-D/2} \frac{\Gamma(a - D/2)}{\Gamma(a)} \left(\frac{\mathcal{F}}{\mathcal{U}} \right)^{D/2-a} = \frac{\Gamma(a - D/2)}{\Gamma(a)} \mathcal{U}^{a-D} \mathcal{F}^{D/2-a} \quad (185)$$

Put back the Feynman-parameter prefactor $\Gamma(a)/\prod \Gamma(a_i)$. The $\Gamma(a)$ cancels:

$$\int [d\ell] \frac{1}{\prod_i D_i^{a_i}} = \frac{\Gamma(a - D/2)}{\prod_i \Gamma(a_i)} \int \prod_i dx_i x_i^{a_i-1} \delta(1 - \sum x_i) \frac{\mathcal{U}^{a-D}}{\mathcal{F}^{a-D/2}} \quad (186)$$

What $\mathcal{F}$ looks like. Write out $\mathcal{U}J - Q^2 = \sum_{i,j} x_i x_j (q_i^2 + m_i^2) - \sum_{i,j} x_i x_j q_i \cdot q_j$. Symmetrising in $i \leftrightarrow j$, $q_i^2 - q_i \cdot q_j \rightarrow \frac{1}{2}(q_i - q_j)^2$, so

$$\mathcal{F} = \sum_{i < j} x_i x_j (q_i - q_j)^2 + \mathcal{U} \sum_i x_i m_i^2, \quad \mathcal{U} = \sum_i x_i \quad (187)$$

Reading them off the graph. The two polynomials have a pretty graphical meaning, which is how they are written for any number of loops.

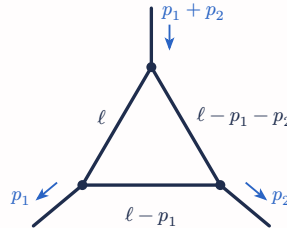
- $\mathcal{U}$ is a sum over **spanning trees**: cut lines until the graph is a tree that still touches every vertex and multiply the x_i of the lines you cut. For one loop, cutting any single line gives a tree, so $\mathcal{U} = x_1 + x_2 + \dots + x_N$.
- $\mathcal{F}_0 = \sum_{i < j} x_i x_j (q_i - q_j)^2$ is a sum over **spanning 2-forests**: cut lines until the graph falls into exactly two pieces, multiply the x_i of the cut lines and the square of the momentum that flows from one piece to the other. Then $\mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum x_i m_i^2$.

For the bubble:

$$\mathcal{U} : \text{---} \bigcirc \text{---} \rightarrow x_1, \quad \text{---} \bigcirc \text{---} \rightarrow x_2, \quad \mathcal{F}_0 : \text{---} \bigcirc \text{---} \rightarrow x_1 x_2 p_E^2 \quad (188)$$

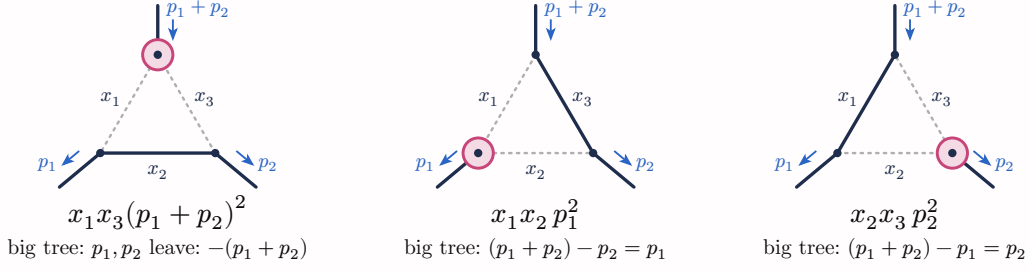
so $\mathcal{U} = x_1 + x_2$ and $\mathcal{F} = x_1 x_2 p_E^2 + (x_1 + x_2)^2 m^2$. For $N = a = 2$ the general formula gives $\Gamma(2 - D/2) \mathcal{U}^{2-D} \mathcal{F}^{D/2-2}$ and on $x_1 + x_2 = 1$ this is Equation 178.

For the triangle, route the momenta like this: $p_1 + p_2$ comes in at the top vertex, p_1 leaves at the bottom left and p_2 at the bottom right. If line 1 carries ℓ , momentum conservation at each vertex fixes the other two lines:



In the notation above, line i carries $\ell - q_i$ with $q_1 = 0$, $q_2 = p_1$, $q_3 = p_1 + p_2$. Cutting two of the three lines gives a 2-forest in three ways. The rule for $\mathcal{F}_0$ (the one used in the lecture) is to **pick one of the two trees, add up all the external momenta that enter it (counting outgoing ones with a minus sign) and square the total**. Momentum conservation guarantees that the other tree gives the same square. In each picture the

pink circle marks the small tree (a single vertex) and the text below gives the sum for the **big** tree, as in the lecture:



For example, in the middle picture the big tree contains the top vertex, where $p_1 + p_2$ comes in and the bottom-right vertex, where p_2 goes out. The total momentum entering it is $(p_1 + p_2) - p_2 = p_1$, so the term is $x_1 x_2 p_1^2$. The small tree (the circled vertex) has only p_1 going out and $(-p_1)^2 = p_1^2$: the same answer, as it must be. In the language of the q_i this is $(q_i - q_j)^2$ for the two removed lines i, j . Together, $\mathcal{U} = x_1 + x_2 + x_3$ and $\mathcal{F}_0 = x_1 x_2 p_1^2 + x_2 x_3 p_2^2 + x_1 x_3 (p_1 + p_2)^2$, which the Mathematica box confirms (there $s_1 = p_1^2$, $s_2 = p_2^2$, $s_{12} = p_1 \cdot p_2$).

MATHEMATICA CHECK: ONE-LOOP GRAPH POLYNOMIALS BY COMPLETING THE SQUARE

One-loop graph polynomials from completing the square. $D_i = (\ell - q_i)^2 + m_i^2$, $\sum x_i D_i = \mathcal{U} \ell^2 - 2\ell \cdot Q + J$.

In [1]: triangle: $q_1 = 0$, $q_2 = p_1$, $q_3 = p_1 + p_2$. $p_1^2 = s_1$, $p_2^2 = s_2$, $p_1 \cdot p_2 = s_{12}$. Q^2 with $Q = \sum x_i q_i$. $\mathcal{U}$

```
q = {0, p1, p1 + p2};
sq[a_] := Expand[a^2] /. {p1^2 -> s1, p2^2 -> s2, p1 p2 -> s12};
xs = {x1, x2, x3};
ms = {m1^2, m2^2, m3^2};
U = Total[xs];
Qsq = sq[Total[xs q]];
J = Total[xs (sq /@ q + ms)];
F = Expand[U J - Qsq];
U
```

Out[1]: $x_1 + x_2 + x_3$

$$\mathcal{F} = \mathcal{U} J - Q^2$$

In [2]: `Collect[F, {x1, x2, x3}, Simplify]`

Out[2]: $x_1 (x_2 (m_1^2 + m_2^2 + s_1) + x_3 (m_1^2 + m_3^2 + s_1 + 2 s_{12} + s_2)) + x_2 x_3 (m_2^2 + m_3^2 + s_2) + m_1^2 x_1^2 + m_2^2 x_2^2 + m_3^2 x_3^2$

the 2-forest rule $\mathcal{F}_0 = \sum_{i < j} x_i x_j (q_i - q_j)^2$, $\mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum x_i m_i^2$. $\mathcal{F} - (\mathcal{F}_0 + \mathcal{U} \sum x_i m_i^2)$

In [3]: `F0 = Sum[xs[[i]] xs[[j]] sq[q[[i]] - q[[j]]], {i, 1, 3}, {j, i + 1, 3}];
Simplify[F - (F0 + U Total[xs ms])]`

Out[3]: 0

the rescaled Gaussian integral. $\int [d\ell] (\mathcal{U} \ell^2 + \mathcal{F}/\mathcal{U})^{-a}$

In [4]: `Omega[D_] := 2 Pi^(D/2)/Gamma[D/2];
g = Omega[D]/Pi^(D/2) Integrate[l^(D - 1) (Uc l^2 + Fc/Uc)^(-a),
{l, 0, Infinity},
Assumptions -> Uc > 0 && Fc > 0 && 0 < D < 2 a];
FullSimplify[g, Uc > 0 && Fc > 0]`

Out[4]:
$$\frac{\mathcal{U}^{a-D} \mathcal{F}^{\frac{1}{2}(D-2a)} \Gamma(a - \frac{D}{2})}{\Gamma(a)}$$

The last line is the rescaled Gaussian integral: $\Gamma(a - D/2)/\Gamma(a) \mathcal{U}^{a-D} \mathcal{F}^{D/2-a}$, as derived. FeynCalc builds the same objects directly from the propagators (in Minkowski form, so $p_E^2 \rightarrow -s$):

MATHEMATICA CHECK: FEYNMAN PARAMETRISATION IN FEYNCALC

FeynCalc: Feynman parametrisation and graph (Symanzik) polynomials. bubble: integrand, prefactor, Feynman

In [1]: parameters

```
fp = FCFeynmanParametrize[FAD[{l, m}, {l - p, m}], {l}, Names -> x,
  FinalSubstitutions -> {SPD[p] -> s},
  FeynmanIntegralPrefactor -> "Unity"];
fp
```

Out[1]: $\{(x_1 + x_2)^{2-D} (m^2 x_1^2 + m^2 x_2^2 + 2 m^2 x_1 x_2 - s x_1 x_2)^{\frac{D}{2}-2}, i \pi^{D/2} \Gamma(2 - \frac{D}{2}), \{x_1, x_2\}\}$

tadpole with power 2

```
In [2]: fpt = FCFeynmanParametrize[FAD[{l, m}, {l, m}], {l}, Names -> x,
  FeynmanIntegralPrefactor -> "Unity"];
fpt
```

Out[2]: $\{1, \frac{i \pi^{D/2} (m^2)^{D/2} \Gamma(2 - \frac{D}{2})}{m^4}, \{\}$

The output reads $\mathcal{U}^{2-D} \mathcal{F}^{D/2-2}$ with $\mathcal{U} = x_1 + x_2$ and $\mathcal{F} = m^2(x_1 + x_2)^2 - s x_1 x_2$ and prefactor $i \pi^{D/2} \Gamma(2 - D/2)$. The i is the Wick factor $i(-1)^2$ of Equation 46. Graph polynomials are the starting point of sector decomposition (pySecDec, FIESTA), of Mellin-Barnes representations (Section 16) and of much of the modern mathematics of Feynman integrals. At L loops the same derivation gives $\mathcal{U}^{a-(L+1)D/2} \mathcal{F}^{LD/2-a}$ with $\Gamma(a - LD/2)$ in front.

12 Bubbles the Rauh way: general powers, one massive line and a trick

Rauh's lecture notes (Section 2.2) do the bubble in a way that is worth copying. Keep the propagator powers general, get a closed formula once and read off every special case from it. Here we follow him for the bubble with one **massless** and one **massive** line (for example a photon and an electron). The notation $[d\ell] = d^D \ell / \pi^{D/2}$ (Euclidean) keeps the formulas short.

$$F(a, b) = \int [d\ell] \frac{1}{(\ell^2)^a ((\ell + q)^2 + m^2)^b} = \text{---} \bigcirc \text{---} \quad (189)$$


In the picture the thick line is the massive propagator (mass m) and the thin line the massless one.

Rauh keeps a and b general, like Kakashi copying a jutsu once with the Sharingan and then using it on every enemy. One formula, every special case.

12.1 The closed formula

DERIVATION, STEP BY STEP

Feynman parameters, general powers. Using the formula of Section 11 with $A = \ell^2$ and $B = (\ell + q)^2 + m^2$, m^2 ,

$$\frac{1}{A^a B^b} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 dx \frac{x^{a-1} (1-x)^{b-1}}{[xA + (1-x)B]^{a+b}} \quad (190)$$

Combine and shift. $x\ell^2 + (1-x)[(\ell + q)^2 + m^2] = \ell^2 + 2(1-x)\ell \cdot q + (1-x)(q^2 + m^2)$. Shift $\ell \rightarrow \ell - (1-x)q$:

$$\ell^2 + (1-x)(q^2 + m^2) - (1-x)^2 q^2 = \ell^2 + x(1-x)q^2 + (1-x)m^2 = \ell^2 + (1-x)(m^2 + xq^2) \quad (191)$$

The loop integral is a tadpole with power $a + b$ and “mass” $M^2 = (1 - x)(m^2 + xq^2)$:

$$\int [d\ell] \frac{1}{(\ell^2 + M^2)^{a+b}} = \frac{\Gamma(a+b-D/2)}{\Gamma(a+b)} (M^2)^{D/2-a-b} \quad (192)$$

The $\Gamma(a+b)$ cancels. Writing $(M^2)^{D/2-a-b} = (1-x)^{D/2-a-b} (m^2)^{D/2-a-b} (1+xz)^{D/2-a-b}$ with $z = q^2/m^2$ and combining with $(1-x)^{b-1}$:

$$F(a, b) = \frac{\Gamma(a+b-D/2)}{\Gamma(a)\Gamma(b)} (m^2)^{D/2-a-b} \int_0^1 dx x^{a-1} (1-x)^{D/2-a-1} (1+xz)^{D/2-a-b} \quad (193)$$

Euler’s integral (Section 9) with $\beta = a$, $\gamma = D/2$, $\alpha = a+b-D/2$ and argument $-z$ gives $B(a, D/2-a) {}_2F_1(a+b-D/2, a; D/2; -z)$ and $B(a, D/2-a) = \Gamma(a)\Gamma(D/2-a)/\Gamma(D/2)$.

RESULT: THE BUBBLE WITH ONE MASSIVE LINE, ANY POWERS

$$F(a, b) = \frac{\Gamma(a+b-D/2)\Gamma(D/2-a)}{\Gamma(b)\Gamma(D/2)} (m^2)^{D/2-a-b} {}_2F_1\left(a+b-\frac{D}{2}, a; \frac{D}{2}; -\frac{q^2}{m^2}\right) \quad (194)$$

(In Minkowski space $q^2 \rightarrow -q^2 - i\epsilon$ and the argument becomes q^2/m^2 , as in Rauh’s eq. (2.24). The first parameter of the hypergeometric function is $a+b-D/2$. Rauh’s eq. (2.24) prints $D/2-a-b$, a sign slip. His own representation (2.26) applied to the line before (2.24) gives $a+b-D/2$, which is also what Mathematica confirms below.)

12.2 The simple case $F(1, 1)$: expanding before integrating

For $a = b = 1$ and $D = 4 - 2\epsilon$ the Feynman-parameter integrand is $(1-x)^{-\epsilon}(1+xz)^{-\epsilon}$ and the prefactor is $\Gamma(\epsilon)(m^2)^{-\epsilon}$. The x integral is finite at both ends, so we may **expand the integrand in ϵ first** and integrate afterwards:

$$(1-x)^{-\epsilon}(1+xz)^{-\epsilon} = 1 - \epsilon[\ln(1-x) + \ln(1+xz)] + O(\epsilon^2) \quad (195)$$

The two integrals are $\int_0^1 \ln(1-x) dx = -1$ and $\int_0^1 \ln(1+xz) dx = (1+1/z)\ln(1+z) - 1$. So

$$\begin{aligned} F(1, 1) &= \Gamma(\epsilon)(m^2)^{-\epsilon} \left[1 + \epsilon \left(2 - \left(1 + \frac{m^2}{q^2} \right) \ln \left(1 + \frac{q^2}{m^2} \right) \right) \right] + O(\epsilon) \\ &= \frac{1}{\epsilon} - \gamma_E - \ln m^2 + 2 - \left(1 + \frac{m^2}{q^2} \right) \ln \left(1 + \frac{q^2}{m^2} \right) + O(\epsilon) \end{aligned} \quad (196)$$

With $q^2 \rightarrow -q^2 - i\epsilon$ this is Rauh’s eq. (2.27) and above threshold ($q^2 > m^2$ in Minkowski) the log produces the imaginary part, as in Section 15.

12.3 The trick: when the Feynman-parameter integral itself diverges

Now take $a = 2$, $b = 1$: the massless line is squared. The formula before Equation 194 gives, at $D = 4 - 2\epsilon$,

$$F(2, 1) = \Gamma(1+\epsilon) \int_0^1 dx x(1-x)^{-1-\epsilon} (m^2 + xq^2)^{-1-\epsilon} \quad (197)$$

At $x \rightarrow 1$ the factor $(1-x)^{-1-\epsilon}$ makes the integral diverge. We can **not** expand in ϵ first, because the ϵ in the exponent is what keeps the integral finite. This is an infrared divergence. The squared massless propagator $1/(\ell^2)^2$ at small ℓ .

DERIVATION: THE SUBTRACTION TRICK AND WHAT IT GIVES

Call $g(x) = x(m^2 + xq^2)^{-1-\epsilon}$, the smooth part. **Add and subtract its value at the dangerous end point $x = 1$:**

$$\int_0^1 (1-x)^{-1-\epsilon} g(x) dx = \int_0^1 (1-x)^{-1-\epsilon} [g(x) - g(1)] dx + g(1) \int_0^1 (1-x)^{-1-\epsilon} dx \quad (198)$$

The second piece holds the pole, exactly. Put $u = 1 - x$: $\int_0^1 u^{-1-\varepsilon} du = [u^{-\varepsilon}/(-\varepsilon)]_0^1 = -1/\varepsilon$, computed for $\varepsilon < 0$ (where $u^{-\varepsilon} \rightarrow 0$ at $u = 0$) and then continued. So this piece is $-g(1)/\varepsilon = -(m^2 + q^2)^{-1-\varepsilon}/\varepsilon$.

The first piece is finite. $g(x) - g(1)$ vanishes linearly at $x = 1$, which cancels one power of $(1 - x)$. Now we may set $\varepsilon = 0$ inside:

$$\begin{aligned} \int_0^1 \frac{g(x) - g(1)}{1 - x} dx &= \int_0^1 \frac{1}{1 - x} \left[\frac{x}{m^2 + xq^2} - \frac{1}{m^2 + q^2} \right] dx \\ &= -\frac{m^2}{m^2 + q^2} \int_0^1 \frac{dx}{m^2 + xq^2} = -\frac{m^2}{q^2(m^2 + q^2)} \ln \left(1 + \frac{q^2}{m^2} \right) \end{aligned} \quad (199)$$

(Over a common denominator, $x(m^2 + q^2) - (m^2 + xq^2) = -m^2(1 - x)$, which is where the factor $1 - x$ cancels.)

Put together. Expand $-(m^2 + q^2)^{-1-\varepsilon}/\varepsilon = -(m^2 + q^2)^{-1} [1/\varepsilon - \ln(m^2 + q^2)] + O(\varepsilon)$:

$$F(2, 1) = \frac{\Gamma(1 + \varepsilon)}{m^2 + q^2} \left[-\frac{1}{\varepsilon} + \ln(m^2 + q^2) - \frac{m^2}{q^2} \ln \left(1 + \frac{q^2}{m^2} \right) \right] + O(\varepsilon) \quad (200)$$

So **the trick gives two things**. One is the pole, isolated exactly and multiplied by the value of the integrand at the end point $(-g(1)/\varepsilon)$. The other is a remainder that is an ordinary, finite integral one can expand in ε and do in closed form. The same idea (subtract the singular behaviour at the boundary, add it back in a form that can be integrated exactly) is behind sector decomposition and the plus-distributions of QCD.

MATHEMATICA CHECK: THE CLOSED FORMULA, F(1,1) AND THE SUBTRACTION TRICK

Bubble with a massless line (power a) and a massive line (power b): after the shift the x -integrand is

In [1]: $x^{a-1}(1-x)^{D/2-a-1}(m^2+xq^2)^{D/2-a-b} \cdot \int_0^1 x^{a-1}(1-x)^{D/2-a-1}(1+xz)^{D/2-a-b} dx$

```
fpx = Integrate[x^(a - 1) (1 - x)^(Dd/2 - a - 1) (1 + x z)^(Dd/2 - a - b),
  {x, 0, 1},
  Assumptions -> a > 0 && Dd/2 - a > 0 && z > 0];
fpx
```

Out[1]: $\Gamma(a) \Gamma\left(\frac{D}{2} - a\right) {}_2\tilde{F}_1\left(a, a + b - \frac{D}{2}; \frac{D}{2}; -z\right)$

minus $B(a, D/2 - a) {}_2F_1(a + b - D/2, a; D/2; -z)$

In [2]:

```
euler = Gamma[a] Gamma[Dd/2 - a]/Gamma[Dd/2] Hypergeometric2F1[a + b - Dd/2,
  a, Dd/2, -z];
FullSimplify[fpx - euler]
```

Out[2]: 0

$F(1, 1)$ at $D = 4 - 2\varepsilon$: order- ε coefficient of the bracket, summed. $\sum_{k \geq 1} (-z)^k/(k(k+1))$

In [3]:

```
c1 = Sum[(-z)^(k/(k + 1)), {k, 1, Infinity}];
c1
```

Out[3]: $\frac{z + z(-\log(z+1)) - \log(z+1)}{z}$

1 + that, minus $2 - (1 + 1/z) \ln(1 + z)$

In [4]:

```
FullSimplify[1 + c1 - (2 - (1 + 1/z) Log[1 + z]), z > 0]
```

Out[4]: 0

$F(2, 1)$ from the subtraction trick, $\Gamma(1 + \varepsilon)/(1 + z)[-1/\varepsilon + \ln(1 + z) - \ln(1 + z)/z]$, against the exact result ($m = 1, z = q^2$). compare Taylor coefficients in z up to z^5 . exact minus trick, coefficients of $z^0, \dots, z^5$

In [5]:

```
trick = Gamma[1 + ep] (1/(1 + z)) (-1/ep + Log[1 + z] - Log[1 + z]/z);
exact = Gamma[1 + ep] Gamma[-ep]/Gamma[2 - ep] Hypergeometric2F1[1 + ep,
  2, 2 - ep, -z];
```

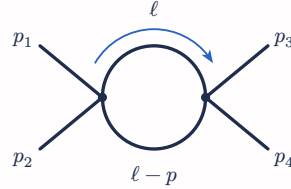
```
ex1 = Table[Normal@Series[Gamma[1 + ep] Gamma[-ep]/Gamma[2 - ep] Pochhammer[1 + ep,
  k] Pochhammer[2, k]/(Pochhammer[2 - ep, k] k!) (-1)^k,
  {ep, 0, 0}], {k, 0, 5}];
tr1 = Table[Normal@Series[SeriesCoefficient[trick,
  {z, 0, k}], {ep, 0, 0}], {k, 0, 5}];
Simplify[ex1 - tr1]
```

Out[5]: {0, 0, 0, 0, 0, 0}

The first two lines confirm Equation 194 (Mathematica writes the regularised hypergeometric function, which is ${}_2F_1/\Gamma(\gamma)$, with its two upper parameters in the other order). The next two check the order- ε bracket of $F(1, 1)$ by summing the exact hypergeometric series. The last line compares the trick with the **exact** $F(2, 1)$ from Equation 194. The Taylor coefficients in $z = q^2/m^2$, each expanded in ε , agree to all orders shown.

13 The fish diagram of $\lambda\varphi^4$

The one-loop correction to $2 \rightarrow 2$ scattering in $\lambda\varphi^4$ has three diagrams, one for each channel s, t, u . Each looks like a fish:



Take the s channel, where the total momentum $p = p_1 + p_2$ flows through the loop.

Luffy would eat this fish raw. We will cook it properly: Feynman parameters first, then an expansion in ε , then one integral over x .

DERIVATION, STEP BY STEP

The diagram in Euclidean space. Two vertices give $(-\lambda)^2$. The symmetry factor is $1/2$, because swapping the two internal lines gives the same diagram. In $D = 2\omega$ the coupling is written $\lambda(\mu^2)^{2-\omega}$ at each vertex, so two vertices give $(\mu^2)^{4-2\omega}$:

$$\mathcal{V}_s = \frac{1}{2}(-\lambda)^2(\mu^2)^{4-2\omega} \int \frac{d^{2\omega}\ell}{(2\pi)^{2\omega}} \frac{1}{(\ell^2 + m^2)((\ell - p)^2 + m^2)} = \frac{\lambda^2}{2}(\mu^2)^{4-2\omega}(4\pi)^{-\omega} \text{---}\bigcirc\text{---} \quad (201)$$

where the bubble picture means $J(1, 1)$ with our measure $d^D\ell/\pi^{D/2}$ and $(4\pi)^{-\omega}$ converts to $d^D\ell/(2\pi)^D$.

Feynman parameters. The loop integral is the bubble of Equation 178, but with the measure $d^D\ell/(2\pi)^D$, which is $(4\pi)^{-\omega}$ times the measure $d^D\ell/\pi^{D/2}$:

$$\mathcal{V}_s = \frac{\lambda^2}{2}(\mu^2)^{4-2\omega} \frac{\Gamma(2-\omega)}{(4\pi)^\omega} \int_0^1 dx [m^2 + x(1-x)p^2]^{\omega-2} \quad (202)$$

Expand in $\varepsilon = 2 - \omega$. Then $\Gamma(2-\omega) = \Gamma(\varepsilon)$, $(4\pi)^{-\omega} = (4\pi)^{\varepsilon-2} = (4\pi)^\varepsilon/(16\pi^2)$ and $[M^2]^{\omega-2} = [M^2]^{-\varepsilon}$ with $M^2 = m^2 + x(1-x)p^2$. Keep one factor $(\mu^2)^\varepsilon$ outside (it carries the dimension of the four-point coupling) and combine the rest into one power:

$$\Gamma(\varepsilon)(4\pi)^\varepsilon(\mu^2)^\varepsilon \frac{[M^2]^{-\varepsilon}}{16\pi^2} = \frac{\Gamma(\varepsilon)}{16\pi^2} \left(\frac{4\pi\mu^2}{M^2} \right)^\varepsilon \quad (203)$$

Now $\Gamma(\varepsilon) = 1/\varepsilon + \psi(1) + O(\varepsilon)$ and $(4\pi\mu^2/M^2)^\varepsilon = 1 + \varepsilon \ln(4\pi\mu^2/M^2) + O(\varepsilon^2)$. Their product, to order ε^0 , is $1/\varepsilon + \psi(1) + \ln(4\pi\mu^2/M^2)$. Finally split the log, $\ln(4\pi\mu^2/M^2) = \ln(4\pi\mu^2/m^2) - \ln(1 + x(1-x)p^2/m^2)$:

$$= \frac{1}{16\pi^2} \left[\frac{1}{\varepsilon} + \psi(1) + \ln \frac{4\pi\mu^2}{m^2} - \ln \left(1 + x(1-x) \frac{p^2}{m^2} \right) \right] + O(\varepsilon) \quad (204)$$

The first three terms do not depend on x , so $\int_0^1 dx$ of them is just themselves.

The x integral. Call $z = p^2/m^2 > 0$ and $L(z) = \int_0^1 \ln(1 + x(1-x)z) dx$. Integrate by parts with $u = \ln(1 + x(1-x)z)$ and $v = x - 1/2$ (a handy choice, symmetric about the middle):

$$L = \left[\left(x - \frac{1}{2} \right) \ln(1 + x(1-x)z) \right]_0^1 - \int_0^1 \left(x - \frac{1}{2} \right) \frac{z(1-2x)}{1 + x(1-x)z} dx \quad (205)$$

The boundary term vanishes because $\ln 1 = 0$ at both ends. Since $-(x - 1/2)(1 - 2x) = (1 - 2x)^2/2$, put $y = 1 - 2x$: then $x(1-x) = (1 - y^2)/4$, $dx = -dy/2$ and $x : 0 \rightarrow 1$ is $y : 1 \rightarrow -1$:

$$L = \int_{-1}^1 \frac{dy}{2} \frac{zy^2/2}{1 + z(1-y^2)/4} = \int_{-1}^1 \frac{zy^2}{4 + z - zy^2} dy = \int_{-1}^1 \frac{y^2}{\tilde{\beta}^2 - y^2} dy, \quad \tilde{\beta}^2 = \frac{4+z}{z} \quad (206)$$

Write $y^2/(\tilde{\beta}^2 - y^2) = -1 + \tilde{\beta}^2/(\tilde{\beta}^2 - y^2)$ and use partial fractions, $\tilde{\beta}^2/(\tilde{\beta}^2 - y^2) = (\tilde{\beta}/2) [1/(\tilde{\beta} - y) + 1/(\tilde{\beta} + y)]$:

$$L = -2 + \frac{\tilde{\beta}}{2} [-\ln(\tilde{\beta} - y) + \ln(\tilde{\beta} + y)]_{-1}^1 = -2 + \tilde{\beta} \ln \frac{\tilde{\beta} + 1}{\tilde{\beta} - 1} \quad (207)$$

Put together. Multiply by $\lambda^2/2$ and use $1/(2 \cdot 16\pi^2) = 1/(32\pi^2)$. Since $-L = 2 - \tilde{\beta} \ln(\dots)$,

$$\mathcal{V}_s = (\mu^2)^{2-\omega} \frac{\lambda^2}{32\pi^2} \left[\frac{1}{2-\omega} + \psi(1) + 2 + \ln \frac{4\pi\mu^2}{m^2} - \sqrt{1 + \frac{4m^2}{p^2}} \ln \frac{\sqrt{1 + 4m^2/p^2} + 1}{\sqrt{1 + 4m^2/p^2} - 1} + O(2-\omega) \right] \quad (208)$$

This is Ramond's result, written at the end of the lecture. Two remarks on signs. The value of the diagram is $+\lambda^2/2\dots$ as above. The one-particle irreducible four-point function is $\Gamma^{(4)} = \lambda - (\mathcal{V}_s + \mathcal{V}_t + \mathcal{V}_u) + \dots$, so inside $\Gamma^{(4)}$ the diagrams appear with a minus sign.

MATHEMATICA CHECK: RAMOND'S EXPANSION OF THE FISH

Fish diagram of $\lambda\varphi^4$, Euclidean $p^2 > 0$, $D = 2\omega = 4 - 2\varepsilon$. exact expansion minus Ramond's form

```
In [1]: fish = lam^2/2 mu2^(4 - 2 w) Gamma[2 - w]/(4 Pi)^w *
         Integrate[(m2 + y (1 - y) pp)^(w - 2), {y, 0, 1},
         Assumptions -> 1 < w < 2 && pp > 0 && m2 > 0];
ser = Normal@Series[fish /. w -> 2 - eps, {eps, 0, 0}];
b = Sqrt[1 + 4 m2/pp];
ramond = mu2^eps lam^2/(32 Pi^2) (1/eps + PolyGamma[1] + 2 + Log[4 Pi mu2/m2] -
         b Log[(b + 1)/(b - 1)]);
FullSimplify[ser - Normal@Series[ramond,
         {eps, 0, 0}], pp > 0 && m2 > 0 && mu2 > 0]
```

Out[1]: 0

the x integral in closed form and the $y = 1 - 2x$ step of the text. $\int_0^1 \ln(1 + x(1-x)z) dx$

```
In [2]: FullSimplify[Integrate[Log[1 + x (1 - x) z],
         {x, 0, 1}, Assumptions -> z > 0], z > 0]
```

Out[2]: $2 \sqrt{\frac{z+4}{z}} \tanh^{-1}\left(\sqrt{\frac{z}{z+4}}\right) - 2$

$\int_{-1}^1 y^2/(\tilde{\beta}^2 - y^2) dy, \tilde{\beta} > 1$

In [3]:

```
FullSimplify[Integrate[y^2/(bt^2 - y^2),
  {y, -1, 1}, Assumptions -> bt > 1], bt > 1]
```

Out[3]: $2\tilde{\beta} \coth^{-1}(\tilde{\beta}) - 2$

The first line expands the exact D -dimensional expression and subtracts Ramond's form. The difference is zero. The second does $L(z)$ in closed form $(2\sqrt{(4+z)/z} \operatorname{arctanh} \sqrt{z/(4+z)} = \tilde{\beta} \ln((\tilde{\beta}+1)/(\tilde{\beta}-1))$ since $\operatorname{arctanh} w = \frac{1}{2} \ln((1+w)/(1-w))$ with $w = 1/\tilde{\beta}$. The third is the y integral of our derivation, with $\operatorname{arccoth} \tilde{\beta} = \operatorname{arctanh}(1/\tilde{\beta})$.

PHYSICAL PICTURE: THE RUNNING COUPLING

The pole $1/\varepsilon$ multiplies λ^2 and does not depend on p . So it can be removed by a counterterm $\delta_\lambda \propto \lambda^2/\varepsilon$ in the coupling and since the three channels contribute equally to the pole, $\delta_\lambda = 3\lambda^2/(32\pi^2\varepsilon) + \dots$. At large p^2 the finite part behaves like $-\ln(p^2/m^2)$, so the effective coupling at momentum p changes logarithmically with p . To get the beta function, write the bare coupling as $\lambda_0 = \mu^{2\varepsilon}(\lambda + 3\lambda^2/(32\pi^2\varepsilon) + \dots)$. It does not depend on μ , so $\mu d\lambda_0/d\mu = 0$ gives $2\varepsilon(\lambda + 3\lambda^2/(32\pi^2\varepsilon)) + \beta(1 + 6\lambda/(32\pi^2\varepsilon)) = 0$ with $\beta = \mu d\lambda/d\mu$. Solving order by order, $\beta = -2\varepsilon\lambda - 6\lambda^2/(32\pi^2) + 12\lambda^2/(32\pi^2) + \dots = -2\varepsilon\lambda + 3\lambda^2/(16\pi^2)$. At $\varepsilon \rightarrow 0$ this is the beta function $\beta(\lambda) = 3\lambda^2/(16\pi^2)$ of $\lambda\varphi^4$ theory (Peskin, Section 12.2).

14 Part 2: graph polynomials for any diagram

The second lecture started from the fish of Section 13, written in the order one actually does it:

1. Write down the expression and combine the denominators with one Feynman parameter:

$$\frac{\lambda^2}{2}(\mu^2)^{4-2\omega} \int_0^1 dx \int \frac{d^{2\omega}\ell}{(2\pi)^{2\omega}} \frac{1}{(\ell^2 + m^2 + p^2x(1-x))^2} \quad (209)$$

2. Do the loop integral. It is a tadpole with power 2 and a shifted mass:

$$\frac{\lambda^2}{2}(\mu^2)^{4-2\omega} \int_0^1 dx \frac{\Gamma(2-\omega)}{(4\pi)^\omega} \frac{1}{(m^2 + p^2x(1-x))^{2-\omega}} \quad (210)$$

3. What survives is only an integral over the Feynman parameters. The loop momentum is gone.

So the real work is always the Feynman-parameter integral and the natural question is whether we can write down that integrand **directly from the picture**, without ever doing the loop integral. We can. The integrand is built from two polynomials, $\mathcal{U}$ and $\mathcal{F}$, that can be read off the graph by cutting lines. This chapter derives them for any number of loops and then does three examples from the lecture, following Weinzierl's and Smirnov's books (Smirnov writes the Feynman parameters as α_i and calls $\mathcal{F}_0$ “ V ”, Weinzierl writes them as a_i . Here they are x_i and $\mathcal{F}_0$).

Hashirama's Wood Release can grow a whole forest in one move. We need something more modest: spanning trees and 2-forests. And the one rule of our forest is the rule of the Hidden Leaf: a tree may not contain a loop.

14.1 Keeping the delta function general (Cheng-Wu)

On the board the bubble was first written “without imposing $\delta(1 - \alpha_1 - \alpha_2)$ ”. This is allowed and useful.

RESULT: THE CHENG-WU THEOREM

In a Feynman-parameter integral the delta function $\delta(1 - \sum_{i=1}^N x_i)$ may be replaced by $\delta(1 - \sum_{i \in S} x_i)$ for **any** non-empty subset S of the lines. The parameters in S then run over $[0, 1]$ as before and all the others run from 0 to ∞ .

WHY IT WORKS

Go back to the moment in Section 11 where the overall scale was separated. There we inserted $1 = \int_0^\infty d\lambda \delta(\lambda - \sum_{i=1}^N \alpha_i)$. Nothing forces us to sum over all lines. Insert instead $1 = \int_0^\infty d\lambda \delta(\lambda - \sum_{i \in S} \alpha_i)$ and again write $\alpha_i = \lambda x_i$ for every i . The Jacobian is the same, $\prod d\alpha_i = \lambda^N \prod dx_i$ and $\delta(\lambda - \lambda \sum_{i \in S} x_i) = \lambda^{-1} \delta(1 - \sum_{i \in S} x_i)$. Since the integrand is homogeneous in the α_i , the λ integral goes through as before. The only change is in the delta function and in the ranges.

For the bubble, choosing $S = \{1\}$ fixes $x_1 = 1$ and lets x_2 run to infinity. The Mathematica check below does both versions at $D = 3$ (where the bubble is finite) and finds the same function.

MATHEMATICA CHECK: TWO CHOICES OF THE DELTA FUNCTION GIVE THE SAME ANSWER

Cheng-Wu for the bubble at $D = 3$: integrand $\mathcal{U}^{2-D} \mathcal{F}^{D/2-2}$ with $\mathcal{U} = x_1 + x_2$, $\mathcal{F} = x_1 x_2 p^2 + (x_1 + x_2)^2 m^2$.

In [1]: $\delta(1 - x_1 - x_2): \int_0^1 dx_1$

```
Dd = 3;
integrand = (x1 + x2)^(2 - Dd) (x1 x2 pp + (x1 + x2)^2 m2)^(Dd/2 - 2);
full = Integrate[integrand /. x2 -> 1 - x1,
  {x1, 0, 1}, Assumptions -> pp > 0 && m2 > 0];
cw = Integrate[integrand /. x1 -> 1,
  {x2, 0, Infinity}, Assumptions -> pp > 0 && m2 > 0];
FullSimplify[full, pp > 0 && m2 > 0]
```

Out[1]:
$$\frac{\pi - 2 \tan^{-1}\left(2 \sqrt{\frac{m^2}{p^2}}\right)}{\sqrt{p^2}}$$

$\delta(1 - x_1)$ only: $\int_0^\infty dx_2$

In [2]: FullSimplify[cw, pp > 0 && m2 > 0]

Out[2]:
$$\frac{2 \cot^{-1}\left(2 \sqrt{\frac{m^2}{p^2}}\right)}{\sqrt{p^2}}$$

difference

In [3]: FullSimplify[full - cw, pp > 0 && m2 > 0]

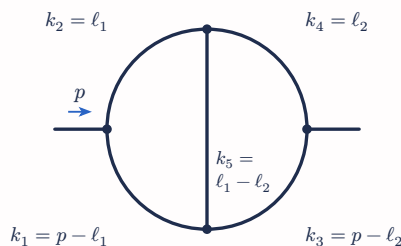
Out[3]: 0

14.2 The general formula, derived

We use the notation of the board: L is the number of loops, N the number of propagators, ν_i the power of propagator i and $N_\nu = \sum_{i=1}^N \nu_i$ the sum of all powers. (In the one-loop chapters the powers were called a_i and their sum a . For $\nu_i = 1$ we simply have $N_\nu = N$.)

Take an L -loop Feynman integral with N propagators. Line i carries momentum $k_i = \sum_{r=1}^L c_{ir} \ell_r + q_i$, where the $c_{ir} \in \{0, \pm 1\}$ say which loops run through the line and q_i is a combination of external momenta. In Euclidean space $D_i = k_i^2 + m_i^2$.

For example, in the two-loop kite (one of the examples below) the five lines carry



so line 5 has $k_5 = \ell_1 - \ell_2$: $c_{51} = 1$, $c_{52} = -1$, $q_5 = 0$. Line 1 has $k_1 = p - \ell_1$: $c_{11} = -1$, $c_{12} = 0$, $q_1 = p$. Line 2 has only $c_{21} = 1$. The numbers c_{ir} just record which loop momenta flow through which line and in which direction.

DERIVATION: $\mathcal{U}$ AND $\mathcal{F}$ AT ANY LOOP ORDER

Step 1, Schwinger parameters. For each propagator, $1/D_i^{\nu_i} = \Gamma(\nu_i)^{-1} \int_0^\infty d\alpha_i \alpha_i^{\nu_i-1} e^{-\alpha_i D_i}$. So the integral becomes

$$I = \frac{1}{\prod_i \Gamma(\nu_i)} \int_0^\infty \prod_i d\alpha_i \alpha_i^{\nu_i-1} \int \prod_{r=1}^L \frac{d^D \ell_r}{\pi^{D/2}} e^{-\sum_i \alpha_i D_i} \quad (211)$$

Step 2, the exponent is quadratic in the loop momenta. Expand $\sum_i \alpha_i k_i^2$:

$$\sum_i \alpha_i D_i = \sum_{r,s} \ell_r \cdot \ell_s M_{rs} - 2 \sum_r \ell_r \cdot Q_r + J \quad (212)$$

$$M_{rs} = \sum_i \alpha_i c_{ir} c_{is}, \quad Q_r = - \sum_i \alpha_i c_{ir} q_i, \quad J = \sum_i \alpha_i (q_i^2 + m_i^2) \quad (213)$$

M is an $L \times L$ symmetric matrix, linear in the α 's.

Step 3, complete the square. Shift $\ell_r \rightarrow \ell_r + \sum_s (M^{-1})_{rs} Q_s$. The linear term disappears and the exponent becomes

$$\sum_{r,s} \ell_r \cdot \ell_s M_{rs} + J - \sum_{r,s} Q_r \cdot (M^{-1})_{rs} Q_s \quad (214)$$

Step 4, the Gaussian integral. Each of the D components gives an L -dimensional Gaussian, $\int d^L y e^{-y^T M y} = \pi^{L/2} (\det M)^{-1/2}$. So

$$\int \prod_r \frac{d^D \ell_r}{\pi^{D/2}} e^{-\ell^T M \ell} = (\det M)^{-D/2} \quad (215)$$

Define

$$\mathcal{U} = \det M, \quad \mathcal{F} = \det M (J - Q^T M^{-1} Q) \quad (216)$$

($\mathcal{F}$ is a polynomial because $\det M \cdot M^{-1}$ is the adjugate matrix, whose entries are polynomials.) Then

$$I = \frac{1}{\prod_i \Gamma(\nu_i)} \int_0^\infty \prod_i d\alpha_i \alpha_i^{\nu_i-1} \mathcal{U}^{-D/2} e^{-\mathcal{F}/\mathcal{U}} \quad (217)$$

Step 5, homogeneity. M is linear in the α 's and $L \times L$, so $\mathcal{U}(\lambda\alpha) = \lambda^L \mathcal{U}(\alpha)$. J and Q are linear, so $\mathcal{F}(\lambda\alpha) = \lambda^{L+1} \mathcal{F}(\alpha)$. $\mathcal{U}$ is homogeneous of degree L and $\mathcal{F}$ of degree $L+1$.

Step 6, separate the scale. Insert $1 = \int d\lambda \delta(\lambda - \sum \alpha_i)$, put $\alpha_i = \lambda x_i$. With $N_\nu = \sum \nu_i$ the powers of λ are: λ^{N-1} from the measure, $\lambda^{N_\nu-N}$ from $\prod \alpha_i^{\nu_i-1}$, $\lambda^{-LD/2}$ from $\mathcal{U}^{-D/2}$ and the exponent is $-\lambda \mathcal{F}(x)/\mathcal{U}(x)$. The λ integral is a Gamma function:

$$\int_0^\infty d\lambda \lambda^{N_\nu-LD/2-1} e^{-\lambda \mathcal{F}/\mathcal{U}} = \Gamma\left(N_\nu - L\frac{D}{2}\right) \left(\frac{\mathcal{U}}{\mathcal{F}}\right)^{N_\nu-LD/2} \quad (218)$$

Collect the powers of $\mathcal{U}$: $\mathcal{U}^{-D/2} \mathcal{U}^{N_\nu-LD/2} = \mathcal{U}^{N_\nu-(L+1)D/2}$.

RESULT: THE FEYNMAN-PARAMETER FORMULA FOR ANY DIAGRAM (EUCLIDEAN)

$$\int \prod_{r=1}^L \frac{d^D \ell_r}{\pi^{D/2}} \frac{1}{\prod_{i=1}^N D_i^{\nu_i}} = \frac{\Gamma(N_\nu-LD/2)}{\prod_i \Gamma(\nu_i)} \int_0^\infty \prod_{i=1}^N dx_i x_i^{\nu_i-1} \delta\left(1 - \sum x_i\right) \frac{\mathcal{U}^{N_\nu-(L+1)D/2}}{\mathcal{F}^{N_\nu-LD/2}} \quad (219)$$

with $N_\nu = \sum_i \nu_i$, $\mathcal{U}$ homogeneous of degree L and $\mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum_i x_i m_i^2$ of degree $L+1$. For $L=1$ this is the formula of Section 11.

14.3 Reading $\mathcal{U}$ and $\mathcal{F}$ off the graph

Doing the determinant by hand is tedious. Kirchhoff's matrix-tree theorem (1847, from electrical circuits) says it can be replaced by counting trees:

- A **1-tree** (spanning tree) is what is left after removing lines so that the graph is still connected, touches every vertex and has **no loop**. For an L -loop graph one must remove exactly L lines.

$$\mathcal{U} = \sum_{\text{1-trees removed lines}} \prod x_i \quad (220)$$

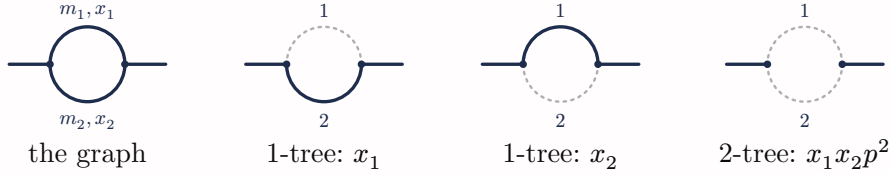
- A **2-tree** (spanning 2-forest) is what is left after removing lines so that the graph falls into exactly **two** pieces, each without a loop. One must remove $L + 1$ lines. Pick one of the two pieces and add up all external momenta entering it (outgoing ones with a minus sign). Call the total P . By momentum conservation the other piece has total $-P$, so P^2 does not depend on which piece you pick. Then

$$\mathcal{F}_0 = \sum_{\text{2-trees}} \left(\prod_{\text{removed lines}} x_i \right) P^2, \quad \mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum_i x_i m_i^2 \quad (221)$$

Two practical rules follow. A 2-tree in which one piece receives no external momentum (or only a massless on-shell leg, $p_i^2 = 0$) contributes nothing. And since every term of $\mathcal{U}$ has L factors and every term of $\mathcal{F}_0$ has $L + 1$, the degrees are automatic.

14.4 Example 1: the bubble with two masses

This is the example in Weinzierl's book (Section 2.5, eqs. (2.172) to (2.176)), with masses m_1, m_2 on the two lines. He works in Minkowski space with a scale μ , so his $-p^2/\mu^2$ is our Euclidean p^2 at $\mu = 1$.



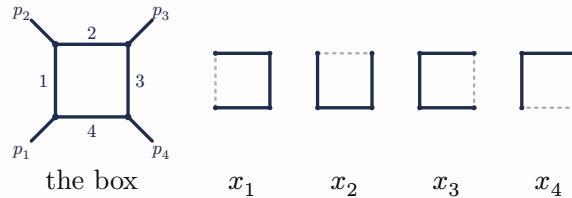
Removing the top line leaves a connected tree and so does removing the bottom one: $\mathcal{U} = x_1 + x_2$. Removing both lines splits the graph into the left vertex and the right vertex and the momentum between them is p : $\mathcal{F}_0 = x_1 x_2 p^2$. So

$$\mathcal{U} = x_1 + x_2, \quad \mathcal{F} = x_1 x_2 p^2 + (m_1^2 x_1 + m_2^2 x_2)(x_1 + x_2) \quad (222)$$

For $m_1 = m_2 = m$ and $x_1 + x_2 = 1$ this is $m^2 + x(1-x)p^2$, the effective mass of Section 11, so the graphical rule gives the same integral as doing the loop by hand.

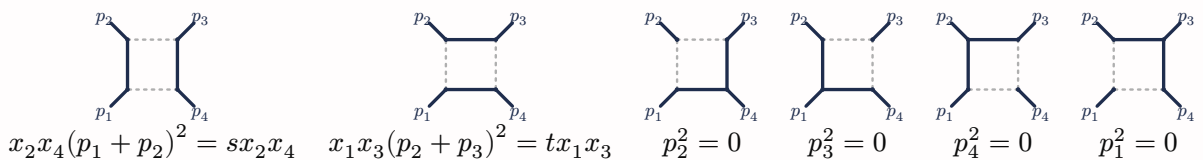
14.5 Example 2: the massless box (after Smirnov, Fig. 3.6)

All internal lines massless, all external legs on shell and massless, $p_i^2 = 0$. Number the lines **cyclically** around the box (this matters, see below): 1 on the left, 2 on top, 3 on the right, 4 at the bottom. Smirnov numbers them in a different cyclic order. The warning below compares the two. The Mandelstam variables are $s = (p_1 + p_2)^2$ and $t = (p_2 + p_3)^2$ (Euclidean squares here).



1-trees. One loop, so remove one line. Any single line works: $\mathcal{U} = x_1 + x_2 + x_3 + x_4$, degree 1.

2-trees. Remove two lines, $\binom{4}{2} = 6$ ways:



Removing the top and bottom lines (2 and 4) splits the box into a left half and a right half. The left tree contains the vertices where p_1 and p_2 enter, so its total is $P = p_1 + p_2$ and the term is $s x_2 x_4$. Removing the two vertical lines (1 and 3) splits it into top and bottom. The top tree has p_2 and p_3 , so $P = p_2 + p_3$ and the term is $t x_1 x_3$. The other four choices remove two neighbouring lines and cut off a single corner, which receives only p_i with $p_i^2 = 0$. They “do not contribute”, as the board says. So

$$\mathcal{U} = x_1 + x_2 + x_3 + x_4, \quad \mathcal{F} = s x_2 x_4 + t x_1 x_3 \quad (223)$$

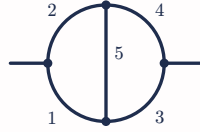
degrees 1 and 2 as they must be.

A warning about labels. Which pair of x ’s multiplies s depends entirely on how the lines are numbered. If the lines are not numbered in cyclic order around the box (for example 1 on top, 2 on the left, 3 on the right, 4 at the bottom), the same physics reads $s x_1 x_4 + t x_2 x_3$. Computer programs also renumber the lines on their own. FeynCalc sorts the propagators, so its x_1 is our line 4. This is how “ s and t exchanged” slips appear in books and notes. The safe rule is always to look at the picture. **s goes with the pair of lines whose removal separates p_1, p_2 from p_3, p_4 .**

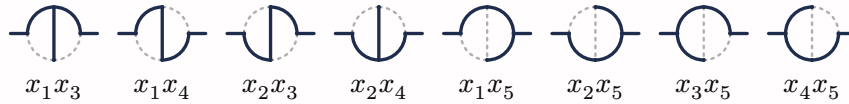
Smirnov’s Fig. 3.6 (2006) puts p_1 and p_2 on the left and p_3 at the top right, with $t = (p_1 + p_3)^2$. His lines are 1 on top, 2 on the left, 3 at the bottom and 4 on the right, again cyclic. Removing 1 and 3 separates p_1, p_2 from p_3, p_4 , so his polynomial is $V = s \alpha_1 \alpha_3 + t \alpha_2 \alpha_4$. This is what was written on the board and what his eq. (3.28) and his Example 4.6 use. His eq. (3.27) prints $V = t \alpha_1 \alpha_3 + s \alpha_2 \alpha_4$ with s and t exchanged, a misprint of exactly this kind. It does no harm in eq. (3.29), where all powers are 1 and the integral is symmetric.

14.6 Example 3: the two-loop kite (Smirnov, Fig. 3.10)

This is a two-point function at two loops. Momentum p enters at the left vertex L and leaves at the right vertex R, the top and bottom vertices are T and B and all five lines are massless. Lines 1 (L-B), 2 (L-T), 3 (B-R), 4 (T-R) run around the outside, line 5 (T-B) is the chord. This is Smirnov’s Example 3.5 with the lines numbered exactly as in his Fig. 3.10. The $\mathcal{U}$ and $\mathcal{F}$ below are his eqs. (3.44) and (3.45).



1-trees. Two loops, so remove two lines and what is left must be connected and loop-free. Of the $\binom{5}{2} = 10$ pairs, eight work:

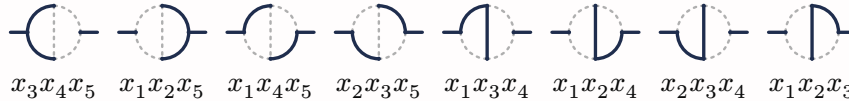


The two missing pairs, (1, 2) and (3, 4), cut off the vertex L (or R) completely and leave a closed loop T-R-B (or L-T-B) behind, so they are not trees. So

$$\mathcal{U} = (x_1 + x_2)(x_3 + x_4) + x_5(x_1 + x_2 + x_3 + x_4) \quad (224)$$

a homogeneous polynomial of degree 2.

2-trees. Remove three lines so that two loop-free pieces remain. Of the $\binom{5}{3} = 10$ triples, only those that put L and R in **different** pieces carry the momentum p between them. Eight triples do:



each times p^2 . The remaining two triples, (1, 3, 5) and (2, 4, 5), leave L and R connected (through T or through B) and an isolated vertex that receives no momentum, so they give zero. Collecting,

$$\mathcal{F} = \mathcal{F}_0 = p^2 [x_1 x_2 (x_3 + x_4) + x_3 x_4 (x_1 + x_2) + x_5 (x_1 + x_3)(x_2 + x_4)] \quad (225)$$

homogeneous of degree 3 and the formula Equation 219 gives, with $N = N_\nu = 5$, $L = 2$:

$$\text{Diagram} = \Gamma(5 - D) \int \prod_{i=1}^5 dx_i \delta(1 - \sum x_i) \frac{\mathcal{U}^{5-3D/2}}{\mathcal{F}^{5-D}} \quad (226)$$

14.6.1 How $\mathcal{U}$ and $\mathcal{F}$ are used and why they save work

Once the two polynomials are written down, **the loop momenta are finished with**. Every integral over $\ell_1, \dots, \ell_L$ has already been done once and for all in the derivation of Section 14.2 (the Gaussian integral) and the graph rules just tell us what the answer of that Gaussian is for this particular graph. So the recipe for any diagram is:

1. Draw the graph and label the lines.
2. Read off $\mathcal{U}$ from the spanning trees and $\mathcal{F}$ from the 2-forests.
3. Plug into the formula Equation 219. What is left is an integral over the Feynman parameters only, with no vectors, no scalar products, no shifts.
4. Choose the delta function (Cheng-Wu) to make $\mathcal{U}$ and $\mathcal{F}$ as simple as possible and do the parameter integrals: (as below and in Section 24), or by expanding in ε , sector decomposition (Section 21) or Mellin-Barnes (Section 16).

What happens without the graph rules. To appreciate step 2, do the kite the long way, one loop at a time. The loop momentum ℓ_1 runs through lines 1, 2 and 5. With Schwinger parameters the exponent is $-\sum_i x_i D_i$ and its ℓ_1 -dependent part is

$$x_1(p - \ell_1)^2 + x_2 \ell_1^2 + x_5(\ell_1 - \ell_2)^2 = A_1 \ell_1^2 - 2\ell_1 \cdot (x_1 p + x_5 \ell_2) + x_1 p^2 + x_5 \ell_2^2, \quad A_1 = x_1 + x_2 + x_5 \quad (227)$$

Complete the square and do the ℓ_1 Gaussian. It gives a factor $A_1^{-D/2}$ and leaves behind a new, awkward expression that still depends on ℓ_2 :

$$(x_1 + x_2)x_5 \ell_2^2 - 2x_1 x_5 p \cdot \ell_2 + x_1(x_2 + x_5)p^2, \quad \text{all divided by } A_1 \quad (228)$$

Now add the two lines that carry only ℓ_2 (lines 3 and 4), collect the coefficient a_2 of ℓ_2^2 , complete the square again and do the ℓ_2 Gaussian, which gives a factor $a_2^{-D/2}$ and a final remainder. After a page of algebra one finds

$$A_1 \cdot a_2 = \mathcal{U}, \quad (\text{final remainder}) = \mathcal{F}/\mathcal{U} \quad (229)$$

the two polynomials that the trees gave in two lines (the Mathematica box below does this algebra and checks both statements). So the graph rules are not a new approximation or a trick. They are a shortcut that gives the **result** of all the loop integrations without doing them. For more loops, the loop-by-loop route needs one square completion per loop and the intermediate expressions grow quickly, while reading trees off a picture stays easy.

MATHEMATICA CHECK: THE KITE ONE LOOP AT A TIME GIVES THE SAME $\mathcal{U}$ AND $\mathcal{F}$

The kite one loop at a time: exponent $-\sum x_i D_i$; $\ell_2 \ell_2$, $p \ell_2$, p^2 . step 1: the ℓ_1 part is

In [1]: $A_1 \ell_1^2 - 2\ell_1 \cdot B + C_1$ with $B = x_1 p + x_5 \ell_2$. $B \cdot B$. what is left after the ℓ_1 Gaussian. remainder after the ℓ_1 Gaussian (it still depends on ℓ_2)

```
A1 = x1 + x2 + x5;
C1 = x1 pp + x5 l2l2;
Bsq = x1^2 pp + 2 x1 x5 pl2 + x5^2 l2l2;
rest1 = Expand[C1 - Bsq/A1];
Collect[Together[rest1], {l2l2, pl2}, Factor]
```

Out[1]: $\frac{l2l2 (x_1 + x_2) x_5}{x_1 + x_2 + x_5} + \frac{p^2 x_1 (x_2 + x_5)}{x_1 + x_2 + x_5} - \frac{2 pl2 x_1 x_5}{x_1 + x_2 + x_5}$

step 2: add lines 3 and 4, complete the square in ℓ_2 . what is left after the ℓ_2 Gaussian. the two Gaussians give

In [2]: $A_1^{-D/2}$ and $a_2^{-D/2}$: so $A_1 a_2$ must be $\mathcal{U}$ and the final remainder $\mathcal{F}/\mathcal{U}$. $A_1 a_2 - \mathcal{U}$

```
tot2 = rest1 + x3 (pp - 2 pl2 + l2l2) + x4 l2l2;
a2 = Coefficient[tot2, l2l2];
b2 = -Coefficient[tot2, pl2]/2;
c2 = tot2 /. {l2l2 -> 0, pl2 -> 0};
rest2 = c2 - b2^2 pp/a2;
U = (x1 + x2) (x3 + x4) + x5 (x1 + x2 + x3 + x4);
F = pp (x1 x2 (x3 + x4) + x3 x4 (x1 + x2) + x5 (x1 + x3) (x2 + x4));
Simplify[A1 a2 - U]
```

Out[2]: 0

final remainder minus $\mathcal{F}/\mathcal{U}$

In [3]: `Simplify[rest2 - F/U]`

Out[3]: 0

14.6.2 The value of the kite: $6\zeta(3)$

Rauh (eq. 2.41 to 2.45) goes one step further and evaluates the kite. With the shorthand $x_{ij\dots k} = x_i + x_j + \dots + x_k$ the polynomials read

$$\mathcal{U} = x_{12}x_{34} + x_{1234}x_5, \quad \mathcal{F} = q^2[x_{12}x_3x_4 + x_1x_2x_{34} + x_{13}x_{24}x_5] \quad (230)$$

which is what we found above (with $p \rightarrow q$). In $D = 4 - 2\varepsilon$ with all powers 1: $N_\nu = 5$, $L = 2$, so $\Gamma(N_\nu - LD/2) = \Gamma(1 + 2\varepsilon)$, $\mathcal{U}^{N_\nu - 3D/2} = \mathcal{U}^{-1+3\varepsilon}$ and $\mathcal{F}^{-(N_\nu - D)} = \mathcal{F}^{-1-2\varepsilon}$. The kite has no UV or IR divergence in four dimensions, so we may set $\varepsilon = 0$:

$$-\bigcirc = \int \prod_{i=1}^5 dx_i \delta(1 - \sum x_i) \frac{1}{\mathcal{U}\mathcal{F}} + O(\varepsilon) \quad (231)$$

Here Cheng-Wu pays off. Choose $\delta(1 - x_5)$: then $x_5 = 1$ and $x_1, \dots, x_4$ run from 0 to ∞ . Now $\mathcal{U}$ and $\mathcal{F}$ are **linear** in each of the remaining parameters, so each integration gives logarithms and dilogarithms but never anything worse and the four integrals can be done one after the other:

$$-\bigcirc = \frac{1}{q^2} \int_0^\infty dx_1 dx_2 dx_3 dx_4 \frac{1}{[x_{12}x_{34} + x_{1234}][x_{12}x_3x_4 + x_1x_2x_{34} + x_{13}x_{24}]} = \frac{6\zeta(3)}{q^2} + O(\varepsilon) \quad (232)$$

where $\zeta(3) = \sum_{n \geq 1} 1/n^3 = 1.2020\dots$ is Apéry's constant. Numbers like $\zeta(3)$ are typical of multi-loop integrals. The one-loop answers were logarithms, the two-loop ones already contain values of the Riemann zeta function. If the zeta function is new to you, my video [Riemann Zeta Function and Riemann's Hypothesis](#) is a gentle introduction.

MATHEMATICA CHECK: THE FOUR CHENG-WU INTEGRALS, ONE AFTER THE OTHER

Two-loop kite at $D = 4$, Cheng-Wu with $\delta(1 - x_5)$: $x_5 = 1$, $x_1, \dots, x_4 \in [0, \infty)$, integrand $1/(\mathcal{U}\mathcal{F})$ with $\mathcal{U} =$

In [1]: $x_{12}x_{34} + x_{1234}x_5$, $\mathcal{F} = q^2[x_{12}x_3x_4 + x_1x_2x_{34} + x_{13}x_{24}x_5]$. $x_5 = 1$, $q^2 = 1$. after the x_1 integral

```
x12 = x1 + x2;
x34 = x3 + x4;
x13 = x1 + x3;
x24 = x2 + x4;
x1234 = x1 + x2 + x3 + x4;
U = x12 x34 + x1234;
F = x12 x3 x4 + x1 x2 x34 + x13 x24;
i1 = Integrate[1/(U F),
  {x1, 0, Infinity}, Assumptions -> x2 > 0 && x3 > 0 && x4 > 0];
Simplify[i1, x2 > 0 && x3 > 0 && x4 > 0]
```

Out[1]:
$$\frac{\log\left(\frac{(x_3+x_4+x_2)(x_3+x_4+1)((x_3+1)x_4+x_2(x_3+x_4+1))}{x_3(x_3+x_4+1)(x_4x_2+x_2+x_4)}\right)}{(x_4+x_2(x_3+x_4+1))^2}$$

after the x_2 integral

In [2]: `i2 = Integrate[i1, {x2, 0, Infinity}, Assumptions -> x3 > 0 && x4 > 0];`
`Simplify[i2, x3 > 0 && x4 > 0]`

Out[2]:
$$\frac{-x_4 \log(x_4) + \log\left(\frac{(x_3+1)(x_4+1)}{x_3+x_4+1}\right) + x_3 \log\left(\frac{(x_3+1)(x_3+x_4)}{x_3(x_3+x_4+1)}\right) + x_4 \log\left(\frac{x_3}{x_3+x_4+1} + x_4\right)}{x_3 x_4 (x_3 + x_4 + 1)}$$

after the x_3 integral

In [3]: `i3 = Integrate[i2, {x3, 0, Infinity}, Assumptions -> x4 > 0];`
`Simplify[i3, x4 > 0]`

Out[3]:
$$-\frac{x_4 \operatorname{Li}_2\left(-\frac{1}{x_4}\right) + \operatorname{Li}_2(-x_4) + (x_4 + 1) \log\left(\frac{x_4}{x_4+1}\right) \log(x_4 + 1)}{x_4 (x_4 + 1)}$$

$q^2 F(1, 1, 1, 1, 1)$ at $\varepsilon = 0$

In [4]: `i4 = Integrate[i3, {x4, 0, Infinity}];
FullSimplify[i4]`

Out[4]: $6\zeta(3)$

minus $6\zeta(3)$

In [5]: `FullSimplify[i4 - 6 Zeta[3]]`

Out[5]: 0

Mathematica does the four integrals in sequence, symbolically. After x_1 and x_2 the intermediate results contain logarithms. After x_3 dilogarithms Li_2 appear. The last integral gives exactly $6\zeta(3)$.

One Piece has the Poneglyphs and multi-loop physics has $\zeta(3)$: an ancient number that keeps showing up in places where nobody wrote it down.

MATHEMATICA CHECK: U AND F FROM THE MATRIX DETERMINANT, COMPARED WITH THE TREE RULES

UF builds M, Q, J and returns $\mathcal{U} = \det M$, $\mathcal{F} = \det M(J - Q^T M^{-1} Q)$ (full code in the file).

In [1]: 1. bubble with masses m_1, m_2 : $k_1 = \ell$, $k_2 = \ell - p$

. bubble $\mathcal{U}$

```
{Ub, Fb} = UF[{{1}, {1}}, {{0}, {-1}}, {mm1, mm2}, {{pp}}, {x1, x2}];
Ub
```

Out[1]: $x_1 + x_2$

bubble $\mathcal{F}$

In [2]: `Collect[Fb, {pp}, Factor]`

Out[2]: $(x_1 + x_2)(m_1^2 x_1 + m_2^2 x_2) + p^2 x_1 x_2$

2. massless box with cyclic labels, $p_i^2 = 0$, $s = (p_1 + p_2)^2$, $t = (p_2 + p_3)^2$, so $p_1 \cdot p_3 = -(s + t)/2$

In [3]: . box $\mathcal{U}$

```
gramBox = {{0, s/2, -(s + t)/2}, {s/2, 0, t/2}, {-(s + t)/2, t/2, 0}};
{Ubox, Fbox} = UF[{{1}, {1}, {1}, {1}}, {{1, 0, 0}, {1, 1, 0}, {1,
  1, 1}, {0, 0, 0}}, {0, 0, 0, 0}, gramBox, {x1, x2, x3, x4}];
Ubox
```

Out[3]: $x_1 + x_2 + x_3 + x_4$

box $\mathcal{F}$

In [4]: `Factor[Fbox]`

Out[4]: $s x_2 x_4 + t x_1 x_3$

3. two-loop kite: $k_1 = p - \ell_1$, $k_2 = \ell_1$, $k_3 = p - \ell_2$, $k_4 = \ell_2$, $k_5 = \ell_1 - \ell_2$

In [5]: . kite $\mathcal{U}$

```
{Uk, Fk} = UF[{{-1, 0}, {1, 0}, {0, -1}, {0, 1}, {1, -1}}, {{1}, {0},
  {1}, {0}, {0}}, {0, 0, 0, 0, 0}, {{pp}}, {x1, x2, x3, x4, x5}];
Uk
```

Out[5]: $x_1 x_3 + x_2 x_3 + x_5 x_3 + x_1 x_4 + x_2 x_4 + x_1 x_5 + x_2 x_5 + x_4 x_5$

kite $\mathcal{F}$ In [6]: `Factor[Fk]`Out[6]: $p^2 (x_1 x_2 x_3 + x_1 x_4 x_3 + x_2 x_4 x_3 + x_2 x_5 x_3 + x_4 x_5 x_3 + x_1 x_2 x_4 + x_1 x_2 x_5 + x_1 x_4 x_5)$ the tree rules read off the pictures. kite: matrix method minus tree rules, for $\mathcal{U}$ and $\mathcal{F}$

In [7]: `Utree = (x1 + x2) (x3 + x4) + x5 (x1 + x2 + x3 + x4);
 Ftree = pp (x3 x4 x5 + x1 x2 x5 + x1 x4 x5 + x2 x3 x5 + x1 x3 x4 + x1 x2 x4 + x2 x3 x4
 + x1 x2 x3);
 Simplify[{Uk - Utree, Fk - Ftree}]`

Out[7]: `{0, 0}`box: $\mathcal{F} - (sx_2x_4 + tx_1x_3)$ In [8]: `Simplify[Fbox - (s x2 x4 + t x1 x3)]`Out[8]: `0`degrees of $(\mathcal{U}, \mathcal{F})$ for the box and the kite

In [9]: `{Exponent[Ubox /. Thread[{x1, x2, x3, x4} -> la {x1,
 x2, x3, x4}], la], Exponent[Fbox /. Thread[{x1, x2, x3, x4} -> la {x1,
 x2, x3, x4}], la],
 Exponent[Uk /. Thread[{x1, x2, x3, x4, x5} -> la {x1,
 x2, x3, x4, x5}], la], Exponent[Fk /. Thread[{x1,
 x2, x3, x4, x5} -> la {x1, x2, x3, x4, x5}], la]}`

Out[9]: `{1, 2, 2, 3}`

The matrix method of Section 14.2, done by brute force, reproduces every tree-rule polynomial, namely those of the bubble with two masses, the box ($tx_1x_3 + sx_2x_4$ with our cyclic labels) and the kite (difference zero). The last line confirms the degrees (1, 2) for the box and (2, 3) for the kite.

MATHEMATICA CHECK: THE SAME POLYNOMIALS FROM FEYNCALC, WITH ITS OWN LABELS

FeynCalc (Minkowski): $\mathcal{U}$ and $\mathcal{F}$ of the box and the kite; it also tells which propagator got which x_i . box $\mathcal{U}$

In [1]: `FCClearScalarProducts[];
 SPD[p1] = 0;
 SPD[p2] = 0;
 SPD[p3] = 0;
 SPD[p1, p2] = s/2;
 SPD[p2, p3] = t/2;
 SPD[p1, p3] = -(s + t)/2;
 box = FCFeynmanPrepare[FAD[{l + p1}, {l + p1 + p2}, {l + p1 + p2 + p3},
 {l}], {l}, Names -> x];
 box[[1]]`

Out[1]: $x_1 + x_2 + x_3 + x_4$ box $\mathcal{F}$ In [2]: `box[[2]]`Out[2]: $-s x_1 x_3 - t x_2 x_4$ box: which x_i belongs to which propagatorIn [3]: `{#[[1]], FCE[#[[2]]]} & /@ box[[3]]`

Out[3]:
$$\begin{pmatrix} x_1 & \frac{1}{\ell^2} \\ x_2 & \frac{1}{(\ell+p_1)^2} \\ x_3 & \frac{1}{(\ell+p_1+p_2)^2} \\ x_4 & \frac{1}{(\ell+p_1+p_2+p_3)^2} \end{pmatrix}$$

kite $\mathcal{U}$

In [4]: `kite = FCFeynmanPrepare[FAD[{p - l1}, {l1}, {p - l2}, {l2},
{l1 - l2}], {l1, l2}, Names -> x,
FinalSubstitutions -> {SPD[p] -> pp}];
box[[1]] /. box[[1]] -> kite[[1]]`

Out[4]: $x_1 x_2 + x_3 x_2 + x_4 x_2 + x_5 x_2 + x_1 x_3 + x_3 x_4 + x_1 x_5 + x_4 x_5$

kite $\mathcal{F}$

In [5]: `kite[[2]]`

Out[5]: $-p^2 (x_1 x_2 x_4 + x_1 x_3 x_4 + x_2 x_3 x_4 + x_1 x_5 x_4 + x_3 x_5 x_4 + x_1 x_2 x_5 + x_1 x_3 x_5 + x_2 x_3 x_5)$

kite: which x_i belongs to which propagator

In [6]: `{#[[1]], FCE#[[2]]} & /@ kite[[3]]`

Out[6]:
$$\begin{pmatrix} x_1 & \frac{1}{l_1^2} \\ x_2 & \frac{1}{(l_1-l_2)^2} \\ x_3 & \frac{1}{l_2^2} \\ x_4 & \frac{1}{(p-l_1)^2} \\ x_5 & \frac{1}{(p-l_2)^2} \end{pmatrix}$$

translate the tree-rule polynomials of the text into FeynCalc's labels and compare. kite: FeynCalc minus text,

for $\mathcal{U}$ and $\mathcal{F}$

In [7]: `toFC = {y1 -> x[4], y2 -> x[1], y3 -> x[5], y4 -> x[3], y5 -> x[2]};
Utext = (y1 + y2) (y3 + y4) + y5 (y1 + y2 + y3 + y4);
F0text = pp (y1 y2 (y3 + y4) + y3 y4 (y1 + y2) + y5 (y1 + y3) (y2 + y4));
Simplify[{kite[[1]] - (Utext /. toFC), kite[[2]] + (F0text /. toFC)}]`

Out[7]: {0, 0}

box: FeynCalc's $\mathcal{F}$ plus $s x_2 x_4 + t x_1 x_3$ (Minkowski sign)

In [8]: `boxMap = {y1 -> x[2], y2 -> x[3], y3 -> x[4], y4 -> x[1]};
Simplify[box[[2]] + ((s y2 y4 + t y1 y3) /. boxMap)]`

Out[8]: 0

FeynCalc rennumbers the lines (the lists show which x_i belongs to which momentum) and it works in Minkowski space, so $\mathcal{F}$ comes with an overall minus sign. After translating its labels into ours, both polynomials agree (last two lines).

14.7 The formula in Minkowski space

On the board the general formula was written in Smirnov's Minkowski form (his eq. (3.36), with the measure divided by $(i\pi^{D/2})^L$):

$$\int \prod_{r=1}^L \frac{d^D \ell_r}{i\pi^{D/2}} \frac{1}{\prod_i (k_i^2 - m_i^2 + i\epsilon)^{\nu_i}} = (-1)^{N_\nu} \frac{\Gamma(N_\nu - LD/2)}{\prod_i \Gamma(\nu_i)} \int_0^\infty \prod_i dx_i x_i^{\nu_i-1} \delta(1 - \sum x_i) \frac{\mathcal{U}^{N_\nu - (L+1)D/2}}{\mathcal{F}^{N_\nu - LD/2}} \quad (233)$$

with $\mathcal{F} = -\mathcal{F}_0^{\text{Mink}} + \mathcal{U} \sum_i x_i m_i^2 - i\epsilon \mathcal{U} \sum_i x_i$ (the small imaginary part is written this way so that $\mathcal{F}$ stays homogeneous of degree $L + 1$. On the delta function $\sum_i x_i = 1$ it is just $-i\epsilon \mathcal{U}$), where $\mathcal{F}_0^{\text{Mink}}$ uses Minkowski squares of the momenta.

WHERE THE $(-1)^{N_\nu}$ AND THE SIGN OF $\mathcal{F}_0$ COME FROM

Wick rotate every loop momentum. Each loop gives $d^D \ell_r = i d^D \ell_{r,E}$, which cancels the $1/i$ in the measure. Each propagator becomes $k_i^2 - m_i^2 = -(k_{i,E}^2 + m_i^2)$, so $(k_i^2 - m_i^2)^{-\nu_i} = (-1)^{\nu_i} (k_{i,E}^2 + m_i^2)^{-\nu_i}$ and all together this is $(-1)^{\sum \nu_i} = (-1)^{N_\nu}$. What remains is the Euclidean integral, with the Euclidean $\mathcal{F}_0$. External momenta satisfy $P_E^2 = -P^2$, so $\mathcal{F}_0^{\text{Eucl}} = -\mathcal{F}_0^{\text{Mink}}$. The $-i\epsilon$ follows the $+i\epsilon$ of the propagators, as in Section 15.

FeynCalc's output above is of this form: $\mathcal{F} = -sx_1x_3 - tx_2x_4$ (in its labels) for the box and $-p^2(\dots)$ for the kite.

14.8 The three diagrams of the lecture notebook

In the lecture a Mathematica notebook was shown on the projector. It used Package-X's `LoopIntegrate` on three diagrams, a **triangle** with massless internal lines and two incoming momenta p_1, p_2 . The **bubble with two unequal masses** m_1, m_2 ("equivalent to the bubble in Ramond, up to conventions for Euclidean versus Minkowski"). And a **box** with massless external lines and two massive horizontal lines. Its internal masses are those of Smirnov's Fig. 4.4, but Smirnov's Example 4.6 for that figure also makes the external legs massive, $p_i^2 = m^2$. The notebook also warns that `LoopIntegrate` uses a non-standard sign convention for V and that Smirnov's Eq. 3.27 has s and t exchanged. Below, all three are first done by hand with the tools of these notes and at the end the notebook's own Package-X calls are run and compared.

14.8.1 The massless triangle

Take on-shell massless legs, $p_1^2 = p_2^2 = 0$ and $Q^2 = (p_1 + p_2)^2$ (Euclidean), all internal lines massless. The graph rules: $\mathcal{U} = x_1 + x_2 + x_3$ and of the three 2-trees only the one that separates $p_1 + p_2$ from the rest carries momentum, so $\mathcal{F} = x_1x_3Q^2$ (the other two cut off a single on-shell leg, $p_i^2 = 0$). With $a = 3, L = 1$:

$$\triangle = \Gamma(1 + \varepsilon)(Q^2)^{-1-\varepsilon} \int dx_1 dx_2 dx_3 \delta(1 - \sum x_i) (x_1 + x_2 + x_3)^{-1+2\varepsilon} (x_1x_3)^{-1-\varepsilon} \quad (234)$$

On the delta function $\mathcal{U} = 1$ and the x_2 integration just enforces $x_1 + x_3 \leq 1$. What is left is a **Dirichlet integral**, $\int_{x_1+x_3 < 1} x_1^{\alpha-1} x_3^{\beta-1} = \Gamma(\alpha)\Gamma(\beta)/\Gamma(1 + \alpha + \beta)$ with $\alpha = \beta = -\varepsilon$:

$$\triangle = \frac{\Gamma(1 + \varepsilon)\Gamma(-\varepsilon)^2}{\Gamma(1 - 2\varepsilon)}(Q^2)^{-1-\varepsilon} = \frac{e^{-\gamma_E \varepsilon}}{Q^2} \left[\frac{1}{\varepsilon^2} - \frac{\ln Q^2}{\varepsilon} + \frac{1}{2} \ln^2 Q^2 - \frac{\pi^2}{12} + O(\varepsilon) \right] \quad (235)$$

This is the $1/\varepsilon^2$ of Section 4.5, a soft and a collinear divergence at the same time, which is only possible because the external legs are massless and on shell.

MATHEMATICA CHECK: THE MASSLESS ON-SHELL TRIANGLE

Massless one-loop triangle, on-shell legs $p_1^2 = p_2^2 = 0, Q^2 = (p_1 + p_2)^2$ (Euclidean), measure $d^D D /$

In [1]: `pi^(D/2). Graph polynomials: U = x1+x2+x3, F = x1 x3 Q^2 (the only 2-tree that carries momentum). a = 3, L = 1.. On delta(1 - x1 - x2 - x3), U = 1 and the integral is Gamma(1+eps) (Q^2)^(-1-eps) Int (x1 x3)^(-1-eps) over the simplex. Dirichlet: Int over x1 + x3 < 1 of x1^(al-1) x3^(be-1)`

```
dir = Integrate[x1^(al - 1) x3^(be - 1), {x1, 0, 1}, {x3, 0, 1 - x1},
  Assumptions -> al > 0 && be > 0];
FullSimplify[dir]
```

Out[1]: $\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)}$

triangle times $(Q^2)^{1+\varepsilon}$

In [2]: `tri = Gamma[1 + ep] Gamma[-ep]^2/Gamma[1 - 2 ep];
tri`

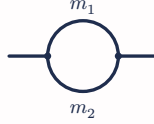
Out[2]: $\frac{\Gamma(-\epsilon)^2 \Gamma(\epsilon + 1)}{\Gamma(1 - 2\epsilon)}$

expansion, times $e^{\epsilon\gamma_E}$

In [3]: `Collect[Normal@Series[Exp[ep EulerGamma] tri, {ep, 0, 0}], ep, Simplify]`

Out[3]: $\frac{1}{\epsilon^2} - \frac{\pi^2}{12}$

14.8.2 The bubble with two unequal masses



From Example 1 above, $\mathcal{F} = x_1 x_2 Q^2 + (m_1^2 x_1 + m_2^2 x_2)(x_1 + x_2)$. On $x_1 = x$, $x_2 = 1 - x$ the finite part of the bubble is

$$f = - \int_0^1 dx \ln[x m_1^2 + (1-x) m_2^2 + x(1-x) Q^2] \quad (236)$$

The argument is a quadratic in x . Factorising it with its two roots and integrating each logarithm (the same steps as in Section 13) gives, in Minkowski notation $s = -Q^2$ and with the Källén function $\lambda(s, m_1^2, m_2^2) = s^2 + m_1^4 + m_2^4 - 2sm_1^2 - 2sm_2^2 - 2m_1^2 m_2^2$,

$$f = 2 - \ln(m_1 m_2) + \frac{m_1^2 - m_2^2}{s} \ln \frac{m_2}{m_1} + \frac{\sqrt{\lambda}}{s} \ln \frac{m_1^2 + m_2^2 - s + \sqrt{\lambda}}{2m_1 m_2} \quad (237)$$

For $m_1 = m_2 = m$ the middle term drops out and $\sqrt{\lambda} = \sqrt{s(s - 4m^2)}$. What is left is the equal-mass result of Section 15 plus the term $-\ln m^2$. In Section 15 that term was kept apart, inside the $\ln(\mu^2/m^2)$ piece and m was set to 1. The two-particle threshold is now at $s = (m_1 + m_2)^2$, where λ vanishes. The pseudo-threshold is at $s = (m_1 - m_2)^2$.

MATHEMATICA CHECK: THE UNEQUAL-MASS BUBBLE, SYMBOLICALLY

Bubble with two masses, finite part $f = - \int_0^1 \ln(x m_1^2 + (1-x) m_2^2 + x(1-x) Q^2) dx$. closed form with the Källén

In [1]: function $\lambda(s, m_1^2, m_2^2)$, $s = -Q^2$. f - closed form at $m_1^2 = 1$, $m_2^2 = 3$, $Q^2 = 5$

```
f = -Integrate[Log[x m1s + (1 - x) m2s + x (1 - x) Q2],
  {x, 0, 1}, Assumptions -> m1s > 0 && m2s > 0 && Q2 > 0];
s = -Q2;
lam = s^2 + m1s^2 + m2s^2 - 2 s m1s - 2 s m2s - 2 m1s m2s;
closed = 2 - Log[Sqrt[m1s m2s]] + (m1s - m2s)/s Log[Sqrt[m2s/m1s]] +
  Sqrt[lam]/s Log[(m1s + m2s - s + Sqrt[lam])/(2 Sqrt[m1s m2s])];
FullSimplify[(f - closed) /. {m1s -> 1, m2s -> 3, Q2 -> 5}]
```

Out[1]: 0

f - closed form at $m_1^2 = 2$, $m_2^2 = 7$, $Q^2 = 1/3$

In [2]: `FullSimplify[(f - closed) /. {m1s -> 2, m2s -> 7, Q2 -> 1/3}]`

Out[2]: 0

equal masses: closed form minus $2 - \tilde{\beta} \ln((\tilde{\beta} + 1)/(\tilde{\beta} - 1))$

In [3]: `FullSimplify[(closed /. {m1s -> 1, m2s -> 1}) - (2 - Sqrt[1 + 4/Q2] Log[(Sqrt[1 + 4/Q2] + 1)/(Sqrt[1 + 4/Q2] - 1)]), Q2 > 0]`

Out[3]: 0

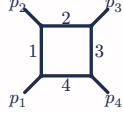
general masses: f – closed form

In [4]: `FullSimplify[f - closed, m1s > 0 && m2s > 0 && Q2 > 0]`

Out[4]: 0

14.8.3 The box with two massive horizontal lines

Put mass m on the top (2) and bottom (4) lines of the box of Example 2, the rest as before. The 1-trees are unchanged and so are the 2-trees (a corner with $p_i^2 = 0$ still gives nothing). Only the mass term is new:



$$: \quad \mathcal{U} = x_1 + x_2 + x_3 + x_4, \quad \mathcal{F} = s x_2 x_4 + t x_1 x_3 + m^2 (x_2 + x_4)(x_1 + x_2 + x_3 + x_4) \quad (238)$$

The full evaluation of this box needs the Mellin-Barnes technique of Section 16, which is what the notebook did with Package-X. Smirnov's Example 4.6 (2006) evaluates a close relative in this way, the same box with the external legs massive as well, $p_i^2 = m^2$. Then the corner 2-trees also contribute and the mass term becomes $-m^2(\alpha_1 + \alpha_3)^2$ in his labels. Here we check the parametric representation against FeynCalc, which again renumbers the lines:

MATHEMATICA CHECK: U AND F OF THE BOX WITH TWO MASSIVE LINES

Box with two massive horizontal lines: tree rule $\mathcal{F} = s x_2 x_4 + t x_1 x_3 + m^2 (x_2 + x_4) \mathcal{U}$, compared with FeynCalc.

In [1]: $\mathcal{U}$

```
FCClearScalarProducts[];
SPD[p1] = 0;
SPD[p2] = 0;
SPD[p3] = 0;
SPD[p1, p2] = s/2;
SPD[p2, p3] = t/2;
SPD[p1, p3] = -(s + t)/2;
bx = FCFeynmanPrepare[FAD[{l + p1}, {l + p1 + p2, m}, {l + p1 + p2 + p3},
  {l, m}], {l}, Names -> x];
bx[[1]]
```

Out[1]: $x_1 + x_2 + x_3 + x_4$

$\mathcal{F}$

In [2]: `Collect[bx[[2]], m, Factor]`

Out[2]: $m^2 (x_1 + x_3) (x_1 + x_2 + x_3 + x_4) - s x_1 x_3 - t x_2 x_4$

which x_i belongs to which propagator

In [3]: `{#[[1]], FCE#[[2]]} & /@ bx[[3]]`

Out[3]:
$$\begin{pmatrix} x_1 & \frac{1}{\ell^2 - m^2} \\ x_2 & \frac{1}{(\ell + p1)^2} \\ x_3 & \frac{1}{(\ell + p1 + p2)^2 - m^2} \\ x_4 & \frac{1}{(\ell + p1 + p2 + p3)^2} \end{pmatrix}$$

our line $i \rightarrow$ FeynCalc's label. Minkowski form of the tree rule. FeynCalc minus the tree rule, after relabelling

```
map = {y1 -> x[2], y2 -> x[3], y3 -> x[4], y4 -> x[1]};
Utext = y1 + y2 + y3 + y4;
Ftext = -(s y2 y4 + t y1 y3) + m^2 (y2 + y4) Utext;
Simplify[bx[[2]] - (Ftext /. map)]
```

Out[4]: 0

14.8.4 The same diagrams with Package-X

Now the notebook's own calls. `LoopIntegrate` writes the integral and `LoopRefine` evaluates it (Minkowski metric, with Package-X's normalisation, which includes $e^{\varepsilon\gamma_E}$. Below $\mu = 1$). The triangle and the unequal-mass bubble come out as derived above. For the box with two massive lines Package-X returns the general four-point function `ScalarD0` unevaluated. That box needs the Mellin-Barnes method of Section 17, which is what the lecture did next.

MATHEMATICA CHECK: TRIANGLE AND UNEQUAL-MASS BUBBLE WITH PACKAGE-X

The lecture notebook's diagrams with Package-X (Minkowski, $\mu = 1$).

In [1]: 1. massless triangle
 . the triangle from Package-X

```
Get["X`"];
triPX = LoopRefine[LoopIntegrate[1, l, {l, 0, 1}, {l + p1,
  0, 1}, {l + p1 + p2, 0, 1}] /. {LDot[p1, p1] -> 0,
  LDot[p2, p2] -> 0, LDot[p1, p2] -> s/2}];
triPX
```

Out[1]:
$$\frac{1}{\varepsilon^2} + \frac{\log\left(-\frac{\mu^2}{s}\right)}{\varepsilon} + \frac{1}{2} \log^2\left(-\frac{\mu^2}{s}\right) - \frac{\pi^2}{12s}$$

minus our triangle at $s = -1$

In [2]: `Simplify[Normal[triPX /. {s -> -1, Mu -> 1}] - Normal[Series[-Exp[Eps EulerGamma] Gamma[1 + Eps] Gamma[-Eps]^2/Gamma[1 - 2 Eps], {Eps, 0, 0}]]]`

Out[2]: 0

2. bubble with two masses at $s = -5$, $m_1^2 = 1$, $m_2^2 = 3$

In [3]: . Package-X minus $(1/\varepsilon + f)$

```
bubPX = LoopRefine[LoopIntegrate[1, l, {l, m1, 1}, {l + p,
  m2, 1}] /. {LDot[p, p] -> s}];
fclosed[s_, a_, b_] := Module[{lam = s^2 + a^2 + b^2 - 2 s a - 2 s b - 2 a b},
  2 - Log[Sqrt[a b]] + (a - b)/s Log[Sqrt[b/a]] + Sqrt[lam]/s Log[(a + b - s +
  Sqrt[lam])/(2 Sqrt[a b])]];
FullSimplify[Normal[bubPX /. {s -> -5, m1 -> 1, m2 -> Sqrt[3],
  Mu -> 1}] - (1/Eps + fclosed[-5, 1, 3])]
```

Out[3]: 0

14.9 Sir's UF notebook: completing the square

Sir also shared a short notebook, `uf-new-short.nb`. Its routine does the derivation of Section 14.2 on the computer. It writes the sum over propagators as

$$\sum_i x_i D_i = \sum_{r,s} \ell_r \cdot \ell_s M_{rs} - 2 \sum_r \ell_r \cdot Q_r + J \quad (239)$$

completes the square one loop momentum at a time and reads off

$$\mathcal{U} = \det M, \quad \mathcal{F} = \det M (J - Q^T M^{-1} Q) \quad (240)$$

The notebook works in Minkowski space, so $\mathcal{F} = -\mathcal{F}_0^{\text{Mink}} + \mathcal{U} \sum_i x_i m_i^2$ as in the formula above. Its four examples are

diagram	$\mathcal{U}$	$\mathcal{F}$
massless triangle	$x_1 + x_2 + x_3$	$-p_1^2 x_1 x_2 - Q^2 x_1 x_3 - p_2^2 x_2 x_3$
bubble, masses m_1, m_2	$x_1 + x_2$	$m_1^2 x_1^2 + (m_1^2 + m_2^2 - p^2) x_1 x_2 + m_2^2 x_2^2$
massless box	$x_1 + x_2 + x_3 + x_4$	$-s x_1 x_3 - t x_2 x_4$

diagram	$\mathcal{U}$	$\mathcal{F}$
two-loop sunset, $p^2 = xm^2$	$x_1x_2 + x_1x_3 + x_2x_3$	$m^2(x_1 + x_2 + x_3)\mathcal{U} - xm^2x_1x_2x_3$

with $Q = p_1 + p_2$ for the triangle. For the sunset the last column is $\mathcal{U} \sum_i x_i m^2 - p^2 x_1 x_2 x_3$: the only 2-forest that carries momentum cuts all three lines.

THE SAME IN FEYNSAGE

`family(...).UF()` is this routine (the matrix M , then $\mathcal{U} = \det M$ and $\mathcal{F} = \mathcal{U}J - Q \cdot \text{adj}(M) \cdot Q$, so no division is needed). `graph(...)` gets the same polynomials from spanning trees and 2-forests:

```
from feynsage import family, graph
sun = family([("k", "m"), ("k - l", "m"), ("l + p", "m")], kin={"p^2": "x*m^2"})
sun.UF()      # completing the square
# (x1*x2 + x1*x3 + x2*x3,
#  m^2*x1^2*x2 + m^2*x1*x2^2 + ... + (-m^2*x + 3*m^2)*x1*x2*x3 + ...)
g = graph("A-B:m, A-B:m, A-B:m", {"A": "p", "B": "-p"}, kin={"p^2": "x*m^2"})
g.U(), g.F()  # the same two polynomials, read off the graph
```

All four examples of sir's notebook come out the same both ways (`tests/test_professor_uf.sage` in the feynsage repository). The check also works well beyond two loops. For the three-loop triple box feynsage finds $\mathcal{U}$ with 56 terms and $\mathcal{F}$ with 65 in about 10 ms, three ways (trees, Kirchhoff, $\det M$) and FeynCalc's `FCFeynmanPrepare` gives the same polynomials.

Two notebooks, two conventions, one minus sign hiding between them. Light and L would both spend three episodes on a single missing minus sign. We spent one Mathematica box.

15 Back to Minkowski space: thresholds, the optical theorem and the logarithm

The Euclidean bubble is a real number. A real scattering amplitude above the two-particle threshold is not. It has an imaginary part and that imaginary part is physics (the probability that the particles really scatter). This chapter explains where it comes from, why it has the size it has and why the answer is built from logarithms.

15.1 The continuation and its $i\epsilon$

In Minkowski space the external momentum has $s = p^2 = p_0^2 - \mathbf{p}^2$ and the Wick rotation gives $p_E^2 = -s$ (Section 3.2). The finite part of the bubble (the part of the bracket without the $1/\epsilon$ and $\ln(\mu^2/m^2)$ pieces) is then

$$f(s) = - \int_0^1 dx \ln \left(1 - x(1-x) \frac{s}{m^2} - i\epsilon \right) \quad (241)$$

Where does the $-i\epsilon$ come from? Do the Feynman-parameter combination directly with Minkowski propagators, each carrying its $+i\epsilon$:

$$x(\ell^2 - m^2 + i\epsilon) + (1-x)((\ell - p)^2 - m^2 + i\epsilon) = \ell'^2 - [m^2 - x(1-x)s] + i\epsilon \quad (242)$$

after the same shift as in Section 11. So the effective mass squared is $M^2(x) = m^2 - x(1-x)s - i\epsilon$. The small negative imaginary part tells us on which side of the logarithm's cut (Section 2.5) we sit whenever $M^2(x)$ turns negative.

15.2 When does the effective mass turn negative?

$x(1-x)$ is largest at $x = 1/2$, where it is $1/4$. So $M^2(x) = m^2 - x(1-x)s \geq m^2 - s/4$.

- If $s < 4m^2$, then $M^2(x) > 0$ for every x , the log has a positive argument and f is real.
- If $s > 4m^2$, then $M^2(x) < 0$ between the two roots of $x(1-x) = m^2/s$, namely $x_{\pm} = (1 \pm \beta)/2$ with $\beta = \sqrt{1 - 4m^2/s}$. The width of this window is $x_+ - x_- = \beta$.

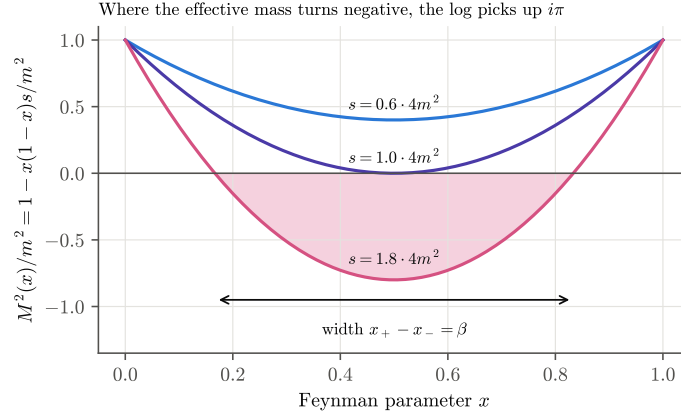


Figure 9: The effective mass $M^2(x)/m^2 = 1 - x(1-x)s/m^2$ for three energies. Below threshold ($s < 4m^2$) it stays positive. At $s = 4m^2$ it just touches zero at $x = 1/2$. Above threshold it is negative in a window of width β (shaded) and there $\ln(M^2 - i\epsilon) = \ln|M^2| - i\pi$.

$s = 4m^2$ is the energy needed to make two real particles of mass m at rest. Below it, the two particles in the loop can only be virtual. Above it, they can be real and this is where the imaginary part switches on.

15.3 One derivation for every value of s

DERIVATION: $f(s)$ BELOW, AT AND ABOVE THRESHOLD

Set $m = 1$. The integration by parts of Section 13 (with $v = x - 1/2$ and then $y = 1 - 2x$) never used that $z = p_E^2$ was positive. With $z = -s - i\epsilon$ it gives

$$f(s) = - \int_{-1}^1 \frac{y^2}{b^2 - y^2} dy, \quad b^2 = 1 + \frac{4}{z} = 1 - \frac{4}{s + i\epsilon} \quad (243)$$

Since $4/(s + i\epsilon) = 4/s - i\epsilon$ for $s > 0$, we have $b^2 = 1 - 4/s + i\epsilon$. Always split $y^2/(b^2 - y^2) = -1 + b^2/(b^2 - y^2)$, so

$$f(s) = 2 - b^2 \int_{-1}^1 \frac{dy}{b^2 - y^2} \quad (244)$$

(a) **Euclidean region, $s < 0$.** Then $b^2 = 1 + 4/|s| > 1$, the integrand has no pole on $[-1, 1]$ and (partial fractions as in Section 13)

$$\int_{-1}^1 \frac{dy}{b^2 - y^2} = \frac{1}{b} \ln \frac{b+1}{b-1} \Rightarrow f(s) = 2 - b \ln \frac{b+1}{b-1}, \quad b = \sqrt{1 + \frac{4}{|s|}} \quad (245)$$

This is Ramond's Euclidean form, with $b = \tilde{\beta} = \sqrt{1 + 4m^2/p_E^2}$.

(b) **Below threshold, $0 < s < 4$.** Now $b^2 = 1 - 4/s < 0$. Write $b^2 = -r^2$ with $r = \sqrt{4/s - 1}$. Then

$$\int_{-1}^1 \frac{dy}{-r^2 - y^2} = -\frac{1}{r} \left[\arctan \frac{y}{r} \right]_{-1}^1 = -\frac{2}{r} \arctan \frac{1}{r} \Rightarrow f(s) = 2 - 2r \arctan \frac{1}{r} \quad (246)$$

Still real and still no $i\epsilon$ needed.

(c) **Above threshold, $s > 4$.** Now $0 < b^2 < 1$ and the integrand has poles at $y = \pm b$, **inside** the range. Here the $+i\epsilon$ matters. By the Sokhotski-Plemelj formula Equation 26 with $X = b^2 - y^2$,

$$\frac{1}{b^2 - y^2 + i\epsilon} = \mathcal{P} \frac{1}{b^2 - y^2} - i\pi \delta(b^2 - y^2) \quad (247)$$

The principal value gives the same log as before, $\mathcal{P} \int_{-1}^1 \frac{dy}{b^2 - y^2} = (1/b) \ln((1+b)/(1-b))$. The delta function has two roots $y = \pm b$, each contributing $1/|d(b^2 - y^2)/dy| = 1/(2b)$, so $\int \delta(b^2 - y^2) dy = 1/b$. Therefore

$$\int_{-1}^1 \frac{dy}{b^2 - y^2 + i\epsilon} = \frac{1}{b} \ln \frac{1+b}{1-b} - \frac{i\pi}{b} \quad (248)$$

and multiplying by $-b^2$ and adding 2,

$$f(s) = 2 - \beta \ln \frac{1+\beta}{1-\beta} + i\pi\beta, \quad \beta = b = \sqrt{1 - \frac{4}{s}} \quad (249)$$

RESULT: THE FINITE PART OF THE BUBBLE ON THE WHOLE REAL AXIS

$$f(s) = \begin{cases} 2 - \tilde{\beta} \ln \frac{\tilde{\beta}+1}{\tilde{\beta}-1} & s < 0 \text{ (Euclidean)}, \quad \tilde{\beta} = \sqrt{1 - 4m^2/s} \\ 2 - 2r \arctan \frac{1}{r} & 0 < s < 4m^2, \quad r = \sqrt{4m^2/s - 1} \\ 2 - \beta \ln \frac{1+\beta}{1-\beta} + i\pi\beta & s > 4m^2, \quad \beta = \sqrt{1 - 4m^2/s} \end{cases} \quad (250)$$

and $\text{Im } f = \pi\beta$ above threshold. In the lecture this last step was phrased as “ $\ln z = \ln|z| + i\pi$ for $z < 0$ ”. The side of the cut is fixed by the $+i\epsilon$ of the propagators and it gives $\text{Im } f = +\pi\beta > 0$.

MATHEMATICA CHECK: THE THREE REGIONS, SYMBOLICALLY

Minkowski: $f(s) = -\int_0^1 \ln(1 - x(1-x)s/m^2 - i\epsilon) dx$, $m = 1$. below threshold, $0 < s < 4$: everything is real. $f(s)$

In [1]: for $0 < s < 4$

```
fBelow = -Integrate[Log[1 - x (1 - x) s],
  {x, 0, 1}, Assumptions -> 0 < s < 4];
FullSimplify[fBelow, 0 < s < 4]
```

Out[1]: $2 - 2 \sqrt{\frac{4}{s} - 1} \csc^{-1}\left(\frac{2}{\sqrt{s}}\right)$

minus $2 - 2r \arctan(1/r)$, $r = \sqrt{4/s - 1}$

In [2]: `FullSimplify[fBelow - (2 - 2 Sqrt[4/s - 1] ArcTan[1/Sqrt[4/s - 1]]), 0 < s < 4]`

Out[2]: 0

above threshold the y integral splits into a principal value and a delta function (Sokhotski). $\mathcal{P} \int_{-1}^1 dy/(b^2 - y^2)$,

In [3]: $0 < b < 1$

```
Integrate[1/(b^2 - y^2), {y, -1, 1}, PrincipalValue -> True,
  Assumptions -> 0 < b < 1]
```

Out[3]: $\frac{2 \tanh^{-1}(b)}{b}$

$\int_{-1}^1 dy \delta(b^2 - y^2)$

In [4]: `Integrate[DiracDelta[b^2 - y^2], {y, -1, 1}, Assumptions -> 0 < b < 1]`

Out[4]: $\frac{1}{b}$

the Euclidean closed form, continued to $s + i\epsilon$ at $s = 6$. $\lim_{\delta \rightarrow 0^+} f(6 + i\delta)$

In [5]: `fE[s_] := 2 - Sqrt[1 - 4/s] Log[(Sqrt[1 - 4/s] + 1)/(Sqrt[1 - 4/s] - 1)];
FullSimplify[Limit[fE[6 + I dd], dd -> 0, Direction -> "FromAbove"]]`

Out[5]: $2 + \frac{i\pi}{\sqrt{3}} - \frac{\log(2 + \sqrt{3})}{\sqrt{3}}$

$2 - \beta \ln((1 + \beta)/(1 - \beta)) + i\pi\beta$ at $s = 6$

In [6]: `FullSimplify[With[{bb = Sqrt[1 - 4/6]},
2 - bb Log[(1 + bb)/(1 - bb)] + I Pi bb]]`

Out[6]: $2 + \frac{i\pi}{\sqrt{3}} - \frac{\log(2 + \sqrt{3})}{\sqrt{3}}$

ε -expansion of the DE solution: since $(\varepsilon)_k = \varepsilon(k-1)! + O(\varepsilon^2)$, the order- ε coefficient of ${}_2F_1(\varepsilon, 1; 3/2; z)$ is $\sum_k (k-1)!/(3/2)_k z^k$. order- ε coefficient of ${}_2F_1$

In [7]: `c1 = Sum[(k - 1)!/Pochhammer[3/2, k] z^k, {k, 1, Infinity}];
c1`

Out[7]: $\frac{2(z - \sqrt{-(z-1)z}) \sin^{-1}(\sqrt{z})}{z}$

that coefficient minus $f(4z)$, $0 < z < 1$

In [8]: `FullSimplify[c1 - (fBelow /. s -> 4 z), 0 < z < 1]`

Out[8]: 0

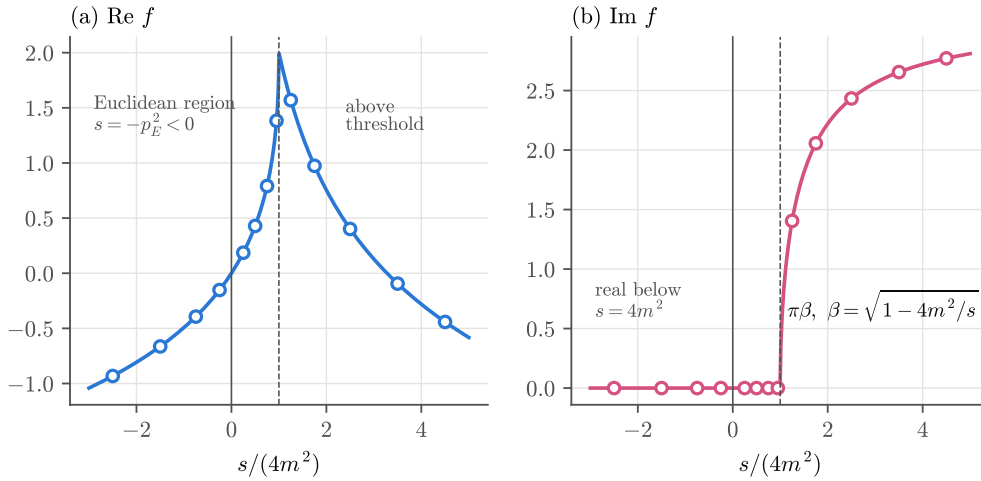


Figure 10: Real and imaginary part of f along the real s axis. Left of $s = 0$ is the Euclidean region. At the threshold $s = 4m^2$ (dashed) the real part has a cusp and the imaginary part switches on as $\pi\beta$. The curves are the three closed forms of the Result box, the circles an independent direct evaluation.

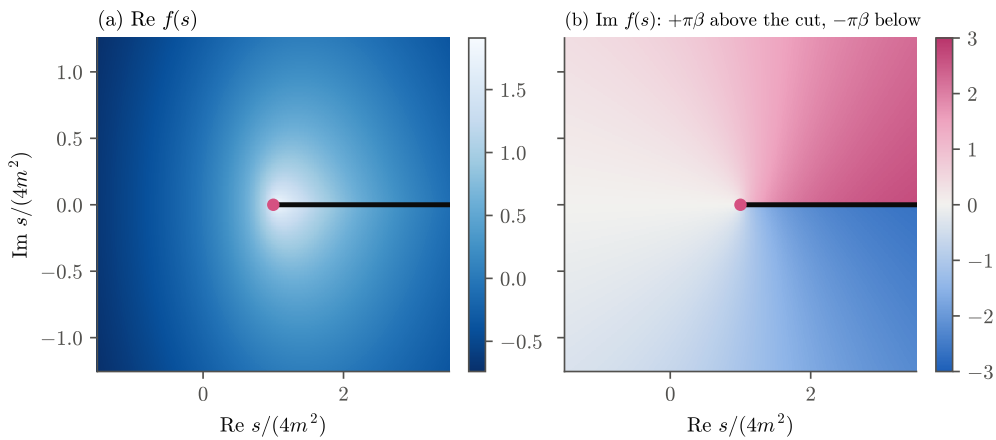


Figure 11: The same function over the complex s plane, (a) its real part and (b) its imaginary part. It is smooth everywhere except on the cut $s \geq 4m^2$ (thick line). The imaginary part is $+\pi\beta$ just above the cut and $-\pi\beta$ just below: the physical amplitude lives on the upper lip, at $s + i\epsilon$.

The cut is the Gate of Truth from Fullmetal Alchemist. Walk around the branch point once and you come back changed: the log has paid a toll of $2\pi i$. Stay on the upper lip ($s + i\epsilon$) and Truth only takes $i\pi\beta$.

15.4 The optical theorem, in detail

Why must the imaginary part be positive and why $\pi\beta$? Because of conservation of probability.

DERIVATION: THE OPTICAL THEOREM FROM UNITARITY

Unitarity. The S-matrix takes an initial state to a final state. Probability is conserved, so $S^\dagger S = 1$. Write $S = 1 + iT$ (the 1 is “nothing happens”, T is the scattering). Then

$$(1 - iT^\dagger)(1 + iT) = 1 \Rightarrow -i(T - T^\dagger) = T^\dagger T \quad (251)$$

Forward matrix element. Take the expectation value in the initial state $|a\rangle = |p_1 p_2\rangle$. On the left, $\langle a|T|a\rangle - \langle a|T^\dagger|a\rangle = \langle a|T|a\rangle - \langle a|T|a\rangle^* = 2i \operatorname{Im} \langle a|T|a\rangle$. On the right, insert a complete set of final states, $1 = \sum_f |f\rangle\langle f|$:

$$2 \operatorname{Im} \langle a|T|a\rangle = \sum_f |\langle f|T|a\rangle|^2 \quad (252)$$

Strip off momentum conservation. Every matrix element contains a delta function, $\langle f|T|a\rangle = (2\pi)^4 \delta^4(P_f - P_a) \mathcal{M}(a \rightarrow f)$. The same $(2\pi)^4 \delta^4(0)$ appears on both sides and cancels and the sum over final states becomes an integral over their momenta (the phase space):

$$2 \operatorname{Im} \mathcal{M}(a \rightarrow a) = \sum_f \int d\Pi_f |\mathcal{M}(a \rightarrow f)|^2 \quad (253)$$

The right side is a sum of squares, so it is positive. The forward amplitude must have a positive imaginary part.

Two-body phase space. For two particles of mass m ,

$$\int d\Pi_2 = \int \frac{d^3 k_1}{(2\pi)^3 2E_1} \frac{d^3 k_2}{(2\pi)^3 2E_2} (2\pi)^4 \delta^4(P - k_1 - k_2) \quad (254)$$

Work in the centre of mass frame, $P = (\sqrt{s}, \mathbf{0})$. The δ^3 sets $\mathbf{k}_2 = -\mathbf{k}_1$, so $E_1 = E_2 = E = \sqrt{k^2 + m^2}$ and

$$\int d\Pi_2 = \frac{1}{4\pi^2} \int \frac{d^3 k}{4E^2} \delta(\sqrt{s} - 2E) = \frac{1}{4\pi^2} \int_0^\infty \frac{4\pi k^2 dk}{4E^2} \delta(\sqrt{s} - 2E) \quad (255)$$

Change variable to E : $E dE = k dk$, so $k^2 dk/E^2 = (k/E) dE$ and $\delta(\sqrt{s} - 2E) = \frac{1}{2} \delta(E - \sqrt{s}/2)$:

$$\int d\Pi_2 = \frac{1}{4\pi^2} \cdot \pi \cdot \frac{k}{E} \cdot \frac{1}{2} = \frac{1}{8\pi} \frac{k}{E}, \quad \frac{k}{E} = \sqrt{1 - 4m^2/s} = \beta \quad (256)$$

So $\int d\Pi_2 = \beta/(8\pi)$: β is the speed of each outgoing particle and the phase space vanishes at threshold, where they are produced at rest.

Apply it to the fish. The tree amplitude is $\mathcal{M}_{\text{tree}} = -\lambda$. For identical particles in the final state we include $1/2$. So the right side is $\frac{1}{2} \cdot \beta/(8\pi) \cdot \lambda^2 = \lambda^2 \beta/(16\pi)$. On the left, with Peskin's convention $i\mathcal{M}$ = (sum of diagrams), the s -channel fish is $i\mathcal{M} = (\lambda^2/2) \int d^4 \ell / (2\pi)^4 1/(D_1 D_2)$ and by the Wick rule this is $(\lambda^2/2) \cdot i/(16\pi^2)(\dots + f(s))$. So $\mathcal{M}_{1\text{-loop}} = \lambda^2/(32\pi^2)(\dots + f(s))$ and

$$2 \operatorname{Im} \mathcal{M}_{1\text{-loop}} = 2 \cdot \frac{\lambda^2}{32\pi^2} \cdot \pi\beta = \frac{\lambda^2 \beta}{16\pi} \quad (257)$$

The two sides agree.

PHYSICAL PICTURE: CUTTING THE DIAGRAM (CUTKOSKY)

The imaginary part comes from the window of x where both loop particles can be real. Cutkosky's rule makes this literal. To get the imaginary part, cut the two lines and replace each propagator by the on-shell condition $2\pi\delta(\ell^2 - m^2)\theta(\ell^0)$:



$$\xrightarrow{\text{cut}} \int \frac{d^4\ell}{(2\pi)^4} (2\pi)^2 \delta_+(\ell^2 - m^2) \delta_+((p - \ell)^2 - m^2) = \frac{(2\pi)^2}{(2\pi)^4} \cdot \frac{\pi\beta}{2} = \frac{\beta}{8\pi} \quad (258)$$

which is the two-body phase space. The cut loop is the phase space of the two real particles. Every imaginary part of every Feynman diagram arises this way.

MATHEMATICA CHECK: PHASE SPACE, OPTICAL THEOREM, THE CUT AND A DISPERSION RELATION

Optical theorem and dispersion relation for the s -channel fish, $m = 1$. two-body phase space: $\int d\Pi_2 =$

In [1]: $1/(4\pi^2) \int d^3\ell/(4E^2) \delta(\sqrt{s} - 2E)$, $E = \sqrt{\ell^2 + 1}$. $\int d\Pi_2$

```
beta[s_] := Sqrt[1 - 4/s];
dPi2 = 1/(4 Pi^2) Integrate[4 Pi l^2/(4 (l^2 + 1)) DiracDelta[rs - 2 Sqrt[l^2 + 1]],
  {l, 0, Infinity},
  Assumptions -> rs > 2] /. rs -> Sqrt[s];
FullSimplify[dPi2, s > 4]
```

Out[1]: $\frac{\sqrt{\frac{s-4}{s}}}{8\pi}$

$\int d\Pi_2 - \beta/(8\pi)$

In [2]: FullSimplify[dPi2 - beta[s]/(8 Pi), s > 4]

Out[2]: 0

unitarity: $2 \operatorname{Im} \mathcal{M}_{1\text{-loop}} = \frac{1}{2} \int d\Pi_2 |\mathcal{M}_{\text{tree}}|^2$ with $\operatorname{Im} f = \pi\beta$. $2 \operatorname{Im} \mathcal{M} - \frac{1}{2} \int d\Pi_2 \lambda^2$

In [3]: FullSimplify[2 lam^2/(32 Pi^2) Pi beta[s] - 1/2 dPi2 lam^2, s > 4]

Out[3]: 0

Cutkosky cut: $\int d^4\ell \delta_+(\ell^2 - 1) \delta_+((p - \ell)^2 - 1)$ in the rest frame of p . the cut integral

```
cut = Integrate[4 Pi l^2/(2 Sqrt[l^2 + 1]) DiracDelta[s - 2 Sqrt[s] Sqrt[l^2 + 1]],
  {l, 0, Infinity},
  Assumptions -> s > 4];
FullSimplify[cut, s > 4]
```

Out[4]: $\frac{1}{2} \pi \sqrt{\frac{s-4}{s}}$

dispersion relation $f(s) = s/\pi \int_{-q}^{\infty} dt \operatorname{Im} f(t)/(t(t-s))$, checked at $s = -q < 0$. right side minus $f(s)$

```
fE[s_] := 2 - beta[s] Log[(beta[s] + 1)/(beta[s] - 1)];
disp = Integrate[Pi beta[t]/(t (t + q)),
  {t, 4, Infinity}, Assumptions -> q > 0];
FullSimplify[-q/Pi disp - fE[-q], q > 0]
```

Out[5]: 0

15.5 Why the answer is a logarithm

Logarithms are everywhere in loop results: $\ln(\mu^2/m^2)$, $\ln(p^2/m^2)$, $\beta \ln((1+\beta)/(1-\beta))$. This is not a coincidence. There are three ways to see why they are natural.

1. Scale invariance. A logarithmic divergence comes from $\int d\ell/\ell$. This integral does not care about scale: every factor of 10 in ℓ contributes the same amount, $\ln 10$. Figure 12 shows it for $\int d^4\ell/(\ell^2 + m^2)^2$: per unit of $\ln \ell$ the integrand is flat above $\ell \sim m$, so the integral up to a cutoff grows like $\ln(\Lambda/m)$. The logarithm is the only function whose growth is the same for every decade, so it is what a scale-free integrand must give.

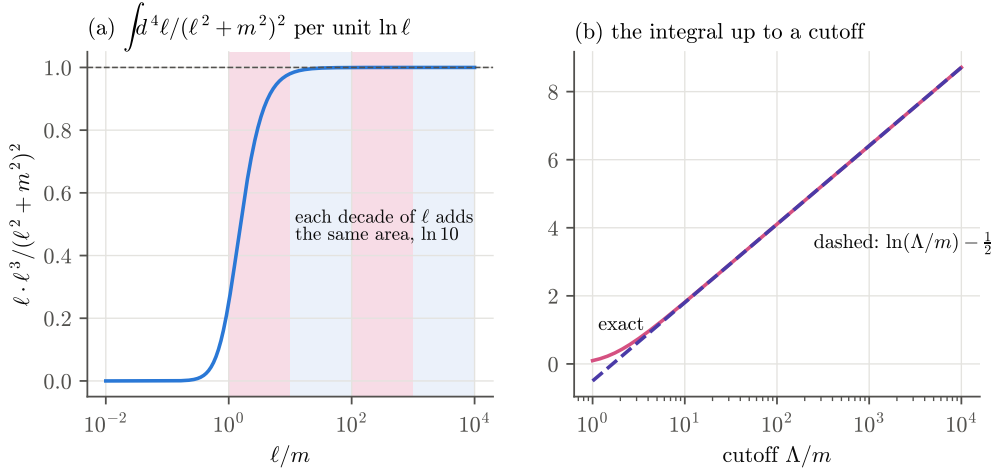


Figure 12: (a) The integrand of $\int d^4\ell/(\ell^2 + m^2)^2$ per unit of $\ln \ell$ (that is, $\ell \cdot \ell^3/(\ell^2 + m^2)^2$). Above $\ell \sim m$ it is flat, so each decade adds the same area. The pink and blue bands are the decades m to $10m$, $10m$ to $100m$, $100m$ to $1000m$ and $1000m$ to 10^4m . The two colours only alternate so that the neighbouring decades can be told apart. Every band has the same width on the log axis and the same height (1), so each one adds the area $\ln 10$. (b) The integral up to a cutoff Λ , which quickly becomes $\ln(\Lambda/m) - 1/2$.

2. Dimensional regularisation turns the log into a pole and back. Compare the same scale-free integral with the two regulators. With a cutoff in four dimensions, $\int_m^\Lambda d\ell/\ell = \ln \Lambda - \ln m$. In $D = 4 - 2\varepsilon$ with no cutoff, the power changes to $\ell^{-1-2\varepsilon}$ and

$$\int_m^\infty \ell^{-1-2\varepsilon} d\ell = \left[\frac{\ell^{-2\varepsilon}}{-2\varepsilon} \right]_m^\infty = \frac{m^{-2\varepsilon}}{2\varepsilon} = \frac{1}{2\varepsilon} - \ln m + O(\varepsilon) \quad (259)$$

The $-\ln m$ is the same in both and $1/(2\varepsilon)$ plays the role of $\ln \Lambda$. The pole is the logarithm in disguise. And whenever a pole meets a factor $a^\varepsilon = 1 + \varepsilon \ln a + \dots$, a finite $\ln a$ is left over (Equation 69). So every $1/\varepsilon$ comes with its logarithms of the masses and momenta.

3. Analyticity plus unitarity (the dispersion relation). $f(s)$ is analytic in the whole complex s plane except the cut $s \geq 4m^2$ (Figure 11). A theorem of complex analysis (Cauchy's integral formula, with the contour wrapped around the cut) then lets us rebuild the whole function from its imaginary part on the cut:

$$f(s) = \frac{s}{\pi} \int_{4m^2}^\infty \frac{dt \operatorname{Im} f(t)}{t(t-s)} \quad (260)$$

(One subtraction at $s = 0$ is used, since $f(0) = 0$.) The imaginary part is $\pi\beta(t)$, the phase space, which tends to a **constant** π at high energy. A constant spread over the cut, weighted by dt/t , is $\int dt/t$ again, a logarithm. So unitarity (the imaginary part is phase space) plus analyticity (the function is determined by its cut) forces the real part to be a logarithm. The last line of the Mathematica box checks this dispersion relation for $s < 0$.

L from Death Note would reconstruct the whole case from one clue. The dispersion relation does the same: from the imaginary part on the cut alone, it rebuilds $f(s)$ everywhere.

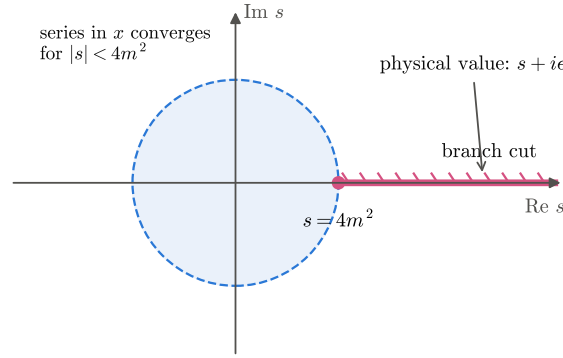


Figure 13: Summary of the analytic structure. The bubble is analytic except for a cut along $s \geq 4m^2$. The hypergeometric series in $x = -s/4m^2$ converges in the disc $|s| < 4m^2$, whose edge touches the branch point. Physical amplitudes are evaluated just above the cut, at $s + i\epsilon$.

16 A first look at Mellin-Barnes representations

Near the end of the second half of the lecture, Mellin-Barnes (MB) representations came up briefly. This section builds them from zero and tests every step on the bubble, whose answer we already know.

Luffy's rubber arm can stretch around any obstacle. An MB contour is the same: a straight line that we may bend to the left or to the right, as long as it never crosses a pole.

16.1 The problem MB solves

After Feynman parameters (Section 11) a one-loop integral always looks like

$$\int dx (A + B)^{-\lambda} \quad (261)$$

where A and B are simple pieces such as m^2 and $x(1-x)p^2$. If it were a **product** $A^{-\lambda_1} B^{-\lambda_2}$, every x integral would be a Beta function and we would be done. The **sum** $A + B$ is what spoils it.

We already know one way to break a power of a sum into powers. It is the binomial series,

$$(A + B)^{-\lambda} = A^{-\lambda} \sum_{k=0}^{\infty} \binom{-\lambda}{k} \left(\frac{B}{A}\right)^k \quad (262)$$

Every term is a product $B^k A^{-\lambda-k}$, which is exactly what we want. But the series converges only for $B < A$ and in a Feynman integral the ratio B/A runs over all values as x and p^2 change. MB is the binomial series rewritten as an integral, so that it works for all $A, B > 0$.

PHYSICAL PICTURE: A FOURIER TRANSFORM WITH POWERS

Fourier analysis writes a signal as a sum of pure waves e^{ikt} . Each wave passes through a linear circuit in a simple way, so we send the waves through one by one and add up. MB does the same thing with pure powers $B^z A^{-\lambda-z}$ in place of pure waves. Each power passes through the x integral in a simple way (it gives a Beta function) and at the end we add up over z .

16.2 From a sum to a contour integral

The trick uses one fact from Section 2.3. $\Gamma(w)$ has a pole at every $w = -k$ ($k = 0, 1, 2, \dots$) and near it

$$\Gamma(w) \approx \frac{(-1)^k}{k! (w + k)} \quad (263)$$

Put $w = -z$. Then $w + k = -(z - k)$, so near $z = k$

$$\Gamma(-z) \approx \frac{(-1)^{k+1}}{k! (z-k)}, \quad \text{Res}_{z=k} \Gamma(-z) = \frac{(-1)^{k+1}}{k!} \quad (264)$$

So $\Gamma(-z)$ is a comb. It has one pole at each $z = 0, 1, 2, \dots$ and the residue at $z = k$ is $(-1)^{k+1}/k!$. The clockwise contour below adds one more minus sign, which turns these into the $(-1)^k/k!$ of Equation 265. Now multiply it by any smooth function $g(z)$ and integrate around a closed loop that encloses the poles $z = 0, 1, \dots, N$ **clockwise**. The residue theorem gives $-2\pi i$ times the sum of residues,

$$\frac{1}{2\pi i} \oint_{\text{clockwise}} dz \Gamma(-z) g(z) = \sum_{k=0}^N \frac{(-1)^k}{k!} g(k) \quad (265)$$

The contour integral picks the values $g(0), g(1), g(2), \dots$ one by one. It is a sum in disguise.

Now write the binomial coefficient with Gamma functions, $\binom{-\lambda}{k} = (-1)^k \Gamma(\lambda + k) / (k! \Gamma(\lambda))$. The k -th term of the series is then $(-1)^k/k!$ times

$$g(k) = \frac{\Gamma(\lambda + k)}{\Gamma(\lambda)} \frac{B^k}{A^{\lambda+k}} \quad (266)$$

This g makes sense for any complex k , so put $g(z) = \Gamma(\lambda + z) B^z A^{-\lambda-z} / \Gamma(\lambda)$ into Equation 265. That is the whole MB formula.

RESULT: SPLITTING A SUM INTO A PRODUCT

For $\lambda > 0$ and $A, B > 0$,

$$\frac{1}{(A+B)^\lambda} = \frac{1}{\Gamma(\lambda)} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} dz \Gamma(-z) \Gamma(\lambda + z) \frac{B^z}{A^{\lambda+z}}, \quad -\lambda < c < 0 \quad (267)$$

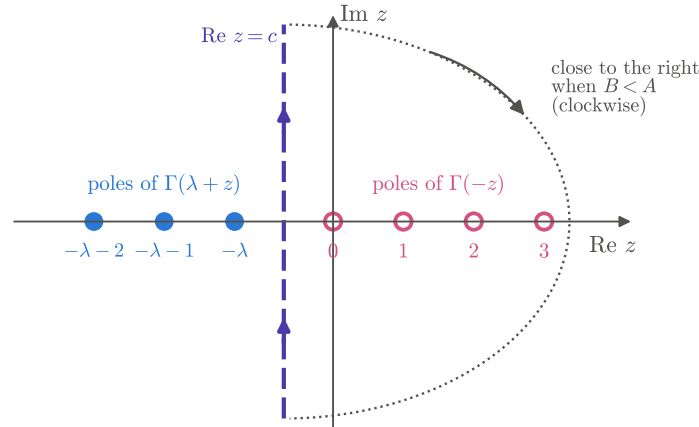


Figure 14: The complex z plane of Equation 267, drawn for $\lambda = 1.4$. Open circles: poles of $\Gamma(-z)$. Filled circles: poles of $\Gamma(\lambda + z)$. The MB contour is the dashed vertical line $\text{Re } z = c$, run upwards, with $-\lambda < c < 0$ so that it separates the two families. For $B < A$ we close it with a large arc on the right (dotted). The loop then goes round the open circles clockwise.

Read the formula piece by piece, using Figure 14.

- z is a complex variable. It runs along the vertical straight line $\text{Re } z = c$, from the bottom $(c - i\infty)$ to the top $(c + i\infty)$. On the line $z = c + iy$, so $dz = i dy$ and the i cancels the i in $1/(2\pi i)$.
- $\Gamma(-z)$ is the comb. Its poles at $z = 0, 1, 2, \dots$ (open circles) produce the terms of the binomial series.
- $\Gamma(\lambda + z)$ gives the $\Gamma(\lambda + k)$ of the binomial coefficient. Its own poles sit at $z = -\lambda, -\lambda - 1, \dots$ (filled circles).
- $B^z A^{-\lambda-z}$ is the pure power, the “wave” of the picture above.
- $-\lambda < c < 0$ says where the line must go: **between** the two families of poles, with every open circle on its right and every filled circle on its left.
- On the line the integral converges for every $A, B > 0$. By Stirling’s formula, $|\Gamma(x + iy)|$ falls like $e^{-\pi |y|/2}$ for large $|y|$, so $\Gamma(-z)\Gamma(\lambda + z)$ falls like $e^{-\pi |y|}$. The power has constant size on the line, $|B^z A^{-z}| = (B/A)^c$. So nothing limits B/A and this is the gain over the series.

WHY IT IS TRUE: CLOSE THE CONTOUR

Take $B < A$. Go up the line and come back down along a large half circle on the right (Figure 14). This loop encloses the open circles $z = 0, 1, 2, \dots$ and runs round them clockwise.

On the far part of the arc the integrand is tiny, because $|(B/A)^z| = (B/A)^{\text{Re } z}$ and $B/A < 1$, so it dies exponentially as $\text{Re } z$ grows. As the arc is pushed out, its contribution goes to zero and the loop integral becomes the line integral. By Equation 265, with $g(z) = \Gamma(\lambda + z)B^z A^{-\lambda-z}/\Gamma(\lambda)$,

$$\text{RHS} = \frac{1}{\Gamma(\lambda)} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \Gamma(\lambda + k) \frac{B^k}{A^{\lambda+k}} = A^{-\lambda} \sum_k \binom{-\lambda}{k} \left(\frac{B}{A}\right)^k = (A+B)^{-\lambda} \quad (268)$$

The two minus signs, one from the residue $(-1)^{k+1}/k!$ and one from the clockwise direction, cancel. That is why the terms come out with $(-1)^k$ and not $(-1)^{k+1}$.

Check with $\lambda = 1$. The residues give $1/A - B/A^2 + B^2/A^3 - \dots$, which is the geometric series of $1/(A+B)$.

The other side. For $B > A$ close to the left instead. Now the loop is anticlockwise and encloses the filled circles $z = -\lambda - n$. The same steps give $B^{-\lambda} \sum_n \binom{-\lambda}{n} (A/B)^n$, the binomial series in the other variable. One integral holds both expansions and it works just as well at $B = A$, which sits on the edge of both series.

The point of Equation 267 is that the right side is a **product** of powers, $B^z A^{-\lambda-z}$. After a Feynman-parameter representation, the troublesome object is a power of a sum (like $\mathcal{F}^{-a}$). MB turns it into powers of the individual terms and then the Feynman-parameter integrals become Beta functions.

16.3 The bubble through MB**DERIVATION, STEP BY STEP**

Start from Equation 178 with $a = 2 - D/2$, $A = m^2$ and $B = x(1-x)p^2$:

$$\begin{aligned} J(1,1) &= \Gamma(a) \int_0^1 dx (m^2 + x(1-x)p^2)^{-a} \\ &= \frac{1}{2\pi i} \int dz \Gamma(-z) \Gamma(a+z) (p^2)^z (m^2)^{-a-z} \int_0^1 x^z (1-x)^z dx \end{aligned} \quad (269)$$

The x integral is $B(1+z, 1+z) = \Gamma(1+z)^2/\Gamma(2+2z)$. So the bubble is a single contour integral of Gamma functions:

$$J(1,1) = \frac{(m^2)^{-a}}{2\pi i} \int dz \Gamma(-z) \Gamma(a+z) \frac{\Gamma(1+z)^2}{\Gamma(2+2z)} \left(\frac{p^2}{m^2}\right)^z \quad (270)$$

Small p^2 : close to the right. The residues at $z = k$ give

$$J = (m^2)^{-a} \sum_k \frac{(-1)^k}{k!} \Gamma(a+k) B(1+k, 1+k) (4x)^k, \quad x = \frac{p^2}{4m^2} \quad (271)$$

which is the power series of $\Gamma(a) {}_2F_1(a, 1; 3/2; -x)$, term by term (the same algebra as in Section 11, with $(3/2)_k = (2k+1)!/(4^k k!)$). In pictures,

$$\text{---} \bigcirc \text{---} = \frac{1}{2\pi i} \int dz \Gamma(-z) \Gamma(a+z) \frac{\Gamma(1+z)^2}{\Gamma(2+2z)} \left(\frac{p^2}{m^2}\right)^z (m^2)^{-a} \quad (272)$$

one line integral replaces the loop integral and the Feynman parameter together.

MATHEMATICA CHECK: THE MB INTEGRAL AND ITS RESIDUES

Mellin-Barnes: $1/(A+B)^\lambda = 1/\Gamma(\lambda) 1/(2\pi i) \int dz \Gamma(-z) \Gamma(\lambda+z) B^z A^{-\lambda-z}$. closing to the right is clockwise:

In [1]: minus the residues at $z = k$. minus the residues at $z = 0, 1, 2$

```
integrand = Gamma[-z] Gamma[lam + z] B^z A^(-lam - z)/Gamma[lam];
Simplify@Table[-Residue[integrand, {z, k}], {k, 0, 2}]
```

Out[1]: $\left\{A^{-\lambda}, -\frac{B A^{-\lambda-1} \Gamma(\lambda+1)}{\Gamma(\lambda)}, \frac{B^2 A^{-\lambda-2} \Gamma(\lambda+2)}{2 \Gamma(\lambda)}\right\}$

general term of the residue sum

```
In [2]: Simplify[(-1)^k/k! Pochhammer[lam, k] B^k A^(-lam - k)]
```

Out[2]: $\frac{(-1)^k B^k A^{-k-\lambda} (\lambda)_k}{k!}$

sum over all k

```
In [3]: Simplify[Sum[(-1)^k/k! Pochhammer[lam, k] B^k A^(-lam - k),
  {k, 0, Infinity}],
  A > 0 && B > 0]
```

Out[3]: $(A+B)^{-\lambda}$

bubble: residues at $z = k$ of $\Gamma(-z)\Gamma(a+z)\Gamma(1+z)^2/\Gamma(2+2z)(p^2/m^2)^z$, $m = 1$. minus the residues at $z = 0, 1, 2$, divided by $\Gamma(a)$

```
In [4]: bub = Gamma[-z] Gamma[a + z] Gamma[1 + z]^2/Gamma[2 + 2 z] (4 x)^z/Gamma[a];
Simplify@Table[-Residue[bub, {z, k}], {k, 0, 2}]
```

Out[4]: $\left\{1, -\frac{2 x \Gamma(a+1)}{3 \Gamma(a)}, \frac{4 x^2 \Gamma(a+2)}{15 \Gamma(a)}\right\}$

sum of all residues

```
In [5]: Sum[(-1)^k/k! Pochhammer[a, k] Beta[k + 1, k + 1] (4 x)^k,
  {k, 0, Infinity}]
```

Out[5]: ${}_2F_1\left(1, a; \frac{3}{2}; -x\right)$

The first line computes the residues at $z = 0, 1, 2$ symbolically (with the minus sign from the clockwise contour). The next two lines show the general term and sum all of them to $(A+B)^{-\lambda}$ exactly. The last two lines do the same for the bubble. The residues are the terms of the series and the full sum is ${}_2F_1(a, 1; 3/2; -x)$.

PHYSICAL PICTURE: TWO ENDS OF THE SAME CONTOUR

Closing the contour to the right gives an expansion in p^2/m^2 , useful at low energy. Closing it to the left, we pick up the poles of $\Gamma(a+z)$ and of $\Gamma(1+z)^2/\Gamma(2+2z)$ and get an expansion in m^2/p^2 , useful at high energy. As $\varepsilon \rightarrow 0$ some of these left poles run into each other and a double pole produces a $\ln(p^2/m^2)$. So the high energy logarithms that drive the running coupling come from colliding poles in the MB plane. This is the germ of the method of regions and of the MB codes (MB.m, MBsums) used at two loops and beyond.

17 Mellin-Barnes at work: the massless box

Section 16 showed the idea on the bubble, where we already knew the answer. Now we use it where it is needed. The one-loop box with massless lines and massless on-shell legs. This is the diagram behind $2 \rightarrow 2$ scattering of massless particles (gluons, photons at high energy) and it is the first integral in these notes whose answer contains a **trilogarithm**. The lecture did it in the notebook MB_Box. Here it is written out step by step.

The box is the Colossal Titan of one-loop integrals: four propagators, two scales and poles everywhere. Mellin-Barnes is the Thunder Spear: it does not fight the whole thing at once, it turns it into Gamma functions one piece at a time.

17.1 The integral and its graph polynomials

Take the four propagators, in Minkowski space and with all legs incoming,

$$F(a_1, a_2, a_3, a_4) = \int \frac{d^D \ell}{i\pi^{D/2}} \frac{1}{[\ell^2]^{a_1} [(\ell + p_1)^2]^{a_2} [(\ell + p_1 + p_2)^2]^{a_3} [(\ell + p_3)^2]^{a_4}} \quad (273)$$

with $p_i^2 = 0$, $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$ and $u = (p_2 - p_3)^2 = -s - t$ (the labelling of the course notebook: p_1, p_2 incoming, p_3 outgoing, so $p_1 \cdot p_3 = -t/2$). From Section 14 (or from FeynCalc) the graph polynomials are

$$\mathcal{U} = x_1 + x_2 + x_3 + x_4, \quad \mathcal{F} = -s x_1 x_3 - t x_2 x_4 \quad (274)$$

and for all powers equal to 1 ($N_\nu = 4$, $L = 1$) the parametric formula gives $\Gamma(2 + \varepsilon) \int d^4 x \delta(1 - \sum x_i) \mathcal{U}^{2\varepsilon} \mathcal{F}^{-2-\varepsilon}$.

17.2 Step 1: start from the parametric integral

DERIVATION: THE BOX AS A FEYNMAN-PARAMETER INTEGRAL

Use the general formula of Section 14 in its Minkowski form with $N = N_\nu = 4$ (four propagators, all powers 1), $L = 1$ and $D = 4 - 2\varepsilon$. The exponents are

$$N_\nu - L \frac{D}{2} = 4 - (2 - \varepsilon) = 2 + \varepsilon, \quad N_\nu - (L + 1) \frac{D}{2} = 4 - (4 - 2\varepsilon) = 2\varepsilon \quad (275)$$

and $(-1)^{N_\nu} = 1$. With $\mathcal{U} = x_1 + x_2 + x_3 + x_4$ and $\mathcal{F} = -s x_1 x_3 - t x_2 x_4$:

$$F(1, 1, 1, 1) = \Gamma(2 + \varepsilon) \int_0^\infty dx_1 dx_2 dx_3 dx_4 \delta(1 - x_1 - x_2 - x_3 - x_4) \times (x_1 + x_2 + x_3 + x_4)^{2\varepsilon} (-s x_1 x_3 - t x_2 x_4)^{-2-\varepsilon} \quad (276)$$

In the Euclidean region $s, t < 0$ both terms of $\mathcal{F}$ are positive, so the power is well defined.

17.3 Step 2: a change of variables that factorises everything

DERIVATION, STEP BY STEP

New variables. In $\mathcal{F}$ the parameter x_1 always appears together with x_3 and x_2 with x_4 . So write the four parameters through two “sizes” u_1, u_2 and two “shares” v_1, v_2 :

$$x_1 = u_1 v_1, \quad x_2 = u_1(1 - v_1), \quad x_3 = u_2 v_2, \quad x_4 = u_2(1 - v_2), \quad u_i \in [0, \infty), v_i \in [0, 1] \quad (277)$$

(So $u_1 = x_1 + x_2$ and $v_1 = x_1/(x_1 + x_2)$ and the same for the second pair.)

Jacobian. The variables (x_1, x_2) depend only on (u_1, v_1) and (x_3, x_4) only on (u_2, v_2) , so the Jacobian is a product of two 2×2 determinants:

$$\det \begin{pmatrix} \partial x_1 / \partial u_1 & \partial x_1 / \partial v_1 \\ \partial x_2 / \partial u_1 & \partial x_2 / \partial v_1 \end{pmatrix} = \det \begin{pmatrix} v_1 & u_1 \\ 1 - v_1 & -u_1 \end{pmatrix} = -u_1 v_1 - u_1(1 - v_1) = -u_1 \quad (278)$$

Taking absolute values, $dx_1 dx_2 dx_3 dx_4 = u_1 u_2 du_1 du_2 dv_1 dv_2$.

The polynomials. $\mathcal{U} = u_1 v_1 + u_1(1 - v_1) + u_2 v_2 + u_2(1 - v_2) = u_1 + u_2$ and

$$\mathcal{F} = -s(u_1 v_1)(u_2 v_2) - t(u_1(1 - v_1))(u_2(1 - v_2)) = u_1 u_2 G(v_1, v_2), \quad G = -s v_1 v_2 - t(1 - v_1)(1 - v_2) \quad (279)$$

The sizes and the shares have separated.

Put it together. The delta function is now $\delta(1 - u_1 - u_2)$, so on it $\mathcal{U} = 1$ and $\mathcal{U}^{2\varepsilon} = 1$. The integrand becomes

$$u_1 u_2 \cdot (u_1 u_2)^{-2-\varepsilon} G^{-2-\varepsilon} = (u_1 u_2)^{-1-\varepsilon} G^{-2-\varepsilon} \quad (280)$$

The u integral. Use the delta function to set $u_2 = 1 - u_1$. What is left is a Beta function (Section 2.4) with $a = b = -\varepsilon$:

$$\int_0^1 u_1^{-1-\varepsilon} (1 - u_1)^{-1-\varepsilon} du_1 = B(-\varepsilon, -\varepsilon) = \frac{\Gamma(-\varepsilon)^2}{\Gamma(-2\varepsilon)} \quad (281)$$

It converges for $\varepsilon < 0$ and the answer is then continued, as always in dimensional regularisation. So far

$$F(1, 1, 1, 1) = \Gamma(2 + \varepsilon) \frac{\Gamma(-\varepsilon)^2}{\Gamma(-2\varepsilon)} \int_0^1 dv_1 \int_0^1 dv_2 [-sv_1 v_2 - t(1 - v_1)(1 - v_2)]^{-2-\varepsilon} \quad (282)$$

17.4 Step 3: one Mellin-Barnes integral splits the sum

DERIVATION, STEP BY STEP

Split the power of a sum. The bracket is $A + B$ with $A = -sv_1 v_2$ and $B = -t(1 - v_1)(1 - v_2)$. By Equation 267 with $\lambda = 2 + \varepsilon$,

$$(A + B)^{-2-\varepsilon} = \frac{1}{\Gamma(2 + \varepsilon)} \frac{1}{2\pi i} \int dz \Gamma(-z) \Gamma(2 + \varepsilon + z) B^z A^{-2-\varepsilon-z} \quad (283)$$

The $1/\Gamma(2 + \varepsilon)$ cancels the $\Gamma(2 + \varepsilon)$ in front.

Separate the v 's. Write out the powers:

$$A^{-2-\varepsilon-z} = (-s)^{-2-\varepsilon-z} (v_1 v_2)^{-2-\varepsilon-z}, \quad B^z = (-t)^z ((1 - v_1)(1 - v_2))^z \quad (284)$$

The kinematics combine to $(-s)^{-2-\varepsilon} ((-t)/(-s))^z = (-s)^{-2-\varepsilon} x^z$ with $x = t/s$. Each v_i now appears only in $v_i^{-2-\varepsilon-z}(1 - v_i)^z$.

The v integrals. Each is a Beta function with $a = -1 - \varepsilon - z$ and $b = 1 + z$:

$$\int_0^1 v^{-2-\varepsilon-z} (1 - v)^z dv = B(-1 - \varepsilon - z, 1 + z) = \frac{\Gamma(-1 - \varepsilon - z) \Gamma(1 + z)}{\Gamma(-\varepsilon)} \quad (285)$$

using $a + b = -\varepsilon$. It converges when $\text{Re}(1 + z) > 0$ and $\text{Re}(-1 - \varepsilon - z) > 0$, that is $-1 < \text{Re } z < -1 - \varepsilon$. Such a strip exists only for $\varepsilon < 0$. Keep this in mind. It is the origin of the pinch below.

Collect. There are two v integrals, so the Beta functions come squared and their $1/\Gamma(-\varepsilon)^2$ cancels the $\Gamma(-\varepsilon)^2$ from the u integral. What remains:

RESULT: THE MASSLESS BOX AS ONE MELLIN-BARNES INTEGRAL

$$F(1, 1, 1, 1) = \frac{(-s)^{-2-\varepsilon}}{\Gamma(-2\varepsilon)} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} dz x^z \Gamma(-1 - \varepsilon - z)^2 \Gamma(-z) \Gamma(1 + z)^2 \Gamma(2 + \varepsilon + z), \quad x = \frac{t}{s} \quad (286)$$

A two-scale, four-propagator loop integral has become a single contour integral of Gamma functions.

PHYSICAL PICTURE: WHY THERE IS A POLE STRUCTURE AT ALL

Each Gamma function records one way the integral can become singular. $\Gamma(-\varepsilon)^2$ and $1/\Gamma(-2\varepsilon)$ carry the soft and collinear divergences (the box is infrared divergent, like the triangle of Section 14). The z -dependent ones decide how the answer depends on t/s : poles on the right give powers of x , poles on the left give powers of $1/x$ and double poles give logarithms of x .

17.5 Step 4: the pinch and how to live with it

A straight contour must separate the “left” poles (from $\Gamma(1 + z)^2$ at $z = -1, -2, \dots$ and $\Gamma(2 + \varepsilon + z)$ at $z = -2 - \varepsilon, \dots$) from the “right” poles (from $\Gamma(-z)$ at $z = 0, 1, \dots$ and $\Gamma(-1 - \varepsilon - z)^2$ at $z = -1 - \varepsilon, -\varepsilon, \dots$). For small ε the right pole at $z = -1 - \varepsilon$ sits just to the left of the left pole at $z = -1$: no straight line fits between them and as $\varepsilon \rightarrow 0$ they pinch the contour (Figure 15).

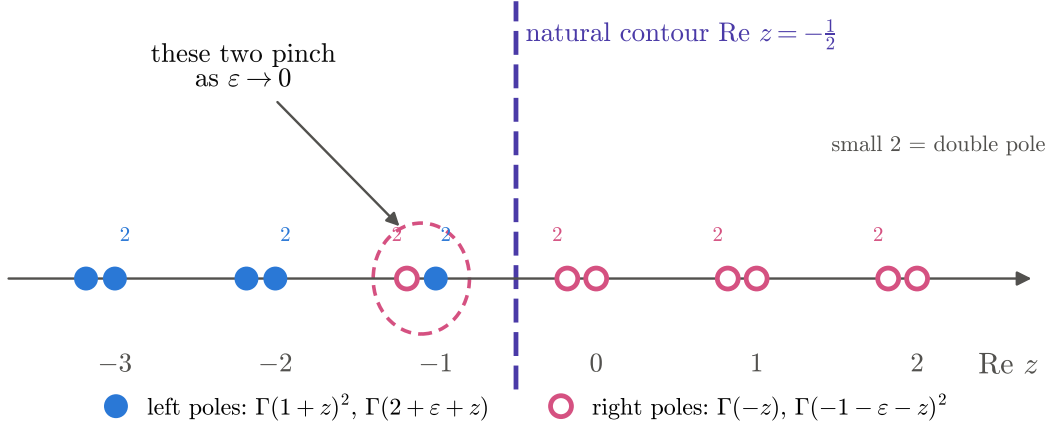


Figure 15: Poles of the Mellin-Barnes integrand of the massless box for a small $\varepsilon > 0$. Filled: poles that must stay on the left. Open: poles that must stay on the right. A small 2 marks a double pole, from a squared Gamma function. The right pole at $z = -1 - \varepsilon$ has moved to the left of the natural contour $\text{Re } z = -1/2$ and nearly touches the left pole at $z = -1$.

DERIVATION: RESOLVING THE PINCH

Start where it is fine. For $-3/2 < \varepsilon < -1/2$ (unphysical but allowed, since everything is analytic in ε) the straight contour $\text{Re } z = -1/2$ separates all poles correctly. The right pole at $z = -1 - \varepsilon$ lies to the right of $-1/2$ and the left pole of $\Gamma(2 + \varepsilon + z)$ at $z = -2 - \varepsilon$ lies to the left of it.

Continue to $\varepsilon \rightarrow 0$. As ε increases towards 0, the pole at $z = -1 - \varepsilon$ moves to the left and crosses the contour at $\varepsilon = -1/2$. To keep the integral the same function, the contour must be deformed around it. That is the same as keeping the straight contour and **adding the residue** of the crossed pole (with a minus sign, since a right pole is enclosed clockwise):

$$F = -\text{Res}_{z=-1-\varepsilon}[\dots] + \frac{1}{2\pi i} \int_{\text{Re } z=-1/2} dz[\dots] \quad (287)$$

The residue term, by hand. Call the integrand $g(z)\Gamma(-1 - \varepsilon - z)^2$, with

$$g(z) = \frac{(-s)^{-2-\varepsilon}}{\Gamma(-2\varepsilon)} x^z \Gamma(-z) \Gamma(1+z)^2 \Gamma(2+\varepsilon+z) \quad (288)$$

Put $z = z_0 + \delta$ with $z_0 = -1 - \varepsilon$. From Section 2.3, $\Gamma(-\delta) = -1/\delta - \gamma_E + O(\delta)$, so

$$\Gamma(-1 - \varepsilon - z)^2 = \Gamma(-\delta)^2 = \frac{1}{\delta^2} + \frac{2\gamma_E}{\delta} + O(1) \quad (289)$$

A double pole. Its residue is read off from $g(z_0 + \delta) = g(z_0) + \delta g'(z_0) + \dots$:

$$\text{Res}_{z=z_0} [g\Gamma(-1 - \varepsilon - z)^2] = g'(z_0) + 2\gamma_E g(z_0) = g(z_0) \left[\frac{g'(z_0)}{g(z_0)} + 2\gamma_E \right] \quad (290)$$

At z_0 : $\Gamma(-z_0) = \Gamma(1 + \varepsilon)$, $\Gamma(1 + z_0)^2 = \Gamma(-\varepsilon)^2$, $\Gamma(2 + \varepsilon + z_0) = \Gamma(1) = 1$ and $x^{z_0} = x^{-1-\varepsilon}$. The logarithmic derivative g'/g is a sum of digamma functions plus $\ln x$ (from $dx^z/dz = x^z \ln x$):

$$\frac{g'(z_0)}{g(z_0)} = \ln x - \psi(1 + \varepsilon) + 2\psi(-\varepsilon) + \psi(1) \quad (291)$$

Now count powers of ε : $\Gamma(-\varepsilon)^2 \approx 1/\varepsilon^2$, $1/\Gamma(-2\varepsilon) \approx -2\varepsilon$ and $\psi(-\varepsilon) \approx 1/\varepsilon$. So $g(z_0) \sim 1/\varepsilon$ and the bracket $\sim 1/\varepsilon$: the residue starts at $1/\varepsilon^2$ and the $\ln x$ in the bracket produces the $-2 \ln x/\varepsilon$. Expanding every Gamma and digamma function to the needed order (Mathematica does this in the box below) gives

$$-e^{\varepsilon\gamma_E} (-s)^{2+\varepsilon} x \text{Res}_{z_0} = \frac{4}{\varepsilon^2} - \frac{2 \ln x}{\varepsilon} - \frac{4\pi^2}{3} + O(\varepsilon) \quad (292)$$

All the poles and the whole finite part, come from this one residue.

The integral term is now finite and starts only at order ε (because of the overall $1/\Gamma(-2\varepsilon) \approx -2\varepsilon$), so it can be expanded in ε under the integral sign. Close the contour to the right, pick up the residues at $z = n = 0, 1, 2, \dots$ (double and triple poles give $\ln x$ and $\ln^2 x$) and sum them. The sums are of the type $\sum_n (-x)^n/n^k$, which are $-\text{Li}_k(-x)$.

RESULT: THE MASSLESS ONE-LOOP BOX

$$e^{\varepsilon\gamma_E}(-s)^\varepsilon \text{st} F(1, 1, 1, 1) = \frac{4}{\varepsilon^2} - \frac{2 \ln x}{\varepsilon} - \frac{4\pi^2}{3} + \varepsilon \left[\frac{7\pi^2}{6} \ln x + \frac{1}{3} \ln^3 x - (\pi^2 + \ln^2 x) \ln(1+x) \right. \\ \left. - 2 \ln x \text{Li}_2(-x) + 2 \text{Li}_3(-x) - \frac{34}{3} \zeta(3) \right] + O(\varepsilon^2) \quad (293)$$

MATHEMATICA CHECK: THE NOTEBOOK MB_BOX, RUN STEP BY STEP

Massless box by Mellin-Barnes: $p_i^2 = 0$, $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $D = 4 - 2\varepsilon$. Feynman-parameter integrand

```
In [1]: integrand
Get[FileNameJoin[{$snipDir, "..", "course_nb", "GFP.m"}]];
FP = FeynmanParametrization[{{l1,0,a1},{l1+p1,0,a2},{l1+p1+p2,
0,a3},{l1+p3,0,a4}},{l1},{p1,p2,p3},
Replacements->{p1^2->0,p2^2->0,p3^2->0,p1*p2->s/2,
p1*p3->-t/2,p2*p3->(s+t)/2}];
FP
```

```
Out[1]: 1
Gamma(a1) Gamma(a2) Gamma(a3) Gamma(a4) x1^a1-1 x2^a2-1 x3^a3-1 x4^a4-1 (-1)^a1+a2+a3+a4
(x1 + x2 + x3 + x4)^a1+a2+a3+a4+2 epsilon-4 Gamma(a1 + a2 + a3 + a4 + 1/2 (2 epsilon - 4)) (-s x1 x3 - t x2 x4)^-a1-a2-a3-a4-epsilon+2
```

Jacobian of $x_1 = u_1 v_1$, $x_2 = u_1(1 - v_1)$, $x_3 = u_2 v_2$, $x_4 = u_2(1 - v_2)$. Jacobian

```
In [2]: jac = Det[Table[D[{u1 v1, u1 (1 - v1), u2 v2, u2 (1 - v2)}][[i]],
{u1, u2, v1, v2}][[j]]], {i, 4}, {j, 4}];
Simplify[jac]
```

```
Out[2]: -u1 u2
```

substitute, impose $\delta(1 - u_1 - u_2)$ and do the u_1 Beta integral (full code in the file). after the u_1 integral

```
In [3]: FP
```

```
Out[3]: (v1^a1-1 (1 - v1)^a2-1 v2^a3-1 (1 - v2)^a4-1 (-1)^a1+a2+a3+a4 Gamma(-a1 - a2 - epsilon + 2)
Gamma(-a3 - a4 - epsilon + 2) Gamma(a1 + a2 + a3 + a4 + epsilon - 2) (t (v1 (-v2) + v1 + v2 - 1) - s v1 v2)^-a1-a2-a3-a4-epsilon+2) /
(Gamma(a1) Gamma(a2) Gamma(a3) Gamma(a4) Gamma(-a1 - a2 - a3 - a4 - 2 epsilon + 4))
```

one Mellin-Barnes integral splits $-s v_1 v_2 - t(1 - v_1)(1 - v_2)$; $x = t/s$. Mellin-Barnes integrand of $F(1, 1, 1, 1)$

```
In [4]: MB = Simplify[FP*(Gamma[-2+a1+a2+a3+a4+eps+z] Gamma[-z] /
(Gamma[-2+a1+a2+a3+a4+eps]))*x^z*(1-v1)^z (1-v2)^z*(-s)^(2-a1-a2-a3-a4-eps)*v1^(2-a1-
a2-a3-a4-eps-z)*v2^(2-a1-a2-a3-a4-eps-z)/((-s v1 v2+t (-1+v1+v2-v1 v2))^(2-a1-a2-a3-a4-
eps))];
MB = Integrate[MB,{v1,0,1},{v2,0,1},GenerateConditions->False];
MB1111 = MB /. {a1->1,a2->1,a3->1,a4->1};
MB1111
```

```
Out[4]: (-s)^-epsilon-2 x^z Gamma(-z) Gamma(z+1)^2 Gamma(-epsilon-z-1)^2 Gamma(epsilon+z+2)
Gamma(-2 epsilon)
```

the pinch: residue at $z = -1 - \varepsilon$. that residue, times $e^{\varepsilon\gamma_E}(-s)^{2+\varepsilon}x$

```
In [5]: res = FullSimplify[-Residue[MB1111,{z,-1-eps}]];
resExp = FullSimplify[Series[res*Exp[eps*EulerGamma]*(-s)^(2+eps)*x,
```

```
{eps,0,1}]]];
resExp
```

Out[5]: $\frac{4}{\epsilon^2} - \frac{2 \log(x)}{\epsilon} - \frac{4 \pi^2}{3} + \frac{1}{6} \epsilon (2 \log^3(x) + 7 \pi^2 \log(x) - 68 \zeta(3)) + O(\epsilon^2)$

$e^{\epsilon \gamma_E} (-s)^\epsilon st F(1, 1, 1, 1)$ through $O(\epsilon)$

```
In [6]: int = FullSimplify[-Residue[Normal[Series[MB1111,
{eps,0,1}]]],{z,n},Assumptions->n\[Element]Integers&& n>=0],
Assumptions->n\[Element]Integers&& n>=0];
int = Apart[int/.{n->n-1},n]/.{n->n+1};
intExp = Series[Sum[int,
{n,0,Infinity}]*Exp[eps*EulerGamma]*(-s)^(2+eps)*x,{eps,0,1}];
tot = FullSimplify[resExp + intExp];
tot
```

Out[6]: $\frac{4}{\epsilon^2} - \frac{2 \log(x)}{\epsilon} - \frac{4 \pi^2}{3} + \epsilon \left(\frac{1}{6} (12 \text{Li}_3(-x) + 2 \log^3(x) - 6 (\log^2(x) + \pi^2) \log(x+1) + 7 \pi^2 \log(x) - 68 \zeta(3)) - 2 \text{Li}_2(-x) \log(x) \right) + O(\epsilon^2)$

the same function from the canonical differential equation. difference, orders $\epsilon^{-2}, \dots, \epsilon^1$

```
In [7]: known = 4/eps^2 - 2 Log[x]/eps - 4 Pi^2/3 + (7 Pi^2/6 Log[x] + Log[x]^3/3 - (Pi^2 +
Log[x]^2) Log[1 + x] - 2 Log[x] PolyLog[2,
-x] + 2 PolyLog[3, -x] - 34 Zeta[3]/3) eps;
FullSimplify[Normal[tot] - known, x > 0]
```

Out[7]: 0

The last line compares with the result of the canonical differential equation of Section 20, obtained in a completely different way. The difference is zero through $O(\epsilon)$. Section 21 gets the same number at $s = t = -1$ from sector decomposition and Package-X gives it too.

PHYSICAL PICTURE: EXPANSIONS FROM THE TWO SIDES

Closing the contour to the right expands in $x = t/s$ (forward scattering, small t). Closing it to the left expands in s/t . The notebook does both. The leading term for small t is the residue at $z = -1 - \epsilon$ alone, the $1/t$ singularity of forward scattering. This is how Mellin-Barnes gives asymptotic expansions (Regge limits, high-energy limits) almost for free.

17.6 The box with one off-shell leg

Let one leg be massive, $p_1^2 = M^2$ (for example a virtual photon or a Z boson). The same steps now need **two** Mellin-Barnes integrals, z_2 and z_4 and the integrand is

$$\frac{(-M^2)^{z_2} (-t)^{z_4}}{(-s)^{z_2+z_4}} \Gamma(2 + \epsilon + z_2 + z_4) \Gamma(1 + z_2 + z_4) \Gamma(1 + z_4) \times \Gamma(-1 - \epsilon - z_2 - z_4) \Gamma(-1 - \epsilon - z_4) \Gamma(-z_2) \Gamma(-z_4) \quad (294)$$

(over $\Gamma(-2\epsilon)$ and the prefactor). The notebook finds straight contours at $\epsilon = -1$ ($\text{Re } z_2 = -1/2$, $\text{Re } z_4 = -1/4$), then continues to $\epsilon \rightarrow 0$. Three poles cross on the way and each crossing adds a residue term, one with no Mellin-Barnes integral left, one with a single integral and a two-fold integral that is already $O(\epsilon)$. This is the algorithm that the programs MB.m and MBresolve automate.

MATHEMATICA CHECK: THE NOTEBOOK MB_OFFSHELLBOX

Box with $p_1^2 = M^2$: two-fold Mellin-Barnes integral, straight contours at $\epsilon = -1$, continued to $\epsilon \rightarrow 0$. the pole of

```
In [1]:  $\Gamma(1/4 - \tilde{\epsilon} - \tilde{z}_4)$  crosses. the pole of  $\Gamma(1/2 - \tilde{\epsilon} + \tilde{z}_2)$  crosses. the pole of  $\Gamma(3/4 - \tilde{\epsilon} - \tilde{z}_2 - \tilde{z}_4)$  crosses. the term with no Mellin-Barnes integral left
```

```

OSBint = (-M2)^z2*(-t)^z4/(-s)^(z2+z4)*Gamma[2+eps+z2+z4]*Gamma[1+z2+z4]*
Gamma[1+z4]*Gamma[-1-eps-z2-z4]*Gamma[-1-eps-z4]*Gamma[-z2]*
Gamma[-z4];
OSBintTilde = OSBint /. {eps->epsTil-1, z2->-1/2+z2Til, z4->-1/4+z4Til};
R = {OSBintTilde};
AppendTo[R, -Residue[R[[1]], {z4Til, 1/4 - epsTil}]];
AppendTo[R, +Residue[R[[2]], {z2Til, -1/2 + epsTil}]];
AppendTo[R, -Residue[R[[1]], {z4Til, 3/4 - epsTil - z2Til}]];
R = Simplify[R /. {epsTil -> 1 + eps,
z2Til -> 1/2 + z2, z4Til -> 1/4 + z4}];
R[[3]]

```

Out[1]:
$$\frac{s(-M2)^\epsilon (-t)^{-\epsilon} \Gamma(-\epsilon)^3 \Gamma(\epsilon+1)^2}{t}$$

$e^{\epsilon\gamma_E} stF(1, 1, 1, 1)$ through ϵ^0

In [2]:

```

Part1 = FullSimplify[Series[(s t Exp[eps*EulerGamma])/((-s)^(2+eps)Gamma[-2eps]) ((-
M2)^eps s (-t)^-eps Gamma[-eps]^3 Gamma[1+eps]^2)/t,
{eps, 0, 0}]];
term = FullSimplify[-Residue[Normal[Series[R[[2]]+R[[4]],
{eps, 0, -1}]], {z2, n}, Assumptions->n\{Element}Integers&&n>0]];
Part2 = FullSimplify[FullSimplify[Series[(s t Exp[eps*EulerGamma])/((-
s)^(2+eps)Gamma[-2eps])*(-Residue[Normal[Series[R[[2]]+R[[4]],
{eps, 0, -1}]], {z2, 0}]+Sum[term, {n, 1, Infinity}]), {eps, 0, 0}]] /. {
PolyLog[2, M2/s]->-PolyLog[2,
1-M2/s]+Pi^2/6-(Log[-M2]-Log[-s])Log[1-M2/s],
PolyLog[2, M2/t]->-PolyLog[2,
1-M2/t]+Pi^2/6-(Log[-M2]-Log[-t])Log[1-M2/t}]];
res = Normal[Part1 + Part2];
Collect[Simplify[res], eps, Simplify]

```

Out[2]:
$$-2 \operatorname{Li}_2\left(1 - \frac{M^2}{s}\right) - 2 \operatorname{Li}_2\left(1 - \frac{M^2}{t}\right) + \frac{2(\log(-M2) - \log(-s) - \log(-t))}{\epsilon} - \log^2(-M2) + 2 \log(-s) \log(-t) + \frac{2}{\epsilon^2} - \frac{\pi^2}{2}$$

the one-mass box formula (Bern, Dixon, Kosower) with r_Γ , in the Euclidean region. Mellin-Barnes result minus the formula

In [3]:

```

rG = Gamma[1 + eps] Gamma[1 - eps]^2/Gamma[1 - 2 eps];
bdk = 2/eps^2 (Exp[-eps lS] + Exp[-eps lT] - Exp[-eps lM]) - 2 PolyLog[2,
1 - M2/s] - 2 PolyLog[2, 1 - M2/t] - (lS - lT)^2 - Pi^2/3;
bdkN = Normal@Series[Exp[eps EulerGamma] rG bdk, {eps, 0, 0}];
Simplify[(res /. {Log[-M2] -> lM, Log[-s] -> lS, Log[-t] -> lT}) - bdkN]

```

Out[3]: 0

The result agrees with the well known closed form of the one-mass box (Bern, Dixon and Kosower),

$$e^{\epsilon\gamma_E} stF = e^{\epsilon\gamma_E} r_\Gamma \left\{ \frac{2}{\epsilon^2} [(-s)^{-\epsilon} + (-t)^{-\epsilon} - (-M^2)^{-\epsilon}] - 2 \operatorname{Li}_2\left(1 - \frac{M^2}{s}\right) - 2 \operatorname{Li}_2\left(1 - \frac{M^2}{t}\right) - \ln^2 \frac{s}{t} - \frac{\pi^2}{3} \right\} + O(\epsilon) \quad (295)$$

with $r_\Gamma = \Gamma(1 + \epsilon)\Gamma(1 - \epsilon)^2/\Gamma(1 - 2\epsilon)$. (As a check of the normalisation, take the massless limit. Drop the $(-M^2)^{-\epsilon}$ term, which becomes scaleless and let $M^2 \rightarrow 0$ in the dilogarithms. Each $\operatorname{Li}_2(1 - M^2/s) \rightarrow \operatorname{Li}_2(1) = \pi^2/6$, so the constant becomes $-4\pi^2/6 - \pi^2/3 = -\pi^2$ and the formula reproduces Equation 293 exactly.)

18 IBP beyond one loop: Laporta's algorithm by hand

In Section 8 the IBP identities of the bubble were solved by hand and Kira and LiteRed did the same by machine. What do the programs actually do? The course notebooks `IBP_Ex1` and `IBP_Toy_Laporta` answer this with a small reduction program written in about a hundred lines of Mathematica. This chapter walks through it.

18.1 Shift operators

Write the index-raising and index-lowering operations as operators. For a family $F(a_1, a_2)$ let $\mathbf{1}^+$ raise a_1 by one and $\mathbf{1}^-$ lower it and the same for $\mathbf{2}^\pm$. In the notebook these are written `op[i, j]`, meaning “shift a_1 by i and a_2 by j ”. A dot on a line is one $\mathbf{1}^+$. Removing a line altogether is $\mathbf{1}^-$ applied until the index is zero.

DERIVATION: THE TWO IBP OPERATORS OF THE BUBBLE WITH ONE MASSIVE LINE

Take (Minkowski) $D_1 = \ell^2 - m^2$ (massive) and $D_2 = (\ell + q)^2$ (massless), $q^2 = q_2$ in the notebook's notation.

Vector ℓ^μ : differentiate. By the product rule,

$$\frac{\partial}{\partial \ell^\mu} \frac{\ell^\mu}{D_1^{a_1} D_2^{a_2}} = \frac{d}{D_1^{a_1} D_2^{a_2}} - a_1 \frac{\ell \cdot \partial D_1}{D_1^{a_1+1} D_2^{a_2}} - a_2 \frac{\ell \cdot \partial D_2}{D_1^{a_1} D_2^{a_2+1}} \quad (296)$$

Write the numerators through denominators. $\ell \cdot \partial D_1 = 2\ell^2 = 2(D_1 + m^2)$. For the second, $D_2 - D_1 = 2\ell \cdot q + q^2 + m^2$ gives $2\ell \cdot q = D_2 - D_1 - q^2 - m^2$, so

$$\ell \cdot \partial D_2 = 2\ell \cdot (\ell + q) = 2(D_1 + m^2) + (D_2 - D_1 - q^2 - m^2) = D_1 + D_2 + m^2 - q^2 \quad (297)$$

Integrate term by term. A factor D_1 upstairs cancels one power of D_1 and so on:

$$\begin{aligned} 0 &= d F(a_1, a_2) - 2a_1 [F(a_1, a_2) + m^2 F(a_1 + 1, a_2)] \\ &\quad - a_2 [F(a_1 - 1, a_2 + 1) + F(a_1, a_2) + (m^2 - q^2) F(a_1, a_2 + 1)] \end{aligned} \quad (298)$$

In operator form this is

$$\text{ibp}_1 = (d - 2a_1 - a_2) - 2a_1 m^2 \mathbf{1}^+ - a_2 \mathbf{1}^- \mathbf{2}^+ + a_2 (q^2 - m^2) \mathbf{2}^+ \quad (299)$$

Vector q^μ . Now $\partial_\mu q^\mu = 0$ and $q \cdot \partial D_1 = 2\ell \cdot q = D_2 - D_1 - q^2 - m^2$, $q \cdot \partial D_2 = 2(\ell + q) \cdot q = D_2 - D_1 + q^2 - m^2$. The same three steps give

$$\begin{aligned} 0 &= -a_1 [F(a_1 + 1, a_2 - 1) - F(a_1, a_2) - (q^2 + m^2) F(a_1 + 1, a_2)] \\ &\quad - a_2 [F(a_1, a_2) - F(a_1 - 1, a_2 + 1) + (q^2 - m^2) F(a_1, a_2 + 1)] \end{aligned} \quad (300)$$

Multiplying by -1 (allowed, it equals zero) gives the form used in the notebook:

$$\text{ibp}_2 = (a_2 - a_1) - a_1 (q^2 + m^2) \mathbf{1}^+ + a_1 \mathbf{1}^+ \mathbf{2}^- - a_2 \mathbf{1}^- \mathbf{2}^+ + a_2 (q^2 - m^2) \mathbf{2}^+ \quad (301)$$

Eliminate $\mathbf{1}^+$. The combination $(q^2 + m^2)\text{ibp}_1 - 2m^2\text{ibp}_2$ has no $\mathbf{1}^+$ term. Solving it for the highest a_2 gives a rule that **lowers a_2** . Solving ibp_1 for $\mathbf{1}^+$ gives a rule that **lowers a_1** (moving powers from line 1 to line 2). Together with “ $F(a_1 \leq 0, a_2) = 0$ ” (a massless tadpole is scaleless) these rules reduce every $F(a_1, a_2)$ to two masters,

$$F(1, 1) = \text{thick circle with } m \text{ above it} \quad \text{and} \quad F(1, 0) = \text{thin circle with } m \text{ above it} \quad (302)$$

where the thick line labelled m is the massive propagator D_1 and the thin line the massless one D_2 .

MATHEMATICA CHECK: THE NOTEBOOK `IBP_EX1` AND A COMPARISON WITH `LITERED`

Shift operators for $F(a_1, a_2) = \int [d\ell] / ([\ell^2 - m^2]^{a_1} [(\ell + q)^2]^{a_2})$; `op[i, j]` shifts $a_1 \rightarrow a_1 + i$, $a_2 \rightarrow a_2 + j$. the

In [1]: combination in which $\mathbf{1}^+$ cancels

```
ibp1 = (d-2a1-a2)*op[0,0] - 2a1*m^2*op[1,0] - a2*op[-1,1] + a2*(q2-m^2)*
      op[0,1];
ibp2 = (a2-a1)*op[0,0] - a1*(q2+m^2)*op[1,0] + a1*
      op[1,-1] - a2*op[-1,1] + a2*(q2-m^2)*op[0,1];
```

```
ibpLC = Collect[(q2+m^2)*ibp1 - 2m^2*ibp2, {op[0,1],op[1,0],op[-1,
1],op[1,-1],op[0,0],q2}, Simplify];
ibpLC
```

Out[1]: $\text{op}(0, 0) (q2 (-2 a1 - a2 + d) + m^2 (d - 3 a2)) - 2 a1 m^2 \text{op}(1, -1) +$
 $\text{op}(-1, 1) (a2 m^2 - a2 q2) + \text{op}(0, 1) (a2 m^4 - 2 a2 m^2 q2 + a2 q2^2)$

$F(3, 2)$ in terms of $F(1, 1)$ and the tadpole $F(1, 0)$

```
In [2]: ReduceA2 = {Fm[a1_, a2_] /;
TrueQ[a1>=1 && a2>1] -> (-(-1+a2) (m^2-q2) Fm[-1+a1,
a2]+((-3+3 a2-d) m^2+(-1+2 a1+a2-d) q2) Fm[a1,-1+a2]+2 a1 m^2 Fm[1+a1,
-2+a2])/((-1+a2) (m^2-q2)^2)};
ShuffleA1 = {Fm[a1_, a2_] /; TrueQ[a1>1] -> (-a2 Fm[-2+a1,
1+a2]+(2-2 a1-a2+d) Fm[-1+a1,a2]+a2 (-m^2+q2) Fm[-1+a1,
1+a2])/(2 (-1+a1) m^2)};
BC = {Fm[a1_, a2_] /; TrueQ[a1\{Element\Integers && a1<=0] :> 0};
ReduceAll = Union[ShuffleA1, ReduceA2, BC];
r32 = Collect[Fm[3,2] /. ReduceAll, {Fm[1,0],Fm[1,1]}, Simplify];
r32
```

Out[2]:
$$\frac{(d-2) Fm(1, 0) ((4 d^2 - 39 d + 96) m^4 + (28 - 6 d) m^2 q2 + (d - 4) q2^2)}{8 (m^3 - m q2)^4} -$$

$$\frac{(d^2 - 8 d + 15) Fm(1, 1) ((d - 4) m^2 + (d - 8) q2)}{2 (m^2 - q2)^4}$$

Load LiteRed and solve the family $\{\ell^2 - m^2, (\ell + q)^2\}$ (full code in the file). minus LiteRed's reduction

```
In [3]: lr = IBPReduce[j[bm, 3, 2]];
Simplify[(r32 /. Fm[a_, b_] :> j[bm, a, b]) - lr]
```

Out[3]: 0

The reduction of $F(3, 2)$ found with these two rules agrees with LiteRed.

18.2 The two-loop propagator family

The toy model of the notebook is the family of massless two-loop propagator integrals,

$$F(a_1, \dots, a_5) = \int [d\ell_1][d\ell_2] \frac{1}{D_1^{a_1} D_2^{a_2} D_3^{a_3} D_4^{a_4} D_5^{a_5}} \quad (303)$$

$$D_1 = \ell_1^2, \quad D_2 = (\ell_1 + q)^2, \quad D_3 = (\ell_1 + \ell_2)^2, \quad D_4 = (\ell_1 + \ell_2 + q)^2, \quad D_5 = \ell_2^2 \quad (304)$$

With all $a_i = 1$ this is the kite of Section 14 (with a different labelling of the lines). There are 5 denominators and exactly 5 scalar products involving loop momenta $(\ell_1^2, \ell_1 \cdot q, \ell_1 \cdot \ell_2, \ell_2^2, \ell_2 \cdot q)$, so every scalar product can be written through denominators. The notebook checks these five relations first.

The identities. Two loop momenta to differentiate, three vectors to contract with (ℓ_1, ℓ_2, q) : $2 \times 3 = 6$ IBP identities for every choice of the five indices.

Boundary conditions. An integral vanishes when it is scaleless. For example if $a_1, a_2 \leq 0$, only D_3, D_4 and D_5 are left in the denominator. Shift $\ell_1 \rightarrow \ell_1 - \ell_2$: then $D_3 = \ell_1^2$, $D_4 = (\ell_1 + q)^2$ and the ℓ_1 integral is a massless bubble in q , while $D_5 = \ell_2^2$ is the only propagator left for ℓ_2 . So the ℓ_2 integral is a massless tadpole, which is scaleless and vanishes. The notebook encodes all such cases as rules.

Symmetries. The graph has symmetries: $\ell_1 \leftrightarrow \ell_2$ type relabellings swap $(a_1, a_2) \leftrightarrow (a_3, a_4)$ and a left-right reflection swaps $a_1 \leftrightarrow a_2$ together with $a_3 \leftrightarrow a_4$. The rules map each integral to a standard representative.

18.3 Laporta's ordering

The algorithm rests on one choice. Integrals are ordered by “complexity”:

1. first by the number of lines present (positive indices): more lines is more complicated
2. then by the total number of dots and numerator powers, $\sum |a_i|$.

Then one generates identities for a list of **seeds** and, one identity at a time, solves each for its **most complicated** integral and stores the result as a rule that rewrites that integral in terms of simpler ones. An identity that becomes $0 = 0$ after inserting the known rules is dependent and is thrown away. The notebook's **ProcessIBP** does this and **ReduceSector** runs it over all seeds of one sector with up to three extra powers.

WHY THE ORDER OF SECTORS MATTERS

Suppose we start in the top sector (all five lines) and meet an integral of a subsector that is not yet reduced. Solving for the most complicated integral is still possible, but the subsector integrals in the answer are not in terms of masters yet, so more identities have to be generated later and substituted back. Starting from the **simplest** sectors and working upwards (Laporta's order) guarantees that whenever we arrive at a sector, everything below it is already reduced. The notebook compares the running times of both strategies on $F(2, 2, 1, 2, 2)$.

18.4 Counting: why a few masters are enough

Argeri and Mastrolia (Section 3 of their review: the count of integrals is their eq. (4) in Section 3.1, the count of IBP identities is in Section 3.2 and the argument that equations grow faster than integrals is in Section 3.4.5) make the Laporta idea quantitative with a little combinatorics. We do it here because it explains why the method works at all.

DERIVATION: HOW MANY INTEGRALS, HOW MANY IDENTITIES

Scalar products. With L loop momenta and E independent external momenta, the scalar products that contain a loop momentum are $\ell_i \cdot \ell_j$ ($i \leq j$, that is $L(L+1)/2$ of them) and $\ell_i \cdot p_k$ (LE of them):

$$N_{\text{sp}} = \frac{L(L+1)}{2} + LE \quad (305)$$

A family with t denominators can absorb t of them. The other $q = N_{\text{sp}} - t$ are irreducible numerators. (For the bubble $L = E = 1$: $N_{\text{sp}} = 2 = t$, $q = 0$. For the two-loop kite $L = 2$, $E = 1$: $N_{\text{sp}} = 5 = t$.)

Integrals in a class. Call $I_{t,r,s}$ the integrals with all t denominators present, r extra powers of denominators in total ($r = \sum (m_j - 1)$) and s powers of numerators in total ($s = \sum n_i$). Counting them is the “stars and bars” problem. Put r identical balls (the extra powers) into t boxes (the denominators). Write the r balls in a row and put $t - 1$ bars between them to separate the boxes. Every arrangement of r balls and $t - 1$ bars is one way, so there are $\binom{r+t-1}{t-1}$ of them. Doing the same for the numerators:

$$N(I_{t,r,s}) = \binom{r+t-1}{t-1} \binom{s+q-1}{q-1} \quad (q \geq 1) \quad (306)$$

If there are no irreducible numerators ($q = 0$), there is exactly one way to have $s = 0$ and none for $s > 0$, so the second factor is simply 1 (for $s = 0$).

Identities. Each integral (a “seed”) gives $L(L+E)$ IBP identities: L derivatives times $L+E$ vectors. Take the kite ($t = 5$, $q = 0$, $L(L+E) = 6$) and all seeds with up to R extra powers. There are $\sum_{r=0}^R \binom{r+4}{4} = \binom{R+5}{5}$ seeds, so $6 \binom{R+5}{5}$ identities. They contain integrals with up to $R+1$ extra powers and there are $\binom{R+6}{5}$ of those. The ratio

$$\frac{6 \binom{R+5}{5}}{\binom{R+6}{5}} = \frac{6(R+1)}{R+6} \quad (307)$$

equals 1 at $R = 0$ and grows towards 6.

So once the seeds carry more than a couple of dots, there are **several times more equations than unknowns**. Most of them are not independent (that is why “ $0 = 0$ ” appears so often in the reduction), but there are always enough independent ones to eliminate all but a handful of integrals, the masters. Mathematica checks the counting formula by brute-force enumeration and prints the table.

MATHEMATICA CHECK: COUNTING INTEGRALS AND IDENTITIES

Counting integrals and identities (stars and bars). $N(I_{t,r,s}) = \binom{r+t-1}{t-1} \binom{s+q-1}{q-1}$. brute force: list every index

In [1]: vector and count. formula and brute force side by side, $t = 5, q = 2, r = 0, \dots, 3, s = 0, 1, 2$

```
Nclass[t_, r_, q_, s_] := Binomial[r + t - 1, t - 1] Binomial[s + q - 1,
  q - 1];
brute[t_, r_, q_, s_] := Length[Select[Tuples[Range[0,
  r], t], Total[#] == r &]] *
  Length[Select[Tuples[Range[0, s], q], Total[#] == s &]];
Table[{Nclass[5, r, 2, s], brute[5, r, 2, s]}, {r, 0, 3}, {s, 0, 2}]
```

Out[1]: $\begin{pmatrix} \{1, 1\} & \{2, 2\} & \{3, 3\} \\ \{5, 5\} & \{10, 10\} & \{15, 15\} \\ \{15, 15\} & \{30, 30\} & \{45, 45\} \\ \{35, 35\} & \{70, 70\} & \{105, 105\} \end{pmatrix}$

$N_{sp} = L(L+1)/2 + LE$. N_{sp} for the bubble, the box, the kite and the two-loop box

In [2]: `Nsp[L_, E_] := L (L + 1)/2 + L E;`
`{Nsp[1, 1], Nsp[1, 3], Nsp[2, 1], Nsp[2, 3]}`

Out[2]: {2, 4, 5, 9}

the kite family: $t = 5, q = 0, L(L+E) = 6$ identities per seed; seeds with $r \leq R$ produce integrals up to $r = R +$

In [3]: 1. R , number of identities, number of integrals

```
ints[R_] := Sum[Nclass[5, r, 1, 0], {r, 0, R}];
table = Table[{R, 6 ints[R], ints[R + 1]}, {R, 0, 6}];
table
```

Out[3]: $\begin{pmatrix} 0 & 6 & 6 \\ 1 & 36 & 21 \\ 2 & 126 & 56 \\ 3 & 336 & 126 \\ 4 & 756 & 252 \\ 5 & 1512 & 462 \\ 6 & 2772 & 792 \end{pmatrix}$

identities divided by integrals

In [4]: `Table[6 ints[R]/ints[R + 1], {R, 0, 6}]`

Out[4]: $\left\{1, \frac{12}{7}, \frac{9}{4}, \frac{8}{3}, 3, \frac{36}{11}, \frac{7}{2}\right\}$

the ratio for $R \rightarrow \infty$

In [5]: `Limit[6 ints[R]/ints[R + 1], R -> Infinity]`

Out[5]: 6

The same stars-and-bars thinking counts the **sectors**. A family with t denominators has $\binom{t}{k}$ subsets with k lines present, 2^t sectors in all. Many of them are zero (scaleless) or equal to each other by a relabelling of momenta, which is why the programs of Section 23 first find the unique non-zero sectors.

Laporta's order is the Hidden Leaf's mission ranking: D-rank sectors first, then C, B, A and the S-rank top sector last. Send a genin to an S-rank mission and they keep coming back for help.

RESULT: THE MASTERS OF THE TWO-LOOP PROPAGATOR FAMILY

After running through the sectors, every integral of the family is a combination of two masters:

$$F(0, 1, 1, 0, 1) = \text{Sunset Diagram}, \quad F(1, 1, 1, 1, 0) = \text{Product of two Bubbles Diagram} \quad (308)$$

The first is the massless sunset, the second the product of two one-loop bubbles. For example

$$F(2, 2, 1, 2, 2) = c_1(d, q^2) F(0, 1, 1, 0, 1) + c_2(d, q^2) F(1, 1, 1, 1, 0) \quad (309)$$

with the rational coefficients shown in the Mathematica box.

Note that the kite itself, $F(1, 1, 1, 1, 1)$, is not a master. It reduces to the two simpler ones. Its reduction is

$$\text{Kite} = \frac{2(3d-10)(3d-8)}{(d-4)^2 q^4} \text{Bubble} - \frac{2(d-3)}{(d-4)q^2} \text{TwoBubbles} \quad (310)$$

Both masters are products of Gamma functions (the product of two bubbles and a bubble inside a bubble) and the coefficients have poles at $d = 4$. When the $1/\varepsilon^2$ of the coefficient meets the ε -expansion of the Gamma functions, all poles and all rational terms cancel and exactly $6\zeta(3)/q^2$ is left. The value found with Cheng-Wu in Section 14, now obtained by pure algebra. This is checked in the last lines of the box below.

MATHEMATICA CHECK: THE NOTEBOOK IBP_TOY_LAPORTA, COMPARED WITH LITERED

Laporta by hand for $F(a_1, \dots, a_5)$ with $D_1 = \ell_1^2$, $D_2 = (\ell_1 + q)^2$, $D_3 = (\ell_1 + \ell_2)^2$, $D_4 = (\ell_1 + \ell_2 + q)^2$, $D_5 = \ell_2^2$.
check: scalar products in terms of denominators

```
In [1]: Get[FileNameJoin[{$snipDir, "..", "course_nb", "Laporta.m"}]];
verifyDens
```

```
Out[1]: {True, True, True, True, True}
```

the IBP $\partial/\partial\ell_1 \cdot \ell_1$ at $(1, 1, 1, 1, 1)$

```
In [2]: Collect[IBP[1, {1, 1, 1, 1, 1}], F[_], Simplify]
```

```
Out[2]: (d-4) F(1, 1, 1, 1, 1) + 2 q^2 F(1, 1, 1, 2, 1) - 2 F(0, 1, 2, 1, 1) - 2 F(0, 2, 1, 1, 1) + 2 F(1, 1, 1, 2, 0)
```

Laporta order: simplest sectors first, each with up to 3 extra powers. $F(2, 2, 1, 2, 2)$ in masters

```
In [3]: ReduceSector[{1, -1, -1, 1, 1}, 3, PrintInfo -> False];
ReduceSector[{1, 1, 1, 1, -1}, 3, PrintInfo -> False];
ReduceSector[{1, 1, 1, -1, 1}, 3, PrintInfo -> False];
ReduceSector[{1, 1, -1, 1, 1}, 3, PrintInfo -> False];
ReduceSector[{1, 1, 1, 1, 1}, 3, PrintInfo -> False];
red = Collect[F[2, 2, 1, 2, 2], F[_], Factor];
red
```

```
Out[3]: - 9 (d-7) (3 d-16) (3 d-14) (3 d-10) (3 d-8) (d^2-11 d+20) F(0, 1, 1, 0, 1) -
          (d-10) (d-8)^2 (d-4) q^6
          4 (d-9) (d-6) (d-5) (d-3) F(1, 1, 1, 1, 0)
          (d-8) q^5
```

the masters that appear

```
In [4]: Union[Cases[red, _F, Infinity]]
```

```
Out[4]: {F(0, 1, 1, 0, 1), F(1, 1, 1, 1, 0)}
```

Load LiteRed and solve the same family (full code in the file). masters found by LiteRed

```
In [5]: lr = IBPReduce[j[kite, 2, 2, 1, 2, 2]];
MIs[kite]
```

```
Out[5]: {j(kite, 0, 1, 1, 0, 1), j(kite, 1, 1, 1, 1, 0)}
```

rewrite the hand reduction in LiteRed's masters and subtract LiteRed's own. difference

```
In [6]: courseInLR = red /. F[a_] :> IBPReduce[j[kite, a]];
Simplify[courseInLR - lr]
```

```
Out[6]: 0
```

the kite: reduce $F(1,1,1,1,1)$ and insert the exact masters, built from the one-loop bubble $G(a,b)(q^2)^{D/2-a-b}$.
 In [7]: two bubbles: $G(1,1)^2(q^2)^{D-4}$. sunset: a bubble inside a bubble. $F(1,1,1,1,1)$ in terms of the masters

```

G[a_, b_] := Gamma[a + b - d/2] Gamma[d/2 - a] Gamma[d/2 - b]/(Gamma[a] Gamma[b]
Gamma[d - a - b]);
masterValues = {j[kite, 1, 1, 1, 1, 0] -> G[1, 1]^2 q2^(d - 4),
  j[kite, 0, 1, 1, 0, 1] -> G[1, 1] G[1, 2 - d/2] q2^(d - 3)};
kiteRed = IBPReduce[j[kite, 1, 1, 1, 1, 1]];
Collect[kiteRed, _j, Factor]

```

Out[7]:
$$\frac{2(3d-10)(3d-8)j(\text{kite}, 0, 1, 1, 0, 1)}{(d-4)^2 q2^2} - \frac{2(d-3)j(\text{kite}, 1, 1, 1, 1, 0)}{(d-4) q2}$$

its value at $D = 4 - 2\varepsilon$, times $(q^2)^{1+2\varepsilon}$

In [8]: `FullSimplify[FunctionExpand[Normal@Series[q2^(1 + 2 eps) (kiteRed /. masterValues) /. d -> 4 - 2 eps, {eps, 0, 0}]]]`

Out[8]: $6\zeta(3)$

The reduction built by hand, sector by sector and the reduction by LiteRed are identical (difference zero) and both find the same two masters. Programs like FIRE and Kira use the same idea, organised much more efficiently (sparse linear algebra, finite-field arithmetic), which is why they can handle millions of identities.

19 Master integrals

The words “master integral” have come up many times already, one for the tadpole (Section 5), two for the bubble (Section 8), two for the two-loop propagator family (Section 18). This chapter puts the idea together in one place, what a master is, why there are only a few, how to count them and what we do with them.

19.1 A one-line warm-up

Before any loop, look at an ordinary integral with the same structure,

$$I(n) = \int_0^\infty dx x^n e^{-x^2}, \quad n = 0, 1, 2, \dots \quad (311)$$

There are infinitely many of them. Integrate a total derivative, exactly as in IBP:

$$0 = \int_0^\infty dx \frac{d}{dx} (x^{n-1} e^{-x^2}) = (n-1)I(n-2) - 2I(n) \quad (n \geq 2) \quad (312)$$

The boundary terms vanish at both ends. So $I(n) = (n-1)/2 I(n-2)$ and every $I(n)$ comes down to one of two numbers:

$$I(0) = \frac{\sqrt{\pi}}{2}, \quad I(1) = \frac{1}{2} \quad (313)$$

For example $I(4) = 3/2 \cdot 1/2 \cdot I(0) = 3\sqrt{\pi}/8$ and $I(5) = 2 \cdot 1 \cdot I(1) = 1$. Even n never reach odd n , so neither of the two can be written through the other. $I(0)$ and $I(1)$ are the **master integrals** of this family. The relation never computes them. We need them from somewhere else ($I(0)$ is a Gaussian, $I(1)$ is elementary). Loop integrals work in exactly this way.

19.2 The family and its relations

A **family** is a fixed list of propagators $D_1, \dots, D_t$ together with all integrals

$$I(a_1, \dots, a_t) = \int [d\ell_1] \dots [d\ell_L] \frac{1}{D_1^{a_1} \dots D_t^{a_t}}, \quad a_i \in \mathbb{Z} \quad (314)$$

A positive a_i is a propagator with $a_i - 1$ dots. A negative a_i is a numerator. For the bubble of Section 8 the family is $J(a_1, a_2)$ with $D_1 = \ell^2 + m^2$ and $D_2 = (\ell - p)^2 + m^2$. If the loop momenta have more scalar products than there are propagators, the extra ones are added to the list as numerators that never appear downstairs (Section 18 counts them).

Three kinds of relation connect the members of a family:

- IBP identities, from the integral of a total derivative (Section 5).
- Symmetries, from a change of loop momentum that permutes the propagators, such as $J(a_1, a_2) = J(a_2, a_1)$.
- Zero integrals. A scaleless integral vanishes in dimensional regularisation, such as a massless tadpole.

All three are **linear** and their coefficients are **rational functions** of D , of the masses and of the external invariants. No logarithm, no Gamma function, no square root ever appears in them.

RESULT: WHAT A SET OF MASTER INTEGRALS IS

A set of master integrals of a family is a list of members $M_1, \dots, M_n$ such that

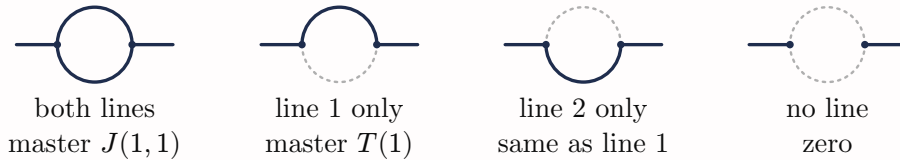
1. every member of the family can be written as $I = \sum_{i=1}^n c_i M_i$, with coefficients c_i that are rational functions of D and the kinematics
2. no such combination of the M_i is zero, so none of them can be written through the others

The number n is fixed by the family. The choice of the M_i is not.

It helps to think of linear algebra. The members of the family are vectors. The “numbers” we may multiply them by are rational functions of D , m^2 , p^2 . The relations say that these infinitely many vectors live in a space of finite dimension n and the masters are a basis of that space. The coefficients c_i are the components of an integral in this basis. IBP reduction (by hand, Laporta, LiteRed, Kira, feynsage) is nothing but finding these components.

19.3 Sectors: the masters come in layers

Which lines are present (which $a_i > 0$) is called the **sector** of an integral. $J(2, 1)$ and $J(1, 3)$ are in the top sector of the bubble, where both lines are present. $J(3, 0)$ has only line 1, so it is in a subsector. It is a tadpole. A family with t propagators has 2^t sectors. For the bubble:



The sector with line 2 only is equal to the one with line 1 by the shift $\ell \rightarrow \ell + p$, so it gives nothing new. The empty sector has no propagator at all and is zero.

The rules of the game are:

- IBP never moves an integral **up** to a higher sector. A derivative can add a dot or a numerator and a numerator can cancel a propagator, but no identity creates a propagator that was not there.
- So each sector is reduced to **its own** masters plus the masters of the sectors below it. This is why Laporta’s algorithm (Section 18) works from the bottom up.
- Every master belongs to one sector. The bubble has one master in the top sector, $J(1,1)$ and one in the tadpole sector, $T(1)$.
- A sector can have **no** master even when its integrals are not zero. The massless two-loop kite of Section 18 is the example: the kite $F(1, 1, 1, 1)$ reduces completely to the sunset and the product of two bubbles, so its top sector has no master of its own.

The examples we have met:

family	masters	n	where
tadpole $T(a)$	$T(1)$	1	Section 5
bubble, equal masses	$J(1, 1)$, $T(1)$	2	Section 8
bubble, one massless line	$F(1, 1)$, $F(1, 0)$	2	Section 18
bubble, masses $m_1 \neq m_2$	bubble, $T(m_1)$, $T(m_2)$	3	Section 14

massless box	box, s -bubble, t -bubble	3	Section 20
massless two-loop propagator	sunset, bubble $\times$ bubble	2	Section 18

With two different masses the two tadpoles are different integrals, so both survive. In the bubble with one massless line the massless tadpole is zero, so only one tadpole is left. In the massless box every tadpole and every triangle reduces away or vanishes and only the two bubbles and the box remain.

19.4 Why the number is finite and how to count it

The counting of Section 18 shows that the identities soon outnumber the integrals. That makes a finite answer plausible but it is not a proof, because most identities are dependent. The proof that every family has finitely many masters was given by Smirnov and Petukhov (2010).

There is also a way to count them without doing any reduction, found by Lee and Pomeransky (2013). Take the graph polynomials of Section 14 and form their sum (Weinzierl's book, Section 2.5.4, eq. (2.213), where the whole integral is written with this one polynomial as $\int \prod_i x_i^{\nu_i-1} dx_i G^{-D/2}$)

$$G = \mathcal{U} + \mathcal{F} \quad (315)$$

In each sector set the x_i of the missing lines to zero and count the points where all derivatives of G vanish, with every remaining x_i different from zero and G itself different from zero (the **proper** critical points). A simple point counts once. A degenerate point counts with its multiplicity, the Milnor number. In the words of Lee and Pomeransky, the number of masters in the sector is “the sum of the Milnor numbers of the proper critical points”.

Two cautions come with this rule.

- It counts masters **before** any symmetry is used. For the two-loop sunset with three equal masses it finds 4 points in the sunset sector, but the symmetries that swap the three lines leave only 2 masters there (and the three tadpole-product sectors become one). With three different masses there is no symmetry and the count, $4 + 3 = 7$, is exact.
- Points where $G = 0$ must be left out. This matters in massless sectors. In the top sector of the massless kite all derivatives of G vanish on a whole two-dimensional set, but $\mathcal{U} = \mathcal{F} = 0$ there, so $G = 0$. With those points removed nothing is left. The kite's top sector has no master, which agrees with its reduction to the two simpler masters. The general statement behind the rule is that the number of masters is the Euler characteristic of the complement of $G = 0$ (Bitoun, Bogner, Klausen and Panzer, 2019). They also note that a count of critical points can miss points at infinity.

DERIVATION: THE BUBBLE HAS TWO MASTERS

Top sector. With equal masses $\mathcal{U} = x_1 + x_2$ and $\mathcal{F} = x_1 x_2 p^2 + (x_1 + x_2)^2 m^2$, so

$$G = x_1 + x_2 + x_1 x_2 p^2 + m^2 (x_1 + x_2)^2 \quad (316)$$

The two derivatives are

$$\partial_1 G = 1 + x_2 p^2 + 2m^2 (x_1 + x_2), \quad \partial_2 G = 1 + x_1 p^2 + 2m^2 (x_1 + x_2) \quad (317)$$

Their difference is $(x_2 - x_1)p^2$, so $x_1 = x_2 = x$ and then $1 + x(p^2 + 4m^2) = 0$. There is exactly one point, $x = -1/(p^2 + 4m^2)$. One master in the top sector.

Tadpole sector. Put $x_2 = 0$: $G = x_1 + m^2 x_1^2$ and $\partial_1 G = 1 + 2m^2 x_1 = 0$ has one solution. One master. The sector of line 2 is the same tadpole and is not counted again.

In total $1 + 1 = 2$, the two masters $J(1, 1)$ and $T(1)$ found by IBP.

For the massless box ($\mathcal{U} = x_1 + \dots + x_4$ and $\mathcal{F} = s x_2 x_4 + t x_1 x_3$) the top sector gives one point. The two bubble sectors give one each. A triangle sector, for example lines 1, 2, 3, has $G = x_1 + x_2 + x_3 + t x_1 x_3$ and $\partial_2 G = 1$ never vanishes, so it has no master. The total is 3. The Mathematica box counts all these cases.

MATHEMATICA CHECK: COUNTING MASTERS WITH THE CRITICAL POINTS OF $G = U + F$

In [1]: Lee-Pomeransky count of master integrals: in each sector (the lines that are present) count the points where all derivatives of $G = U + F$ vanish, with every present x_i non-zero and G non-zero. Generic numbers for the kinematics. Sectors that are equal by a relabelling are counted once.. tadpole. tadpole: sector $\rightarrow$ number of critical points

```
crit[G_, xs_] := Length@Select[NSolve[Thread[D[G,
  {xs}] == 0], xs], (And @@ Thread[{xs /. #] != 0}) && Chop[G /. #] != 0 &];
count[U_, F_, xs_, sectors_] := Table[With[{r = Thread[Complement[xs,
  s] -> 0]}, s -> crit[(U + F) /. r, s]], {s, sectors}];
count[x1, m2 x1^2 /. m2 -> 7/3, {x1}, {{x1}}]
```

Out[1]: $\{\{x_1\} \rightarrow 1\}$

equal-mass bubble, Euclidean: $F = x_1 x_2 p^2 + (x_1 + x_2)^2 m^2$. equal-mass bubble

```
In [2]: num = {pp -> 11/5, m2 -> 7/3, mm1 -> 7/3, mm2 -> 13/4,
  s -> -11/5, t -> -17/6};
count[x1 + x2, x1 x2 pp + (x1 + x2)^2 m2 /. num, {x1, x2}, {{x1, x2},
  {x1}}]
```

Out[2]: $\{\{x_1, x_2\} \rightarrow 1, \{x_1\} \rightarrow 1\}$

unequal masses: the two tadpoles are different, so both sectors count. unequal-mass bubble

```
In [3]: count[x1 + x2, x1 x2 pp + (x1 + x2) (mm1 x1 + mm2 x2) /. num,
  {x1, x2}, {{x1, x2}, {x1}, {x2}}]
```

Out[3]: $\{\{x_1, x_2\} \rightarrow 1, \{x_1\} \rightarrow 1, \{x_2\} \rightarrow 1\}$

massless on-shell box: $U = x_1 + \dots + x_4$, $F = s x_2 x_4 + t x_1 x_3$ (cyclic labels), all non-empty sectors up to symmetry. massless box

```
In [4]: boxsec = {{x1, x2, x3, x4}, {x1, x2, x3}, {x2, x4}, {x1, x3}, {x1, x2},
  {x1}};
count[x1 + x2 + x3 + x4, s x2 x4 + t x1 x3 /. num,
  {x1, x2, x3, x4}, boxsec]
```

Out[4]: $\{\{x_1, x_2, x_3, x_4\} \rightarrow 1, \{x_1, x_2, x_3\} \rightarrow 0, \{x_2, x_4\} \rightarrow 1, \{x_1, x_3\} \rightarrow 1, \{x_1, x_2\} \rightarrow 0, \{x_1\} \rightarrow 0\}$

19.5 The choice of masters is free

Any n members that are independent will do. From Equation 134,

$$J(2, 1) = -\frac{D-3}{p^2 + 4m^2} J(1, 1) + \frac{1}{p^2 + 4m^2} T(2), \quad T(2) = -\frac{D-2}{2m^2} T(1) \quad (318)$$

so as long as $D \neq 3$ we can solve for $J(1, 1)$ and use the pair $J(2, 1)$, $T(1)$ (or $J(2, 1)$, $T(2)$) as masters instead. Both choices are correct. They differ in how convenient they are:

- **Simple coefficients.** A bad basis can give coefficients with poles in ε . The kite reduction of Section 18 has $1/(d-4)^2$ in front of the sunset. To get the kite to order ε^0 we then need the sunset two orders deeper in ε than the final answer. All those extra terms cancel at the end, so the work is wasted.
- **Quasi-finite masters.** One can choose masters whose Feynman-parameter integrals are finite, for example by adding dots and moving to $D = 6 - 2\varepsilon$ (von Manteuffel, Panzer and Schabinger, 2015). At worst a single overall $1/\varepsilon$ is left, from the Gamma function in front. Then a numerical method like sector decomposition (Section 21) has an easy job.
- **A simple differential equation.** The derivative of a master, reduced again by IBP, is a combination of masters. This gives the differential equation of Section 10. With a clever choice the equation takes the canonical form of Section 20, where ε factors out and the solution is written order by order in polylogarithms.

19.6 The whole workflow

Every multi-loop calculation in practice follows the same four steps:

1. Do the Dirac and Lorentz algebra and write the amplitude as a sum of members of a few families.

2. Reduce every member to masters by IBP. This is pure linear algebra with rational coefficients and is exact. Laporta's algorithm (Section 18) or a program like Kira (Section 23) does it.
3. Compute the masters, once. Here every method of this note is used: Gamma functions for the tadpole and the massless bubble, the differential equation (Section 10), Feynman parameters (Section 11), Mellin-Barnes (Section 16), sector decomposition (Section 21) or expansion by regions (Section 22).
4. Insert the masters into the reduction and expand in ε .

The hard analysis is all in step 3 and there it is needed only for a handful of integrals. A two-loop process can have hundreds of thousands of members but only tens or hundreds of masters.

The masters are the Infinity Stones. The family has endless integrals but every one of them is built from the same few stones. IBP is the gauntlet that brings them together and computing the stones themselves is where all the real fighting happens.

20 Differential equations in canonical form

For the bubble (Section 10) the differential equation could be solved by hand and the answer was a hypergeometric function. For harder integrals this does not scale. The modern method (Henn, 2013) is to choose the master integrals so that the equation takes a **canonical form**,

$$\frac{d\vec{g}}{dx} = \varepsilon A(x) \vec{g}, \quad A(x) = \sum_k \frac{a_k}{x - x_k} \quad (319)$$

where ε appears only as an overall factor and A has only simple poles with constant matrices a_k . Then the solution can be written down order by order in ε with no further thinking. The course notebook DE_Box does this for the massless box. Here the reduction is done with LiteRed instead of FIRE5 and everything is checked against Section 17.

A canonical basis is the Philosopher's Stone of differential equations: once you have it, every order in ε is transmuted from the previous one by a single integration. And like the Stone, finding it is the hard part.

20.1 The masters and the naive equation

The box family of Section 17 has three masters, the t -channel bubble, the s -channel bubble and the box itself,

$$F(0, 1, 0, 1) = \text{bubble}_t, \quad F(1, 0, 1, 0) = \text{bubble}_s, \quad F(1, 1, 1, 1) = \text{box} \quad (320)$$

Multiplying by powers of $(-s)$ makes them dimensionless, so they depend only on $x = t/s$:

$$f_1 = (-s)^\varepsilon F(0, 1, 0, 1), \quad f_2 = (-s)^\varepsilon F(1, 0, 1, 0), \quad f_3 = (-s)^{2+\varepsilon} F(1, 1, 1, 1) \quad (321)$$

DERIVATION: HOW TO DIFFERENTIATE WITH RESPECT TO x

As for the bubble, we cannot differentiate under the integral with respect to a scalar like t directly, but we can differentiate with respect to an external **vector**. The notebook builds the operator

$$\partial_x = (c_1 p_1^\mu + c_2 p_2^\mu + c_3 p_3^\mu) \frac{\partial}{\partial p_1^\mu} \quad (322)$$

and fixes the three numbers by demanding that it acts on $x = t/s$ as $\partial_x x = 1$ and it keeps the on-shell conditions, $\partial_x p_1^2 = 0$ and $\partial_x p_4^2 = 0$. The solution is

$$c_1 = \frac{1-x}{2x(1+x)}, \quad c_2 = -\frac{1}{2(1+x)}, \quad c_3 = -\frac{1}{2x(1+x)} \quad (323)$$

Acting on the integrands produces integrals with dots and numerators and IBP reduction brings them back to the three masters. (LiteRed does the same thing with its `Dinv` function, which differentiates with respect to an invariant at fixed others. With $\partial_x = s \partial_t$ at fixed s we get the same equation.) Why is that the same? The vector operator does change s : acting on $s = (p_1 + p_2)^2$ it gives $-s/(1+x)$. But each f_i is

dimensionless and depends on the momenta only through x , so the operator acting on f_i is just (df_i/dx) times its action on x , which is 1.

The result is $\partial_x \vec{f} = A(\varepsilon, x) \vec{f}$ with

$$A = \begin{pmatrix} -\varepsilon/x & 0 & 0 \\ 0 & 0 & 0 \\ \frac{2-4\varepsilon}{x^2(1+x)} & -\frac{2-4\varepsilon}{x(1+x)} & -\frac{1+\varepsilon+x}{x(1+x)} \end{pmatrix} \quad (324)$$

The first two rows can be understood without any IBP. The t -channel bubble depends only on t and by dimensions $F(0, 1, 0, 1) = c(\varepsilon)(-t)^{-\varepsilon}$. Then $f_1 = (-s)^\varepsilon c(-t)^{-\varepsilon} = c x^{-\varepsilon}$, so $df_1/dx = -\varepsilon x^{-\varepsilon-1} c = -(\varepsilon/x)f_1$: the entry $A_{11} = -\varepsilon/x$. The s -channel bubble depends only on s , so f_2 is a constant and its row is zero. Only the third row needs the IBP reduction of the differentiated box.

It is not canonical: ε is mixed with numbers and there is a double pole $1/x^2$.

20.2 Finding the canonical basis

A change of basis $\vec{f} = T\vec{g}$ changes the matrix as

$$A \rightarrow A' = T^{-1}(AT - \partial_x T) \quad (325)$$

(Insert $\vec{f} = T\vec{g}$ into $\partial_x \vec{f} = A\vec{f}$: $T\partial_x \vec{g} + (\partial_x T)\vec{g} = AT\vec{g}$.)

DERIVATION: THREE SMALL STEPS TO THE CANONICAL FORM, ENTRY BY ENTRY

For a diagonal $T = \text{diag}(t_1, t_2, t_3)$ the rule $A' = T^{-1}(AT - \partial_x T)$ reads, entry by entry,

$$A'_{ij} = \frac{t_j}{t_i} A_{ij} - \delta_{ij} \frac{t'_i}{t_i} \quad (326)$$

Only the third row of A is complicated, so follow it.

Step 1: remove the double pole. Take $t_3 = 1/x$ (and $t_1 = t_2 = 1$), so the new box integral is $stF(1, 1, 1, 1)$ instead of $s^2F(1, 1, 1, 1)$. Then $t_{3'}/t_3 = -1/x$ and

$$A'_{31} = xA_{31} = \frac{2-4\varepsilon}{x(1+x)}, \quad A'_{32} = xA_{32} = -\frac{2-4\varepsilon}{1+x} \quad (327)$$

$$A'_{33} = A_{33} + \frac{1}{x} = \frac{-(1+\varepsilon+x) + (1+x)}{x(1+x)} = -\frac{\varepsilon}{x(1+x)} \quad (328)$$

The $1/x^2$ is gone and the diagonal entry is already proportional to ε .

Step 2: make the box row proportional to ε . Take $t_3 = 1/(x\varepsilon)$. The extra $1/\varepsilon$ is a constant, so the diagonal entry does not change and the two off-diagonal entries get multiplied by ε :

$$A'_{31} = \frac{\varepsilon(2-4\varepsilon)}{x(1+x)}, \quad A'_{32} = -\frac{\varepsilon(2-4\varepsilon)}{1+x} \quad (329)$$

Step 3: remove the $(1-2\varepsilon)$. Take $t_1 = t_2 = -1/(1-2\varepsilon)$. Then A'_{31} and A'_{32} get multiplied by t_1/t_3 , which contains $-1/(1-2\varepsilon)$ and since $2-4\varepsilon = 2(1-2\varepsilon)$:

$$A'_{31} = -\frac{2\varepsilon}{x(1+x)} = \varepsilon \left(-\frac{2}{x} + \frac{2}{1+x} \right), \quad A'_{32} = \frac{2\varepsilon}{1+x} \quad (330)$$

(Partial fractions: $1/(x(1+x)) = 1/x - 1/(1+x)$.) The diagonal entry is $-\varepsilon/(x(1+x)) = \varepsilon(-1/x + 1/(1+x))$ and the first row stays $-\varepsilon/x$.

A constant normalisation. Finally multiply all three basis integrals by the constant $\varepsilon e^{\varepsilon\gamma_E}$. A constant factor does not change the equation at all (it cancels between the two sides), but it makes the expansion of every g_i start at ε^0 and removes all γ_E 's. This is the normalisation of the course notebook and the one used for g_1, g_2 and $g_3 = \varepsilon^2 e^{\varepsilon\gamma_E} (-s)^\varepsilon stF(1, 1, 1, 1)$ below.

Read off a and b . Every entry is now ε times (number)/ x plus (number)/($1+x$):

$$\frac{d\vec{g}}{dx} = \varepsilon \left(\frac{a}{x} + \frac{b}{1+x} \right) \vec{g}, \quad a = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & 2 & 1 \end{pmatrix} \quad (331)$$

The two **letters** are x and $1+x$. They tell us where the integral can be singular: $x=0$ ($t=0$), $x=-1$ ($u=-s-t=0$) and $x=\infty$ ($s=0$). These are the physical thresholds of the three channels.

MATHEMATICA CHECK: THE NOTEBOOK DE_BOX, WITH LITERED DOING THE REDUCTION

Load LiteRed with the box basis $\{\ell^2, (\ell+p_1)^2, (\ell+p_1+p_2)^2, (\ell+p_3)^2\}$ and solve all IBP relations (full code in the file). master integrals

In [1]:

```
MIs[box]
```

Out[1]: {j(box, 0, 1, 0, 1), j(box, 1, 0, 1, 0), j(box, 1, 1, 1, 1)}

naive basis $\vec{f} = ((-s)^\varepsilon F_{0101}, (-s)^\varepsilon F_{1010}, (-s)^{2+\varepsilon} F_{1111})$. $\partial/\partial t$ at fixed s . $d/dx = s \partial/\partial t$. A in $d\vec{f}/dx = A\vec{f}$

In [2]:

```
m = {j[box, 0, 1, 0, 1], j[box, 1, 0, 1, 0], j[box, 1, 1, 1, 1]};
dt = IBPReduce[Dinv[#, t]] & /@ m;
toF = Thread[m -> {f1 (-s)^-eps, f2 (-s)^-eps, f3 (-s)^(-2 - eps)}];
eqs = {(-s)^eps, (-s)^eps,
  (-s)^(2 + eps)} s (dt /. toF /. d -> 4 - 2 eps);
Amat = Table[Coefficient[Expand[Simplify[eqs[[i]] /. t -> x s]],
  {f1, f2, f3}][[k]], {i, 3}, {k, 3}] // Simplify;
Amat
```

Out[2]:

$$\begin{pmatrix} -\frac{\varepsilon}{x} & 0 & 0 \\ 0 & 0 & 0 \\ \frac{2-4\varepsilon}{x^3+x^2} & -\frac{2-4\varepsilon}{x^2+x} & -\frac{x+\varepsilon+1}{x^2+x} \end{pmatrix}$$

basis change $\vec{f} = T\vec{g}$ towards the canonical form. the new matrix, divided by ε

In [3]:

```
T = {{-1/(1 - 2 eps), 0, 0}, {0, -1/(1 - 2 eps), 0}, {0, 0, 1/(x eps)}};
Ap = Apart[Simplify[Inverse[T] . Amat . T - Inverse[T] . D[T, x]], x];
Simplify[Ap/eps]
```

Out[3]:

$$\begin{pmatrix} -\frac{1}{x} & 0 & 0 \\ 0 & 0 & 0 \\ -\frac{2}{x^2+x} & \frac{2}{x+1} & -\frac{1}{x^2+x} \end{pmatrix}$$

a (coefficient of ε/x) and b (coefficient of $\varepsilon/(1+x)$)

In [4]:

```
amat = Simplify[Limit[x Ap/eps, x -> 0]];
bmat = Simplify[Limit[(1 + x) Ap/eps, x -> -1]];
{amat, bmat}
```

Out[4]:

$$\begin{pmatrix} \{-1, 0, 0\} & \{0, 0, 0\} & \{-2, 0, -1\} \\ \{0, 0, 0\} & \{0, 0, 0\} & \{2, 2, 1\} \end{pmatrix}$$

check $d\vec{g}/dx = \varepsilon(a/x + b/(1+x))\vec{g}$

In [5]:

```
Simplify[Ap - eps (amat/x + bmat/(1 + x))]
```

Out[5]:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

solve order by order: $\vec{g}^{(n)}(x) = \int (a/x' + b/(1+x')) \vec{g}^{(n-1)} + \text{boundary value. bubbles, from Feynman}$

In [6]:

parameters. boundary condition from the missing u -channel cut: $g_3(-1) = -2(g_1(-1) + g_2(-1))$. $g_3(-1)$, orders $\varepsilon^0, \dots, \varepsilon^3$

```
g1 = -x^(-eps) Exp[eps EulerGamma] Gamma[1 - eps]^2 Gamma[1 + eps]/Gamma[1 - 2 eps];
g2 = -Exp[eps EulerGamma] Gamma[1 - eps]^2 Gamma[1 + eps]/Gamma[1 - 2 eps];
gE = Normal[Series[{g1, g2}, {eps, 0, 3}]];
```

```
g3bc = Series[-2 g1 - 2 g2, {eps, 0, 3}] /. Log[x] -> I Pi;
Normal[g3bc]
```

Out[6]: $\frac{1}{6} \epsilon^3 (28 \psi^{(2)}(1) + 3 i \pi^3) - \frac{4 \pi^2 \epsilon^2}{3} - 2 i \pi \epsilon + 4$

integrate $g_3 = \epsilon[(-2g_1 - g_3)/x + (2g_1 + 2g_2 + g_3)/(1+x)]$ order by order. g_3 at orders $\epsilon^0, \dots, \epsilon^3$

```
In [7]: ord[f_, k_] := Coefficient[f, eps, k];
g3 = 0;
sol = {};
Do[Module[{rhs, prim, c},
  rhs = If[k == 0, 0,
    (-2 ord[gE[[1]], k - 1] - sol[[k]])/x + (2 ord[gE[[1]],
      k - 1] + 2 ord[gE[[2]], k - 1] + sol[[k]])/(1 + x)];
  prim = If[k == 0, 0, Integrate[Expand[rhs], x]];
  c = ord[Normal[g3bc], k] - Limit[prim /. Log[x] -> Log[-x] + I Pi,
    x -> -1, Direction -> "FromAbove"];
  AppendTo[sol, FullSimplify[prim + c]], {k, 0, 3}];
sol
```

Out[7]: $\left\{4, -2 \log(x), -\frac{4 \pi^2}{3}, \frac{1}{6} (12 \text{Li}_3(-x) - 12 \text{Li}_2(-x) \log(x) + 2 \log^3(x) - 6 (\log^2(x) + \pi^2) \log(x+1) + 7 \pi^2 \log(x) - 68 \zeta(3))\right\}$

minus the course result

```
In [8]: known = {4, -2 Log[x],
  -4 Pi^2/3, 7 Pi^2/6 Log[x] + Log[x]^3/3 - (Pi^2 + Log[x]^2) Log[1 + x] - 2 Log[x]
  PolyLog[2,
    -x] + 2 PolyLog[3, -x] - 34 Zeta[3]/3};
FullSimplify[sol - known, x > 0]
```

Out[8]: {0, 0, 0, 0}

The matrices a and b come out as in Rauh's notes (his eq. 6.22) and the check line confirms the canonical form.

20.3 Solving it: iterated integrals

Expand $\vec{g} = \sum_n \epsilon^n \vec{g}^{(n)}$. The canonical equation gives, order by order,

$$\vec{g}^{(n)}(x) = \int_{x_0}^x dx' \left(\frac{a}{x'} + \frac{b}{1+x'} \right) \vec{g}^{(n-1)}(x') + \vec{g}^{(n)}(x_0) \quad (332)$$

where x_0 is a point where the boundary value is known. For the box, $x = -1$ is itself a singular point of the equation, so in practice one integrates the right-hand side (an indefinite integral) and fixes the constant by the **limit** $x \rightarrow -1$, where the boundary condition below is known. This is what the Mathematica box does. Each order is one integration of the previous one against dx/x or $dx/(1+x)$. Such repeated integrals define the **Goncharov polylogarithms**,

$$G(a_1, a_2, \dots, a_n; x) = \int_0^x \frac{dt}{t - a_1} G(a_2, \dots, a_n; t), \quad G(; x) = 1, \quad G(0, \dots, 0; x) = \frac{1}{n!} \ln^n x \quad (333)$$

which contain the logarithms, dilogarithms and trilogarithms we have met. The number n of integrations is called the **weight**. At order ϵ^n the box has weight n ($\ln$, Li_2 , Li_3 , $\zeta(3)$ all have the weight their subscript or power says).

DERIVATION: THE BOUNDARY CONDITIONS

The first two basis integrals are one-scale bubbles. The differential equation says nothing new about them ($g_1 \propto x^{-\epsilon}$, $g_2 = \text{constant}$) and their values come from Feynman parameters:

$$g_1 = -x^{-\varepsilon} e^{\varepsilon \gamma_E} \frac{\Gamma(1-\varepsilon)^2 \Gamma(1+\varepsilon)}{\Gamma(1-2\varepsilon)}, \quad g_2 = -e^{\varepsilon \gamma_E} \frac{\Gamma(1-\varepsilon)^2 \Gamma(1+\varepsilon)}{\Gamma(1-2\varepsilon)} \quad (334)$$

For the box, the equation itself supplies the boundary condition. Near $x = -1$ ($u \rightarrow 0$) the solution behaves as $(1+x)^{\varepsilon b} \tilde{g}(-1)$. Since $b^2 = b$, this is $[1 + b((1+x)^\varepsilon - 1)]\tilde{g}(-1)$, which develops a $\ln(1+x)$ and hence an imaginary part for $x < -1$ unless $b\tilde{g}(-1) = 0$. But an imaginary part means a cut in the u -channel and the massless box has none. Cutting two of its lines always separates either $p_1 + p_2$ (the s -channel) or $p_1 - p_3$ (the t -channel), never $p_2 - p_3$. Hence $b\tilde{g}(-1) = 0$, that is

$$g_3(-1) = -2(g_1(-1) + g_2(-1)) \quad (335)$$

to all orders in ε . Physics (unitarity, which cuts exist) fixes the constant that mathematics left open.

With these boundary values, four integrations give g_3 through ε^3 :

$$g_3 = 4 - 2\varepsilon \ln x - \frac{4\pi^2}{3}\varepsilon^2 + \varepsilon^3 \left[\frac{7\pi^2}{6} \ln x + \frac{1}{3} \ln^3 x - (\pi^2 + \ln^2 x) \ln(1+x) - 2 \ln x \text{Li}_2(-x) + 2 \text{Li}_3(-x) - \frac{34}{3} \zeta(3) \right] + O(\varepsilon^4) \quad (336)$$

Since $g_3 = \varepsilon^2 e^{\varepsilon \gamma_E} (-s)^\varepsilon st F(1, 1, 1, 1)$, this is Equation 293 multiplied by ε^2 . The Mathematica box above does the four integrations and finds zero difference with the course result and the Mellin-Barnes box of Section 17 gives the same function.

PHYSICAL PICTURE: WHY CANONICAL FORMS HELP

In canonical form every order in ε has a definite weight and the whole answer is fixed by a finite list of letters (here x and $1+x$) plus boundary values fixed by physics. Most of the multi-loop results used at the LHC today are computed this way. The hard creative step is to find the canonical basis. Programs such as epsilon, Fuchsia, CANONICA and INITIAL help with that.

20.4 Outlook: a two-loop problem with a mass (Bhabha scattering)

The course notebook DE_Bhabha applies the same method to a real two-loop family with a massive electron line (from Henn and Smirnov, arXiv:1307.4083). Two tricks appear that the box did not need:

- Variables that remove square roots. The thresholds bring square roots like $\sqrt{-s(4m^2 - s)}$. Writing $-s/m^2 = (1-x)^2/x$ and $-t/m^2 = (1-y)^2/y$ makes them rational: $\sqrt{-s(4m^2 - s)} = m^2(1-x)(1+x)/x$.
- Two variables and a larger alphabet. The canonical matrix is $d\tilde{A} = \sum_k a_k d \ln L_k$ with seven letters,

$$\{x, 1+x, y, 1+y, 1-y, x+y, 1+xy\} \quad (337)$$

and the solution is built from Goncharov polylogarithms in x and y , integrating first along $y = 0$ from $(0, 0)$ to $(x, 0)$ and then up to (x, y) .

There are five basis integrals (two tadpole- and bubble-type ones with known closed forms, one hypergeometric one and two genuinely two-scale ones). The notebook fixes the boundary values at $x = y = 1$ (where $s = t = 0$) and integrates order by order.

MATHEMATICA CHECK: THE NOTEBOOK DE_BHABHA, THROUGH SECOND ORDER

Basis integrals $f_1, \dots, f_5$, boundary values at $x = y = 1$, the seven letters with their matrices $a_1, \dots, a_7$ and the

In [1]: solver (full code in the file). order ε : the five basis integrals

```
fVec1 /. GPtoLogLi
```

Out[1]: {0, log(y) - 2 log(1 - y), -log(x), 0, 4 log(x)}

check of the boundary values. order ε : boundary values at $x = y = 1$

In [2]: Simplify[Coefficient[fVecBC11exp, eps] - fVec1 /. {x->1, y->1} /. GPSpecialValues]

Out[2]: {0, 0, 0, 0, 0}

rewrite in $\ln$ and Li_n . Checks. order ε : f_1, f_2, f_3 against their exact forms, f_5 against arXiv:1307.4083

```
In [3]: fVec1Alt = fVec1 /. GPtoLogLi;
{Simplify[fVec1Alt[[1]] - Coefficient[f1exp,
eps]], Simplify[fVec1Alt[[2]] - Coefficient[f2exp,
eps]], FullSimplify[fVec1Alt[[3]] - Coefficient[Series[f3,
{eps, 0, 1}], eps], Assumptions -> 0 < x < 1], Simplify[Coefficient[f5HS,
eps, 1] - fVec1[[5]]]}
```

Out[3]: {0, 0, 0, 0}

result at order ε^2 . check of the boundary values. order ε^2 : boundary values at $x = y = 1$

```
In [4]: fVec2 = fsol[2, fVec1] /. {fVec2BC[1, 0, 0] -> \[Pi]^2/12, fVec2BC[2,
0, 0] -> -\[Pi]^2/12, fVec2BC[3, 0, 0] -> \[Pi]^2/6, fVec2BC[4,
0, 0] -> -4*\[Pi]^2/3, fVec2BC[5, 0, 0] -> 0};
Simplify[Coefficient[fVec2BC1exp, eps, 2] - fVec2 /. {x -> 1,
y -> 1} /. GPSpecialValues]
```

Out[4]: {0, 0, 0, 0, 0}

rewrite in $\ln$ and Li_n . Checks. order ε^2 : f_1, f_2 exact and f_5 against arXiv:1307.4083

```
In [5]: fVec2Alt = Simplify[fVec2 /. GPtoLogLi;
{Simplify[fVec2Alt[[1]] - Coefficient[f1exp,
eps, 2]], Simplify[fVec2Alt[[2]] - Coefficient[f2exp,
eps, 2]], Simplify[Coefficient[f5HS, eps, 2] - fVec2[[5]]]}
```

Out[5]: {0, 0, 0}

Every check is zero. The boundary values are reproduced at orders ε and ε^2 , the first three basis integrals agree with their exact forms and the fifth (the two-loop integral) agrees with the published result of 1307.4083.

All Might needed a successor who could carry One For All. The canonical basis is the successor of the naive basis: same integrals, rearranged, but now strong enough to carry every order of ε .

21 Sector decomposition

Every method so far needed some luck, a good change of variables, a closed Mellin-Barnes sum, a canonical basis. Sector decomposition needs none. It is a purely mechanical procedure that takes **any** Feynman-parameter integral, pulls out all its poles in ε explicitly and leaves finite integrals behind. In practice those are done numerically (pySecDec, FIESTA), but for the examples of the course notebook `SectorDecomposition` they can be done exactly and that is what we do here.

Sector decomposition is Kage Bunshin no Jutsu: one hard integral becomes many shadow clones, each weak enough to defeat on its own. The price is the same as Naruto's: you have to deal with every single clone.

21.1 The problem: overlapping singularities

After Feynman parametrisation (Section 14) the integrand is $\mathcal{U}^{N_\nu - (L+1)D/2} \mathcal{F}^{LD/2 - N_\nu}$. When $\mathcal{F}$ (or $\mathcal{U}$) sits in a denominator, the integrand blows up where $\mathcal{F} = 0$. For Euclidean kinematics that only happens on the **boundary**, when some $x_i \rightarrow 0$. The trouble is that several x_i can go to zero at once and the singularities **overlap**.

The massless triangle of Section 14 shows this: $\mathcal{F} = -sx_1x_3$ vanishes when $x_1 \rightarrow 0$ **or** $x_3 \rightarrow 0$. Near the corner where both vanish we cannot simply expand in ε and we cannot subtract the singularity in one variable at a time either. The idea of sector decomposition is to split the integration region so that in each piece only **one** variable at a time makes things singular, in the simple factorised form $t^{-1-k\varepsilon} \times (\text{something finite})$.

21.2 Primary sectors

DERIVATION, STEP BY STEP

Split the simplex $\sum x_i = 1$ into N regions, one for each i , where x_i is the **largest** parameter. In sector i put

$$x_j = x_i t_j \quad (j \neq i), \quad t_j \in [0, 1] \quad (338)$$

The Jacobian is x_i^{N-1} . Since $\mathcal{U}$ is homogeneous of degree L and $\mathcal{F}$ of degree $L+1$ (Section 14.2),

$$\mathcal{U} = x_i^L \mathcal{U}_i(t), \quad \mathcal{F} = x_i^{L+1} \mathcal{F}_i(t) \quad (339)$$

where $\mathcal{U}_i$ and $\mathcal{F}_i$ are $\mathcal{U}$ and $\mathcal{F}$ with $x_i = 1$ and $x_j = t_j$. Counting all powers of x_i , they add up to x_i^{-1} , which together with the delta function integrates to 1. What is left in each primary sector is an integral over the unit cube $[0, 1]^{N-1}$:

$$\int_0^1 \prod_{j \neq i} dt_j t_j^{\nu_j-1} \mathcal{U}_i^{N-(L+1)D/2} \mathcal{F}_i^{LD/2-N_\nu} \quad (340)$$

(The factors $t_j^{\nu_j-1}$ come from $x_j^{\nu_j-1}$ in the formula. For unit powers, $\nu_j = 1$, they are absent.)

21.3 Iterated sectors

After the primary split, a polynomial may still vanish at a corner where **several** t 's are zero at once. The smallest example shows what goes wrong and how to fix it.

WORKED EXAMPLE: WHY ONE MORE SPLIT IS NEEDED

Take the toy integral

$$K = \int_0^1 dt_1 \int_0^1 dt_2 (t_1 + t_2)^{-2-\varepsilon} \quad (341)$$

The denominator vanishes only at the corner $t_1 = t_2 = 0$, not when just one of them is zero. So neither t_1 nor t_2 alone carries the singularity. It is “overlapping” and we cannot pull out a factor t_1 .

Split the square along its diagonal. In the half where $t_1 > t_2$, write $t_2 = t_1 t'_2$ with $t'_2 \in [0, 1]$ (so $dt_2 = t_1 dt'_2$). Then $t_1 + t_2 = t_1(1 + t'_2)$ and

$$K_1 = \int_0^1 dt_1 \int_0^1 dt'_2 t_1 \cdot t_1^{-2-\varepsilon} (1 + t'_2)^{-2-\varepsilon} = \int_0^1 t_1^{-1-\varepsilon} dt_1 \int_0^1 (1 + t'_2)^{-2-\varepsilon} dt'_2 \quad (342)$$

Now the singularity sits in one variable, $t_1^{-1-\varepsilon}$ and the other factor $(1 + t'_2)$ never vanishes (it has a constant term). The other half, $t_2 > t_1$, is the mirror image.

Result. Each half is a product of two one-dimensional integrals: $\int_0^1 t_1^{-1-\varepsilon} dt_1 = -1/\varepsilon$ and $\int_0^1 (1 + t)^{-2} dt = 1/2$ at leading order. So $K = 2 \times (-1/\varepsilon)(1/2) + O(\varepsilon^0) = -1/\varepsilon + \dots$

THE GENERAL RULE

Find the set S . In a sector, look for the **smallest** set of variables $S = \{t_{k_1}, \dots, t_{k_r}\}$ such that $\mathcal{U}$ or $\mathcal{F}$ vanishes when all of them are set to zero (the notebook's `FindSets`). In the toy example $S = \{t_1, t_2\}$. If there is no such set, both polynomials have a constant term and the sector is finished.

Split the cube. Split the r -dimensional cube of the variables in S into r pieces, one for each t_{k_l} that is the **largest** of them. In piece l rescale the others, $t_{k_m} \rightarrow t_{k_l} t_{k_m}$ for $m \neq l$. Each rescaling brings a factor t_{k_l} into the measure, so the Jacobian is $t_{k_l}^{r-1}$.

Factor out. Now the polynomial vanishes when $t_{k_l} = 0$, so it contains a factor t_{k_l} : pull it out, together with its power (the exponent of $\mathcal{U}$ or $\mathcal{F}$).

Repeat until every sector is finished. At the end each sector looks like

$$\int_0^1 \prod_j dt_j t_j^{A_j+B_j\epsilon} \cdot \mathcal{U}_{\text{sec}}^{N_\nu-(L+1)D/2} \mathcal{F}_{\text{sec}}^{LD/2-N_\nu} \quad (343)$$

where $\mathcal{U}_{\text{sec}}$ and $\mathcal{F}_{\text{sec}}$ are what is left of the two polynomials after all the factors of t have been pulled out. They have a constant term, so they never vanish in the cube. All the singular behaviour is in the explicit powers $t_j^{A_j+B_j\epsilon}$, where A_j and B_j are numbers collected from the Jacobians and the pulled-out factors.

21.4 Extracting the poles: the subtraction trick again

When $A_j = -1$ the t_j integral diverges logarithmically at $t_j = 0$ and gives a pole. This is the situation of Section 12 and the same subtraction works. With $g(t)$ the smooth rest of the integrand,

$$\int_0^1 dt t^{-1+B\epsilon} g(t) = \int_0^1 dt t^{-1+B\epsilon} [g(t) - g(0)] + \frac{g(0)}{B\epsilon} \quad (344)$$

The pole is now explicit and the first integral is finite and can be expanded in ϵ . (In the language of distributions, $t^{-1+B\epsilon} = \delta(t)/(B\epsilon) + \sum_n (B\epsilon)^n/n! [\ln^n t/t]_+$.) The notebook's `IntegrateSectors` applies this to every singular variable in every sector. Here its last step does the finite integrals with `Integrate` instead of numerically.

21.5 Example 1: the massless triangle, step by step

DERIVATION: FROM THE PARAMETRIC INTEGRAL TO THE POLES

The integral. For the massless on-shell triangle (Minkowski, $N = N_\nu = 3$, $L = 1$) $\mathcal{U}$ has the exponent $N_\nu - (L+1)D/2 = 3 - (4 - 2\epsilon) = -1 + 2\epsilon$ and $\mathcal{F}$ the exponent $LD/2 - N_\nu = -1 - \epsilon$, with $(-1)^3 = -1$:

$$F_\Delta = -\Gamma(1+\epsilon) \int dx_1 dx_2 dx_3 \delta(1-x_1-x_2-x_3) (x_1+x_2+x_3)^{-1+2\epsilon} (-sx_1x_3)^{-1-\epsilon} \quad (345)$$

The integrand blows up when $x_1 \rightarrow 0$ and when $x_3 \rightarrow 0$: two overlapping singularities.

Primary sector 1 (x_1 largest). Put $x_2 = x_1 t_2$, $x_3 = x_1 t_3$ with $t_2, t_3 \in [0, 1]$, so that $dx_2 dx_3 = x_1^2 dt_2 dt_3$. Then $\mathcal{U} = x_1(1+t_2+t_3)$ and $\mathcal{F} = -sx_1^2 t_3$. The powers of x_1 :

$$x_1^2 \cdot x_1^{-1+2\epsilon} \cdot (x_1^2)^{-1-\epsilon} = x_1^{-1} \quad (346)$$

as promised in general and the x_1 integral with the delta function gives 1. What is left:

$$S_1 = (-s)^{-1-\epsilon} \int_0^1 dt_2 \int_0^1 dt_3 t_3^{-1-\epsilon} (1+t_2+t_3)^{-1+2\epsilon} \quad (347)$$

The only singular variable is t_3 and it is already factorised. This is the sector labelled 13 in the box below.

Primary sector 3 (x_3 largest) is the mirror image, with $t_1^{-1-\epsilon}$ (sector 31).

Primary sector 2 (x_2 largest). Now $x_1 = x_2 t_1$, $x_3 = x_2 t_3$ and $\mathcal{F} = -sx_2^2 t_1 t_3$, so

$$S_2 = (-s)^{-1-\epsilon} \int_0^1 dt_1 \int_0^1 dt_3 (t_1 t_3)^{-1-\epsilon} (1+t_1+t_3)^{-1+2\epsilon} \quad (348)$$

Both singular factors are already separated, each in its own variable (sector 213).

Extract the poles. Take S_1 and call $h(t_3) = \int_0^1 dt_2 (1+t_2+t_3)^{-1+2\epsilon}$, a smooth function. The subtraction trick gives

$$\int_0^1 dt_3 t_3^{-1-\epsilon} h(t_3) = -\frac{h(0)}{\epsilon} + \int_0^1 dt_3 t_3^{-1-\epsilon} [h(t_3) - h(0)] \quad (349)$$

The first term is the pole. The second is finite and may be expanded in ϵ . In S_2 the same is done twice, once in t_1 and once in t_3 , which produces the double pole:

$$\int_0^1 \int_0^1 (t_1 t_3)^{-1-\varepsilon} H(t_1, t_3) = \frac{H(0,0)}{\varepsilon^2} - \frac{1}{\varepsilon} \int_0^1 \frac{H(t_1,0) - H(0,0)}{t_1^{1+\varepsilon}} dt_1 - \frac{1}{\varepsilon} \int_0^1 \frac{H(0,t_3) - H(0,0)}{t_3^{1+\varepsilon}} dt_3 + (\text{finite}) \quad (350)$$

with $H(0,0) = 1$. So the $1/\varepsilon^2$ of the triangle comes entirely from sector 2, the corner where both x_1 and x_3 are small compared with x_2 : the soft region, where the loop momentum is small and both lines next to the massless legs go on shell.

The single poles also check out by hand: $h(0) = \int_0^1 (1+t_2)^{-1} dt_2 + O(\varepsilon) = \ln 2 + O(\varepsilon)$, so sectors 1 and 3 together give $-2\ln 2/\varepsilon$. In sector 2, $H(t_1,0) - H(0,0) = (1+t_1)^{-1} - 1 = -t_1/(1+t_1)$, so each of its two single-pole terms is $+(1/\varepsilon) \int_0^1 dt/(1+t) = \ln 2/\varepsilon$. The $\ln 2/\varepsilon$ pieces cancel and only the $1/\varepsilon^2$ is left.

Sum. Adding the three sectors (the box below does every remaining integral exactly) at $s = -1$ and multiplying by $-\Gamma(1+\varepsilon)e^{\varepsilon\gamma_E}$:

$$-\frac{1}{\varepsilon^2} + \frac{\pi^2}{12} + O(\varepsilon) \quad (351)$$

This is minus the triangle of Section 14 at $s = -1$ (the Minkowski sign). Mathematica leaves the finite part as a combination of $\ln^2 2$, $\text{Li}_2(-2)$ and $\text{Li}_2(-1/2)$. The dilogarithm inversion formula $\text{Li}_2(-z) + \text{Li}_2(-1/z) = -\pi^2/6 - \frac{1}{2} \ln^2 z$ turns it into $\pi^2/12$.

21.6 Example 2: the massless box at the symmetric point

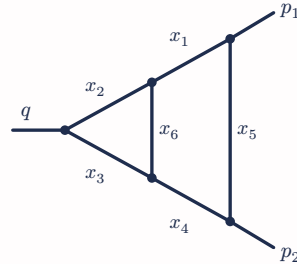
$\mathcal{F} = -sx_1x_3 - tx_2x_4$ decomposes into 12 sectors. At $s = t = -1$ their sum is

$$e^{\varepsilon\gamma_E} F(1,1,1,1) = \frac{4}{\varepsilon^2} - \frac{4\pi^2}{3} + O(\varepsilon) \quad (352)$$

Equation 293 at $x = 1$. So the massless box has now been computed four independent ways, with Mellin-Barnes (Section 17), a canonical differential equation (Section 20), sector decomposition and Package-X.

21.7 Example 3: the leading pole of a two-loop vertex

The planar two-loop vertex with two on-shell massless legs,



has six propagators. Line i carries the Feynman parameter x_i as in the picture. These labels follow the propagator order of the course notebook, not Smirnov's Fig. 5.3 (2006). His lines 1 and 2 (from the q vertex), 3 and 4 (outer) and 5 and 6 (right and middle rung) are our $x_2, x_3, x_1, x_4, x_5, x_6$. The two lines from the q vertex are x_2 (upper) and x_3 (lower), the two outer lines are x_1 and x_4 , the rung on the right is x_5 and the middle rung is x_6 . The rules of Section 14 give

- spanning trees: remove two lines so that both loops are open. For example removing x_2 and x_6 leaves a tree and gives the term x_2x_6 in $\mathcal{U}$
- 2-forests that carry momentum: remove three lines so that the q vertex sits in one tree and both p_1, p_2 vertices in the other. For example removing x_1, x_3, x_4 gives $x_1x_3x_4q^2$. A forest that cuts off only the p_1 or the p_2 vertex gives $p_1^2 = 0$ or $p_2^2 = 0$ and drops out

Collecting all of them,

$$\begin{aligned} \mathcal{U} &= (x_2 + x_3)(x_1 + x_4 + x_5) + (x_1 + \dots + x_5)x_6, \\ \mathcal{F} &= -q^2[x_1x_3x_4 + x_1x_2(x_3 + x_4) + x_2x_3(x_4 + x_5) + (x_1 + x_2)(x_3 + x_4)x_6] \end{aligned} \quad (353)$$

Its worst pole is $1/\varepsilon^4$: two loops, each with a soft and a collinear divergence (Section 4.6). The notebook's decomposition finds that only two sectors carry an ε^{-4} pole. In each, four of the variables appear with a power $-1 + O(\varepsilon)$, for example $(t_2 t_3)^\varepsilon (t_1 t_2 t_3 t_4)^{-1-2\varepsilon}$ and each such variable contributes one $1/\varepsilon$. At leading order the rest of the integrand is evaluated at those variables equal to zero. Their leading coefficients add up to

$$\frac{1}{4\varepsilon^4} \quad (354)$$

the leading term in Smirnov (2006), eq. (5.45) (his Example 5.7 and Fig. 5.3). The comment in the course notebook gives this as “(6.22)”, which is not the numbering of the 2006 book. (The full two-loop vertex needs the integration of all sectors to finite order. The notebook's simple implementation gives up on that, which is why dedicated programs such as pySecDec and FIESTA exist.)

MATHEMATICA CHECK: THE NOTEBOOK SECTORDECOMPOSITION, INTEGRATED EXACTLY

Sector decomposition, with each sector integrated exactly.

In [1]: 1. massless triangle: $\mathcal{U} = x_1 + x_2 + x_3$, $\mathcal{F} = -sx_1x_3$
 . the sectors: label, factor, $\mathcal{U}$, $\mathcal{F}$

```
Get[FileNameJoin[{$snipDir, "..", "course_nb", "GFP.m"}]];
Get[FileNameJoin[{$snipDir, "..", "course_nb", "SD.m"}]];
Off[Limit::alimv];
secs = SectorDecompose[1, {x1 + x2 + x3, -1 + 2 eps},
  {-s x1 x3, -1 - eps}, 3, 1];
secs
```

Out[1]: $\begin{pmatrix} 13 & \{t_3^{-\epsilon-1}, t_2 + t_3 + 1, -s\} \\ 31 & \{t_1^{-\epsilon-1}, t_1 + t_2 + 1, -s\} \\ 213 & \{(t_1 t_3)^{-\epsilon-1}, t_1 + t_3 + 1, -s\} \end{pmatrix}$

triangle at $s = -1$ from the sectors

In [2]: tri = IntegrateSectors[secs, {-1 + 2 eps, -1 - eps},
 {t1, t2, t3}, {s -> -1}];
 triSD = Series[-Gamma[1 + eps] Exp[eps EulerGamma] tri, {eps, 0, 0}];
 triSD

Out[2]: $-\frac{1}{\epsilon^2} + \frac{1}{12} \left(-12 \text{Li}_2(-2) - 12 \text{Li}_2\left(-\frac{1}{2}\right) - \pi^2 - 6 \log^2(2) \right) + O(\epsilon^1)$

inversion formula $\text{Li}_2(-z) + \text{Li}_2(-1/z) = -\pi^2/6 - \frac{1}{2} \ln^2 z$ at $z = 2$. check of the inversion formula

In [3]: triExact = Series[-Exp[eps EulerGamma] Gamma[1 + eps] Gamma[-eps]^2/Gamma[1 - 2 eps],
 {eps, 0, 0}];
 inv = PolyLog[2, -2] -> -Pi^2/6 - Log[2]^2/2 - PolyLog[2, -1/2];
 FullSimplify[PolyLog[2, -2] + PolyLog[2, -1/2] + Pi^2/6 + Log[2]^2/2]

Out[3]: 0

minus the exact triangle

In [4]: Simplify[(Normal[triSD] - Normal[triExact]) /. inv]

Out[4]: 0

2. massless box at $s = t = -1$

In [5]: . number of sectors

```
secsB = SectorDecompose[1, {x1 + x2 + x3 + x4, 2 eps},
  {-s x1 x3 - t x2 x4, -2 - eps}, 4, 1];
Length[secsB]
```

Out[5]: 12

box at $s = t = -1$ from the sectors

In [6]:

```
boxSD = Series[Gamma[2 + eps] Exp[eps EulerGamma] IntegrateSectors[secsB,
  {2 eps, -2 - eps}, {t1, t2, t3, t4}, {s -> -1,
  t -> -1}], {eps, 0, 0}];
FullSimplify[boxSD]
```

Out[6]: $\frac{4}{\epsilon^2} - \frac{4\pi^2}{3} + O(\epsilon^1)$

minus $4/\epsilon^2 - 4\pi^2/3$

In [7]: `FullSimplify[Normal[boxSD] - (4/eps^2 - 4 Pi^2/3)]`

Out[7]: 0

3. planar two-loop vertex: the ϵ^{-4} pole from its two sectors

In [8]: . leading pole

```
p4 = Integrate[(t2 t3)^eps (t1 t2 t3 t4)^(-1 - 2 eps),
  {t1, 0, 1}, {t2, 0, 1}, {t3, 0, 1}, {t4, 0, 1},
  GenerateConditions -> False]*
Integrate[(1 + t1 + t1 t3 + t4 + t2 t4 + (1 + t2) (1 + t3) t6)^(-2) /. {t1 -> 0,
  t2 -> 0, t3 -> 0, t4 -> 0}, {t5, 0, 1}, {t6, 0, 1}] +
Integrate[(t3 t6)^eps (t1 t3 t4 t6)^(-1 - 2 eps),
  {t1, 0, 1}, {t3, 0, 1}, {t4, 0, 1}, {t6, 0, 1},
  GenerateConditions -> False]*
Integrate[(1 + t3 + t4 + t2 (1 + t1 + t1 t3 + (1 + t3 + t4) t6))^(-2) /. {t1 -> 0,
  t3 -> 0, t4 -> 0, t6 -> 0}, {t2, 0, 1}, {t5, 0, 1}];
Normal[Series[p4, {eps, 0, -4}]]
```

Out[8]: $\frac{1}{4\epsilon^4}$

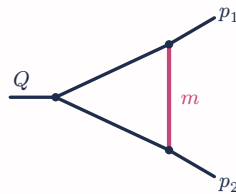
22 Expansion by regions: the Sudakov form factor

Often we do not need an integral exactly, only its expansion in a small ratio, a light mass over a large energy, a small momentum over a large one. The **expansion by regions** (Beneke and Smirnov, 1998) gives such expansions directly, term by term, without ever computing the full integral. The course notebook `EBR_Sudakov` does the classic example.

Eren sees the future in pieces through the Attack Titan, never the whole thing at once. Expansion by regions is the same: it never computes the full integral, only the piece that lives in each momentum region and the pieces add up to the whole.

22.1 The integral

The one-loop vertex with two massless on-shell legs, $p_1^2 = p_2^2 = 0$ and a **massive** line of mass m exchanged between them:



$$F(n_1, n_2, n_3) = \int \frac{d^D \ell}{i\pi^{D/2}} \frac{1}{[(\ell + p_1)^2]^{n_1} [(\ell - p_2)^2]^{n_2} [\ell^2 - m^2]^{n_3}}, \quad Q^2 = -(p_1 + p_2)^2 > 0 \quad (355)$$

This is the building block of the **Sudakov form factor**, the probability that a charged particle scatters off a hard probe (momentum transfer Q) without radiating, in a theory with a massive exchanged boson (think of a W or Z at energies far above its mass). We want the regime $m^2 \ll Q^2$, that is small

$$z = \frac{m^2}{Q^2} \quad (356)$$

For $n_1 = n_2 = n_3 = 1$ the exact answer is known:

$$F(1, 1, 1) = -\frac{1}{Q^2} \left[\frac{1}{2} \ln^2 z - \ln z \ln(1 - z) - \text{Li}_2(z) + \frac{\pi^2}{3} \right] \quad (357)$$

22.2 The idea

Split the loop-momentum space into **regions** according to how the components of ℓ scale with the small parameter. Use light-cone components along p_1 and p_2 , with $\lambda = m/Q$:

- hard: all components of ℓ of order Q
- collinear to p_1 : ℓ nearly parallel to p_1 , with $\ell^2 \sim m^2$
- collinear to p_2 : the same along p_2 .

In each region, Taylor-expand the integrand in the quantities that are small **there** and then integrate each term over **all** of momentum space, in dimensional regularisation. The claim of the method is that the sum over regions reproduces the expansion of the full integral. Other regions (for example soft, all components of order m) give scaleless integrals here and vanish (Section 3.3).

DERIVATION: WHAT EACH REGION DOES TO THE INTEGRAND

Hard region. $\ell^2 \sim Q^2 \gg m^2$, so expand the massive propagator in m^2 (a geometric series, as for the dots of Section 5.1):

$$\frac{1}{(\ell^2 - m^2)^{n_3}} = \sum_{j \geq 0} \frac{\binom{n_3}{j}}{j!} \frac{(m^2)^j}{(\ell^2)^{n_3+j}} \quad (358)$$

The $j = 0$ term is the integral with $m = 0$: the massless on-shell triangle, done with Feynman parameters in Section 14. With the Minkowski sign $(-1)^3$ and in units $Q^2 = 1$,

$$F_{\text{hard}}^{(0)} = -\Gamma(1 + \varepsilon) \frac{\Gamma(-\varepsilon)^2}{\Gamma(1 - 2\varepsilon)} (Q^2)^{-1-\varepsilon} \quad (359)$$

Expand: $\Gamma(-\varepsilon)^2 = 1/\varepsilon^2 + \dots$ and after multiplying by $e^{\varepsilon\gamma_E}$ all γ_E drop out, leaving

$$e^{\varepsilon\gamma_E} Q^2 F_{\text{hard}}^{(0)} = -\frac{1}{\varepsilon^2} + \frac{\pi^2}{12} + O(\varepsilon) \quad (360)$$

The higher terms $j = 1, 2, \dots$ are the same massless triangle with the third line raised to a higher power. They carry $(m^2/Q^2)^j = z^j$.

Collinear to p_1 . Here $(\ell + p_1)^2 \sim m^2$ is small, while $(\ell - p_2)^2 \sim Q^2$ is large. Since $p_2^2 = 0$, $(\ell - p_2)^2 = \ell^2 - 2\ell \cdot p_2$ and here $\ell^2 \sim m^2$ while $\ell \cdot p_2 \sim Q^2$ (a momentum along p_1 has a large product with p_2). So at leading order the propagator becomes linear in ℓ and the region integral is

$$F_{\text{cl}}^{(0)} = \int \frac{d^D \ell}{i\pi^{D/2}} \frac{1}{[(\ell + p_1)^2]^{n_1} [-2\ell \cdot p_2]^{n_2} [\ell^2 - m^2]^{n_3}} \quad (361)$$

a one-scale integral in m^2 (with Q^2 appearing only through $2p_1 \cdot p_2$). Feynman parameters work here too. A linear propagator is combined with the others in the same way and after completing the square the loop integral is again a tadpole (the steps of Section 12). For general powers this gives

$$F_{\text{cl}}^{(0)} = (-1)^{n_1+n_2+n_3} \frac{(m^2)^{2-\varepsilon-n_1-n_3}}{(Q^2)^{n_2}} \frac{\Gamma(n_1 - n_2) \Gamma(2 - n_1 - \varepsilon) \Gamma(n_1 + n_3 - 2 + \varepsilon)}{\Gamma(n_1) \Gamma(n_3) \Gamma(2 - \varepsilon - n_2)} \quad (362)$$

For $n_1 = n_2 = n_3 = 1$ this has $\Gamma(n_1 - n_2) = \Gamma(0)$: infinite and dimensional regularisation cannot help, because ε does not appear in it. This is the rapidity divergence discussed next. With $n_1 = 1 + \delta$, $n_2 = 1 - \delta$ it becomes $\Gamma(2\delta) = 1/(2\delta) - \gamma_E + \dots$, a pole in δ .

Collinear to p_2 . The mirror image, with the roles of lines 1 and 2 exchanged.

There is one subtlety. The collinear integrals have divergences that dimensional regularisation does not regulate (they come from the rapidity of the loop momentum and the D -dimensional measure does not see them). The notebook adds an **analytic regulator**. The powers become $n_1 = 1 + \delta$, $n_2 = 1 - \delta$. Individual regions then have poles in δ and they must cancel in the sum.

RESULT: THE THREE REGIONS AT LEADING POWER

With the normalisation $e^{\varepsilon\gamma_E}$ and the natural scale of each region pulled out,

$$\begin{aligned} \text{hard:} & \quad -\frac{1}{\varepsilon^2} + \frac{\pi^2}{12}, \\ \text{collinear 1:} & \quad \frac{1}{2\varepsilon^2} - \frac{1}{2\delta\varepsilon} - \frac{5\pi^2}{24}, \\ \text{collinear 2:} & \quad \frac{1}{2\varepsilon^2} + \frac{1}{2\delta\varepsilon} - \frac{5\pi^2}{24} \end{aligned} \tag{363}$$

The $1/\delta$ poles cancel between the two collinear regions. The $1/\varepsilon^2$ poles cancel between the hard region and the two collinear ones, **but** they come with different scales ($(Q^2)^{-\varepsilon}$ against $(m^2)^{-\varepsilon}$) and expanding $(m^2/Q^2)^\varepsilon/\varepsilon^2$ leaves logarithms behind. The sum is finite:

$$Q^2 F(1, 1, 1) = -\frac{1}{2} \ln^2 z - \frac{\pi^2}{3} + O(z) \tag{364}$$

This agrees with the exact result. For $z \rightarrow 0$, $\ln(1 - z) \rightarrow 0$ and $\text{Li}_2(z) \rightarrow 0$. The notebook also computes the next two orders, $j = 1, 2$, of every region and the sum agrees with the Taylor expansion of the exact function through z^2 .

MATHEMATICA CHECK: THE NOTEBOOK EBR_SUDAKOV

Expansion by regions for the Sudakov form factor, $z = m^2/Q^2$; δ is the analytic regulator. hard region, leading

In [1]: term

```
Fjh[n1_,n2_,n3_,j_] := m1Pow[n1+n2+n3] Pochhammer[n3,
j]/j!*((-1)^j m2^j)/Q2^(n1+n2+n3+j-2+eps)*
(Gamma[n1+n2+n3+j-2+eps]Gamma[2-n1-n3-j-eps]Gamma[2-n2-n3-j-eps])/
(Gamma[n1]Gamma[n2]Gamma[4-2eps-n1-n2-n3-j])/.{m1Pow[n_] /;
IntegerQ[n]:>(-1)^n};
Fjc1[n1_,n2_,n3_,j_] := m1Pow[n1+n2+n3] Pochhammer[n2,
j]/j!*((-1)^j m2^(2-eps-n1-n3+j))/Q2^(n2+j)*
(Gamma[n1-n2-j]Gamma[2-n1+j-eps]Gamma[n1+n3-j-2+eps])/(Gamma[n1]Gamma[n3]Gamma[2-
eps-n2-j])/.{m1Pow[n_] /;
IntegerQ[n]:>(-1)^n};
Fjc2[n1_,n2_,n3_,j_] := Fjc1[n2,n1,n3,j];
Normal@Series[Q2^(1+eps)*Exp[eps*EulerGamma]*Fjh[1,1,1,0],{eps,0,0}]
```

Out[1]: $\frac{\pi^2}{12} - \frac{1}{\varepsilon^2}$

collinear-1 region, leading term

In [2]: `Normal@Series[Q2^(1-delta)*m2^(eps+delta)*Exp[eps*EulerGamma]*Fjc1[1+delta, 1-delta,1,0],{delta,0,0},{eps,0,0}]`

Out[2]: $-\frac{1}{2\delta\varepsilon} + \frac{1}{2\varepsilon^2} - \frac{5\pi^2}{24}$

collinear-2 region, leading term

```
In [3]: Normal@Series[Q2^(1+del)*m2^(eps-del)*Exp[eps*EulerGamma]*Fjc2[1+del,
1-del,1,0],{del,0,0},{eps,0,0}]
```

```
Out[3]: 1/(2 del ε) + 1/(2 ε^2) - 5 π^2/24
```

sum of the three regions at leading power

```
In [4]: res0 = Normal@Series[Q2*Exp[eps*EulerGamma]*(Fjh[1+del,
1-del,1,0]+Fjc1[1+del,1-del,1,0]+Fjc2[1+del,1-del,1,0]),{del,0,0},
{eps,0,0}];
res1 = Normal@Series[Q2^2/m2*Exp[eps*EulerGamma]*(Fjh[1+del,
1-del,1,1]+Fjc1[1+del,1-del,1,1]+Fjc2[1+del,1-del,1,1]),{del,0,0},
{eps,0,0}];
res2 = Normal@Series[Q2^3/m2^2*Exp[eps*EulerGamma]*(Fjh[1+del,
1-del,1,2]+Fjc1[1+del,1-del,1,2]+Fjc2[1+del,1-del,1,2]),{del,0,0},
{eps,0,0}];
Simplify[res0]
```

```
Out[4]: 1/6 (6 log(m^2) log(Q^2) - 3 log^2(m^2) - 3 log^2(Q^2) - 2 π^2)
```

the exact result, expanded in z

```
In [5]: Ffull = -1/Q2*(1/2 Log[z]^2 - Log[z]*Log[1-z] - PolyLog[2,z] + Pi^2/3);
Normal@Series[Q2 Ffull, {z,0,2}]
```

```
Out[5]: 1/4 z^2 (1 - 2 log(z)) + 1/6 (-3 log^2(z) - 2 π^2) + z (1 - log(z))
```

exact minus regions, through z^2

```
In [6]: Simplify[Normal@Series[Ffull - (res0 + res1*z + res2*z^2 /. {m2->z,
Q2->1})/Q2, {z,0,2}]]
```

```
Out[6]: 0
```

PHYSICAL PICTURE: SUDAKOV DOUBLE LOGARITHMS

The leading term is $-\frac{1}{2} \ln^2(Q^2/m^2)$: a **double** logarithm. At high energy ($Q \gg m$) it becomes large and in the full theory such terms appear at every order in the coupling, as $(\alpha \ln^2)^n$. They must be summed to all orders (Sudakov resummation) and the result exponentiates. The probability of scattering **without** radiating falls off like $\exp(-c \alpha \ln^2(Q^2/m^2))$.

The regions explain where the two logarithms come from. The hard region knows only the scale Q , the collinear regions only the scale m . Each alone has a $1/\varepsilon^2$ and the double logarithm is what is left when these poles cancel between regions with different scales. This is the structure of **effective field theories** such as SCET, where the hard region becomes a matching coefficient and the collinear regions become collinear fields.

23 Doing the reduction with Kira

For one-loop integrals, LiteRed or FIRE inside Mathematica is enough. For real multi-loop problems people use Kira, a fast C++ program. Here is a complete, tested run on the same equal-mass bubble, step by step with pictures, so the workflow is clear.

Not that Kira. This one does not need a Death Note or a shinigami: it only needs three YAML files and instead of names it writes integrals and they do not die, they get reduced.

23.1 What Kira actually does

Kira uses **Laporta's algorithm**. The idea is simple enough to do by hand for the bubble:

1. Choose a family of integrals (which propagators, which momenta).
2. Pick a set of **seed** integrals: all index choices up to some total power.
3. For each seed, write every IBP identity (one per vector, Section 5.3) and the symmetry relations.
4. This gives a big linear system among integrals. Order the integrals from “simple” to “complicated” (fewer lines first, then fewer dots) and solve by Gaussian elimination.
5. Whatever cannot be eliminated are the master integrals. Everything else is expressed through them.

23.1.1 Step 1: the family

Kira works in Minkowski space. The two propagators of the bubble are

$$\begin{array}{c}
 D_1 = \ell^2 - m^2 \\
 \begin{array}{c} \xrightarrow{p} \text{---} \bigcirc \text{---} \xrightarrow{p} \end{array} \\
 D_2 = (\ell - p)^2 - m^2
 \end{array}
 \quad \text{bub}[a_1, a_2] = \int d^D \ell \frac{1}{D_1^{a_1} D_2^{a_2}}. \quad (365)$$

23.1.2 Step 2: sectors

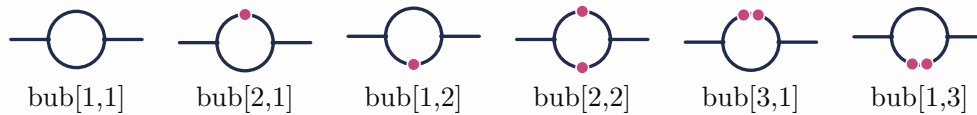
A **sector** says which propagators are present (index > 0). Kira labels it by a binary number, bit 1 for D_1 , bit 2 for D_2 . So there are four sectors:



Sector 0 has no propagator at all, so it is scaleless and vanishes (Section 3.3). Sectors 1 and 2 are the same tadpole (shift $\ell \rightarrow \ell + p$), so Kira finds a symmetry and keeps one of them. The **top sector** is 3, which is what we ask Kira to reduce.

23.1.3 Step 3: seeds

With **r**: 4 Kira takes all integrals in sector 3 whose indices add up to at most 4 and with **s**: 1 it also allows one power of an irreducible numerator (the bubble has none, so this costs nothing). The seeds are therefore



plus the tadpoles they produce.

23.2 Installing

Kira is distributed as a static Linux binary (it runs on any x86-64 Linux, even old ones):

SHELL, ON A LINUX MACHINE

```

mkdir -p ~/local/kira && cd ~/local/kira
# kira-3.1 : static binary from https://kira.hepforge.org (downloads page)
chmod +x kira-3.1 && ./kira-3.1 --version
# Kira version: 3.1 (Git: 3.1)
# Build configuration: release (static); non-MPI; 64 bit weights; with jemalloc;
# with external FireFly with FLINT
# Fermat, the computer algebra engine Kira uses for rational functions:
curl -LO http://home.bway.net/lewis/fermat64/Ferl7.tar.gz && tar xzf Ferl7.tar.gz
export FERMATPATH=$HOME/local/kira/Ferl7/fer64

```

23.3 The three input files

A Kira project is a directory with a `config/` folder and a jobs file. First the integral family. Kira works in Minkowski space. A propagator `["l", "m2"]` means $1/(\ell^2 - m_2)$, where `m2` is the name of a mass-squared symbol.

CONFIG/INTEGRALFAMILIES.YAML

```

integralfamilies:
- name: "bub"
  loop_momenta: [l]
  top_level_sectors: [3]
  propagators:
    - [ "l", "m2" ]
    - [ "l-p", "m2" ]

```

Then the kinematics. The only invariant is $s = p^2$ and we set $m^2 = 1$ to make the problem dimensionless (symbol_to_replace_by_one). The numbers after each invariant are the mass dimensions.

CONFIG/KINEMATICS.YAML

```

kinematics:
  incoming_momenta: [p]
  kinematic_invariants:
    - [s, 2]
    - [m2, 2]
  scalarproduct_rules:
    - [[p,p], s]
  symbol_to_replace_by_one: m2

```

Finally the job. sectors: [3] is the sector where both propagators are present (binary 11 = 3). r is the largest total power of propagators we want to reduce and s the largest power of irreducible numerators (the bubble has none, so s: 1 costs nothing). The integrals we want are listed in the file targets, one per line as bub[a1,a2].

JOBS.YAML

```

jobs:
- reduce_sectors:
  reduce:
    - {topologies: [bub], sectors: [3], r: 4, s: 1}
  select_integrals:
    select_mandatory_list:
      - [bub, targets]
  run_initiate: true
  run_triangular: true
  run_back_substitution: true
- kira2math:
  target:
    - [bub, targets]

```

TARGETS

```

bub[2,1]
bub[1,2]
bub[2,2]
bub[0,2]
bub[3,1]

```

23.3.1 Step 4: run

Run it with kira-3.1 jobs.yaml. On an ordinary Linux workstation it finished in 4.2 seconds and found 2 master integrals, the two we found by hand:

$$\text{masters: } \text{bub}[1,1] = \text{---}\bigcirc\text{---}, \quad \text{bub}[1,0] = \underline{\bigcirc} \quad (366)$$

23.4 The output

The file `results/bub/kira_targets.m` is ready to be read by Mathematica:

RESULTS/BUB/KIRA_TARGETS.M

```
{
bub[2,1] ->
+ bub[1,1]*((-d+3)/(s-4))
+ bub[1,0]*((d-2)/(2*s-8))
,
bub[1,2] ->
+ bub[1,1]*((-d+3)/(s-4))
+ bub[1,0]*((d-2)/(2*s-8))
,
bub[2,2] ->
+ bub[1,1]*(((d^2-9*d+18)*s+4*d-12)/(s^3-8*s^2+16*s))
+ bub[1,0]*(((d-2)*s-2*d^2+10*d-12)/(s^3-8*s^2+16*s))
,
bub[0,2] ->
+ bub[1,0]*((d-2)/2)
,
bub[3,1] ->
+ bub[1,1]*(((d^2-7*d+12)*s-4*d+12)/(2*s^3-16*s^2+32*s))
+ bub[1,0]*(((d^2-6*d+8)*s^2+(-8*d^2+48*d-64)*s+8*d^2-40*d+48)/(8*s^3-64*s^2+128*s))
}
```

23.4.1 Step 5: read the output as diagrams

The first and fourth lines of the output, drawn:

$$\text{---} \bigcirc \text{---} = -\frac{d-3}{s-4} \text{---} \bigcirc \text{---} + \frac{d-2}{2(s-4)} \text{---} \bigcirc \text{---}, \quad \text{---} \bigcirc \text{---} = \frac{d-2}{2} \text{---} \bigcirc \text{---}, \quad (m^2 = 1) \quad (367)$$

and the doubly dotted bubble,

$$\text{---} \bigcirc \text{---} = \frac{(d^2 - 9d + 18)s + 4d - 12}{s(s-4)^2} \text{---} \bigcirc \text{---} + \frac{(d-2)s - 2d^2 + 10d - 12}{s(s-4)^2} \text{---} \bigcirc \text{---} \quad (368)$$

READING THE KIRA OUTPUT IN OUR LANGUAGE

Kira's `bub[1,0]` is the Minkowski tadpole and `bub[1,1]` the Minkowski bubble, with $m^2 = 1$ and $s = p^2$. For example

$$\text{bub}[0,2] = \frac{d-2}{2} \text{bub}[1,0] \quad (369)$$

Compare with our Euclidean Equation 90, $T(2) = -(D-2)T(1)/2$ at $m = 1$. The sign differs by the Wick factor. By Equation 46 the Minkowski integral with n propagators is $i(-1)^n$ times the Euclidean one, so going from $n = 1$ to $n = 2$ flips the sign. In the same way, with $p_E^2 = -s$, our Equation 134 becomes Kira's first line

$$\text{bub}[2,1] = -\frac{d-3}{s-4} \text{bub}[1,1] + \frac{d-2}{2(s-4)} \text{bub}[1,0] \quad (370)$$

Note the denominator $s-4$, which is $s-4m^2$: the threshold appears in the reduction coefficients too.

As an independent test I set up the same Minkowski family in LiteRed and compared all five reductions:

MATHEMATICA CHECK: KIRA AGAINST LITERED

Load LiteRed and solve the Minkowski bubble family $\{\ell^2 - 1, (\ell - p)^2 - 1\}$ (full code in the file). Kira minus

In [1]: LiteRed for $J(2,1), J(1,2), J(2,2), J(0,2), J(3,1)$

```
kira = Get[FileNameJoin[{$snipDir, "..", "kira", "kira_targets.m"}]] /.
bub[a_, b_] := j[bm, a, b];
```

```
lr = # -> IBPReduce[#] & /@ (kira[[All, 1]]);
diff = Simplify[(kira[[All, 2]] - lr[[All, 2]]) /. j[bm,
  0, 1] -> j[bm, 1, 0]];
diff
```

Out[1]: {0, 0, 0, 0, 0}

All five differences are zero.

FeynHelpers also has an interface (KiraCreateConfigFiles, KiraCreateJobFile, KiraRunReduction, KiraImportResults) that writes these files from a FeynCalc topology, runs Kira and reads the result back. For learning, writing the three YAML files by hand as above is clearer.

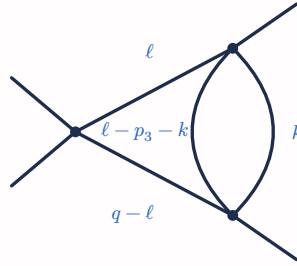
24 Just for fun: a two-loop φ^4 diagram, drawn and solved

To end, one diagram from start to finish, using only what these notes built. Draw it, read off $\mathcal{U}$ and $\mathcal{F}$ from its trees and evaluate it. We stay with massless $\lambda\varphi^4$ theory, where every vertex has exactly four legs.

This is the final exam arc. No new jutsu: only the ones learned so far (trees, forests, Feynman parameters, Cheng-Wu, the bubble), used together.

24.1 The diagram

Two particles p_1, p_2 come in and meet at vertex A . Two lines run from A to B and to C , where the outgoing particles p_3 and p_4 leave and B and C are joined by a pair of lines (a bubble):



Every vertex has four legs. A has p_1, p_2 and lines 1, 2. B has line 1, lines 3, 4 and p_3 . C has line 2, lines 3, 4 and p_4 . It is one of the two-loop corrections to $2 \rightarrow 2$ scattering in $\lambda\varphi^4$.

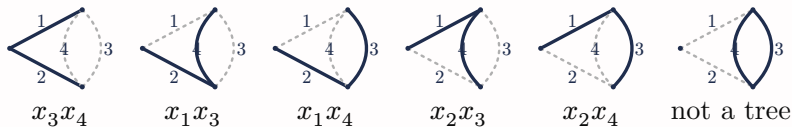
Counting loops. There are $N = 4$ internal lines and $V = 3$ vertices, so $L = N - V + 1 = 2$.

Momenta. With $q = p_1 + p_2$, momentum conservation at each vertex fixes all lines in terms of two loop momenta ℓ and k : line 1 carries ℓ , line 2 carries $q - \ell$ and the two bubble lines carry k and $\ell - p_3 - k$. The outgoing legs are on shell and massless, $p_3^2 = p_4^2 = 0$ and $Q^2 = q^2$ (Euclidean). In Euclidean space

$$\text{Diagram} = \int [d\ell][dk] \frac{1}{\ell^2 (q - \ell)^2 k^2 (\ell - p_3 - k)^2} \quad (371)$$

24.2 Reading $\mathcal{U}$ off the spanning trees

Two loops, so a spanning tree is what is left after removing $L = 2$ lines, provided it still touches all three vertices and has no loop. Of the $\binom{4}{2} = 6$ pairs:



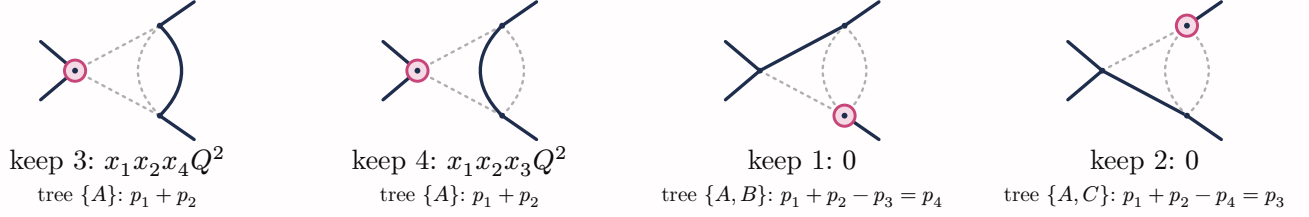
Removing lines 1 and 2 leaves vertex A alone and a loop between B and C : not a tree. The other five work, so

$$\mathcal{U} = x_3x_4 + (x_1 + x_2)(x_3 + x_4) \quad (372)$$

a polynomial of degree $L = 2$ with 5 terms, one per spanning tree.

24.3 Reading $\mathcal{F}$ off the 2-forests

A 2-forest is what is left after removing $L + 1 = 3$ lines, so here exactly one line survives. There are four choices. For each, add up the external momenta entering one of the two trees and square (the pink circle marks the tree made of a single vertex):



When only a bubble line survives, A is alone and its total is $p_1 + p_2 = q$: the term is Q^2 . When line 1 survives, the tree $\{A, B\}$ has $p_1 + p_2$ coming in and p_3 going out, so its total is $p_1 + p_2 - p_3 = p_4$ and $p_4^2 = 0$. The same for line 2, where the total is p_3 . So

$$\mathcal{F} = Q^2 x_1 x_2 (x_3 + x_4) \quad (373)$$

of degree $L + 1 = 3$. The Mathematica box below gets the same two polynomials from the matrix determinant of Section 14.2.

24.4 Evaluating it from $\mathcal{U}$ and $\mathcal{F}$ alone

Now the loop momenta are gone for good. Everything we need is in the two polynomials and the general formula of Section 14.2.

DERIVATION: THE DIAGRAM FROM ITS GRAPH POLYNOMIALS, BY HAND

The parametric integral. With $N = N_\nu = 4$ (four lines, all powers 1), $L = 2$ and $D = 4 - 2\epsilon$ the exponents are

$$N_\nu - L \frac{D}{2} = 4 - D, \quad N_\nu - (L + 1) \frac{D}{2} = 4 - \frac{3D}{2} \quad (374)$$

so

$$\text{Diagram} = \Gamma(4 - D) \int_0^\infty dx_1 dx_2 dx_3 dx_4 \delta(\dots) \mathcal{U}^{4-3D/2} \mathcal{F}^{D-4} \quad (375)$$

Choose the delta function well (Cheng-Wu). x_3 and x_4 appear in $\mathcal{F}$ only as $x_3 + x_4$. So take $\delta(1 - x_3 - x_4)$ and write $x_3 = y$, $x_4 = 1 - y$ with $y \in [0, 1]$, while x_1, x_2 run from 0 to ∞ . Then $x_3 + x_4 = 1$ and

$$\mathcal{U} = y(1 - y) + x_1 + x_2, \quad \mathcal{F} = Q^2 x_1 x_2 \quad (376)$$

Both polynomials are now **linear** in x_1 and in x_2 and $\mathcal{F}$ is a product. That makes every remaining integral elementary:

$$\text{Diagram} = \Gamma(4 - D) (Q^2)^{D-4} \int_0^1 dy \int_0^\infty dx_1 \int_0^\infty dx_2 (x_1 x_2)^{D-4} [y(1 - y) + x_1 + x_2]^{4-3D/2} \quad (377)$$

The one integral we need. For $A > 0$ and $0 < b < c$ (put $x = At$, then $t = u/(1 - u)$ and it becomes a Beta function):

$$\int_0^\infty x^{b-1} (A + x)^{-c} dx = A^{b-c} B(b, c - b) \quad (378)$$

The x_2 integral. Here $A = y(1 - y) + x_1$, $b = D - 3$ and $c = 3D/2 - 4$, so $c - b = D/2 - 1$ and $b - c = 1 - D/2$:

$$\int_0^\infty x_2^{D-4} [y(1 - y) + x_1 + x_2]^{4-3D/2} dx_2 = B\left(D - 3, \frac{D}{2} - 1\right) [y(1 - y) + x_1]^{1-D/2} \quad (379)$$

The x_1 integral. Same formula with $A = y(1 - y)$, $b = D - 3$ and now $c = D/2 - 1$, so $c - b = 2 - D/2$ and $b - c = D/2 - 2$:

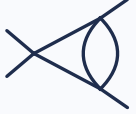
$$\int_0^\infty x_1^{D-4} [y(1 - y) + x_1]^{1-D/2} dx_1 = B\left(D - 3, 2 - \frac{D}{2}\right) [y(1 - y)]^{D/2-2} \quad (380)$$

The y integral is a Beta function directly:

$$\int_0^1 [y(1 - y)]^{D/2-2} dy = B\left(\frac{D}{2} - 1, \frac{D}{2} - 1\right) \quad (381)$$

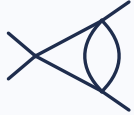
All three steps converge together for $3 < D < 4$ (we need $D - 3 > 0$, $D/2 - 1 > 0$ and $2 - D/2 > 0$). Outside that strip the answer is defined by continuation, as always.

RESULT: THE DIAGRAM AS GAMMA FUNCTIONS



$$= \Gamma(4 - D)(Q^2)^{D-4} B\left(D - 3, \frac{D}{2} - 1\right) B\left(D - 3, 2 - \frac{D}{2}\right) B\left(\frac{D}{2} - 1, \frac{D}{2} - 1\right) \quad (382)$$

and writing each Beta function as $\Gamma\Gamma/\Gamma$, one factor $\Gamma(D/2 - 1)$ cancels:



$$= (Q^2)^{D-4} \frac{\Gamma(4 - D) \Gamma(D - 3)^2 \Gamma(2 - D/2) \Gamma(D/2 - 1)^2}{\Gamma(3D/2 - 4) \Gamma(D - 2)} \quad (383)$$

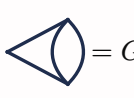
Where the poles come from. At $D = 4 - 2\varepsilon$: $\Gamma(4 - D) = \Gamma(2\varepsilon)$ and $\Gamma(2 - D/2) = \Gamma(\varepsilon)$ each have a simple pole and all other Gamma functions are finite. So the leading pole is $\Gamma(2\varepsilon)\Gamma(\varepsilon) \approx 1/(2\varepsilon^2)$: one pole from the overall two-loop divergence and one from the subdivergence of the bubble between B and C .

24.5 A check: doing the bubble first

The same number can be reached a second way, which is a good check. The two lines between B and C are a massless one-loop bubble carrying $\ell - p_3$ and it equals $G(1, 1)((\ell - p_3)^2)^{D/2-2}$ with

$$G(a, b) = \frac{\Gamma(a + b - D/2) \Gamma(D/2 - a) \Gamma(D/2 - b)}{\Gamma(a) \Gamma(b) \Gamma(D - a - b)} \quad (384)$$

So the bubble acts like a single line from B to C with power $2 - D/2 = \varepsilon$ and what is left is an on-shell triangle with powers $(1, 1, \varepsilon)$:



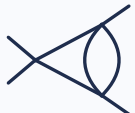
$$= G(1, 1) \times \text{triangle}, \quad \text{powers } (1, 1, \varepsilon) \quad (385)$$

The triangle with any powers a_1, a_2 (lines at A) and a_3 follows from its own $\mathcal{U} = y_1 + y_2 + y_3$, $\mathcal{F} = Q^2 y_1 y_2$ with the same one-integral formula:

$$I(a_1, a_2, a_3) = (Q^2)^{D/2-a} \frac{\Gamma(a - D/2) \Gamma(D/2 - a_1 - a_3) \Gamma(D/2 - a_2 - a_3)}{\Gamma(a_1) \Gamma(a_2) \Gamma(D - a)}, \quad a = a_1 + a_2 + a_3 \quad (386)$$

The product $G(1, 1) I(1, 1, 2 - D/2)$ is identical to the Gamma-function result above (the Mathematica box checks this and also integrates the parametric form directly).

RESULT: ITS EXPANSION NEAR FOUR DIMENSIONS



$$= e^{-2\varepsilon\gamma_E} (Q^2)^{-2\varepsilon} \left[\frac{1}{2\varepsilon^2} + \frac{5}{2\varepsilon} + \frac{114 + \pi^2}{12} + O(\varepsilon) \right] \quad (387)$$

Two loops, two powers of $1/\varepsilon$, the counting of the pole paragraph above.

MATHEMATICA CHECK: THE DIAGRAM FROM U AND F, BY HAND AND BY MACHINE

Two-loop φ^4 diagram: $k_1 = \ell$, $k_2 = q - \ell$, $k_3 = k$, $k_4 = \ell - p_3 - k$, $q = p_1 + p_2$, $p_3^2 = p_4^2 = 0$. external momenta

In [1]: p_3, p_4 with $2p_3 \cdot p_4 = Q^2$. $\mathcal{U}$ from the matrix method

```
UF[c_, q_, m2_, gram_, xs_] := Module[{L = Length[c[[1]]],
  M, Q, J, Minv, dot},
  dot[a_, b_] := a . gram . b;
  M = Table[Sum[xs[[i]] c[[i, ir]] c[[i, jr]], {i, Length[xs]}, {ir, L},
    {jr, L}];
  Q = Table[-Sum[xs[[i]] c[[i, ir]] q[[i]], {i, Length[xs]}, {ir, L}];
  J = Sum[xs[[i]] (dot[q[[i]], q[[i]]] + m2[[i]]), {i, Length[xs]}];
  Minv = Inverse[M];
  {Factor[Det[M]], Factor[Together[Det[M] (J - Sum[Minv[[ir,
    jr]] dot[Q[[ir]], Q[[jr]]], {ir, L}, {jr, L}]]]}];
  gram = {{0, Q2/2}, {Q2/2, 0}};
  cc = {{1, 0}, {-1, 0}, {0, 1}, {1, -1}};
  qq = {{0, 0}, {-1, -1}, {0, 0}, {1, 0}};
  {U, F} = UF[cc, qq, {0, 0, 0, 0}, gram, {x1, x2, x3, x4}];
  U
```

Out[1]: $x_1 x_3 + x_2 x_3 + x_4 x_3 + x_1 x_4 + x_2 x_4$

$\mathcal{F}$ from the matrix method

In [2]: F

Out[2]: $Q^2 x_1 x_2 (x_3 + x_4)$

minus the tree rules $\mathcal{U} = x_3 x_4 + (x_1 + x_2)(x_3 + x_4)$, $\mathcal{F} = Q^2 x_1 x_2 (x_3 + x_4)$

In [3]: Simplify[{U - (x3 x4 + (x1 + x2) (x3 + x4)), F - Q2 x1 x2 (x3 + x4)}]

Out[3]: {0, 0}

route 1: the bubble first, $G(1, 1)((\ell - p_3)^2)^{D/2-2}$, leaving a triangle with powers $(1, 1, \varepsilon)$. route 1

```
G[a_, b_] := Gamma[a + b - d/2] Gamma[d/2 - a] Gamma[d/2 - b]/(Gamma[a] Gamma[b]
  Gamma[d - a - b]);
tri[a1_, a2_, a3_] := With[{a = a1 + a2 + a3},
  Q2^(d/2 - a) Gamma[a - d/2] Gamma[d/2 - a1 - a3] Gamma[d/2 - a2 - a3]/(Gamma[a1]
  Gamma[a2] Gamma[d - a])];
route1 = G[1, 1] tri[1, 1, 2 - d/2];
FullSimplify[route1]
```

Out[4]:
$$\frac{\pi^2 (Q^2)^{d-4} \csc\left(\frac{\pi d}{2}\right) \csc(\pi d) \Gamma\left(\frac{d}{2} - 1\right)}{(d-3) \Gamma\left(\frac{3d}{2} - 4\right)}$$

route 2: $\Gamma(4-D) \int \delta \mathcal{U}^{4-3D/2} \mathcal{F}^{D-4}$ with $\delta(1 - x_3 - x_4)$, $x_3 = y$: $\mathcal{U} = y(1-y) + x_1 + x_2$, $\mathcal{F} = Q^2 x_1 x_2$. route 2

```
inner = Integrate[(y (1 - y) + x1 + x2)^(4 - 3 d/2) (x1 x2)^(d - 4),
  {x1, 0, Infinity}, {x2, 0, Infinity},
  Assumptions -> 0 < y < 1 && 3 < d < 4];
route2 = Gamma[4 - d] Q2^(d - 4) Integrate[inner,
  {y, 0, 1}, Assumptions -> 3 < d < 4];
FullSimplify[route2]
```

Out[5]:
$$\frac{\pi^2 (Q^2)^{d-4} \csc\left(\frac{\pi d}{2}\right) \csc(\pi d) \Gamma\left(\frac{d}{2} - 1\right)}{(d-3) \Gamma\left(\frac{3d}{2} - 4\right)}$$

route 1 minus route 2

In [6]: FullSimplify[route1 - route2, 3 < d < 4]

Out[6]: 0

the hand result: $\Gamma(4-D)(Q^2)^{D-4}B(D-3, D/2-1)B(D-3, 2-D/2)B(D/2-1, D/2-1)$. hand result minus the direct integration

In [7]:

```
hand = Gamma[4 - d] Q2^(d - 4) Beta[d - 3, d/2 - 1] Beta[d - 3,
    2 - d/2] Beta[d/2 - 1, d/2 - 1];
FullSimplify[hand - route2, 3 < d < 4]
```

Out[7]: 0

the one integral used at every step. $\int_0^\infty x^{b-1}(A+x)^{-c} dx$

In [8]:

```
Integrate[x^(b - 1) (A + x)^(-c),
    {x, 0, Infinity}, Assumptions -> A > 0 && 0 < b < c]
```

Out[8]:

$$\frac{\Gamma(b) A^{b-c} \Gamma(c-b)}{\Gamma(c)}$$

expansion at $D = 4 - 2\epsilon$, $Q^2 = 1$, times $e^{2\epsilon\gamma_E}$

In [9]:

```
Collect[Normal@Series[Exp[2 eps EulerGamma] route1 /. {Q2 -> 1,
    d -> 4 - 2 eps}, {eps, 0, 0}], eps, FullSimplify]
```

Out[9]:

$$\frac{1}{2\epsilon^2} + \frac{5}{2\epsilon} + \frac{1}{12}(114 + \pi^2)$$

25 From a loop to a decay rate: the neutral pion

Everything so far has been about how to compute a loop. Now we use one to predict a number that can be measured in a laboratory, the rate at which the neutral pion π^0 decays into two photons. The pion has no electric charge, so a photon cannot attach to it directly. The decay happens only through a loop of charged particles (the quarks). The loop is a triangle. We need three things we already have, a Dirac trace (done by FORM), Feynman parameters (Section 11) and one Feynman-parameter integral.

Zoro fights with three swords. The pion decays with three colours of quark. Take two of them away and, as we will see, the decay rate drops by a factor of nine.

25.1 The model

Take the two light quarks u and d , each in N_c colours. They talk to the photon through the usual QED vertex with charge Qe ($Q_u = 2/3$, $Q_d = -1/3$). They talk to the pion through a pseudoscalar vertex,

$$\mathcal{L}_{\text{int}} = -g \bar{q} i\gamma_5 \tau_3 q \pi^0 - e Q_q \bar{q} \gamma^\mu q A_\mu \quad (388)$$

where $\tau_3 = +1$ for u and -1 for d . The γ_5 is there because the pion is a pseudoscalar. It changes sign under parity. $\bar{q} i\gamma_5 q$ is the simplest quark bilinear that does the same.

PHYSICAL PICTURE: WHERE THE COUPLING COMES FROM

In the quark version of the linear sigma model the quarks get their mass from the vacuum. A field σ has a vacuum value f_π . The term $g \bar{q} q \sigma$ then looks like a mass term with $m = g f_\pi$. The pion is the partner of σ in the same term, so it couples with the same g . Hence

$$g = \frac{m}{f_\pi} \quad (389)$$

with $f_\pi \approx 92$ MeV, the constant that also controls the decay $\pi^+ \rightarrow \mu^+ \nu$. This is the Goldberger-Treiman relation at the quark level. A heavier quark couples more strongly. We will see that this is exactly what makes the answer independent of the quark mass.

There are two diagrams for each quark, one with the photons in each order:

$$(390)$$

The photons have momenta k_1, k_2 with $k_1^2 = k_2^2 = 0$. The pion has $(k_1 + k_2)^2 = m_\pi^2$, so $2k_1 \cdot k_2 = m_\pi^2$.

25.2 Step 1: the trace

First fix the momenta in the first diagram. The quark runs round the triangle from the pion vertex P to the photon vertex A, then to the photon vertex B and back to P. Call the momentum on the line A-B ℓ . At A the photon k_1 leaves, so the line P-A carries $\ell + k_1$. At B the photon k_2 leaves, so the line B-P carries $\ell - k_2$. At P the pion brings in $(\ell + k_1) - (\ell - k_2) = k_1 + k_2$, as it must. Now write the Dirac matrices against the direction of the quark, starting at the pion vertex:

$$\text{Tr}[\gamma_5(\ell - \not{k}_2 + m)\gamma^\nu(\ell + m)\gamma^\mu(\ell + \not{k}_1 + m)] \quad (391)$$

Before computing it, count. A trace with γ_5 is non-zero only if it has an even number of other gamma matrices, at least four. Besides γ_5 there are γ^μ, γ^ν and one slash or one m from each propagator. With no m that is five matrices (odd, zero). With two m 's it is three (odd, zero). With three m 's it is two (too few, zero). Only the terms with exactly **one** mass survive. They have four matrices, which gives the Levi-Civita tensor $\varepsilon^{\mu\nu\rho\sigma}$.

Now put in the Feynman shift that we will need anyway (Step 2), $\ell = q - xk_1 + yk_2$. FORM does the rest (the box below). The answer is surprisingly clean:

$$\text{Tr}[\dots] = 4im\varepsilon^{\mu\nu\rho\sigma}k_{1\rho}k_{2\sigma} \quad (392)$$

Nothing depends on q, x or y . This is true before any integration. The qq terms die because ε is antisymmetric in any two of its slots. The terms linear in q come from three places (the m can sit in any of the three propagators) and they cancel among themselves. FORM finds the same $4m$ term even without the shift, with the unshifted ℓ in the trace. So the whole numerator is one constant and the loop integral that is left is the **scalar** triangle.

SYMMETRIC INTEGRATION AND WHY WE DO NOT NEED IT HERE

Usually a numerator does depend on the loop momentum. After the Feynman shift the denominator depends only on q^2 . Then two rules take care of the numerator (Peskin and Schroeder, eqs. 6.45 and 6.46):

$$\int \frac{d^4q}{(2\pi)^4} \frac{q^\mu}{(q^2 - \Delta)^3} = 0, \quad \int \frac{d^4q}{(2\pi)^4} \frac{q^\mu q^\nu}{(q^2 - \Delta)^3} = \int \frac{d^4q}{(2\pi)^4} \frac{\frac{1}{4}g^{\mu\nu}q^2}{(q^2 - \Delta)^3} \quad (393)$$

The first holds because the integrand is odd under $q \rightarrow -q$. For the second, the answer can only be proportional to $g^{\mu\nu}$, since nothing else carries two indices. Contract both sides with $g_{\mu\nu}$ to fix the constant. In D dimensions the $1/4$ becomes $1/D$. These rules are what one needs for the vacuum polarisation or for the magnetic moment of the electron. For the pion both are unnecessary, because the trace has no loop momentum left in it at all. In the rest of these notes they are not needed either, because a scalar product with the loop momentum is always rewritten as a combination of propagators (Section 5.3) and only scalar integrals appear.

THE SAME TRACE IN FEYNSAGE

The package sends it to FORM in one line (dim = 4 switches on trace4, which knows γ_5):

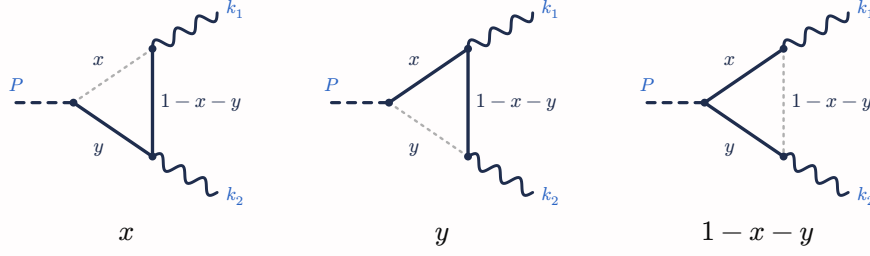
```
form.dirac_trace(['g5', 'q - x*k1 - (1-y)*k2 + m', 'nu', 'q - x*k1 + y*k2 + m', 'mu',
                 'q + (1-x)*k1 + y*k2 + m'], vectors=['q', 'k1', 'k2'])
# -4*m*eps(k1, k2, mu, nu)      (FORM's e_ is -i times the usual epsilon)
```

25.3 Step 2: the triangle integral

Give the Feynman parameter x to the line P-A ($\ell + k_1$), y to the line B-P ($\ell - k_2$) and $1 - x - y$ to the line A-B (ℓ). With Feynman parameters (Section 11), $1/(D_1 D_2 D_3) = 2 \int dx dy 1/(q^2 - \Delta)^3$ over the triangle $x, y \geq 0$, $x + y \leq 1$. The shift above removes the linear terms.

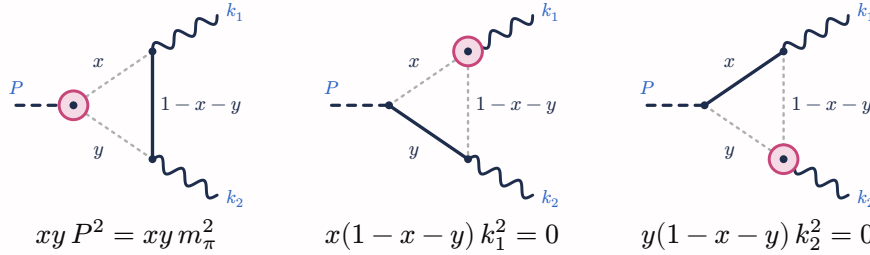
Δ straight from the graph. We do not have to complete the square. The rules of Section 14 give $\mathcal{U}$ and $\mathcal{F}$ by looking at the triangle. Drop the arrows and the γ matrices and keep only the lines with their Feynman parameters: x on P-A, y on B-P and $1-x-y$ on A-B. The dotted line is a removed line.

- Spanning trees: remove any one line. There are three and each gives the parameter of the removed line:



So $\mathcal{U} = x + y + (1 - x - y) = 1$ on the simplex.

- 2-forests: remove two lines, which cuts off one vertex (the pink circle). The term is the product of the two removed parameters times the square of the momentum that flows into the small tree:



Cutting off P leaves the pion momentum in that tree, $P^2 = m_\pi^2$. Cutting off A or B leaves a photon momentum, $k_1^2 = k_2^2 = 0$, so those two forests do not count.

- So $\mathcal{F}_0 = xy m_\pi^2$ and, with the Minkowski sign and the mass term $\mathcal{U} \sum_i x_i m^2 = m^2$,

$$\Delta = \frac{\mathcal{F}}{\mathcal{U}} = m^2 - xy m_\pi^2 \quad (394)$$

Only the two lines that meet at the pion vertex appear in the m_π^2 term, because only the 2-forest that isolates that vertex carries the pion momentum. The Mathematica box at the end of this section finds the same $\mathcal{U}$ and $\mathcal{F}$ with the *UF* routine from the notebook of the lecture, which completes the square by loop. The q integral is the Minkowski version of the tadpole of Section 3,

$$\int \frac{d^4 q}{(2\pi)^4} \frac{1}{(q^2 - \Delta)^3} = \frac{-i}{32\pi^2 \Delta} \quad (395)$$

so

$$\int \frac{d^4 \ell}{(2\pi)^4} \frac{1}{D_1 D_2 D_3} = \frac{-i}{16\pi^2 m^2} I(r), \quad I(r) = \int_0^1 dx \int_0^{1-x} dy \frac{1}{1 - xy r}, \quad r = \frac{m_\pi^2}{m^2} \quad (396)$$

The Mathematica box below does this integral exactly:

$$I(r) = \frac{2}{r} \arcsin^2\left(\frac{\sqrt{r}}{2}\right) = \frac{1}{2} + \frac{r}{24} + \frac{r^2}{180} + \dots \quad (397)$$

For a heavy quark ($r \rightarrow 0$) it is just the area of the triangle, $1/2$.

25.4 Step 3: the amplitude and why the quark mass drops out

Collect the factors of one diagram: N_c from the colours, $(eQ)^2$ from the photon vertices, g from the pion vertex, $4m$ from the trace and $I/(16\pi^2 m^2)$ from the integral. The crossed diagram gives the same (swapping $k_1 \leftrightarrow k_2$ and $\mu \leftrightarrow \nu$ leaves $\varepsilon^{\mu\nu\rho\sigma} k_{1\rho} k_{2\sigma}$ unchanged), so we double it. The factors of i and the sign of the fermion loop only give an overall phase, which drops out of the rate. Writing the amplitude as $\mathcal{M} = A \varepsilon^{\mu\nu\rho\sigma} \varepsilon_{1\mu}^* \varepsilon_{2\nu}^* k_{1\rho} k_{2\sigma}$, each quark gives

$$A_q = \tau_3 N_c \frac{e^2 Q_q^2 g}{4\pi^2 m} 2I(r) \quad (398)$$

Now put in $g = m/f_\pi$. The m in g cancels the $1/m$ of the loop:

$$A_q = \tau_3 N_c \frac{e^2 Q_q^2}{4\pi^2 f_\pi} 2I(r) \xrightarrow{m \gg m_\pi} \tau_3 N_c \frac{e^2 Q_q^2}{4\pi^2 f_\pi} \quad (399)$$

A heavier quark couples more strongly and propagates less. For $m \gg m_\pi$ the two effects cancel exactly. This is a first glimpse of the **axial anomaly** (Adler 1969). In this limit the answer does not care about the details of the particles in the loop, only about their charges and their number. Adding u and d with $e^2 = 4\pi\alpha$:

$$A = N_c (Q_u^2 - Q_d^2) \frac{e^2}{4\pi^2 f_\pi} = N_c \left(\frac{4}{9} - \frac{1}{9} \right) \frac{\alpha}{\pi f_\pi} = \frac{\alpha}{\pi f_\pi} \quad (N_c = 3) \quad (400)$$

25.5 Step 4: the decay rate

Square the amplitude and sum over the photon polarisations ($\sum \varepsilon_\mu \varepsilon_{\mu'}^* \rightarrow -g_{\mu\mu'}$). The Levi-Civita contraction gives (Mathematica box)

$$\varepsilon^{\mu\nu\rho\sigma} \varepsilon_{\mu\nu\alpha\beta} k_{1\rho} k_{2\sigma} k_1^\alpha k_2^\beta = 2(k_1 \cdot k_2)^2 = \frac{m_\pi^4}{2} \quad (401)$$

For a decay into two massless particles the phase space is $\int d\Phi_2 = 1/(8\pi)$. The two photons are identical, which gives another $1/2$. So

$$\Gamma = \frac{1}{2m_\pi} \cdot \frac{1}{2} \cdot \frac{1}{8\pi} \cdot A^2 \frac{m_\pi^4}{2} = \frac{A^2 m_\pi^3}{64\pi} = \frac{N_c^2 \alpha^2 m_\pi^3}{576\pi^3 f_\pi^2} \quad (402)$$

RESULT: THE DECAY RATE OF THE NEUTRAL PION

With $\alpha = 1/137.036$, $m_{\pi^0} = 134.977$ MeV and $f_\pi = 130.2/\sqrt{2} = 92.07$ MeV,

this triangle	
$\Gamma(\pi^0 \rightarrow \gamma\gamma), N_c = 3$	7.79 eV
the same with $N_c = 1$	0.87 eV

One triangle of quarks, computed with the tools of these notes, gives the rate. With one colour it would be nine times smaller.

WHAT THE NUMBER IS TELLING US

The factor N_c^2 is the reason this decay is famous. With one colour the rate would be nine times smaller. Adler already saw this mismatch in 1969. In his paper the amplitude is proportional to a number $S = \sum_j g_j Q_j^2$, summed over the fermions in the triangle. From the measured lifetime he found $|S| = 0.44$, while the quark model with one colour gives $S = \frac{1}{2}(2/3)^2 - \frac{1}{2}(1/3)^2 = 1/6$, about three times too small. Three colours give $S = 1/2$.

The model has a real quark mass in it, and the prediction depends on it only weakly, through the factor $2I(r)$ that goes to 1 for $m \gg m_\pi$. With a constituent mass $m = 330$ MeV the factor $2I(r)$ is 1.014, so the rate moves by less than three per cent. The exact statement is the anomaly. For massless u and d quarks the amplitude is $\alpha/(\pi f_\pi)$ in **any** theory with these charges and three colours.

MATHEMATICA CHECK: THE PION TRIANGLE: TRACE, INTEGRAL, RATE AND NUMBERS

The numerator of the triangle for $\pi^0 \rightarrow \gamma\gamma$: $\text{Tr}[\gamma_5(\not{\ell} - \not{k}_2 + m)\gamma^\nu(\not{\ell} + m)\gamma^\mu(\not{\ell} + \not{k}_1 + m)]$ with the Feynman shift

In [1]: $\ell \rightarrow q - xk_1 + yk_2$ and on-shell photons. trace

```
FCClearScalarProducts[];
SP[k1, k1] = 0;
SP[k2, k2] = 0;
num = DiracTrace[GA[5] . (GS[q] - x GS[k1] - (1 - y) GS[k2] + m) . GA[nu] . (GS[q] - x
GS[k1] + y GS[k2] + m) . GA[mu] . (GS[q] + (1 - x) GS[k1] + y GS[k2] + m),
DiracTraceEvaluate -> True];
Simplify[DiracSimplify[num]]
```

Out[1]: $4 i m \bar{\epsilon}^{\mu \nu \kappa \Gamma \bar{\kappa} 2}$

The same trace without the shift, with the loop momentum ℓ itself: the ℓ -dependent terms already cancel.

In [2]: unshifted trace

```
num0 = DiracTrace[GA[5] . (GS[l] - GS[k2] + m) . GA[nu] . (GS[l] + m) . GA[mu] . (GS[l]
+ GS[k1] + m),
  DiracTraceEvaluate -> True];
Simplify[DiracSimplify[num0]]
```

Out[2]: $4 i m \bar{\epsilon}^{\mu \nu \kappa \Gamma \bar{\kappa} 2}$

$\mathcal{U}$ and $\mathcal{F}$ of the triangle with the UF routine of the lecture notebook (it completes the square in ℓ). Lines: x_1 for

In [3]: $\ell + k_1$, x_2 for ℓ , x_3 for $\ell - k_2$. U, F, number of loops

```
UF[xx_, yy_, z_] := Module[{degree, coeff, i, t2, t1, t0,
  zz}, zz = Map[Rationalize[#, 0] &, z, {0, Infinity}];
degree = -Sum[yy[[i]] x[i], {i, 1, Length[yy]}]; coeff = 1;
For[i = 1, i <= Length[xx], i++, t2 = Coefficient[degree,
  xx[[i]], 2]; t1 = Coefficient[degree, xx[[i]], 1];
  t0 = Coefficient[degree, xx[[i]], 0]; coeff = coeff t2;
  degree = Together[t0 - t1^2/(4 t2)]];
degree = Together[-coeff degree] /. zz;
coeff = Together[coeff] /. zz; {coeff, Expand[degree],
  Length[xx]};
uf = UF[{l}, {-(l + k1)^2 - m^2), -(l^2 - m^2), -((l - k2)^2 - m^2)},
  {k1^2 -> 0, k2^2 -> 0, k1 k2 -> mpi^2/2}];
uf
```

Out[3]: $\{x_1 + x_2 + x_3, m^2 x_1^2 + m^2 x_2^2 + m^2 x_3^2 + 2 m^2 x_1 x_2 + 2 m^2 x_1 x_3 + 2 m^2 x_2 x_3 - \text{mpi}^2 x_1 x_3, 1\}$

$\mathcal{F}/\mathcal{U}$ on the simplex $x_2 = 1 - x_1 - x_3$

In [4]: `Simplify[uf[[2]]/uf[[1]] /. x[2] -> 1 - x[1] - x[3]]`

Out[4]: $m^2 - \text{mpi}^2 x_1 x_3$

The Feynman-parameter integral over the triangle $0 < y < 1 - x$, with $r = m_\pi^2/m^2$ below the threshold $r = 4$.

In [5]: closed form of the integral

```
tri = Integrate[1/(1 - x y r), {x, 0, 1}, {y, 0, 1 - x},
  Assumptions -> 0 < r < 4];
tri
```

Out[5]:
$$\frac{\text{Li}_2\left(\frac{1}{2}(r - i\sqrt{4-r})r\right) + \text{Li}_2\left(\frac{1}{2}(r + i\sqrt{4-r})r\right)}{r}$$

Mathematica writes it with dilogarithms. Check that it equals $(2/r) \arcsin^2(\sqrt{r}/2)$: the power series in r agree term by term. series of the difference up to r^{12}

In [6]: `Simplify[Normal[Series[tri - 2/r ArcSin[Sqrt[r]/2]^2, {r, 0, 12}]]]`

Out[6]: 0

the arcsin form, expanded

In [7]: `Normal[Series[2/r ArcSin[Sqrt[r]/2]^2, {r, 0, 3}]]`

Out[7]:
$$\frac{r^3}{1120} + \frac{r^2}{180} + \frac{r}{24} + \frac{1}{2}$$

heavy quark limit $r \rightarrow 0$

In [8]: `Limit[tri, r -> 0]`

Out[8]: $\frac{1}{2}$

Sum over photon polarisations of $|\varepsilon^{\mu\nu\rho\sigma} k_{1\rho} k_{2\sigma}|^2$. polarisation sum

```
In [9]: Contract[LC[mu, nu][k1, k2] LC[mu, nu][k1, k2]]
```

```
Out[9]: 2 (k1.k2)^2
```

Amplitude $A = \sum_{q=u,d} \tau_3 N_c Q_q^2 e^2 / (4\pi^2 f_\pi)$ and the rate $\Gamma = A^2 m_\pi^3 / (64\pi)$. rate

```
In [10]: amp = Nc (Qu^2 - Qd^2) e^2/(4 Pi^2 f) /. {Qu -> 2/3,
           Qd -> -1/3, e -> Sqrt[4 Pi alpha]};
rate = Simplify[amp^2 mpi^3/(64 Pi)];
rate
```

```
Out[10]: alpha^2 mpi^3 Nc^2
          576 pi^3 f^2
```

Numbers: alpha, m_pi0 = 134.9768 MeV, f_pi = 130.2/sqrt 2 MeV, $\Gamma(\pi^0 \rightarrow \gamma\gamma)$ in eV, $N_c = 3$

```
In [11]: vals = {alpha -> 1/137.035999, mpi -> 134.9768, f -> 130.2/Sqrt[2]};
gam = 10^6 rate /. vals /. Nc -> 3;
gam
```

```
Out[11]: 7.7855
```

Γ in eV if $N_c = 1$

```
In [12]: 10^6 rate /. vals /. Nc -> 1
```

```
Out[12]: 0.865055
```

With a finite constituent quark mass $m = 330$ MeV the factor $2I(r)$ is slightly above 1. $2I(r)$ at $m = 330$ MeV

```
In [13]: Chop[N[2 tri /. r -> (134.9768/330)^2]]
```

```
Out[13]: 1.01426
```

The first cells are the trace, with the shifted and the unshifted loop momentum. Then the UF routine of the lecture notebook gives $\mathcal{U}$ and $\mathcal{F}$ of the triangle. It has only the $m\varepsilon$ term, as the counting said. The next cells compute $I(r)$ exactly. Mathematica writes it with dilogarithms. The series check shows that it equals $(2/r) \arcsin^2(\sqrt{r}/2)$. Its heavy-quark limit is $1/2$. Then come the polarisation sum, the rate as a formula and the numbers.

26 Summary and a guide to the literature

Here is the whole story in a few paragraphs, with the steps to remember highlighted.

A loop means an integral over an unfixed momentum. We first **Wick rotate to Euclidean space**, where the integrand is positive and spherical coordinates work and then **continue the dimension D away from 4**. The tadpole becomes $\pi^{D/2} \Gamma(n - D/2) / \Gamma(n) (m^2)^{D/2-n}$: **ultraviolet divergences turn into poles of the Gamma function** and when a pole meets a^ε it leaves behind $\ln a$. That is where every logarithm of a mass or momentum comes from.

A dot on a line raises the power of a propagator. Dots come from mass derivatives, from zero-momentum insertions and from integration by parts. **IBP says that the integral of a total derivative vanishes**, because in dimensional regularisation there is no boundary at infinity. It reduces every integral of a family to a few **master integrals**: one for the tadpole, two for the bubble, two for the two-loop propagator family. Laporta's algorithm organises this and Kira, FIRE and LiteRed automate it.

To compute the masters we met several methods. **Differentiating a master and reducing the result gives a differential equation** for it. For the bubble its solution is a hypergeometric function and **the boundary condition comes from regularity at a point where the integral cannot be singular**. **Feynman parameters** give the same answer directly and for any graph the parametric integrand is built from $\mathcal{U}$ (spanning trees) and $\mathcal{F}$ (2-forests,

with the momentum entering one tree squared). Mellin-Barnes turns a power of a sum into a product and the massless box followed from one contour integral. In a canonical basis the differential equation becomes $d\vec{g} = \varepsilon dA \vec{g}$ and is solved order by order in polylogarithms. Sector decomposition works for any integral by splitting the parameter space until the singularities factorise and expansion by regions gives small-mass expansions region by region.

Finally, going back to Minkowski space, the $+i\epsilon$ of the propagators decides the side of the logarithm's cut and the bubble gets an imaginary part $\pi\beta$ above $s = 4m^2$: what the optical theorem requires, because the imaginary part is the probability of producing two real particles.

And the tools give real numbers. One quark triangle, a FORM trace and one Feynman-parameter integral predict the decay rate of the neutral pion into two photons, $\Gamma = 7.79$ eV. The quark mass cancels out of the answer and the rate grows as the square of the number of colours, so a measurement of the rate counts the colours. Adler's 1969 comparison with the measured lifetime already pointed to three.

READING GUIDE

- *T. Rauh, Introduction to Feynman integrals and multiloop techniques* (lecture notes, University of Bern, 2019). Short, modern, with exercises. Section 12 of these notes follows his Section 2.2 and his two-loop kite. Sections 2 (dimensional regularisation, Feynman parameters), 3 (sector decomposition), 4 (Mellin-Barnes), 5 (IBP), 6 (differential equations) and 7 (expansion by regions) match these notes.
- *M. Argeri and P. Mastrolia, Feynman diagrams and differential equations*, Int. J. Mod. Phys. A 22 (2007) 4375. The source of the difference equation (Section 5, in the measure $d^D\ell/\pi^{D/2}$) and of the bubble DE (Section 6, in the measure $d^D\ell/(2\pi)^{D-2}$). Section 3 has the counting of integrals and identities.
- *P. Ramond, Field Theory: A Modern Primer* (2nd ed., revised printing). Sections 4.3 and 4.4 (the $\lambda\varphi^4$ tadpole and fish in $D = 2\omega$) and Appendix B (the Euclidean formulas).
- *M. E. Peskin and D. V. Schroeder, An Introduction to Quantum Field Theory*. Section 6.3 (Feynman parameters and Wick rotation), Section 7.3 (optical theorem), Section 7.5 (dimensional regularisation), Section 10.2 (renormalisation of $\lambda\varphi^4$), Section 12.2 (beta function).
- *Quantum Field Theory: Lectures of Sidney Coleman* (ed. B. G. Chen et al., World Scientific 2019), Chapter 10. A gentle treatment of mass renormalisation and of the logic behind counterterms (the story told in Section 6).
- *A. Das, Field Theory: A Path Integral Approach* (3rd ed., World Scientific 2019). The path-integral route to the Feynman rules (Chapter 9), the effective action and the loop expansion (Chapter 10) and the anomalous Ward identity behind Section 25 (Chapter 13).
- *V. A. Smirnov, Feynman Integral Calculus* (Springer, Berlin 2006). The book of the lecture. Chapter 2 (divergences, alpha representation), Chapter 3 (alpha and Feynman parameters, graph polynomials, Fig. 3.6 and Example 3.5), Chapter 4 (Mellin-Barnes), Chapter 5 (IBP), Chapter 7 (differential equations) and Appendix E (sector decompositions). Every Smirnov equation and figure cited in these notes is from this book. Its later version, *Analytic Tools for Feynman Integrals* (Springer 2012), is the one Rauh follows.
- *S. Weinzierl, Feynman Integrals* (Springer 2022, arXiv:2201.03593), Section 2.5 and Chapter 3 for all parametric representations and the graph polynomials (spanning trees, forests, the matrix-tree theorem). Also for everything beyond one loop, including MB and canonical differential equations.
- *S. L. Adler, Axial-vector vertex in spinor electrodynamics*, Phys. Rev. 177 (1969) 2426. The axial anomaly behind Section 25 and the quark-model value $S = 1/6$ quoted there (Appendix, part B, p. 2438).
- *A. V. Smirnov and A. V. Petukhov, The number of master integrals is finite*, Lett. Math. Phys. 97 (2011) 37 (arXiv:1004.4199). *R. N. Lee and A. A. Pomeransky, Critical points and number of master integrals*, JHEP 11 (2013) 165 (arXiv:1308.6676). *A. von Manteuffel, E. Panzer and R. M. Schabinger, A quasi-finite basis for multi-loop Feynman integrals*, JHEP 02 (2015) 120 (arXiv:1411.7392). *T. Bitoun, C. Bogner, R. P. Klausen and E. Panzer, Feynman integral relations from parametric annihilators*, Lett. Math. Phys. 109 (2019) 497 (arXiv:1712.09215). The results behind Section 19.

- Software papers: V. Shtabovenko, R. Mertig and F. Orellana, *FeynCalc 10*, Comput. Phys. Commun. 306 (2025) 109357 (arXiv:2312.14089). H. H. Patel, *Package-X*, Comput. Phys. Commun. 197 (2015) 276 (arXiv:1503.01469). A. V. Smirnov and M. Zeng, *FIRE 6.5*, Comput. Phys. Commun. 302 (2024) 109261 (arXiv:2311.02370). R. N. Lee, *LiteRed 1.4*, J. Phys. Conf. Ser. 523 (2014) 012059 (arXiv:1310.1145). F. Lange, J. Usovitsch and Z. Wu, *Kira 3*, Comput. Phys. Commun. 322 (2026) 109999 (arXiv:2505.20197).

Plus Ultra! If you made it this far, you have done the one-loop tadpole and bubble in every way the lecture promised. Part 2 already took us to two loops with the kite and from here on the same tools (IBP, differential equations, graph polynomials, Mellin-Barnes) do all the heavy lifting.